Supplementary Evidence: Reusing Defense Evidence in LLM and Agent Side Channels

This supplement preserves the sources, accounting and illustrative computations supporting the mitigation evidence atlas within the C1–C5 empirical attack-evaluation frame. S44 supersedes the earlier measured-utility R8 denominator with outcome-specific applicability; earlier numbered tables remain explicitly archived results. S38 retains the historical 20-family inventory; S39–S41 add the forward-citation pilot, object-join examples and the current 19-family privacy/attribution inventory. These are counts within the selected source frame, not a census of defense literature. Pair counts retain their role as a comparison index; external and internal cases illustrate use.

Artifact and data scope. The research artifact distinguishes source references and locators, derived audit labels, experiment protocols, aggregate results and reconstruction scripts from third-party source documents and model traces. Availability of individual inputs and traces depends on data-holder rights, applicable API terms and privacy review. The paper-source archive alone is not the complete experimental artifact. Stored trajectories support replay without new model requests, while mutable model aliases and differing inference budgets limit fresh-run reproducibility.

Reading the reporting inventory. The strict family-level separation of inference and cost is five of eight under the present record boundaries; accepting partial bindings gives three of twelve. These are projections of source-coding and record-splitting rules, not independent findings about the prevalence of reporting defects. Task or component measurements are distinct from structural claims of unchanged outputs or state: an unbound R8 task/component result does not by itself show a loss of function. The retrospective external application uses visible outcomes and illustrates evidence routing; it does not estimate the procedure's accuracy.

Historical Exp 1–4 names and E-keys identify archived case-study components, not four main-text contributions. Emission maps to E1/E3/E5; state/link to E2/E6/E7; agent trajectories to E9/E10/E12 and the model/calibration/planner extensions; timing controls to E13/E14 and the WAN extension. Existing S1–S31 numbers remain stable. S31 retains detailed grids, and S32 collects the cohort, served-model and calibration tables removed from the main text. That reorganization added no experiments or customers; S35 now adds an external documentary check and a provenance audit of the existing seven judgments.

Shared evidence notation. B means bound, P partial, U unreported in inspected scope, A source unavailable, ? an unverified binding, and – inapplicable. Archived machine fields NR-R/NR-A/NR-V map respectively to U/A/?. Source availability and applicability retain distinct meanings. S1–S10 contain search, coding and literature audit details; S11–S17 contain transport/agent mechanisms and initial analyses; S18–S23 contain source reconciliation, timing controls, stronger attacks, the model extension and sensitivity checks.

S1. Search and Evidence Audit

TABLE S1. HISTORICAL ROUTE ACCOUNTING FOR THE ORIGINAL OCTOBER 8, 2026 SNAPSHOT, INCLUDING ITS LATER SOURCE CHECKS. COUNTS OVERLAP AND DO NOT FORM A COMPLETE PRISMA FLOW; THE OCTOBER 21 SUPPLEMENTARY ROUND IS REPORTED SEPARATELY.

Route / disposition	Count
Broad IEEE Q4 hits; not fully screened	35,743
Amended Q4b export; recall not verified	847
IEEE saved export occurrences / distinct records	2,079 / 2,001
IEEE full-text referrals / background routes	25 / 205
ACM saved occurrences / publication groups	16,687 / 13,201
ACM Oct. 14 provisional exclusions / full-text referrals / unresolved	9,368 / 883 / 2,950
ACM dispute groups / combined check candidates (Oct. 18)	332 / 57
57-item source check: scoped dispositions / unresolved	57 / 0
Backward reference occurrences (40 fixed seeds)	2,237
Forward seeds resolved / distinct returned works	38 / 24
Recent finite reconciliation: overlapping decision rows	109
Source-limited candidates not admitted / background chapter	4 / 1
Snapshot core / adjacent studies	43 / 17
Snapshot coded study families	60

The 57 scoped dispositions are not completed full readings. B069's complete official abstract supports background routing; later-page eligibility remains unchecked. The 109 rows overlap; 883/2,950 are dated global routes. Q4b does not replace unscreened Q4. This snapshot's coded-set cutoff is October 8, 2026; its counts do not describe the later expanded collection.

Targeted discovery and seed-anchored snowballing. Our acquisition combined venue-targeted proceedings screening (USENIX Security, NDSS, IEEE S&P, CCS, PMLR), targeted arXiv discovery, and one-hop backward (2,237 reference occurrences) and forward (24 distinct works across 38/40 resolved seeds) citation tracing from

TABLE S2. EXECUTED NATIVE QUERIES AND SAVED EXPORT OCCURRENCES. IEEE DATES: OCTOBER 8, 2026; ACM QUERY/PARTITION ACQUISITION: OCTOBER 8–9, 2026. COUNTS ARE NOT INCLUDED STUDY FAMILIES.

ID	Exact query	IEEE saved/hits	ACM saved/original hits
Q1	`("large language model" OR LLM) AND ("side channel" OR "side-channel" OR "timing attack")`	63/63	479/479
Q2	`(agent OR agentic) AND ("traffic analysis" OR fingerprinting OR "metadata leakage" OR "side channel" OR "side-channel")`	671/671	6,181/6,181
Q3	`("KV cache" OR "prompt caching" OR "speculative decoding") AND (privacy OR leakage OR attack)`	37/37	511/511
Q4	`("browser agent" OR "research agent" OR "tool calling" OR "model context protocol" OR "multi-agent") AND (timing OR fingerprinting OR leakage)`	0/35,743	Not run
Q4b	`(("browser agent" OR "research agent" OR "tool calling" OR "model context protocol" OR "multi-agent") AND (timing OR fingerprinting OR leakage) AND ("large language model" OR LLM))`	847/847	4,026/4,026
Q5	`("language model" OR LLM) AND ("traffic shaping" OR "cache isolation" OR padding OR "side-channel defense")`	95/95	3,689/3,688
Q6	`("language model" OR chatbot OR "AI assistant" OR "generative AI") AND ("side channel" OR "side-channel" OR "traffic analysis" OR fingerprinting OR "timing leakage")`	366/366	1,801/1,801

IEEE used Command Search/default metadata, without a lower-year or document-type filter; every result page was exported as RIS with Citation and Abstract. Q4's zero means not exported in the October 8 acquisition, not zero hits or a completed screen. Q4b supplements Q4; recall preservation is unverified. The later Q4 prefix does not reconstruct historical membership. ACM used the All field in the ACM Full-Text Collection (not Guide), then native E-Publication Date partitions and All Results/BibTeX export. Partition receipts retain exact dates and URLs. Q4b's June bin has 285 displayed versus 284 saved; agreement of the overall saved/original counts does not repair the missing identity. Q5's partition total differs from its original whole-query count. Final-named export files were byte/hash matched on October 9; these two retrieval discrepancies remain. The October 8 coded-set cutoff is checked separately from acquisition partitions.

40 fixed seeds, followed by the completed 565-record October 21 Semantic Scholar and native arXiv search frame (Section S10).

S2. Family Index, Corpus Maps and Prior Work

The accompanying *Supplementary Evidence* archive is the access route for the full tables and fit-free checks. Its README starts with the current paper, family index, result ledger, utility/cost inventory and claim links. Each result joins a citation-key family, source version/hash and locator to its receiver/intervention fields and supported judgment; Table S4 is a filled example. Separate case-evidence entries identify additional source subconditions, rather than silently pooling them into the 33 scoped inventory records.

The merged index enumerates 82 families: 58 core and 24 adjacent, including five authorized diagnostic components. The original 60-family index is a separately labeled historical snapshot. Mixed semantic attribution remains adjacent under the auxiliary-only rule. The package preserves the snapshot's 33 scoped result records/66 field positions and the complete original empirical and workflow analyses. S10 supplies the 22 added-family rows and expansion sensitivity. Source locators and hashes support inspection; third-party full texts/raw traces are obtained from their named sources.

Roles of the remaining indexed sources. Additional core studies concern content/hidden-property inference [2], [3], [4], [5], [6], [7], identity/provenance attribution [8], [9], [10], [11], and authorized integrity diagnostics [12], [13], [14]. Adjacent audit, execution, browser-data and profiling studies supply boundary comparisons rather than additional auxiliary privacy attacks [15], [16], [17], [18], [19], [20], [21], [22]. The index records each receiver, target relation and evidence scope; the reference list includes these indexed sources, not just the central case papers.

S2.1. RQ sensitivity

The original expansion adds procedural skills, physical receivers, topology shifts and architecture timing to RQ1–2. The latest finite search additionally changes the positive defense evidence: CachePrune and OTRO supply useful mechanism-bound privacy/quality/performance conditions. RQ3 still requires generator-specific bridges; RQ4 separates simulation, component overhead, fixed guardrails and executed protection. S10 reports these changes explicitly. Section S6 retains the original source-deletion analysis; the archived controlled examples illustrate its first two placement mechanisms.

S3. Retained Failure-Inclusive Controls

The 288-trace fixed-policy pilot has call-count-determined labels and saturated coarse F1=1.00, so it supports no sequence-increment claim. In the separate tool-front-end study, the observer sees both tools; the constrained baseline succeeds on 19/24 tasks, below the frozen promotion gate. These local results are not passive-network or deployed-defense evidence.

The retail case retains 148 windows, 49 empty observations and 102 stopped-error episodes. All three frozen loss contrasts have intervals including zero; test goal achievement is 7/40 versus 8/40 and joint utility remains unresolved.

TABLE S3. ITEM-BY-ITEM RECONCILIATION OF THE 565-RECORD OCTOBER 21 SEARCH FRAME (BEFORE AND AFTER THE 204-RECORD EXCLUSION AUDIT) AND THE TRANSITION FROM THE 60-FAMILY SNAPSHOT TO ALL 82 CODED FAMILIES.

Route / family category	Pre-audit	Audit Δ	Final (Indexed)
Excluded: no hidden C2/C3 target	221	-4	217 (0)
Excluded: outside LLM / diff. target	175	0	175 (0)
Existing snapshot core / adjacent	23 / 1	$+1/0$	24 / 1 (in 60)
New source-verified core families	14	$+1$	15 (**+15**)
Adjacent source-reviewed contexts	28	$+2$	30 (**+7**)
Adjacent metadata-only contexts	28	0	28 (0)
Background / proposal metadata	58	0	58 (0)
Missing abstract / unavailable text	17	0	17 (0)
565-frame total / Net added families	**565**	**0 net**	**565 (+22)**
Coded collection: Core / Adjacent (**Total**)	Snapshot: 43 / 17 (**60**) $\rightarrow$ Full: 58 / 24 (**82**)		

The exclusion audit selected 177 high-risk pattern matches plus 27 reason-stratified records (seed 20261021), rather than all 396 initially excluded records. This enriched 204-record subset rerouted 4 records out of the 221 initial C2/C3 exclusions: (1) `SUP-0192` (MIRAGE [1]) moved to new core ($14 + 1 = 15$, all 15 added to the family index); (2) `SUP-0552` (InputSnatch arXiv alias) merged into existing family `inputsnatch2024` ($23 + 1 = 24$, $+0$ new families); and (3–4) `SUP-0191` and `SUP-0424` moved to source-reviewed adjacent contexts ($28 + 2 = 30$, of which 7 boundary-defining studies enter the adjacent index and 23 remain in the 565-record ledger). Thus $60 + 15$ (core) $+ 7$ (adjacent) $= 82$.

Neither this null-compatible estimate nor failed tasks establish absence of leakage or effective defense. Full protocols and results remain in the delivered evidence package.

S4. Public-Trace Reanalysis: Protocol and Boundaries

UI Traces (commit `4f3df18`) supplies 4,200 records/14 declared identities, retaining 1,806 error/absent-result indicators. Labels are declarations, not authenticated backends or task correctness. The supplemental package binds source bytes, code and receipts.

Predictors use JS types, local time and callback order, excluding collector time/navigation, URL/DOM text, questions, answers and diagnostics. Cross-document order assumes an archive receiver, not authenticated website capture. Exact-question-disjoint train/validation/test groups have 2,043/1,022/1,135 rows and 395/202/226 questions (823 distinct questions). The source split identifies FRAMES but has no semantic task-family field. Full identity–dataset/exact-question support and a seven-bin first-word *grammatical proxy*, explicitly not a semantic family, are supplemental; neither semantic independence nor family balance is established.

TABLE S4. FILLED RECORD AND ITS LICENSED JUDGMENT. VALUES ARE SOURCE-REPORTED; NO CROSS-MODE EFFECT IS COMPUTED.

Join key / field	Entry
Family / result	Dangol workflow / `DANGOL-livebinding`.
Version / locator	arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [23].
Target / view	Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation.
Intervention	Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures.
Workload / testbed	Workflow classes with client, relay, gateway and agent co-located on one loopback host.
Outcome / cost	Accuracy 0.30; about $13\times$ bytes and $11\times$ latency in that condition.
Unresolved fields	Objective terminal task utility U; precise paired live-refit comparison ?.
Licensed judgment	These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter.

The original 19-fit comparison selects weighted logistic/forest candidates on validation loss only; selected models are forests. H2 adds type bigrams to coarse counts/page-age. Original intervals use 3,000 paired-question resamples and a nominal 98.333% three-contrast allocation, conditional on fixed fitting/selection and noise realizations. Archived H1's identical selected predictor yields exactly zero difference and interval $[0, 0]$; it is a selection illustration, not equivalence evidence.

For Q questions with n_q records each, loss is $Q^{-1} \sum_q n_q^{-1} \sum_{i \in q} - \log p_{i,y_i}$; accuracy substitutes the correct-argmax indicator. Fit weights $N/(Qn_q)$ equalize question weight. Add $\epsilon = 10^{-6}$ to every class probability, renormalize, then clip true-label probability to $[\epsilon, 1]$. Bootstrap draws Q questions with replacement; original contrasts share draw indices, with percentile endpoints 0.8333% and 99.1667%.

Controls declared with original outcomes known. The additive protocol is frozen before new fits, with original outcomes known. All-14 common support is empty. Select the largest identity subset with at least 20 common training questions, then greatest support and lexical tie-break: Gemini 3.1, Gemma 4-31B, Seed 2-lite, with 32/10/10 questions. Hash order selects one row per question/identity, retaining failures; test has 30 rows. This is a three-class sensitivity result.

The first-Wikipedia-document *proxy* stops at HTTP navigation, beforeunload, unmarked origin/URL change or clock reset, but not SPA history alone. Markers gate membership and never enter features. No frame/document IDs authenticate membership; late callbacks may cross boundaries. Empty views remain and epochs are not pooled. The supplement gives the complete six-event projection example and boundary implementation.

Three row-seeded shuffles preserve counts/timestamps/coarse features in *all partitions*, refitting combined candidates with coarse retained. Fits total 8

TABLE S5. PRIMARY OBSERVABLE/RECEIVER PROFILE × SIGNAL ORIGIN ACROSS ALL 58 CORE FAMILIES (SHOWING SNAPSHOT 43 → FULL 58). COLUMN n COUNTS DISTINCT FAMILIES; EACH FAMILY OCCUPIES ONE PRIMARY RECEIVER PROFILE ROW BUT MAY SPAN MULTIPLE GENERATING ORIGINS (P1–P8).

Observable / receiving permission	n (43→58)	P1	P2	P4	P5	P6	P7	P8	Representative sources
Flow / destination / routing metadata	$12 \rightarrow 16$	0	10	1	7	2	2	3	AgentPrint; WRA; NetEcho; LLM-WFIN; Routing Metadata
Own-site actions / browser / harness	$5 \rightarrow 8$	0	0	1	8	2	6	0	UI Traces; FP-Agent; WebAgents;AgentProv; Lidar
API duration / cache / split-return	$12 \rightarrow 14$	0	14	1	0	0	0	0	Early Bird; LeakyLMs; CachePrune; SplitDecoy
Resource / physical / sparsity / page	$10 \rightarrow 15$	15	6	0	1	0	0	0	TDXRay; OTRO; PAGE404; SparLeak; SparSEEty; PhysicalEdge
Trajectory / procedure / diagnostic log	$4 \rightarrow 5$	0	1	1	5	2	0	0	SIGLEAK; ClawGuard; cycle detection; MIRAGE (diagnostics)
Total distinct core families	$43 \rightarrow \mathbf{58}$	Each family contributes once to n; generating-layer memberships may overlap.							

P1 hardware; P2 serving; P4 application/API; P5 orchestration; P6 tools; P7 environment; P8 multi-agent. P3 denotes transport (coded as a receiving projection). Five Type IV authorized diagnostic components (ClawGuard, cycle detection, LumiMAS, LLM RCA, MIRAGE) are separated from Type I confidentiality claims.

TABLE S6. FROM PRIOR CROSS-LAYER SYSTEMATIZATIONS TO RESULT-LEVEL DEPLOYMENT DECISIONS: THE SPECIFIC COMPARISON OBJECT AND ITS DEMONSTRATED USE.

Prior work	Established perspective	Added result-level question	Demonstrating analysis
Banerjee et al. [24]	Cross-layer compound AI attacks can relax threat assumptions; MITRE organization	Does cover end at the protected interface, get reserialized, or execute as an action?	NetEcho final-hop check and Dangol/AHB transport/action distinction; explicit conditions for permitting transfer.
Forough et al. [25]	Confidential-computing threat model; partial inference-defense transfer; isolation is not side-channel immunity	Which oracle, execution mode and component-cost result licenses the proposed defense transfer?	Early Bird KV/semantic split and KVGov hardware/simulation/analytic separation.
Cai et al. [26]	Traffic features, shaping objectives and bandwidth trade-offs	Can a traffic-cover cost/utility argument be used for a dummy tool action?	Dangol/AHB: a state-machine boundary determines which task/policy outcomes must additionally be measured.
Debenedetti et al. [27]	Auxiliary ML components induce privacy channels	Distinguish an execution-mediated oracle from ordinary semantic attribution	C3 boundary and mixed-output eligibility judgments.
Lahjouji–Colaco [28]	Agent data surfaces, governance, composition and cross-session risks	Which bridge test would license a specific transfer?	Receiver/intervention and generator/holdout result maps.
Kim; Horváth et al. [29], [30]	Broad agent security; physical DNN mechanisms/threat models	Compare source and receiving privileges without a universal strength order	Remote calibration versus own-site, local and privileged diagnostic profiles.

matched-task, 8 document-proxy, 8 clean and 12 shuffled; all 36 finish before test prediction, without convergence warnings. Clean H2 reproduces exactly. Each shuffle selects coarse: zero increment is then an algebraic consequence of selecting the same predictor. This checks the selection pipeline, not whether shuffled sequence features contain no information; a forced-sequence comparison would be a different, unperformed analysis.

New intervals use 3,000 question resamples, seed 20261018 and 98.333% allocation across the three new contrasts, conditional on fits and shuffle streams. They are exploratory, not blind confirmation or global adjustment across all historical analyses. Separate code reconstructs points/intervals from saved probabilities. Matched-task uncertainty and the 509/1,135 test error flags (3.4–91.5% by identity) prevent intrinsic-identity claims.

The saved-result verifier supplies local replay; it does not establish independent replication or live utility/cost.

Failure/activity diagnostics. The configuration was frozen with prior outcomes known: three representations, four candidates each, original weights, seed 20261008; all 12 fits precede test prediction. Failure-only uses error/absent-result bits; empty-only uses no projected events; activity-only uses log-count plus seven page-age summaries, not cross-document duration. Candidate hyperparameters and projections are supplemental. No new confirmatory intervals are supplied. Flags cover 43.00% overall and 44.85% test rows, not authenticated task success. Stratifying 509 flagged/203 questions versus 626 unflagged/216 questions changes support; increments are not a causal failure effect. Semantic-family balance remains unavailable.

Exact H3 perturbation and adaptation. The unchanged perturbation sets $t'_{ire} = t_{ie} + u_{ire}$, $u_{ire} \sim U(0, 5000\,\mathrm{ms})$, independently per valid page-local timestamp in source-array order. Missing times consume no draw; types/order/page keys remain fixed. Three row-seeded re-

TABLE S7. SOURCE–RECEIVER–INTERVENTION MAP. THE TARGET COLUMN DISTINGUISHES CONTENT/PROPERTY INFERENCE, ATTRIBUTION AND HUMAN–AGENT DISTINCTION; NONE ALONE ESTABLISHES AN UNAUTHORIZED PRIVACY VIOLATION. U DENOTES AN UNRESOLVED SOURCE/RUN BINDING.

Source / target	Signal origin → receiver	Intervention / evaluated mode	Surviving path or restriction	Comparison consequence
UI Traces / identity attribution	Agent actions → own-site JS	Recorded timestamp delays; offline D2	Types and order preserved	Test residual action features; no live task-cost estimate.
NetEcho / content inference	Serving → network A/B traces	Upstream padding/batching; protected-data fit	Decoded upstream content/state can be reserialized downstream	Inspect final-hop emission; defended fitting alone does not establish a matched adaptation contrast.
SIGLEAK / skill nondisclosure	Procedure → task/output receiver	Common prompt defense	Procedural signatures remain	Compare same receiver/target; not a paired installation effect.
AHB / human–agent distinction	Touch behavior → app/detector	Executed fake touches; separate transforms	Task state may change; fit binding U	Measure transitions and objective utility; do not import offline delay costs.
WRA / private-intent inference	Real/decoy research requests → visited domains	Concurrent execution; interleaved-domain evaluation	Intent reconstruction falls with one decoy	Preserve domain-view privacy benefit; terminal decoy utility and cost are separately unreported.
Dangol / workflow-property inference	Protocol/traffic → defined link	Stored captures versus live OHTTP (§9.7, OHTTP paragraph after Table 5)	Feature ablation differs from network masking	Keep accuracy 0.17 capture and 0.30 live populations separate.
Early Bird / cache state	Cache reuse → API timing	KV simulation versus GPT-Cache prototype	Distinct matching mechanisms	6ms/4% applies to semantic prototype only.
KVGov / private prompt	Shared cache → probe oracle	Simulated inference; real timing; analytic isolation	Hash-key isolation not semantic-index isolation	Hardware gap does not become deployed defended ASR.
FP-Agent / identity attribution	Policy/browser → visited site	Task/site evaluation, mixed attributes	Static configuration may remain	Task holdout is not independent-visitor holdout.
Traffic benchmark / topology	Multi-agent workflow → selected links	Explicit topology/view shift	Unseen generators differ from hidden links	Cross topology with receiving-view restriction.
LeakyLMs / architecture	Inference → streaming client	Chosen-prompt timing; no measured defense	Same-GPU-class calibration for architecture	API reachability is not hardware-independent recovery.

TABLE S8. DEPENDENCY–HOLDOUT–CLAIM MAP. EACH ROW EXCLUDES ONE EXPLANATION, NOT ALL CONFOUNDING.

Dependency / source	Required holdout	Claim licensed; residual limit
Task / UI Traces	Question/task-family	New task units; fixed harness remains.
Browser / FP-Agent	Visitor/configuration	Configuration transfer; task-only split does not test it.
Workflow / traffic benchmark	Agent topology or composition	Tested topology shift; no all-orchestrator guarantee.
Hardware / LeakyLMs	GPU/serving calibration	Tested architectural transfer given calibration; not unknown-hardware recovery.
Repository / coding fingerprints	Repository and generation pathway	Tested attribution domain; not covert human-mediated use.
Order / worked case	Same units, refit shuffle	Fixed-program order increment; not task-independent identity.

TABLE S9. FIXED-ARCHIVE PREDICTIVE EVIDENCE AND MATCHED SUPPORT. UNITS: NATS OF QUESTION-WEIGHTED LOSS REDUCTION. POSITIVE VALUES FAVOR THE SECOND PROGRAM. INTERVALS CONDITION ON FITTED PROGRAMS; THE ORIGINAL AND ADDITIVE-CONTROL ALLOCATIONS REMAIN SEPARATE.

Contrast	Estimate	98.333% interval
Worked case: coarse vs. combined (H2)	0.3447	[0.2974, 0.3944]
Matched questions, 3 identities[a]	0.2634	[−0.2584, 0.7241]
Reanalysis: clean vs. adapted (H3)[b]	0.1503	[0.1205, 0.1815]

[a] 30 rows/10 test questions, 3 classes, rather than 1,135 rows/226 questions, 14 classes; not a direct effect-size comparison. Its interval uses the additive three-control allocation. Archived identical-predictor H1 remains in the protocol.
[b] Both programs predict the same timestamp-perturbed test rows; three fixed realizations are averaged.
Cross-document order assumes an archive receiver, not verified ordinary-website capture. Intervals vary test-question weights, not training, task families or deployments. No live defense utility or cost is inferred.

alizations transform all partitions. Each adaptation clones the clean-selected representation/architecture and fits once on perturbed training rows with identical weights, without retuning or additional data (three fits). Both programs predict the same perturbed test rows; average paired losses across realizations, rows within question, then questions. The supplemental exact protocol supplies entry points, seed strings and invariants. The transformation edits stored times; it does not execute action delays.

TABLE S10. PUBLIC-TRACE PERMISSION PROJECTION. "SITE" MEANS ITS OWN DOCUMENT ONLY; IT DOES NOT AUTHENTICATE CROSS-DOCUMENT CAPTURE. FULL FIELD-LEVEL CSV AND ONE COMPLETE INGESTED-RECORD PROJECTION ARE SUPPLEMENTAL.

Field	Site JS	Analysis use
Type, page-local time	Yes	Original predictors; activity diagnostic
Within-document order	Yes	Type bigrams
Cross-document order	Unverified	Archive-receiver bigrams
URL/origin	Own page	Origin gate only; no text/page-key feature
HTTP navigation, collector time	No	Boundary proxy only / time excluded
Error/result metadata	No	Analyst-only failure baseline
Task/answer metadata	No	Excluded; question hash splits/weights only
DOM/link text	Own page	Excluded from all predictors
Document/frame ID	Absent	No authenticated document membership

TABLE S11. POST-REVIEW H2 CONTROLS. POSITIVE VALUES DENOTE A QUESTION-MEAN LOG-LOSS REDUCTION (NATS); BRACKETS ARE CONDITIONAL 98.333% RESAMPLING INTERVALS. NOT A NEW AGENT RUN.

Contrast	Test rows / questions	Effect [interval]
Matched tasks, 3 identities: coarse minus sequence	30/10	0.2634 [−0.2584,0.7241]
First-document-epoch *proxy*: coarse minus sequence	1135/226	0.1504 [0.1193,0.1825]
Original clean minus mean of 3 refit shuffle increments	1135/226	0.3447 [0.2961,0.3953]

All 14-class common support is empty. The first-document view is reconstructed, not authenticated. Its 0.1504 feature increment and H3's 0.1503 timestamp-adaptation effect are distinct contrasts on the same archive, not independent replications. Cross-document order assumes an archive receiver. All three shuffles select coarse, so their zero increments make this contrast's 0.3447 point equal to original H2. The intervals use separate frozen bootstrap seeds (20261018 here; 20261008 for original H2), yielding different finite-resampling endpoints. Zero selected increments do not establish information equivalence. No task utility or live defense is measured.

S5. Intervention Inventory and Evidence Dispositions

The archived initial mode-specific table and 33-record ledger distinguish executed inference, offline/simulated evidence and component costs. The completed 39-record, 351-cell audit is in Section S29. All 66 initial utility/cost positions retain the source, scope, state and consequence for a named claim or inventory-only role. The main deployment-question table does not treat these heterogeneous records as a common success/failure denominator.

Four textual defense assertions lack the quantitative binding needed for a matched effect (Table S13). Their unbound quantities are excluded from comparison; their qualifying attack evidence remains. AHB has verified condition-wise refit code, but historical D1-run and paired D2 are recorded as ?, not reconstructed. Its task outcome is not inferred from that code.

D1 denotes protected-condition fitting/calibration; D2 denotes a paired training-procedure comparison on the same defended test condition. Neither entails feasible labeled-data access or matched budgets. Token payload, plaintext padding, wire bytes, model loading, map lookup and added service latency remain distinct cost objects. A scoped U completes the requested extraction, rather than asserting zero cost or preserved utility.

TABLE S12. ADDITIVE FAILURE/ACTIVITY DIAGNOSTICS ON ALL 1,135 TEST ROWS/226 QUESTIONS. POINT SCORES USE THE ORIGINAL QUESTION WEIGHTS; NO NEW CONFIRMATORY INTERVALS. ORIGINAL PROGRAMS ARE READ FROM SAVED PROBABILITIES, NOT REFITTED.

Program	Loss (nats)	Accuracy (%)
Training prior (saved)	2.64267	6.59
Error/result flags only[a]	2.47464	14.25
Projected-empty flag only	2.63832	6.79
Activity/page-age only[b]	1.81732	37.68
Original coarse (saved)	1.44058	51.29
Original coarse+sequence (saved)	1.09591	66.26

[a] Analyst-only diagnostic, not an observer-valid attacker; two binary flags, not error text. [b] Total event count and seven local-page-time summaries; no action types/order, diagnostics, text or session duration. Same four candidate architectures and validation-only selection for each of the three new representations (12 fits). Low-dimensional baselines do not rule out higher-order failure/task confounding.

TABLE S13. QUANTITATIVE ASSERTIONS EXCLUDED FOR MISSING BINDINGS.

Source	Missing binding for quantitative use
AgentPrint	Referenced supplementary mitigation result, procedure and population not located.
Weiss	No matched vendor-defense before/after outcome located; the artifact appendix reproduces an attack table.
Gu	No provider-change result with a matching comparison population located.
PromptEcho	MSS/threshold padding is described; exact threshold, population, cost denominator and defended outcome are unbound.

Sources: [31], [32], [33], [34]. No missing value is imputed; withholding these assertions does not remove the four families.

S6. Conclusion Sensitivity

Table S15 exposes direct-source dependence in the original snapshot. Within that snapshot, removing NetEcho withdraws the only downstream re-emission instance; removing AHB withdraws the measured mobile state-change instance, while WRA's privacy-oriented decoy result remains. Removing WRA withdraws that private-intent decoy result while AHB retains the task-state branch. UI Traces and SIGLEAK remain different surviving-feature instances, not replications. Removing Dangol withdraws its capture/live contrast; Early Bird, KVGov and SpliceDefense retain separate component examples. S9/S10 separately report the later

TABLE S14. SCOPED INTERVENTION DISPOSITIONS ACROSS ALL 58 CORE FAMILIES IN THE FULL COLLECTION (ALONGSIDE THE 43-CORE OCTOBER 8 SNAPSHOT AND 15 NEW CORE ADDITIONS). CATEGORIES ARE MUTUALLY EXCLUSIVE AT THE FAMILY LEVEL.

Source-scoped disposition	Snapshot (43)	Added (+15)	Full (58)
Evaluated mitigation component/condition	16	+4	20
Fixed common guardrail (not paired installation)	2	+0	2
Other intervention / decoy boundary / objective	4	+1	5
No evaluation identified in inspected scope	13	+9	22
Type IV diagnostic component (confidentiality –)	4	+1	5
Unbound assertions excluded from effect comparison	4	+0	4
Total core families (Result records)	43 (33)	+15 (+6)	58 (39)

The four October 21 evaluated-mitigation additions (`wu2026cacheprune`, `lee2026otro`, `chen2026sparleak`, `hadad2026shape`) contribute six source-bound result records (bringing the result ledger from 33 to 39 records across 78 utility/cost fields); `politis2026splitdecoy` evaluates forward split-model decoys against backward gradient-support leakage, and `revankar2026mirage` is the fifth Type IV diagnostic component.

controlled evidence and source expansion; they are not part of this historical ablation denominator.

Starting from the original 60/43/17 total/core/adjacent families, deleting AHB, WildAuth and coding fingerprints leaves 57/42/15; deleting its four diagnostics leaves 56/39/17; joint removal leaves 53/38/15. These are snapshot ablations, not the later 82-family collection. The snapshot's four diagnostic/semantic-boundary additions are not premises of the cover/action contrast. For its five multifeature-trace studies, unresolved bindings bound D1 at three–five and D2 at two–five; removing AHB gives three–four and two–four. Early Bird calibration is a separate stage.

Excluding all four unbound defense quantities leaves the three deployment cases unchanged, because none uses those quantities as a premise. AHB's detector-run unknowns prevent an adaptive-detector claim but do not erase its separately reported task-success points. Its non-Fake online point is an additional Table 1 subcondition in the case ledger, not an extra independent experiment or an imputed value in the combined-condition inventory row. Whole-service costs, training access and missing utility admit no favorable imputation or ranking.

Narrowing H2 to matched tasks leaves task-independent identity unresolved; the document proxy does not authenticate capture. H3 supports only same-record offline adaptation. The original joint workflow endpoint remains negative, including the missing-score bounds and post-outcome sensitivity views. These positive, inconclusive, zero and negative outcomes are preserved with their original endpoint meanings.

S7. Retained Indexing Experiment and Scenario Illustration

The retained model-assisted experiment compares flat checklist R, structured checklist T and raw SCOPE traversal V. There are 512 roots per arm and three query judgments per root (1,536 judgments). The endpoint is the mean of each root's three binary joint outcomes, requiring a correct deployment decision and exact configuration counts. R has 624 joint successes (40.63%); V has 494 (32.16%). T has 693 observed successes and three missing queries belonging to one root, giving the retained 45.12–45.31% missing-outcome range. V is worse on this joint endpoint; this is retained negative evidence.

Decision-correct but count-incorrect judgments number 537 (R), 476 (T) and 695 (V), out of 1,536 query opportunities. Observed queries have no schema/evidence-ID failures. Binary-decision successes are 1,161 (R), 1,169 (T, plus three missing) and 1,189 (V). The 1.11–1.30 percentage-point V–T binary difference is a secondary post-outcome decomposition, not proof of a better workflow. Counts are query judgments, not independent roots; percentages average three judgments per root. We do not infer causality from the error decomposition.

The separate 24-scenario exercise uses author-constructed source-reuse cases and author labels: seven Compatible, twelve Incompatible and five Unresolved. The original rule-application outputs and three-category confusion matrices remain in `experiments/reviewer-audit/scenario_confusions.json`. This archived illustration supplies neither participant data nor an accuracy advantage over ordinary expert reading.

S8. Result-Pair Comparability, Nine-Item Split Audit

The complete audit covers 39 records from 26 families, with 33 mitigation records from 20 families and six contextual records. Its 741 unordered pairs include 133 related-cohort comparisons and 608 outside the declared cohorts. The 15 same-family pairs are nested within the 133. These counts describe the declared record partition, not a rate of avoidable research defects. Section S37 projects the unchanged judgments onto coarser units and shows their dependence on granularity. Eight same-family pairs change the primitive, target or cost object. Of seven aligned-outcome pairs, four change the collection/execution population and three lack a verified paired-test binding. The artifact preserves the original 528 judgments and identifies all 31 added related-pair judgments. Cohort membership, source locators and individual reasons are explicit; outside-cohort pairs are excluded rather than treated as adverse evidence. Section S29 reports all 351 record-local audit cells and positive joint-binding identities.

TABLE S15. LOAD-BEARING SOURCE ABLATION WITHIN THE ORIGINAL 60-FAMILY SNAPSHOT. ENTRIES IDENTIFY ITS ACTUAL DEPENDENCE; S9/S10 REPORT SUBSEQUENT CONTROLLED EVIDENCE AND SOURCE EXPANSION SEPARATELY.

Conclusion / source locator	What is supported	What is lost on source removal / NR dependence
Action preservation / UI Traces 6.1,Fig5	Recorded delays preserve structure and allow adaptation	Action-delay instance and one D1/D2 positive lost; exact release-point binding not needed for protocol claim.
Procedural preservation / SIGLEAK 4/C	Skill inference persists under common extraction defense	Skill-defense instance lost; no paired installation-effect claim.
Re-emission / NetEcho IV-C–E	Downstream reserialization can restore fine granularity	Only direct downstream re-emission instance lost; no multisource replication claimed.
State change / AHB 5.2.3/Table1/E.3	Fake actions can cause unintended UI transitions	Only executed mobile state-change instance lost; detector-run binding remains U.
Private-intent cover / WRA 8.1/Table8	Real/decoy execution with interleaved-domain inference evaluation lowers one-decoy E_{func}	Private-intent decoy instance lost; AHB state-effect instance remains. Same-condition terminal utility/cost are U.
Mode split / Dangol §9.7, OHTTP paragraph after Table 5; 9.8–9.9/Table6	Capture frontier differs from live protocol and timing ablation	Concrete accuracy 0.17/0.30 pair and one D1/D2 positive lost.
Cost boundary / Early Bird V-A/B; Splice VII-C; Speculation 6.1; KVGov 5.9	Separate component and deployment costs	Removing one withdraws its example, not other boundary instances; full service costs remain unknown.
Generator dependence / UI Traces, FP-Agent, traffic benchmark, LeakyLMs	Task, browser, topology and calibration tests differ	Each source's axis withdrawn when removed; no pooled portability estimate.

TABLE S16. COMPLETE RECORD-PAIR DESCRIPTIVE COMPARABILITY.

Subset	Compatible	Incompatible	Unresolved
Same family ($n = 15$)	7	8	0
Related cohort ($n = 133$)	17	105	11

S8.1. Nine-item split reporting audit

Table S17 uses R4 for defended fitting, R5 for a matched adaptation contrast, R7 for inference, R8 for utility and R9 for cost across both the 16 original evaluated-mitigation families (144 cells) and the 20 total evaluated-mitigation families (including `wu2026cacheprune`, `lee2026otro`, `chen2026sparleak` and `hadad2026shape`; 180 cells).

The current `combined_result_records.json` makes the six expansion records explicit alongside the 33 preserved baseline records: 39 records from 26 distinct families, while the reporting audit covers 20 mitigation families.

S8.2. Current source-follow-up corrections

The first source-follow-up pass resolved three of 14 unverified family cells. For the cloud-traffic benchmark, §V-C1/Fig. 9(a) locates execution-trigger instructions before and after the original prompt and reports the resulting reformulation experiment [35], establishing R3 placement. For Shape and Substance, §IX-B1 derives padding cost from unprotected timing measurements without reporting protected fitting or a matched adaptation contrast [36]; R4/R5 therefore become U. The targeted reinspection in Section S34 resolves the remaining 11 family cells as four P and seven U, with no new B. These judgments distinguish reported profiling and defended outcomes from an explicitly linked protected fit or matched adaptation contrast. That reread settled the 11 broader source-scope questions. The current 180-cell matrix is instead projected from the record-level ledger: six record-specific bindings remain unresolved, leaving two family-summary question marks where no record supplies B or P. Broader source descriptions cannot fill a missing execution-record binding. Historical coding columns are preserved; versioned source locators and the correction overlay generate the current tables.

TABLE S17. NINE-ITEM SPLIT REPORTING AUDIT OVER THE 16 ORIGINAL EVALUATED-MITIGATION FAMILIES (AND 20 TOTAL INCLUDING THE FOUR OCTOBER 21 MITIGATION ADDITIONS IN PARENTHESES). R1–R9 DESCRIBE EVIDENCE FOR SCOPED REUSE CLAIMS; THEY ARE SEPARATE FROM DEPLOYMENT CHECKS 1–3.

Audit item	Bound	Partial	U	?	–
R1. Target & protection polarity	16 (20)	0 (0)	0 (0)	0 (0)	0 (0)
R2. Receiver fields & linkage	13 (17)	3 (3)	0 (0)	0 (0)	0 (0)
R3. Placement & action dispatch	14 (17)	2 (3)	0 (0)	0 (0)	0 (0)
R4. Protected-condition fit	4 (4)	4 (5)	1 (2)	0 (0)	7 (9)
R5. Matched adapted test	2 (2)	2 (2)	5 (7)	0 (0)	7 (9)
R6. Workloads & generator shifts	14 (18)	2 (2)	0 (0)	0 (0)	0 (0)
R7. Defended inference effect	14 (17)	0 (1)	2 (2)	0 (0)	0 (0)
R8. Task or component utility	2 (4)	2 (2)	10 (11)	0 (0)	2 (3)
R9. Mechanism or protocol cost	10 (11)	1 (3)	4 (5)	0 (0)	1 (1)

Bound: explicitly established for the scoped claim. Partial: a supported element coexists with a proxy or unlinked operating point (e.g., AHB refit code without verified Table 1 run binding; WRA Table 11 visibility utility separate from Tables 8–9 decoys). Under this historical configuration-grouping rule, CachePrune alone is coded B for R7/R8/R9 in one described package. This retained coding does not establish identical requests or paired protection/utility/cost trials; R47 changes no cells or counts. Current totals include SparLeak's matched component utility and the 11-cell reinspection in Section S34; historical coding columns remain unchanged.

SparLeak's §7.5/Table 4 pairs reconstruction success and ROUGE at the same model and perturbation intensity [37]. R8 accepts component as well as task utility, so the current overlay corrects its partial code to B. Its reported spy overhead is attack cost and is not promoted to defense cost. The R8 distribution becomes 4 B, 2 P, 11 U and 3 inapplicable; historical records remain unchanged.

S9. Controlled Replay Details and Revised Accounting

The current experimental source is `experiments/reviewer-audit/recompute.py`, with the instrumented retail implementation in `retail_replay.py`. Historical inputs and previous summaries are retained unchanged; revised JSON files take precedence for the current manuscript. No fresh model generations were requested.

TLS corpus and length accounting. The 48 saved outputs comprise 24 brief/expanded pairs. Training and testing each contain 12 topics and 24 outputs. T1 uses the original verbosity labels. T2 retains brief responses and takes the first two period-delimited sentences of expanded responses. T3 uses the first two and first three sentences of each expanded response. T4 retains both original outputs per topic and labels their domain. There is one generation per prompt, not repeated model sampling. T4 has 14 positive and 10 negative test outputs (seven versus five topics), so 50% is not its empirical majority-class baseline.

For every regime and framing policy, `tls_records.json` saves topic, split, label, plaintext bytes, application-record count, payload bytes, wire bytes and recovered plaintext hash. `tls_gradient.json` reports absolute means and matched denominators. All 48 texts are used for cost means; only 24 held-out outputs are used for accuracy. A record contains five TLS header bytes plus 24 encryption-overhead bytes under the pinned TLS 1.2 suite. Handshake traffic, TCP/IP and timing are outside this cost scope. Bucket 1024 uses four 256-byte writes per quantum, hence 1,140 rather than 1,082 wire bytes per quantum. Buckets 256/512 use 128-byte writes.

The training-fitted cap rounds the maximum framed training length, 1,194 bytes, up to 1,280 bytes. It emits ten records of 128 bytes (1,570 wire bytes). Overflow adds 128-byte frames, never truncates, and leaks an overflow-dependent length; none of the saved test responses overflows. The loose 16-record, 1,024-byte-per-record control is 16,848 wire bytes and is not a lower bound on protection cost.

Cohen's absolute standardized mean difference, recomputed across the 48 constructed plaintext lengths using pooled sample variance, is 12.264 (T1), 5.723 (T2), 3.418 (T3) and 0.033 (T4). Previous values embedded in regime labels were not produced by that calculation and are withdrawn. These descriptive values are not entropy, independent tests or a model of required framing cost.

Adaptive fitting and bootstrap hierarchy. The clean threshold is fitted to training TLS payload bytes. The defended thresholds use the first four or all twelve training topic pairs. The 1024B defended threshold is 1,680 TLS payload bytes. Stochastic padding has 50 draws; the first is measured on sockets and 49 are calculated using socket-checked record accounting. The independent resampling unit is one of 12 test topics. For each topic, correctness or paired gain is averaged across its two outputs and all draws before 10,000 percentile-bootstrap resamples (seed 20261021). We do not refit the model in that bootstrap or infer uncertainty across fresh model generations. Stored per-topic/per-draw correctness permits exact reconstruction. Both constant caps give a constant specified length sequence without overflow; this says nothing about unobserved timing or metadata.

Retained synthetic tool control. The original 24 saved inventory/calendar programs are a separate synthetic demonstration. Discard, shadow and read-only routes preserve the prescribed states by construction; 9/24 live-write runs meet both the narrow goal and protected-bystander condition. This is not evidence for autonomous-agent planning or safe deployment of destructive cover.

S9.1. Retail source, permissions and full statistical results

The pinned τ-bench commit is 59a200c6d575d595120f1cb70fea53cef0632f6b. The instrumented replay preserves the 115 tasks, nine policies, three seeds and two positions (6,210 rows in `retail_units.json`); 97 tasks are eligible. The target is the presence of an attempted reference cancellation. Counts include rejected dispatches. The allowed observer receives ordered tool names and response byte lengths, not argument IDs, response contents or backend state. The tool view is separate from the TLS size observer.

The fixed split takes the first 48 sorted eligible task IDs for fitting (7 positive, 41 negative) and the remaining 49 for testing (11 positive, 38 negative). IDs are recorded in `retail_diagnostics.json`. Every test task contributes all three seeds. For each task we average accuracy across seeds, then resample 49 tasks 10,000 times for accuracy intervals. Balanced-accuracy intervals independently resample within the 11 positive and 38 negative task strata. These intervals condition on this split, fitted models and saved tasks; policy development was not held out. AUROC uses the uncalibrated ridge linear decision score, including ties. Confusion matrices in `retail_observers.json` count 147 task-seed units; they are not 147 independent tasks. Paired $D_2 - D_0$ differences are averaged within task before the task bootstrap.

The known-position filter scores 100% accuracy/BA for each policy and position. The cancellation-count rule scores 100% for blind and other-account cancellation in both positions. Their percentile bootstrap intervals are [100,100] because every observed task is correct; this degeneracy is not a claim about population-level certainty. No identifiers are needed. We do not evaluate an ID-filter attack because IDs are not in the specified view. The legacy model on the other-account after condition recomputes to 71.43%, correcting

TABLE S18. LEGACY LEARNED OBSERVER ON THE HELD-OUT RETAIL TASKS. INTERVALS ARE CONDITIONAL TASK-BOOTSTRAP INTERVALS, NOT POLICY-GENERALIZATION GUARANTEES. THE MAJORITY BASELINE IS 77.55% ACCURACY AND 50% BA.

Policy / position	D_0 accuracy [95% CI]	D_2 accuracy [95% CI]	D_2 BA [95% CI]	AUROC
Instruction aware before	51.02 [36.73, 65.31]	73.47 [61.22, 85.71]	47.37 [43.42, 50.00]	0.634
Instruction aware after	51.02 [36.73, 65.31]	71.43 [59.18, 83.67]	52.51 [41.37, 66.15]	0.694
Other-account cancel before	22.45 [12.24, 34.69]	71.43 [59.18, 83.67]	46.05 [40.79, 50.00]	0.497
Other-account cancel after	22.45 [12.24, 34.69]	71.43 [59.18, 83.67]	46.05 [40.79, 50.00]	0.497
Blind cancel before	22.45 [12.24, 34.69]	71.43 [59.18, 83.67]	46.05 [40.79, 50.00]	0.597
Blind cancel after	22.45 [12.24, 34.69]	68.03 [54.42, 80.27]	47.09 [38.16, 58.37]	0.622

the previous prose value 68.03%; 68.03% belongs to blind cancellation after insertion.

State metrics and uncertainty. `retail_utility.json` contains task-bootstrap intervals for database equality, full-account equality and equality of reference-changed objects, along with full-census means. The field name `native_reward` is retained only for compatibility with the old script; its calculation is orders/users equality, not an invocation of the full reward/response evaluator. Products are shared read-only by the retail tools. The revised account metric checks all orders owned by the active user, plus the user's record. The reference-changed-object set is constructed from baseline differences, not from all accessed objects. Simulator no-cover database equality is a reference-relative 100%, not proof of user-task success.

Error and denominator reconciliation. Item self-write before insertion succeeds and changes state in 173/291 units; all remaining 118 errors are unavailable-variant checks. After insertion, 120 errors are non-pending status and 77 are unavailable variants, leaving 94 successful mutations. The raw error strings are retained. Price-difference checks over the 173 initially successful selections find zero differences, but the API still appends a refund and changes the status. The coincidence 204/291 = 68/97 = 70.10% combines two distinct quantities: preserved reference-changed objects under item self-write and tasks offering an instruction-unmentioned candidate.

For blind cancellation before insertion, database-equal counts by task are: 16:3, 30:3, 31:2, 32:3, 38:3, 54:3, 55:3, 59:2, 66:3, 69:3, 81:2, 99:3. They sum to 33 across 12 tasks. The error-free sensitivity subset is 82 tasks $\times$ 3 seeds = 246 units: 24/246 before and 94/246 after. Simplifying these fractions to 12/123 or 47/123 does not change the underlying sample size. All-unit counts and task IDs are recorded, so repeated seeds are never represented as new independent tasks.

S10. Completed Finite Retrieval Update and Exact Reconciliation

Four Semantic Scholar bulk queries (`S1R`, `S2`, `S3`, `S4`) and two native arXiv searches (`ARXIV-A1`, `ARXIV-A2`) were completed on October 21, 2026, returning 615 occurrences deduplicated to 565 unique records. Combined with the original 60-family October 8 snapshot, the final merged index contains 82 families: 58 core (53 non-diagnostic and 5 Type IV diagnostic) and 24 adjacent.

S10.1. Executable queries and pre/post-audit disposition accounting

The Semantic Scholar bulk queries (2023–2026) and native arXiv CS searches (January 1, 2023–October 21, 2026) are:

> S1R (116): (agent | workflow) + (LLM | "language model") + ("side channel" | "side-channel" | fingerprinting).
> S2 (120): (LLM | "language model") + (traffic | streaming) + (leakage | side-channel | fingerprinting).
> S3 (15): (LLM | "language model") + (padding | "re-emission") + (privacy | leakage | side-channel).
> S4 (287): (agent | LLM) + (decoy | "cover traffic" | "fake actions" | "cover actions").
> ARXIV-A1 (50): `"side-channel" AND "large language"`; ARXIV-A2 (27): `"side-channel" AND agent`.

Table S19 provides the complete item-by-item reconciliation before and after the 204-record exclusion audit (177 high-risk pattern matches plus 27 reason-stratified samples, seed `20261021`). Before the exclusion audit, screening identified 221 C2/C3 exclusions (`EXCLUDE_NO_HIDDEN_AUXILIARY_TARGET`), 14 new core families, 23 existing core families, and 28 source-reviewed adjacent contexts. The exclusion audit rerouted exactly four records out of the 221 C2/C3 exclusions ($221 - 4 = 217$):

1) `SUP-0192` (MIRAGE [1]): rerouted to `NEW_CORE_SOURCE_VERIFIED` ($14 + 1 = 15$ final new core families, all 15 incorporated into the 82-family index via `addition.csv`).
2) `SUP-0552` (InputSnatch arXiv version [38]): rerouted to `EXISTING_CORE` ($23 + 1 = 24$ existing core recoveries; merged as an alias of the already-indexed snapshot family `inputsnatch2024`, adding $+0$ net new families).
3) `SUP-0191` (LLM Self-Recognition) and `SUP-0424` (Reverse-Engineering Model Editing): rerouted to `ADJACENT_SOURCE_REVIEWED` ($28 + 2 = 30$ final source-reviewed adjacent contexts). Of these 30 source-reviewed adjacent contexts, 7 boundary-defining studies (`addition-adjacent.csv`) are incorporated into the 82-family index ($17 + 7 = 24$ adjacent), while the other 23 (including `SUP-0191` and

SUP−0424) remain in the 565-record disposition ledger as non-indexed context.

Thus the family transition is strictly additive and self-consistent: 60 (snapshot) + 15 (14 pre-audit + 1 audited new core) + 7 (selected adjacent) = 82 families ($43 + 15 = 58$ core; $17 + 7 = 24$ adjacent).

TABLE S19. POST-AUDIT DISPOSITION FLOW FOR THE COMPLETED 565-RECORD SUPPLEMENTARY FRAME (SEE TABLE S3 FOR PRE-AUDIT VS. POST-AUDIT Δ).

Stage or final post-audit route	Count
Complete-query occurrences / duplicates merged	615 / 50
Unique records with an explicit route	565
Excluded: no hidden auxiliary target ($221 - 4$ rerouted)	217
Excluded: outside LLM execution / different target	156 / 19
Existing core ($23 + 1$ alias) / adjacent recovered	24 / 1
New source-bound core families ($14 + 1$ MIRAGE; all indexed)	15
Adjacent source-reviewed ($28 + 2$; 7 indexed) / metadata-only	30 / 28
Other security background / reviewed background / proposal	53 / 4 / 1
Missing abstract / unavailable primary full text	16 / 1

S11. Retained buffered-TLS and fixed-position retail studies

This section preserves, unchanged in substance, the first-revision main-text account of the two controlled studies whose operating points now belong to the appendix case study. Section S12 reports the second-round tests E5–E7.

The two studies test distinct observers. The transport observer sees TLS application-record lengths on a selected link. The tool-dispatch observer sees ordered tool names, write counts and response byte lengths, including failed calls; it is an instrumented dispatch view, not an encrypted-network observer. Neither experiment combines these views or runs an autonomous agent. Saved outputs and reference programs let us vary the specific boundary while holding the prescribed workload fixed.

S11.1. Transport placement and workload-matched framing

For 24 public topics, the saved corpus contains one prompted brief and one prompted expanded gpt−6−luna response (48 outputs; 12 training and 12 disjoint test topics). These are designed prompt conditions, not a sample of natural task lengths. The original placement experiment passes text through two TLS 1.2 connections: the frontend decodes and removes upstream framing before sending new terminal chunks. Exact plaintext hashes are checked. Equalizing the upstream length projection leaves the unpadded terminal projection separable (100% verbosity classification); padding both projections reduces the evaluated classifier to 50%. This reproduces the receiving-boundary mechanism identified by NetEcho rather than discovering a new failure mode.

TABLE S20. CORRECTED TLS APPLICATION-RECORD RESULTS. COSTS USE EACH WORKLOAD'S OWN UNPROTECTED TEXTS (ALL 48 OUTPUTS); ACCURACY USES 24 TEST OUTPUTS IN 12 TOPICS. INTERVALS RESAMPLE TOPICS.

A. Two-link placement (T1; retained experiment)

Protection	Upstream acc.	Terminal acc.	Terminal bytes
None	100.0	100.0	720.96
Upstream only	50.0	100.0	720.96
Both, loose cap	50.0	50.0	16,848

B. Matched workload costs and defended-fit accuracy

Workload / framing	Mean bytes	Ratio	Acc. (%)
T1: unprotected	720.96	1.00	100.0
T1: 256B bucket	856.96	1.19	100.0
T1: 512B bucket	1,085.92	1.51	100.0
T1: 1024B bucket	1,401.25	1.94	70.8
T1: training-fitted cap	1,570.00	2.18	50.0
T1: loose cap	16,848.00	23.37	50.0
T2: unprotected	272.48	1.00	100.0
T2: 256B bucket	438.29	1.61	87.5
T2: 512B bucket	628.00	2.30	50.0
T2: 1024B bucket	1,140.00	4.18	50.0
T3: unprotected	456.33	1.00	91.7
T3: 256B bucket	608.38	1.33	62.5
T3: 512B bucket	654.17	1.43	50.0
T3: 1024B bucket	1,140.00	2.50	50.0
T4: unprotected	720.96	1.00	45.8

C. Adaptation on identical defended T1 tests

Defense	D_0 (%)	D_2 (%)	Gain [95% CI], pp
+512B offset	50.0	100.0	50.0 [50.0, 50.0]
1024B bucket	50.0	70.8	20.8 [8.3, 33.3]
$U(0, 512)$B	86.4	99.9	13.5 [10.7, 16.2]
$U(0, 1024)$B	68.2	89.2	21.1 [17.2, 24.8]
Loose cap	50.0	50.0	0.0 [0.0, 0.0]
Training-fitted cap	50.0	50.0	0.0 [0.0, 0.0]

Panel C uses 12 training pairs. Four-pair accuracy is the same except under $U(0, 1024)$B, where it is 90.2%. Random padding averages 50 draws within each topic; these are not 50 independent corpora. T2/T3 are sentence-prefix constructions. T4 relabels both original response lengths and has a 58.3% test majority baseline. The cap equalizes only the specified record-length projection when no message overflows.

The revised framing replay uses ECDHE−RSA−AES128−GCM−SHA256 sockets and parses application-data records. T1 contrasts brief versus full expanded responses. T2 contrasts the brief with the first two period-delimited sentences of the expanded response. T3 contrasts the first two versus first three sentences of the same expanded response. Thus T2/T3 are derived prefixes, with two examples per topic, not independently generated explanations. T4 relabels both brief and expanded responses by topic domain; it has 24 test outputs from 12 topics, not 12 expanded outputs. Its unpadded accuracy is 45.8% (95% topic-bootstrap interval 25.0–66.7%); its test majority-class baseline is 58.3%. T4 is a weak-signal control, not a defense success.

Every cost ratio divides defended application-record bytes by unpadded application-record bytes for the *same texts*, over all 48 examples in that workload. The count includes TLS record headers and encryption overhead; hand-

shakes, TCP/IP, upstream bytes and latency are excluded. The 4-byte length prefix is included. Buckets of 256/512 bytes use 128-byte writes; 1024-byte buckets use 256-byte writes. Consequently the 1024-byte single-quantum wire size is 1,140 bytes, not 1,082. The workload-matched 512-byte ratios for T2/T3 are 2.30×/1.43×; dividing both by T1's 720.96-byte denominator would yield the invalid earlier ratios below one.

A tighter comparison selects a constant application-frame cap using *training lengths only*: round the largest framed training response (1,194 bytes) up to a multiple of 128, yielding 1,280 bytes. Ten 128-byte records cost 1,570 wire bytes (2.18× T1). Longer messages would emit additional 128-byte records without truncation; no test message overflows. This produces identical length sequences on the saved T1 test set and 50% accuracy for the evaluated classifiers. It is a feasible cap for these data, not a globally optimal padding protocol or a guarantee for future lengths. The retained 16,848-byte cap (23.37×) is only a deliberately loose control.

S11.2. Matched attacker adaptation

We use D_0 for an attacker fitted on unprotected training data and D_2 for one fitted on defended training data; both are evaluated on the identical defended test examples. The threshold uses total TLS record payload bytes. D_2 uses the first four or all twelve training topic pairs; these budgets do not represent multiple independently sampled training sets. At a 1024-byte bucket, D_0 constantly predicts the expanded class and scores 50%, while D_2 scores 70.8% (58.3–83.3%). An offset of 512 bytes permits 100% adapted accuracy. This demonstrates why a shifted D_0 alone is insufficient, without claiming novelty for adaptive evaluation.

Random-tail padding is evaluated over 50 draws per output. The first draw is socket measured and the remaining 49 use the record-length accounting checked against the socket replay. Each topic's paired correctness is averaged over its outputs and draws before 10,000 topic-bootstrap resamples (seed 20261021). Fractional accuracies therefore do not imply additional independent topics; the intervals condition on these saved outputs and fitted attackers. Positive adaptation gains span 13.5–50.0 percentage points; both constant caps have zero gain.

No measured accuracy of 50% proves absence of all leakage. The constant-cap construction makes only the specified record-length sequence constant when there is no overflow. Timing, connection behavior, request metadata and content are outside that projection. No sequence/timing attack, TLS 1.3 native padding, QUIC or live token/SSE stream is evaluated.

S11.3. Retail execution: state scope and a visible-call counterexample

We replay the 115 public reference programs in the pinned τ-bench retail simulator. There is no LLM planner or user simulator in this experiment. Ninety-seven tasks have an initially pending order owned by the task user; 18 have none and receive no cover. Three seeds select orders, and cover is inserted before or after the prescribed program. Utility uses 97 task clusters, each with three repeated units per position. Policies include no cover, discard, shadow execution, read only, address self-write, item self-write, instruction-aware selection, other-account cancellation and blind cancellation. Destructive conditions are local negative controls, not proposed deployment policies.

TABLE S21. RETAIL STATE AND OBSERVER RESULTS. STATE COLUMNS USE 97 ELIGIBLE TASKS (291 UNITS PER POSITION); OBSERVER COLUMNS USE 49 HELD-OUT TASKS (147 UNITS). PERCENTAGES ARE TASK MEANS ACROSS THREE SEEDS. B/A DENOTES BEFORE/AFTER.

A. Distinct state scopes (reference execution as comparator)

Policy / position	DB eq.	Account eq.	Object eq.	Mutated
No cover B/A	100.00	100.00	100.00	0.00
Discard B/A	100.00	100.00	100.00	0.00
Shadow B/A	100.00	100.00	100.00	0.00
Read only B/A	100.00	100.00	100.00	0.00
Address self-write B/A	100.00	100.00	100.00	0.00
Item self-write B	40.55	40.55	70.10	59.45
Item self-write A	67.70	67.70	97.25	32.30
Instruction aware B	35.05	35.05	78.35	70.10
Instruction aware A	46.39	46.39	89.69	53.61
Other-account cancel B	0.00	100.00	100.00	100.00
Other-account cancel A	0.00	100.00	100.00	100.00
Blind cancel B	11.34	11.34	62.54	100.00
Blind cancel A	41.24	41.24	92.44	58.76

B. Legacy learned observer versus allowed-view rules

Policy / position	D_0 acc.	D_2 acc.	D_2 BA	Rule acc.
Majority (all conditions)	77.55	77.55	50.00	—
Instruction aware B	51.02	73.47	47.37	100.00
Instruction aware A	51.02	71.43	52.51	100.00
Other-account cancel B	22.45	71.43	46.05	100.00
Other-account cancel A	22.45	71.43	46.05	100.00
Blind cancel B	22.45	71.43	46.05	100.00
Blind cancel A	22.45	68.03	47.09	100.00

DB eq.: equality of orders/users against reference (products are read-only), the native reward's database criterion, not an output-response/authorization check. Account eq.: all orders owned by the active user plus that user's record. Object eq.: objects changed by the reference program, not all accessed objects. Mutated: cover itself changes state. Other-account calls yield 0% DB equality but 100% active-account equality. BA: balanced accuracy. Rule: remove the known first/last cover call, then inspect cancellation verbs; blind/other-account cancellation also admits the $N_{\text{cancel}} \geq 2$ rule. Both rules attain 100% acc./BA in their stated conditions. Other non-cancellation policies retain 100% verb inference. Full intervals, AUROC and confusion matrices appear in Supplement S9 and the JSON artifact.

Separate the outcome scopes. Table S21 reports reference database equality, full active-account equality and equality on objects *changed* by the reference program. The last scope does not include every read/accessed object. The database criterion corresponds to the state part of the native evaluator; the extension script does not evaluate generated response correctness and we do not relabel its state equality as an end-to-end native task reward. All-account equality now checks every order owned by the active user and that user's record, rather than only initially pending orders.

Address self-writes preserve the reference state in this

simulator. Item self-writes mutate state in 173/291 pre-insertion units (59.45%): the tool changes the status to `pending (item modified)` and appends a refund entry even when the item identifier and price are unchanged. The remaining 118/291 calls fail because the requested variant is unavailable, explaining why this mutation rate is below blind cancellation's 291/291 successful calls. This is a state-preservation counterexample from one API, not a prevalence estimate or a discovery about idempotence. Formally, a no-change requirement $f(s, a) = s$ differs from repeated-call idempotence $f(f(s, a), a) = f(s, a)$.

Instruction-aware cover uses only instruction text and the initial order table for selection, with address self-write as fallback. It preserves reference-changed objects in 78.35%/89.69% of before/after units, but preserves the full account and database in only 35.05%/46.39%. Cancelling an unmentioned order can therefore damage a bystander resource. The heuristic was developed using the available task collection; it has no held-out policy-development evaluation. Its 68/97 availability (70.10%) and the item-self-write object equality of 204/291 (70.10%) are separate counts that happen to yield the same percentage.

The other-account condition selects a pending order belonging to another simulated user. It is not a provisioned dedicated decoy account. It preserves the active user's complete account and reference-changed objects in these tasks, while changing the shared database in every eligible unit (0% database equality). Thus account isolation of an effect cannot be described as whole-database preservation or an authorized production defense.

Observer permissions and simple attacks. The secret label is whether the reference program *attempts* a cancellation, including a call that may fail after cover execution. The observer sees the ordered dispatched tool names, write counts and response lengths; failed calls remain visible. It sees no arguments, user/order identifiers or database state. We do not infer that a passive TLS observer has these permissions. A backend with plaintext state would instead need a different confidentiality target.

The fixed task split contains 48 training tasks (7 positive, 41 negative) and 49 test tasks (11 positive, 38 negative); seeds stay within tasks. Always predicting no cancellation gives 77.55% test accuracy and 50% balanced accuracy. The legacy D_2 linear classifier uses cancellation presence, write count and write-response bytes. Its 68.03–73.47% accuracy under cancellation cover is below the majority baseline, and its balanced accuracy ranges from 46.05% to 52.51%. An increase over D_0 is therefore insufficient evidence for its claimed privacy interpretation.

For blind and other-account cancellation, exactly one extra cancellation is dispatched in every eligible session. The permitted rule $\widehat{y} = \mathbf{1}[N_{\text{cancel}} \geq 2]$ recovers the reference label at 100% accuracy and balanced accuracy on all 49 test tasks and three seeds, before and after insertion. For any policy with one live cover call at a known first/last position, removing that call and testing for a remaining cancellation also gives 100%. It needs no identifiers, no response content and no training labels. These are measured finite-sample results and a structural property of the fixed insertion design, not proof of general attack performance. Identifier filtering is unnecessary in this view and is not claimed as an evaluated attack.

Terminal equality does not establish path properties. Blind pre-insertion cancellation leaves the final database equal to reference in 33/291 units across 12 tasks, even though it preempts the later reference cancellation. The exact per-task counts are in Supplement S9. After insertion, 120/291 cover calls are rejected because reference execution already changed the order's status, and those rejections explain the 41.24% equality rate. Rejection is not successful benign cover. In the 82 tasks with no reference execution error (246 units), before/after equality is 24/246 (9.76%) and 94/246 (38.21%). These results illustrate the limits of endpoint scoring; consent and authorization were not measured.

TABLE S22. SOURCE-LOCAL FINDINGS, CROSS-STUDY SYNTHESIS AND THE RESULTING MEASUREMENT OBLIGATION. THESE ARE REUSE JUDGMENTS, NOT ACCUSATIONS THAT THE SOURCE AUTHORS CLAIMED END-TO-END PROTECTION.

Check	Source-local evidence and locator	Synthesis added by record separation	Changed judgment / controlled evidence
1. Receiver	NetEcho IV-C–E/Fig.5: frontend re-emission; SplitDecoy 5.1–5.3: backward gradient support.	Protected upstream and forward views do not cover newly exposed downstream or backward inputs.	Bind the final observer or prove the postprocessing condition. A buffered cap (2.18×) does not price a token stream: per-event padding leaves verbosity at 100% (E5).
2. Execution	WRA 8.1/Table8: read-only query decoys; AHB 5.2.3/Table1/E.3: live GUI humanization.	Objectives, live action semantics and protected state scopes differ, so their utility outcomes are not contradictory.	Evaluate task objects, full account and shared database separately. Same-item writes change status; randomized and decoy cover stay at 77–100% BA; dispatch cover leaks on the tool link (E6, E7).
3. Adaptation and cost	UI Traces 6.1: delayed-trace refit; Dangol 9.7: capture versus live protocol; Early Bird's KV-prefix versus semantic-cache mechanisms.	A paired attack contrast requires the same defended tests; costs from another mode or mechanism cannot price the claimed protection.	Keep source-local comparisons and mark missing bridges unresolved. D_0 reads 50% where D_2 reads 100% under batching and probabilistic cover (E5, E6).

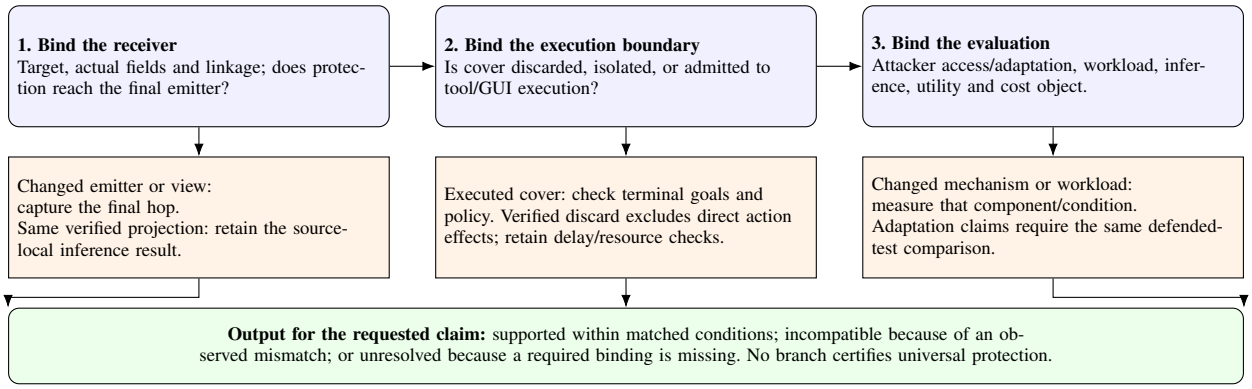


Figure 1. Operational evidence-reuse checks.

S12. Second-round transfer tests (E5–E7)

E5 and individual-attacker E6 results below retain the R2 measurements. Current E6/E7 suite summaries use post-review reanalysis in `experiments/sp2027-r3/`; historical registration and outputs remain in `experiments/sp2027-r2/`. A suite maximum is recomputed in each of 2,000 common task-bootstrap draws. Intervals condition on fixed out-of-fold predictions and omit retraining/search uncertainty. Meets: upper BA bound < 0.60 and maximum point AUROC < 0.65; Above: lower BA bound > 0.60; otherwise inconclusive. This is a finite-suite criterion, not a security guarantee.

TABLE S23. E5 PER ATTACKER WITH 95% TOPIC-BOOTSTRAP INTERVALS (12 TEST TOPICS). WIRE: MEAN APPLICATION-RECORD BYTES INCL. 5-BYTE HEADERS. AMB.: MEAN SIZE OF THE NEAREST-SEQUENCE TIE SET.

TLS	Target	Framing	Wire	D_0 total	D_2 total	D_2 count	D_2 sequence	ID	Amb.
1.2	T1	Buffered, unpadded	721	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	87.5	1.3
1.2	T1	Buffered, training cap	1570	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	2.1	48.0
1.2	T1	Stream, per token	15038	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0	1.0
1.2	T1	+ random event pad	19194	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	62.3	1.0
1.2	T1	+ constant event size	18255	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	58.3	2.3
1.2	T1	8-token batches, const.	3040	50.0 [50.0,50.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	20.8	9.4
1.2	T1	Const. events + count cap	36092	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	6.3	44.1
1.2	T3	Buffered, unpadded	456	91.7 [79.2,100.0]	91.7 [79.2,100.0]	91.7 [79.2,100.0]	91.7 [79.2,100.0]	91.7	1.2
1.2	T3	Buffered, training cap	635	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	4.2	44.2
1.2	T3	Stream, per token	9713	87.5 [75.0,100.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	100.0	1.0
1.2	T3	+ random event pad	12398	71.2 [58.3,83.8]	87.9 [75.8,99.2]	87.5 [75.0,100.0]	87.9 [76.2,98.3]	72.5	1.0
1.2	T3	+ constant event size	11803	79.2 [66.7,91.7]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	70.8	1.8
1.2	T3	8-token batches, const.	2023	50.0 [50.0,50.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	18.8	8.4
1.2	T3	Const. events + count cap	15411	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	12.5	38.6
1.3	T1	Buffered, unpadded	686	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	87.5	1.3
1.3	T1	Buffered, training cap	1500	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	2.1	48.0
1.3	T1	Stream, per token	14294	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0	1.0
1.3	T1	+ random event pad	18450	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	62.3	1.0
1.3	T1	+ constant event size	17511	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	58.3	2.3
1.3	T1	8-token batches, const.	2938	50.0 [50.0,50.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	20.8	9.4
1.3	T1	Const. events + count cap	34626	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	6.3	44.1
1.3	T3	Buffered, unpadded	433	91.7 [79.2,100.0]	91.7 [79.2,100.0]	91.7 [79.2,100.0]	83.3 [70.8,95.8]	91.7	1.2
1.3	T3	Buffered, training cap	606	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	4.2	44.2
1.3	T3	Stream, per token	9230	87.5 [75.0,100.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	100.0	1.0
1.3	T3	+ random event pad	11915	68.8 [56.7,81.7]	87.9 [75.8,99.2]	87.5 [75.0,100.0]	87.5 [75.4,98.3]	72.5	1.0
1.3	T3	+ constant event size	11321	79.2 [66.7,91.7]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	70.8	1.8
1.3	T3	8-token batches, const.	1953	50.0 [50.0,50.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	87.5 [75.0,100.0]	18.8	8.4
1.3	T3	Const. events + count cap	14782	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	50.0 [50.0,50.0]	12.5	38.6

TABLE S24. E6 DISPATCH VIEW: BALANCED ACCURACY (%) WITH TASK-CLUSTER BOOTSTRAP INTERVALS (115 TASKS, 3 SEEDS PER TASK), BEST AUROC, ACTIVE-ACCOUNT EQUALITY AND COVER-CALL ERROR RATE.

Target	Cover	D_0 GB	D_2 LR	D_2 GB	Count rule	AUROC	Acct. eq.	Cover err.
cancel	No cover	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	1.00	100.0	0.0
cancel	Cancel, p=0.5	74.6 [71.6,77.4]	84.5 [75.9,92.1]	86.1 [78.4,93.9]	83.3 [75.0,91.2]	0.95	100.0	0.0
cancel	Cancel, p=1	50.0 [50.0,50.0]	97.1 [92.4,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	1.00	100.0	0.0
cancel	Cancel, p=1, live execution	50.0 [50.0,50.0]	98.1 [95.4,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	1.00	100.0	0.0
cancel	Cancel count $\rightarrow$ 2	50.0 [50.0,50.0]	76.9 [64.8,88.0]	69.4 [57.6,81.7]	50.0 [50.0,50.0]	0.85	100.0	2.8
cancel	One decoy program	91.9 [89.8,94.2]	68.1 [60.2,76.7]	66.0 [60.1,72.2]	91.9 [89.8,94.2]	0.92	100.0	3.0
cancel	Two decoy programs	83.5 [80.5,86.5]	55.9 [48.4,63.6]	57.0 [49.9,64.1]	83.5 [80.5,86.5]	0.86	100.0	4.3
cancel	Shape equalization	50.0 [50.0,50.0]	44.8 [37.6,52.4]	49.2 [44.0,54.8]	50.0 [50.0,50.0]	0.50	100.0	5.7
return	No cover	100.0 [100.0,100.0]	97.8 [94.0,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	1.00	100.0	0.0
return	Cancel, p=0.5	100.0 [100.0,100.0]	98.0 [94.3,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	1.00	100.0	0.0
return	Cancel, p=1	100.0 [100.0,100.0]	97.5 [94.8,99.6]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	1.00	100.0	0.0
return	Cancel, p=1, live execution	100.0 [100.0,100.0]	98.0 [94.3,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	1.00	100.0	0.0
return	Cancel count $\rightarrow$ 2	100.0 [100.0,100.0]	98.0 [95.4,100.0]	100.0 [100.0,100.0]	100.0 [100.0,100.0]	1.00	100.0	2.8
return	One decoy program	85.9 [82.9,89.0]	73.2 [66.6,79.5]	71.1 [65.5,76.8]	85.9 [82.9,89.0]	0.88	100.0	3.0
return	Two decoy programs	78.1 [75.0,81.2]	70.6 [64.1,77.3]	69.1 [62.9,75.0]	78.1 [75.0,81.2]	0.85	100.0	4.3
return	Shape equalization	50.0 [50.0,50.0]	54.5 [48.4,60.3]	50.3 [45.4,55.5]	50.0 [50.0,50.0]	0.53	100.0	5.7

TABLE S25. CORRECTED E6 SUITE MAXIMA AND TASK-BOOTSTRAP INTERVALS. INDIVIDUAL-ATTACKER RESULTS ABOVE ARE UNCHANGED. AUROC IS ALSO MAXIMIZED OVER THE EVALUATED SUITE.

Target	Cover	Max BA [95% CI]	Max AUROC	Criterion
cancel	No cover	100.0 [100.0,100.0]	1.00	Above
cancel	Cancel, p=0.5	86.1 [80.9,94.2]	0.95	Above
cancel	Cancel, p=1	100.0 [100.0,100.0]	1.00	Above
cancel	Cancel, p=1, live	100.0 [100.0,100.0]	1.00	Above
cancel	Cancel count $\rightarrow$ 2	76.9 [67.0,88.1]	0.85	Above
cancel	One decoy	91.9 [89.8,94.2]	0.92	Above
cancel	Two decoys	83.5 [80.5,86.5]	0.86	Above
cancel	Shape	50.0 [50.0,54.8]	0.50	Meets
return	No cover	100.0 [100.0,100.0]	1.00	Above
return	Cancel, p=0.5	100.0 [100.0,100.0]	1.00	Above
return	Cancel, p=1	100.0 [100.0,100.0]	1.00	Above
return	Cancel, p=1, live	100.0 [100.0,100.0]	1.00	Above
return	Cancel count $\rightarrow$ 2	100.0 [100.0,100.0]	1.00	Above
return	One decoy	85.9 [82.9,89.0]	0.88	Above
return	Two decoys	78.1 [75.0,81.2]	0.85	Above
return	Shape	54.5 [50.0,60.4]	0.53	Inconcl.

TABLE S26. CORRECTED E7 SUITE MAXIMA (4,140 SAVED TLS SESSIONS). DECODER: HELD-OUT REQUEST/RESPONSE SIZE-TO-TOOL ACCURACY; IT DIFFERS FROM REQUEST-ONLY IDENTIFICATION. COSTS USE RAW/NO-COVER AS DENOMINATOR. DEFENSE CALIBRATION USES THE FULL PUBLIC TASK POPULATION.

Target	Framing	Cover	Max BA [95% CI]	AUROC	Criterion	Decoder	Bytes$\times$	Calls$\times$
cancel	raw	No cover	100.0 [100.0,100.0]	1.00	Above	92.8	1.00	1.00
cancel	raw	Cancel count $\rightarrow$ 2	72.7 [63.4,84.5]	0.81	Above	94.7	1.38	1.35
cancel	raw	One decoy	91.9 [89.8,94.2]	0.92	Above	99.7	2.03	2.01
cancel	raw	Shape	51.2 [50.0,58.8]	0.54	Meets	100.0	7.56	7.71
cancel	bucket1024	No cover	60.1 [50.0,73.3]	0.67	Inconcl.	35.2	2.42	1.00
cancel	bucket1024	Cancel count $\rightarrow$ 2	67.5 [58.5,78.9]	0.70	Inconcl.	33.5	3.25	1.35
cancel	bucket1024	One decoy	60.0 [53.4,68.7]	0.59	Inconcl.	35.5	4.90	2.01
cancel	bucket1024	Shape	55.3 [52.2,63.6]	0.59	Inconcl.	26.5	18.56	7.71
cancel	const	No cover	59.7 [50.0,72.3]	0.62	Inconcl.	30.4	3.41	1.00
cancel	const	Cancel count $\rightarrow$ 2	64.9 [54.9,76.7]	0.66	Inconcl.	30.0	4.61	1.35
cancel	const	One decoy	59.7 [52.6,67.4]	0.59	Inconcl.	30.5	6.87	2.01
cancel	const	Shape	54.4 [50.0,62.5]	0.50	Inconcl.	21.6	26.32	7.71
return	raw	No cover	95.7 [90.7,99.4]	0.96	Above	93.8	1.00	1.00
return	raw	Cancel count $\rightarrow$ 2	95.7 [90.6,99.4]	0.96	Above	95.4	1.38	1.35
return	raw	One decoy	85.9 [82.9,89.0]	0.88	Above	100.0	2.03	2.01
return	raw	Shape	60.5 [54.9,66.4]	0.61	Inconcl.	100.0	7.56	7.71
return	bucket1024	No cover	73.9 [66.6,83.1]	0.76	Above	35.2	2.42	1.00
return	bucket1024	Cancel count $\rightarrow$ 2	66.0 [59.8,75.8]	0.70	Inconcl.	33.5	3.25	1.35
return	bucket1024	One decoy	59.1 [54.1,66.8]	0.60	Inconcl.	35.5	4.90	2.01
return	bucket1024	Shape	50.6 [50.0,57.3]	0.52	Meets	26.5	18.56	7.71
return	const	No cover	53.6 [50.0,63.9]	0.55	Inconcl.	30.4	3.41	1.00
return	const	Cancel count $\rightarrow$ 2	59.1 [52.3,68.6]	0.61	Inconcl.	30.0	4.61	1.35
return	const	One decoy	56.8 [51.4,63.6]	0.57	Inconcl.	30.5	6.87	2.01
return	const	Shape	50.0 [50.0,55.1]	0.50	Meets	21.6	26.32	7.71

TABLE S27. REQUEST-ONLY TOOL IDENTIFICATION: TRAINING-FOLD SIZE TABLE, UNSEEN-SIZE FALLBACK TO THE TRAINING MAJORITY. EACH OF 582 REFERENCE CALLS IS TESTED ONCE. THE 95.7% RAW IN-SAMPLE PURITY IS EXCLUDED FROM HELD-OUT ACCURACY.

Task stratification	Framing	Held-out accuracy (%)	Training-majority baseline (%)
cancel	raw	89.9	29.4
cancel	bucket1024	29.4	29.4
cancel	const	29.4	29.4
return	raw	91.4	29.4
return	bucket1024	29.4	29.4
return	const	29.4	29.4

Training-only tool-count maxima overflow on five distinct held-out tasks (22, 28, 40, 67, 87) under either target split. This is a feasibility audit, not a rerun of defense efficacy. Full fold membership and overflow records are released. No tested network configuration meets the criterion for both targets.

TABLE S28. E5: THE SAME 48 SAVED RESPONSES EMITTED BY A BUFFERED OR A TOKEN-STREAMING FRONTEND (TLS 1.3 LOOPBACK CAPTURES). BYTES ARE RELATIVE TO THE UNPADDED EMITTER OF THE SAME KIND. TTFB/DONE: FIRST AND LAST DELIVERED BYTE IN TOKEN INTERVALS (ACCOUNTING PROXY). ACCURACIES ON 24 HELD-OUT OUTPUTS (12 TOPICS); ID: CLOSED-WORLD TOP-1 IDENTIFICATION AMONG 48 RESPONSES (CHANCE 2.1%).

				T1 acc.		T3	
Emitter / framing	Bytes×	TTFB	Done	D_0	D_2	D_2	ID
Buffered, unpadded	1.00	105	105	100.0	100.0	91.7	87.5
Buffered, training cap	2.19	105	105	50.0	50.0	50.0	2.1
Stream, per token	1.00	1	105	100.0	100.0	87.5	100.0
+ random event pad	1.29	1	105	100.0	100.0	87.9	62.3
+ constant event size	1.23	1	105	100.0	100.0	87.5	58.3
8-token batches, const.	0.21	8	105	50.0	100.0	87.5	20.8
Const. events + count cap	2.42	1	208	50.0	50.0	50.0	6.3

D_2 is the best of total-byte threshold, record-count threshold and a logistic sequence model, each fitted on defended training outputs. Random padding: 10 socket-measured draws plus an independent profiling run. TLS 1.2 gives identical accuracies and byte ratios within 0.03. Tokens: o200k_base proxy. Count cap: 208 (T1) and 88 (T3) events fitted on training outputs; 2/24 and 5/24 test outputs overflow it.

TABLE S29. E6/E7: RANDOMIZED COVER ON 115 RETAIL REFERENCE PROGRAMS × 3 SEEDS. ENTRIES ARE THE BEST HELD-OUT BALANCED ACCURACY (%) OVER ALL ATTACKERS FOR THE CANCELLATION (C, 18 POSITIVE TASKS) AND RETURN (R, 32) TARGETS. DISPATCH VIEW: TOOL NAMES AND RESPONSE LENGTHS. NETWORK VIEW: TLS 1.2 RECORD LENGTHS ON THE AGENT–TOOL LINK UNDER RAW, 1024-BYTE BUCKET OR CONSTANT FRAMING.

	Dispatch view (E6)				Network view (E7)						
					Raw		Bucket		Constant		
Cover policy	C	R	Calls+	DB eq.	C	R	C	R	C	R	Bytes×
No cover	100.0	100.0	0.00	100.0	100.0	95.7	60.1	73.9	59.7	53.6	3.4
Cancel, p=0.5	86.1	100.0	0.10	100.0	—	—	—	—	—	—	—
Cancel, p=1	100.0	100.0	0.20	100.0	—	—	—	—	—	—	—
Cancel, p=1, live execution	100.0	100.0	0.20	0.0	—	—	—	—	—	—	—
Cancel count → 2	76.9	100.0	0.35	100.0	72.7	95.7	67.5	66.0	64.9	59.1	4.6
One decoy program	91.9	85.9	1.01	100.0	91.9	85.9	60.0	59.1	59.7	56.8	6.9
Two decoy programs	83.5	78.1	1.98	100.0	—	—	—	—	—	—	—
Shape equalization	50.0	54.5	6.71	100.0	51.2	60.5	55.3	50.6	54.4	50.0	26.3

Calls+: cover calls per real call. DB eq.: orders/users equal to reference. Cover is sampled from other users' reference programs and placed at random positions; all policies except "live" run cover in a per-session sandbox. Attackers: D_0/D_2 gradient boosting and logistic regression, a count rule and, on the network view, a size-profiling decoder; task-grouped 5-fold cross-validation. Bytes×: constant framing relative to raw framing without cover. Suite-maximum intervals: Supplement S12. Defense templates/pools/caps use all 115 tasks; attacker fitting uses training folds. No network configuration meets the criterion for both targets.

TABLE S30. CURRENT REASSESSMENT OF THE SEVEN AUTHOR-WRITTEN JUDGMENTS RECORDED BEFORE E5–E7. HISTORICAL OUTCOME LABELS ARE PRESERVED IN THE ARTIFACT, NOT COUNTED AS AUDIT ACCURACY.

ID	Claim under examination	Current evidence and qualification
P1	Buffered cost transfers to a stream	Emitter mismatch supported: per-event padding leaves T1 at 100%; constant-shape stream costs 2.42×.
P2	Low D_0 establishes protection	Counterexample supported: batching and cancellation cover give D_0 50%, D_2 100%.
P3	Per-event padding prevents identification	Counterexample supported: top-1 58–62% versus 2.1% chance in the closed candidate set.
P4	Equal cancellation count hides cancellation	Residual leakage supported: count rule 50%; maximum BA 76.9%.
P5	Dispatch protection transfers to the tool link	Protected-source premise unmet: count/decoy policies already leak at dispatch. Network BA 72.7–95.7% shows persistence.
P6	Framing alone protects intent	Bucketed return remains above threshold; remaining framing-only cells inconclusive.
P7	Joint cover/framing protects both targets	Not established: no configuration meets both target criteria. Joint constant framing: 50.0–54.4% BA; 26.3× bytes, 7.7× calls.

S13. Stand-in planner pipeline (E8) and online cover

`experiments/sp2027-r2/agent/` runs tool-calling agents and a user simulator on the 115 retail tasks, applies *online* cover (decided before each real call from the history only, with a final flush) and replays every session over TLS. Because sandboxed cover and framing never reach the agent, one trajectory per task and seed serves all policies. No LLM was available for this revision; the rows below use stand-in planners and exercise the pipeline. The scripted planner replays reference programs (native reward 1.00); the noisy planner inserts extra reads before each reference call with probability 0.3 (7.2 versus 5.1 real calls per session). These are synthetic results, not agent measurements.

TABLE S31. E8 DIAGNOSTICS REPORT POINT ESTIMATES ONLY; NO SUITE-MAXIMUM INTERVAL WAS RECOMPUTED FOR ONLINE RUNS. CURRENT OFFLINE E6/E7 INTERVALS ARE IN S12. BEST HELD-OUT BALANCED ACCURACY (%) POINT ESTIMATES FOR OFFLINE VERSUS ONLINE COVER. CONST.: CONSTANT FRAMING ON THE AGENT–TOOL LINK. CALLS+: COVER CALLS PER REAL CALL. OVERFLOW: SESSIONS WHOSE REAL CALLS EXCEED THE SHAPE TEMPLATE OR THE CANCELLATION TARGET.

Planner / cover timing	Cover	Dispatch C	Dispatch R	Const. C	Const. R	Calls+	Overflow (%)
Offline (E6/E7)	cancel const2	76.9	100.0	64.9	59.1	0.35	0.0
Offline (E6/E7)	decoy1	91.9	85.9	59.7	56.8	1.01	0.0
Offline (E6/E7)	shape eq	50.0	54.5	54.4	50.0	6.71	0.0
Online, scripted	cancel const2	79.6	100.0	63.0	59.1	0.36	3.8
Online, scripted	decoy1	92.3	88.6	51.2	56.3	0.98	0.0
Online, scripted	shape eq	63.8	55.1	57.5	55.9	6.73	0.0
Online, noisy scripted	cancel const2	79.1	100.0	66.0	60.0	0.25	4.1
Online, noisy scripted	decoy1	92.3	88.6	51.8	56.2	0.69	0.0
Online, noisy scripted	shape eq	55.9	53.9	60.5	56.2	4.52	20.9

The preceding E8 rows and availability statement describe the historical stand-in study. S14 reports the subsequent real-model collection and training-only calibration.

S14. Complete Real-Agent Evaluation with Training-Only Calibration (E9)

S14.1. Cohort and execution

All 690 episodes in the frozen protocol are complete: 115 tasks, two prompt arms and three repetition identifiers. The fixed split assigns 69 tasks from 29 customers to training (414 episodes) and 46 tasks from 24 disjoint customers to testing (276 episodes). Every test task has all six prompt/repetition cells, and all failures and handoffs remain in the analysis. Training finished before defense calibration was frozen and test inference began. This section updates the earlier 663-episode publication snapshot to the full cohort; the immutable snapshot and the previous manuscript remain available in the archived release. No model configuration, prompt, split or defense parameter changed for this completion.

GPT-6.1 Sol with low reasoning effort supplies the planner and a separate user simulator through Codex app-server. The planner receives the retail policy, public tool schemas, interaction history and actual sandbox observations, and emits one structured action at each step. Reference actions and reward targets are used only for scoring. Each decision has a fresh ephemeral context without host tools or a host environment. The harness executes decisions in the pinned public retail simulator, with a 30-step limit. Repetition identifiers are cache namespaces; provider random seeds, temperature and an immutable weight snapshot are unavailable.

S14.2. Prompts and defense calibration

The baseline prompt is the retail policy. The rules prompt adds one customer-details read after identification and before order access, an order read before each order write, and avoidance of unnecessary repeated reads. It preserves authorization and task requirements and prohibits unrelated operations for hiding intent. Both arms are evaluated under raw transport, padding only, online shape cover only, and joint shape cover with constant framing.

Only training trajectories from both prompts determine shared per-tool count maxima, donor pools and size caps. Donor calls exclude the active customer. Request caps are the largest training request plus four bytes rounded to a 256-byte multiple; response caps use the training 95th percentile plus four bytes rounded to a 1024-byte multiple, including training donor responses. Frozen caps are 1,024 and 3,072 bytes. The online dispatcher uses only past real-call counts, inserts a shadow call with probability 0.6 when a deficit exists and flushes remaining deficits at episode end. Overflow calls pass through; oversize messages expand to a 1024-byte bucket. Test data never recalibrate these parameters.

Shadow calls execute on isolated cloned state; their observations are suppressed and real calls/responses pass through unchanged. For deterministic tools with no timing-dependent side effects, projecting away cover gives the same planner-visible history under all program conditions. The replay therefore isolates transport intervention on the same realized actions, with reward preserved by the isolation invariant. All 690 real-call projections agree across the 2,760 saved sessions. Online means scheduling from past calls; interactive wall-clock delay and shared-resource feedback remain outside this replay measurement. A byte-transparent relay captures actual serialized requests and sandbox responses over local TLS 1.2.

S14.3. Attackers and uncertainty

The frozen suite includes a training-majority baseline, clean-trained gradient boosting, defense-trained logistic regression and gradient boosting, a dispatch count rule, and network size-decoding learners and rules. All fitted parameters, thresholds and decoder tables use training data only. Targets are cancellation and return intent from the reference task; native task reward is scored separately using final database equality and required response strings. Points weight tasks equally. Intervals use 2,000 common customer-cluster bootstrap draws, retaining each customer's tasks and repetitions together and recomputing the suite maximum in each draw. Intervals condition on fitted attackers. Prompt contrasts pair task/repetition identifiers, without assuming identical conversations or shared model randomness.

S14.4. Utility, overhead and prompt contrasts

Cancellation has 6 positive tasks from 5 customers; return has 11 positive tasks from 7 customers. Single-class draws excluded from 2,000 replicates: 6 for cancellation and 0 for return.

Baseline native reward is 110/138 (79.7%); rules reward is 105/138 (76.1%). All 24 baseline and 23 rules handoffs remain in the denominator. Joint-program cover-call error rates are 5.3% and 5.0%, respectively; all shape conditions have zero underfill. Overflow rates expose the resource demands of retaining real calls under the frozen training template.

TABLE S32. E9 DISPATCH VIEW: MAXIMUM BA (%) WITH 95% INTERVAL AND MAXIMUM AUROC ACROSS THE FIXED SUITE, ON 138 TEST EPISODES PER PROMPT. BA AND AUROC ARE SEPARATE SUITE MAXIMA.

Prompt	Condition	Cancel BA [CI]	AUC	Return BA [CI]	AUC
baseline	Raw	97.2 [90.0, 100.0]	0.997	97.0 [88.9, 100.0]	0.972
baseline	Padding only	97.2 [90.0, 100.0]	0.997	97.0 [88.9, 100.0]	0.972
baseline	Shape only	55.6 [50.0, 63.0]	0.556	57.7 [50.0, 67.4]	0.620
baseline	Joint	55.6 [50.0, 63.0]	0.556	57.7 [50.0, 67.4]	0.620
rules	Raw	97.2 [90.0, 100.0]	0.972	97.0 [88.9, 100.0]	0.970
rules	Padding only	97.2 [90.0, 100.0]	0.972	97.0 [88.9, 100.0]	0.970
rules	Shape only	56.5 [50.0, 68.3]	0.655	64.7 [54.3, 77.4]	0.693
rules	Joint	56.5 [50.0, 68.3]	0.655	64.7 [54.3, 77.4]	0.693

TABLE S33. E9 NETWORK VIEW: MAXIMUM BA (%) WITH 95% INTERVAL AND MAXIMUM AUROC ACROSS THE FIXED SUITE, ON 138 TEST EPISODES PER PROMPT. BA AND AUROC ARE SEPARATE SUITE MAXIMA.

Prompt	Condition	Cancel BA [CI]	AUC	Return BA [CI]	AUC
baseline	Raw	97.2 [90.0, 100.0]	0.993	94.0 [85.8, 100.0]	0.948
baseline	Padding only	66.1 [50.0, 83.9]	0.634	55.5 [50.0, 73.0]	0.567
baseline	Shape only	60.8 [55.4, 71.3]	0.657	51.9 [50.5, 57.8]	0.579
baseline	Joint	50.0 [50.0, 60.6]	0.500	50.0 [50.0, 65.5]	0.500
rules	Raw	97.2 [90.0, 100.0]	0.978	93.5 [85.0, 99.5]	0.937
rules	Padding only	67.6 [50.0, 85.0]	0.636	57.9 [50.0, 78.9]	0.548
rules	Shape only	52.8 [50.0, 65.5]	0.588	53.2 [50.4, 63.9]	0.544
rules	Joint	50.0 [50.0, 65.1]	0.500	60.6 [50.0, 73.3]	0.641

S14.5. Interpretation and reproducibility

The completed cohort shows strong raw-trace intent inference and a large reduction under joint transport intervention. Rules improve mechanical adherence to the two read-order rules while leaving network BA above 93%. Baseline joint shaping reaches 50% BA for both targets, reducing the raw scores by 47.2 and 44.0 points. The finite-suite criterion requires the BA interval upper bound below 60% and point AUROC below 0.65. Baseline shape-only return meets it for that target; no condition meets it for both targets. Baseline joint upper bounds are 60.6%/65.5%, leaving the absolute-threshold conclusion inconclusive. The full-sample suite maximum equals the 50% majority baseline; several fitted attacks make heterogeneous predictions whose customer-resampled BA can exceed 50%. With only five cancellation-positive and seven return-positive customers, reselecting the maximum within each draw yields wider intervals. This mechanism does not imply a universal 10-point resolution floor or prove absence of residual population signal. The paired reduction intervals above exclude zero. Supplement S16 adds a task-permutation reference; the primary intervals remain clustered by customer. Rules-plus-joint return BA is 60.6%; its rules-minus-baseline contrast includes zero, as does the reward contrast.

The evidence covers one public retail environment and one mutable model alias, which also supplies user simulation. All 115 tasks were available during prior project development. The formal customer holdout therefore provides internal evaluation; external independent replication, broader model/environment transfer, provider-traffic measurement and interactive feedback effects remain follow-up questions.

The full cohort contains 11,586 model decisions, 2,760 TLS sessions and 176 attack-prediction sets. The frozen verifier passed after training-only recalibration, sandbox state replay, cover reconstruction, TLS accounting, metric/interval recomputation and request-provenance checks. Additional checks bind all agent actions and dialogue to actual model outputs and re-score required response strings. Offline attacker refitting reproduces the saved results and predictions exactly.

`experiments/sp2027-r4-full-20261023/` supplies full results, predictions and captures, plus the 27 new episodes and model records needed to extend the immutable 663-run snapshot. The three separately reported snapshot test episodes join the 27 new ones to expand the primary test from 246 to 276. `reproduce.py --reanalyze` reconstructs the full cohort from the base archives and delta, reruns analysis and executes the original frozen verifier without invoking a model. `verify_extension.py` checks decision alignment and provenance. The earlier snapshot, historical R2/R3 studies and pilot archives retain their original denominators and bytes.

TABLE S34. HELD-OUT UTILITY AND OVERHEAD. REWARD AND SESSION OVERFLOW RATES ARE PERCENTAGES; RATIOS USE RAW TRANSPORT WITHIN THE SAME PROMPT.

Prompt	Condition	Reward [CI]	Calls×	Bytes×	Count overflow	Cap overflow	Underfill
baseline	Raw	79.7 [69.0, 89.7]	1.00	1.00	0.0	0.0	0.0
baseline	Padding only	79.7 [69.0, 89.7]	1.00	4.02	0.0	6.5	0.0
baseline	Shape only	79.7 [69.0, 89.7]	3.83	4.16	38.4	0.0	0.0
baseline	Joint	79.7 [69.0, 89.7]	3.83	15.39	38.4	39.9	0.0
rules	Raw	76.1 [63.3, 88.7]	1.00	1.00	0.0	0.0	0.0
rules	Padding only	76.1 [63.3, 88.7]	1.00	4.04	0.0	6.5	0.0
rules	Shape only	76.1 [63.3, 88.7]	3.88	4.27	44.2	0.0	0.0
rules	Joint	76.1 [63.3, 88.7]	3.88	15.72	44.2	38.4	0.0

TABLE S35. MECHANICAL ADHERENCE TO THE TWO PROMPT-SPECIFIED READ-ORDER RULES (% WITH CUSTOMER-CLUSTER INTERVAL).

Prompt	One successful customer read first	Order read before write	Both
baseline	86.2 [74.5, 97.3]	100.0 [100.0, 100.0]	86.2 [74.5, 97.3]
rules	97.1 [91.7, 100.0]	100.0 [100.0, 100.0]	97.1 [91.7, 100.0]

TABLE S36. PAIRED PROMPT CONTRASTS USING COMMON CUSTOMER-BOOTSTRAP DRAWS.

Rules minus baseline	Difference [95% CI], pp
Task reward	-3.6 [-9.5, 1.4]
Raw network, cancel BA	0.0 [-8.3, 8.3]
Raw network, return BA	-0.5 [-2.6, 0.0]
Joint network, cancel BA	0.0 [-1.7, 7.4]
Joint network, return BA	10.6 [-13.3, 22.8]

TABLE S37. POST HOC PAIRED PROGRAM CONTRASTS. THE FIXED ATTACK SUITE IS RE-MAXIMIZED UNDER EACH CONDITION WITHIN EACH COMMON CUSTOMER-BOOTSTRAP DRAW. ORIGINAL PREDICTIONS AND THE PREDECLARED ABSOLUTE CRITERION ARE UNCHANGED.

Prompt	Target	Joint minus raw BA [95% CI], pp
baseline	cancel	-47.2 [-50.0, -33.3]
baseline	return	-44.0 [-48.9, -27.8]
rules	cancel	-47.2 [-50.0, -26.0]
rules	return	-33.0 [-44.3, -18.9]

S15. Cross-Fitting, Operating Points and Leakage Localization (E10–E12)

This section preserves the original-suite results. The expanded 11-attack E10/E12 analyses in S21 supersede those attack maxima in the main paper; the original predictions remain available for direct comparison. E10 retrospectively reuses the E9 defense design and attack suite under customer-grouped stratified 5-fold cross-fitting over all 690 completed episodes (115 tasks, 53 customers; 14 cancellation-positive and 18 return-positive customers). In each fold, the count template, donor pools, size caps and all attackers are fit only on the other folds. Out-of-fold predictions are pooled, and intervals use 2,000 customer-cluster bootstrap draws with the suite maximum recomputed per draw. The six analyzed configurations are specified in `r5lib.py`. E11 selects among them using inner customer folds of the 69 frozen training tasks only (rule in `e11_select.py`), then evaluates it on the 46 original test tasks. These analyses reuse an already analyzed cohort; only the original E9 protocol was frozen before test collection. E12 ablates undefended traces to locate the signal.

Records are computed with an analytic TLS 1.2 AES-GCM model (fragment plus 24 bytes, 16,384-byte fragments). The model is validated against all 2,760 socket captures of the completed cohort and 120 fresh loopback captures of new cross-fitted sessions. `verify_r5.py` also checks byte-identical reproduction of the E9 calibration, all sessions, and all 32 result cells of both the 660-run primary grid of the 663-run snapshot and the completed cohort. The same analysis on the 660-episode snapshot grid is retained in `snapshot-660/`; there the bucket point gives 50.6%/51.6% with upper bounds of 59.2%/58.3%. Cross-fitted templates use more training customers than E9 and are therefore larger, which raises call overhead (5.0× versus 3.8×). A quantile template is a natural next operating point.

TABLE S38. E10 CROSS-FITTED NETWORK VIEW: ALL CONFIGURATIONS AND PROMPTS. CORE = R4 SUITE; EXT. ADDS RANDOM FOREST, k-NN AND A FOREST SIZE DECODER. SA-AUC: SIGN-AGNOSTIC AUROC $\max(A, 1 - A)$.

Configuration	Prompt	Cancel core [CI]	Ext.	SA-AUC	Return core [CI]	Ext.	SA-AUC	Bytes×/Calls×
R4 point (all tools, constant) (padding only)	baseline	68.1 [55.4, 78.9]	68.1	0.63	67.1 [57.2, 77.0]	67.1	0.63	3.9/1.0
R4 point (all tools, constant)	baseline	60.5 [51.5, 70.5]	60.5	0.59	52.1 [50.0, 60.3]	52.7	0.63	19.8/5.0
R4 point (all tools, constant) (padding only)	rules	64.4 [54.6, 75.8]	64.4	0.60	63.3 [52.7, 74.2]	63.3	0.63	3.9/1.0
R4 point (all tools, constant)	rules	58.8 [50.0, 68.6]	58.8	0.57	59.4 [50.5, 68.0]	59.4	0.58	19.9/5.0
All tools, 1024-B buckets	baseline	52.2 [50.0, 60.5]	52.2	0.54	52.7 [50.0, 60.2]	52.7	0.61	12.4/5.0
All tools, 1024-B buckets	rules	53.4 [50.0, 60.6]	53.4	0.57	50.7 [50.0, 57.3]	52.4	0.56	12.4/5.0
All tools, constant, q80 cap	baseline	56.5 [50.5, 66.6]	56.5	0.55	60.2 [55.0, 66.0]	60.2	0.61	15.2/5.0
All tools, constant, q80 cap	rules	51.0 [50.0, 57.8]	51.0	0.54	51.3 [50.0, 56.4]	51.3	0.61	15.3/5.0
Writes only, constant	baseline	53.5 [50.0, 65.8]	56.1	0.60	56.1 [50.0, 67.4]	56.1	0.63	11.3/2.9
Writes only, constant	rules	62.6 [50.0, 74.4]	62.6	0.60	57.6 [50.0, 68.2]	57.6	0.58	11.4/2.9
Writes only, constant, q80 cap	baseline	53.7 [50.0, 66.4]	53.7	0.55	62.0 [51.2, 73.3]	62.0	0.66	8.6/2.9
Writes only, constant, q80 cap	rules	51.7 [50.0, 63.3]	53.4	0.58	58.6 [50.0, 70.3]	58.6	0.64	8.7/2.9
Writes only, 1024-B buckets	baseline	56.4 [50.0, 66.5]	56.4	0.55	58.1 [50.1, 68.6]	59.7	0.66	6.9/2.9
Writes only, 1024-B buckets	rules	52.6 [50.0, 63.8]	52.6	0.60	58.7 [51.6, 69.6]	58.7	0.59	7.0/2.9

TABLE S39. E11 INNER (TRAINING-FOLD) SELECTION STATISTICS. ELIGIBLE: ALL FOUR INNER NETWORK BAS $\leq 60\%$.

Configuration	C/base	R/base	C/rules	R/rules	Bytes×	Eligible
R4 point (all tools, constant)	54.9	53.5	58.4	52.6	15.2	yes
All tools, 1024-B buckets	53.3	58.1	55.3	56.8	9.8	yes
All tools, constant, q80 cap	56.8	56.8	53.1	54.8	11.7	yes
Writes only, constant	67.1	50.0	77.9	57.2	9.4	no
Writes only, constant, q80 cap	71.1	64.9	67.5	71.6	7.1	no
Writes only, 1024-B buckets	56.7	67.4	53.7	64.8	5.9	no

Rules-prompt homogenization: distinct read-only call sequences fall from 74 to 62 (entropy 5.63 to 5.37 bits), whereas state-changing sequences stay at 37 versus 36 (4.08 versus 4.09 bits).

TABLE S40. E12 OBSERVER ABLATION ON UNDEFENDED TRACES, CROSS-FITTED (CORE-SUITE BA, %).

Observer	Prompt	View	Cancel	Return
full	baseline	dispatch	91.7	93.8
full	baseline	network	91.7	85.5
full	rules	dispatch	90.7	93.8
full	rules	network	90.7	85.5
no writes	baseline	dispatch	53.1	65.7
no writes	baseline	network	55.2	59.9
no writes	rules	dispatch	60.8	66.2
no writes	rules	network	55.2	59.6
writes only	baseline	dispatch	91.7	93.8
writes only	baseline	network	91.7	84.9
writes only	rules	dispatch	90.7	93.8
writes only	rules	network	90.7	84.1

S16. Conditional Task-Permutation Reference

This post hoc diagnostic holds the fitted E9 attackers fixed and permutes labels over the 46 test tasks 2,000 times (seed 20261023), keeping each task's three repetitions together. Each draw recomputes the attack-suite maximum. It characterizes a task-label exchangeability reference; it does not preserve dependencies among tasks belonging to one customer. Consequently, the tail fractions below are conditional reference values, not customer-valid population p-values or evidence of equivalence. The original customer-cluster intervals, paired contrasts and frozen absolute criterion remain the primary assessments.

TABLE S41. ALL 32 E9 CELLS: OBSERVED MAXIMUM BA, PERMUTATION MEAN AND 95TH PERCENTILE (%), CONDITIONAL TAIL FRACTION $(1 + \#\{T_b \geq T_{\mathrm{obs}}\})/2001$, AND POSITIVE EXCESS OVER THE 95TH PERCENTILE (PERCENTAGE POINTS).

Prompt	View	Condition	Target	BA	Ref. mean	Ref. 95th	Tail	Excess
baseline	dispatch	Padding	cancel	97.2	54.6	65.3	0.0005	31.9
baseline	network	Padding	cancel	66.1	55.2	70.8	0.0755	0.0
baseline	dispatch	Padding	return	97.0	54.1	65.1	0.0005	31.9
baseline	network	Padding	return	55.5	56.2	67.4	0.4883	0.0
baseline	dispatch	Raw	cancel	97.2	54.6	65.3	0.0005	31.9
baseline	network	Raw	cancel	97.2	55.5	68.5	0.0005	28.7
baseline	dispatch	Raw	return	97.0	54.1	65.1	0.0005	31.9
baseline	network	Raw	return	94.0	55.3	65.4	0.0005	28.7
baseline	dispatch	Joint	cancel	55.6	54.2	62.8	0.3048	0.0
baseline	network	Joint	cancel	50.0	54.7	66.2	1.0000	0.0
baseline	dispatch	Joint	return	57.7	53.0	58.7	0.1034	0.0
baseline	network	Joint	return	50.0	54.1	62.6	1.0000	0.0
baseline	dispatch	Shape	cancel	55.6	54.2	62.8	0.3048	0.0
baseline	network	Shape	cancel	60.8	54.9	63.3	0.0870	0.0
baseline	dispatch	Shape	return	57.7	53.0	58.7	0.1034	0.0
baseline	network	Shape	return	51.9	54.3	60.4	0.7956	0.0
rules	dispatch	Padding	cancel	97.2	54.6	68.5	0.0005	28.7
rules	network	Padding	cancel	67.6	55.7	69.2	0.1034	0.0
rules	dispatch	Padding	return	97.0	54.4	65.1	0.0005	31.9
rules	network	Padding	return	57.9	55.6	67.2	0.3033	0.0
rules	dispatch	Raw	cancel	97.2	54.6	68.5	0.0005	28.7
rules	network	Raw	cancel	97.2	55.1	68.5	0.0005	28.7
rules	dispatch	Raw	return	97.0	54.4	65.1	0.0005	31.9
rules	network	Raw	return	93.5	55.6	65.8	0.0005	27.8
rules	dispatch	Joint	cancel	56.5	53.2	59.7	0.1889	0.0
rules	network	Joint	cancel	50.0	53.8	63.2	1.0000	0.0
rules	dispatch	Joint	return	64.7	53.5	60.7	0.0115	4.0
rules	network	Joint	return	60.6	54.1	60.6	0.0680	0.0
rules	dispatch	Shape	cancel	56.5	53.2	59.7	0.1889	0.0
rules	network	Shape	cancel	52.8	54.3	60.8	0.6122	0.0
rules	dispatch	Shape	return	64.7	53.5	60.7	0.0115	4.0
rules	network	Shape	return	53.2	54.6	59.6	0.7081	0.0

The baseline raw network cells exceed the reference 95th percentiles (68.5%/65.4%), whereas baseline joint cells have no positive excess. This is consistent with the measured paired reductions; failure to exceed a reference threshold is not a population guarantee. The diagnostic uses saved predictions and makes its task-level exchangeability assumption explicit.

TABLE S42. DEPLOYMENT BOUNDARY: THE RELEVANT PASSIVE OBSERVER AND WHAT THE MEASURED MECHANISM CAN PROTECT.

Setting	Visible information	Applicability / trusted point
Same-host sidecar	Host trace/process/IPC metadata, if permitted	Link padding alone does not hide host instrumentation; isolate the monitor's access.
Same-VPC gateway	Ingress record sizes, times and addresses	Tunnel or shadow shaping can protect this link; tunnel horizons must cover path RTT; gateway/markers remain trusted.
Remote SaaS/MCP	IP, unprotected SNI, per-endpoint flows and times	Shared ingress needed for tool-identity cover; ECH alone does not hide destination IP.
Cross-organization	Each hop's routing; provider calls/commit log	No guarantee across all hops. A trusted isolated executor is an explicit added assumption.

S17. Shadow Cover: Deployment Paths and Observer Boundaries

This section translates the measured isolation mechanism into deployment contracts. The evidence is the existing E6/E7 reference-program study and E9–E12 real-agent trajectory analysis. The three paths below are engineering designs grounded in existing API primitives; they have not been evaluated as production integrations.

S17.1. What padding leaves visible, and what a tunnel changes

E7's no-cover bucketed framing leaves return BA at 73.9%. E9's baseline per-call padding leaves cancellation/return BA at 66.1%/55.5%. A padded request and response still contribute observable records: invocation count, direction changes and ordered turns remain available even if individual sizes coincide. The joint E9 intervention reduces the two measured BAs to 50.0% by combining call-profile shaping with constant framing. This identifies a useful intervention boundary; it does not prove that executing dummy tools is necessary for every network threat model.

RFC 9347 describes a different construction: aggregate and fragment inner traffic into fixed-size ESP packets, send at a constant rate and use padding when no inner data is ready [39]. For a passive observer of that tunnel alone, dummy transport traffic can conceal application call boundaries without reaching a tool. Protecting workflow duration also requires a common or otherwise workload-independent transmission horizon in both directions; destination choice, tunnel start/stop and congestion responses remain part of the observation contract. The E7/E9 per-call ablations do not measure this complete construction. E13 separately implements bidirectional fixed-cell TLS encapsulation with a training-sized horizon and measures its security/cost on the same live tool workload; it is not an IPsec implementation.

At the orchestrator-to-tool boundary, a multiparty relay may instead see decoded RPCs, tool names or forwarding events after a transport tunnel terminates. A tunnel's padding disappears at that point. Cover must survive to the relevant observer as plausible calls, or application fields must remain encrypted through the relay. E6/E9 evaluate the specified call-name/length and network projections. They do not establish indistinguishability against unrestricted argument inspection, routing logs or a tool provider's state view.

S17.2. Three execution paths

1. Authorized reads and verified no-ops. Where authorized read operations suffice for the protected observation, issue them against permitted live resources and discard their responses before the planner. This can avoid a replicated backend. The read-only decoy pattern is related to WRA [40]; its ability to conceal write intent depends on what the observer sees. A receiver that recognizes write methods cannot be confused by adding only reads. E12 localizes operation-category information in the write-bearing profile; its action-defined labels do not establish leakage of unrelated attributes.

A no-op needs the invariant $f(s, a) = s$ for the designated protected state. Idempotence alone permits the first call to change state. Reads can also incur billing, access logs, cache changes or rate limits, so the adapter must specify its state scope and budget those effects. Cover selection must respect resource authorization and avoid exposing unrelated users' data. No claim about reads being the majority of production tool traffic is needed for this path.

2. Native noncommit and isolated test environments. Kubernetes server-side dry-run performs admission and validation without persisting the resource; unsafe admission side effects cause rejection, and authorization matches the corresponding real request [41]. This supplies a concrete noncommit primitive. Stripe sandboxes instead keep a separate test environment in which payments do not move real money [42]. They instantiate isolated state, rather than a dry-run flag on a live payment. Neither feature alone establishes cover indistinguishability: an observer that reads the dry-run flag, test credential or separate destination can classify the call.

Database transactions can execute operations and then roll back their transactional updates [43]. This path is suitable only when relevant effects lie within the rollback contract. PostgreSQL sequence updates, for example, are not undone by transaction abort [44]; locks, quotas and external effects also need explicit treatment. A rollback is not a general substitute for isolating irreversible remote actions. Common ingress framing must hide mode metadata from the intended observer, while the provider performing the operation remains trusted for that distinction.

3. A trusted gateway with isolated shadow endpoints. A tool/MCP gateway can receive a common envelope containing an authenticated, encrypted real/cover marker. A

TABLE S43. DEPLOYMENT CONTRACTS BY OBSERVER. THESE ARE ARCHITECTURAL CONDITIONS, NOT NEWLY MEASURED GUARANTEES.

Observer	Required protection boundary
Passive ingress link	Joint call/size shaping or a complete transport tunnel; include timing and session horizon in the threat model.
Call-visible relay	Cover must reach it; marker, mode, routing and identifying arguments must be hidden or matched. E9 covers only its specified projection.
Trusted gateway	Reads the marker and routes branches; it is outside the protected observer set.
Downstream link	Ingress-only cover is insufficient if only real calls traverse this link; add protection on that link.
Tool provider	Sees live effects and mode/credentials; hiding committed intent requires additional assumptions beyond this design.

fixed-length marker field is present in both modes and is readable only by the trusted gateway. It forwards real calls unchanged, routes cover writes to an isolated simulator and suppresses cover responses to the planner. The simulator can maintain only the state needed by supported operations; it need not clone an entire enterprise service. Error handling must fail closed: an unsupported shadow operation must never fall through to a live write.

This gives cover calls real local execution and real response packets, but does not automatically reproduce production latency or size distributions. A shared response scheduler and framing policy must normalize the exposed differences; their delay and resource costs need measurement. Retries, authentication failures, quota exhaustion and overflow must follow the same isolation contract. An encrypted marker alone is insufficient if a relay can infer the branch from destinations or correlate ingress calls with gateway-to-provider traffic.

S17.3. Where the protection ends

The intended topology is an orchestrator, an optional forwarding relay, a trusted gateway, and separate live and shadow executors. Table S43 distinguishes roles that are often conflated as a "proxy." The trust requirement follows the ability to distinguish or commit the two branches, rather than organizational ownership alone.

For a passive client-to-gateway observer, gateway-local cover can supply the measured kind of call-profile intervention. For an honest-but-curious relay, a protection claim additionally requires that its complete view exclude the discriminator and reveal no reliable substitute through arguments, timing or routing. A TLS-terminating relay that can read the marker does not satisfy this condition. The trusted gateway and the provider's commit log are not protected by moving cover into a local simulator.

All three paths preserve the logical planner transcript only when real observations pass unchanged and cover cannot affect live state or planner-visible feedback. That is the invariant checked in the existing replay. E13 adds local live-tool latency, failures, timing-aware attacks and complete tunnel controls at matched workload. Production quotas, WAN effects and fresh timing-dependent LLM interaction require deployment-specific evaluation. The contribution of shadow cover is a concrete way to shape an observable call profile while isolating its state effects, with an explicit choice of where that isolation must be enforced.

S18. Evidence Concordance and Evaluation Boundaries

S18.1. From families to result records

The map has 82 families: 58 core and 24 adjacent. Ledger inclusion is an additional, result-level decision; it does not redefine core eligibility. All 20 audited mitigation families contribute 33 result records. Six contextual families contribute one record each, giving 26 families and 39 records. Multiple records separate mechanisms, execution modes or cost objects within a family. The 78 utility/cost field positions are two fields per result, not 78 independent experiments.

TABLE S44. EXACT FAMILY-TO-RECORD RECONCILIATION. MITIGATION RECORDS TOTAL 33; CONTEXTUAL RECORDS TOTAL 6. AUDIT CELLS ARE 20 MITIGATION FAMILIES × 9 ITEMS.

Family	Records
AHB [45]	2
CachePrune [46]	2
Cloud traffic benchmark [35]	2
Dangol workflow [23]	2
Early Bird [47]	2
I Know What You Said [48]	1
KVGov [49]	3
NetEcho [50]	2
OTRO [51]	2
Remote timing [3]	1
SafeKV [52]	1
Shape and Substance [36]	1
SparLeak [37]	1
Speculative decoding [53]	1
SpliceLeak [54]	2
TDXRay [55]	1
Time Will Tell [56]	2
UI Traces [57]	1
Whisper Leak [58]	3
WRA [40]	1
Contextual records (outside the 20-family audit)	
AgentTell [59]	1
Behind Bars [60]	1
FP-Agent [61]	1
LLMs Have Rhythm [9]	1
ML privacy channels [27]	1
SIGLEAK [62]	1

The contextual record IDs are SIG-common-guardrail (SIGLEAK), AGENTTELL-fixed-instruction (AgentTell), FP-CLOUDFLARE (FP-Agent), BAR-co-running-noise (Behind Bars), RHYTHM-VPN (LLMs Have Rhythm), and DEB-filter-origin (ML privacy channels). Their roles are common-instruction residual leakage, fixed privacy instruction, access control, interference, generic VPN transport, and a privacy filter that creates an auxiliary signal. These contextual conditions do not constitute the audited confidentiality-mitigation claim.

The other 32 core families remain in the map: AgentPrint [31], AgentProv [63], ArchSnoop [11], Cache contention [2], Campaign attribution [64], ClawGuard [12], Cycle detection [13], Detokenization Leaks [5], Edge EM leakage [65], Energon [10], InputSnatch [38], Leaky Language Models [66], Lidar [67], LLM-serving RCA [68], LLM-WFIN [69], LumiMAS [14], MIRAGE [1], MoE prompt stealing [7], MoEcho [6], PAGE404 [70], PersonaFingerprint [4], ProcGrep [8], Prompt Echo [34], Prompt-cache audit [33], PromptPeek [71], Remote keylogging [32], Routing metadata [72], SparSEEty [73], Spill the Beans [74], SplitDecoy [75], Web-agent unmasking [76], Whose Agent? [77]. Their dispositions comprise 22 without a mitigation evaluation in the inspected scope, five authorized diagnostics, four unbound quantitative assertions, and SplitDecoy's distinct split-training boundary case. Thus absence from the result ledger does not imply exclusion from the SoK or that the family has no empirical evaluation. The artifact's family–record concordance gives the group, record IDs, source version and locators for every one of the 82 families.

S18.2. Experiment identifiers and manuscript locations

The archived identifiers resolve protocols, predictions and verifiers. The appendix combines these components into one case study; the table maps the archived keys to their detailed accounts.

TABLE S45. EXPERIMENT CONCORDANCE. HISTORICAL NAMES ARE RETAINED IN CODE AND DATA.

ID	Evidence / question	Current account
E1	Two-link buffered transport; terminal caps	App. E; S9/S11
E2	Saved finite-state tool plans; cover/state effects	S9; retail extension in S11
E3	Clean versus defended fitting on buffered traces	App. E; S9/S11
E5	Token-stream emitters	App. E; S12
E6	Randomized dispatch cover	App. E; S12
E7	Agent–tool TLS composition	App. E; S12
E8	Scripted/noisy stand-in planners	S13; pipeline check
E9	Real agents; pretest-frozen calibration	App. E; S14
E10/E12	Cross-fitting and oracle localization	App. E; original suite S15, expanded suite S21
E11	Retrospective operating-point selection	Main Appendix C; S15
E13	Live tool timing, latency and complete tunnels	App. E; S19
E14	Expanded attacks and classical WF adaptations	App. E; S20
E15	Additional same-provider Luna trajectories	Main Appendix C; S22
E16	Same-operation attribute diagnostic	Main Appendix C; S23.6

S18.3. Absolute thresholds and relative reductions

The E9 suite criterion remains BA upper 95% bound < 0.60 and maximum AUROC < 0.65. Joint shaping under the standard prompt has BA 50.0%/50.0%, with upper

bounds 60.6%/65.5%; the absolute claim is inconclusive. The post hoc paired intervals support a reduction relative to undefended transport, not certification of this absolute target. With 24 held-out customers and only 5/7 positive customers, finite-cluster resampling and reselection of the suite maximum affect uncertainty. This is not a demonstrated universal bootstrap resolution floor or proof that the width is caused only by sample size. More repeats of the same customers do not supply new independent clusters.

The original-suite E10 bucketed standard-prompt upper bounds (60.5%/60.2%) do not meet the two-target absolute criterion; S21 reports the larger expanded-suite maxima and intervals retained in the case-study archive. Customer-grouped fitting avoids training/test mixing within a fold but remains retrospective reuse of the same cohort. E11's training-fold point-estimate selection rule is distinct from the E9 interval criterion; choosing a configuration under the former does not imply passing the latter. A future confirmatory cohort should freeze the defended policy, attacker suite, timing projection and cluster analysis before collection. Sample planning should vary independent positive and negative customers using explicit assumptions about heterogeneity and refitting, rather than infer guaranteed power from a near-chance point estimate.

S18.4. Timing evaluation: protocol and E13 coverage

E7 and E9–E12 evaluate record sizes, directions and sequence order. They omit inter-arrival times, elapsed session duration and request service times. The end flush may form a detectable suffix; serial cover may delay responses, whereas concurrent cover may produce bursts. Logical planner-history equivalence excludes clocks, resource contention and timeout-dependent behavior. The observed byte/call ratios therefore cannot be read as measured interactive latency or timing-channel suppression.

The follow-up protocol calls for capturing timestamped ingress and egress traffic during live tool execution, with stable monotonic clock calibration, and measuring both serial and bounded-concurrency scheduling. Fit joint size/timing attackers, a flush/change-point segmenter and a real-versus-cover detector using only training customers. Keep the customer split, prompt arms and repeats together; evaluate the resulting intent BA/AUROC and paired changes on frozen test customers. Report timing-only, size/order-only and joint views, both with and without the flush suffix, including failures and timeouts.

At the same operating points, report first-response and episode p50/p95 latency, queueing, deadline violations, backend quotas, bytes/calls and native task reward. Compare per-call padding, full-profile cover, a paced/bounded-flush variant and a complete transport tunnel with a stated traffic horizon under matched workloads. Do not label synthetic delays or analytic service-time scenarios as measured production latency. E13 (S19) executes the local tool-chain portion of this specification, including observed TLS arrival timing, learned flush removal, serial/two-in-flight RPCs and complete fixed-rate tunnels. Fresh LLM replanning, WAN deployment, production quotas, an alternative paced-flush policy and a new customer cohort remain separate validation targets.

S19. Live Tool Timing, Latency and Complete Tunnel Controls (E13)

S19.1. Questions and measured boundary

E13 executes the saved 690 real-agent action trajectories again through a live TLS tool server. It asks whether arrival timing restores intent inference after length/count shaping, whether an observer can identify the end flush or cover requests, and what actual waiting time accompanies each intervention. Six conditions yield 4,140 newly measured sessions. These are new tool executions and ciphertext arrival measurements, not new LLM generations or a new customer population.

The client driver waits for each real response before releasing its next saved action. The server invokes the pinned retail tool implementation against a live database clone; cover invokes the same implementation against a separate clone. Each real response is checked against the corresponding saved model observation, and the final live database is compared with unprotected execution of the same actions. This feedback-synchronized tool loop measures real queues and service times while holding the planner's action sequence fixed. Its clocks exclude LLM/user generation and WAN transit. No artificial read/write delay is injected. Reward preservation is inherited when all responses and final live state agree; it is not a new stochastic planner trial.

S19.2. Transport and execution controls

All conditions use a fresh TLS 1.2 connection with ECDHE-RSA-AES128-GCM-SHA256 and TCP_NODELAY through a byte-transparent loopback relay. The relay timestamps ciphertext chunk arrivals using a monotonic high-resolution clock and parses records; a record inherits the timestamp of its final observed chunk. The retained chunk log permits inspection of coalescing. These are relay-observed record times, not kernel packet timestamps. The application projection excludes handshakes and closure alerts. One session is measured at a time; episode and condition order are shuffled using a fixed seed.

The six conditions are raw transport, per-call constant framing, serial shadow cover with constant framing, the same cover with at most two outstanding RPCs, and two complete fixed-rate tunnel settings. The cover scheduler uses E9's frozen training donors, count template and random seed, observes only the executed prefix, and flushes remaining counts after the last real response. Real and shadow states have separate execution locks; neither a marker nor a shadow response is exposed to the passive link observer.

The tunnel fragments and reassembles the complete application stream. Both directions send a 4,096-byte plaintext cell at every slot, including when no payload is ready. An encrypted four-byte field gives the useful payload length; the remainder is padding. Dummy cells never invoke a tool. Every cell produces a 4,120-byte TLS ciphertext record, plus its five-byte header. This implements the fixed-size/constant-rate traffic-flow principle described by RFC 9347 [39] over TLS; it is not an IPsec wire-format implementation.

The common service horizon is fixed using training messages only. For each training trajectory, sum the required request/response payload cells plus four slack cells per real call. The maximum budget, multiplied by 1.5 and rounded up with 16 additional slots, is lower-bounded by 128. This yields 133 slots: 266 ms at 2 ms per cell or 665 ms at 5 ms. Both directions continue through this horizon after the real workload completes. A horizon overflow is a retained failure, never a secret-dependent extension. The per-call framing caps use E9's frozen caps plus 256 bytes for the RPC envelope, with whole-cap overflow buckets. The operating points are fixed before test measurement.

Actual deadlines are rechecked against the measurement clock after sleeping; no ideal timestamps replace late sends or catch-up bursts. Both receivers must consume all 133 cells before closure. The saved outputs include cell deadlines, observed sends, pending queues and all requests/completions. A host scheduler and a local Python gateway remain part of this measured implementation. The two tunnel settings are fixed operating points, not an optimized cost frontier or a measurement of multi-session amortization.

S19.3. Attacks and fitting order

The original customer-disjoint split is preserved: 414 training trajectories from 29 customers and 276 test trajectories from 24 customers. All 2,484 final training measurements precede attack fitting and a model-hash freeze; only then are the 1,656 final test sessions collected. An earlier complete development pass exposed two incomplete tunnel tails at connection closure. We added a receive-drain barrier, restored E9-style profilers to the attacker suite, and recollected the entire cohort; no individual test sessions were selectively replaced. Development records are archived separately. The original trajectories and labels were already known from E9, so this is a post hoc measurement study on the same population.

The intent suite has a training-majority baseline, balanced logistic regression ($C = 1$), gradient boosting with 100 trees, and random forests with 200 trees and a minimum leaf size of two. Its three basic views are size/order, timing/direction and their joint features. Features include direction-specific count, size/gap quantiles, request/response size histograms, exchange-length vectors, duration, gap thresholds at 0.1/1/5/20 ms, the first/last 32 positions, and largest-gap change-point summaries.

For non-tunnel conditions, training-only size-to-tool profilers additionally feed a decoded gradient-boosting attacker and a training-selected count rule; these join both size/order and full-suite evaluations. At test time, all attackers consume observable records only. True tool names, RPC IDs, real/cover markers, true flush boundaries and internal execution times are excluded from test features.

Separate 80-tree forests (depth eight, minimum leaf size five) detect the flush suffix and real versus cover requests. Flush detection uses observable record neighborhoods and relative position/time; cover detection uses the same visible features at request-record positions. True execution annotations supply fitting/scoring labels only. For the flush-stripped intent view, training traces are transformed by three-fold customer-disjoint out-of-fold detectors. The full-training detector transforms test traces; a maximum-likelihood single transition estimates the cut. No oracle test boundary is used. The full adaptive suite unions every applicable view, allowing an observer to ignore timing or retain the unstripped trace.

S19.4. Inference, latency and reproducibility

Intent intervals use 2,000 common customer-cluster bootstrap draws, retaining all tasks and repeats and recomputing the maximum over the suite in each draw. Paired differences reuse these draws. Draws lacking either target class have undefined BA/AUROC and are omitted from interval quantiles; their counts are archived. Intervals condition on fitted predictions and do not supply a new independent population estimate. The E9 absolute criterion remains unchanged: BA upper 95% bound below 0.60 and AUROC below 0.65. Detector BA also uses customer clusters; detector AUROC is reported as a point estimate.

Latency is reported for individual real RPCs, the first real response, the last real response and the complete traffic window. Client scheduling queues, backend service, slot lateness and timeouts are retained separately. The descriptive count of real calls exceeding 100 ms is a reference threshold, not a service-level guarantee. Byte ratios count measured application TLS records and their headers; call ratios count actual tool RPCs, excluding tunnel dummy cells. Source, calibration, model and capture hashes bind the analysis. The saved captures, fitted models and predictions support offline verification without new LLM or network requests.

S19.5. Findings across the measured boundaries

For baseline prompts, serial cover's full-suite cancellation BA exceeds its size/order BA by 11.4 points [3.7,19.6]. The full-suite reduction relative to raw remains 30.3 points [23.1,37.6] for cancellation and 32.9 [19.9,40.7] for return. Thus the measurement supports a substantial reduction while exposing a residual clock signal. Flush and real/cover detectors pool both prompt arms: serial/parallel flush BA is 90.4%/92.3%, and cover BA is 82.5%/96.1%. Successful segmentation does not itself establish successful intent recovery; the separately fitted stripped-trace attacks test that transfer.

All 1,380 tunnel sessions contain exactly 133 records per direction with identical ciphertext lengths. Nevertheless, the measured directions can interleave differently, and scheduling is imperfect. In baseline-prompt tests, 2/5 ms tunnel slot lateness has p95 1.88/1.68 ms and maxima 282.6/307.5 ms. These retained observations characterize the implementation, rather than an ideal zero-jitter clock. The full-suite baseline BAs are 62.5%/52.9% and 53.1%/57.3%; neither setting meets the two-target absolute criterion. Their 125.7× byte cost and one tool call per original action identify a different resource trade-off from shadow cover's 15.1× bytes and 3.83× calls. These fixed operating points do not establish an optimal tunnel or universal superiority of either mechanism.

No formal session fails, loses a real response, changes final live state or sends a cell before its deadline. Twelve of 2,000 cancellation bootstrap draws do not contain both target classes; all return draws are defined. The archived invalid-draw counters distinguish this finite-sample calculation issue from execution failures.

S19.6. Evaluation under WAN transit delay and jitter (Exp 4 Extension)

Local loopback measurement isolates backend execution and tool scheduling from transit variance, but real agent deployments operate over wide-area networks subject to packet propagation delays and jitter. To evaluate whether timing-based flush detection and intent inference hold outside the loopback environment, we extend Exp 4 with network emulation matching standard wide-area link profiles:

- **WAN Mild**: 20 ms one-way delay with 5 ms normal jitter ($\sigma = 5$ ms), representing regional cloud tool hosting (40 ms nominal RTT).
- **WAN Realistic**: 50 ms one-way delay with 10 ms normal jitter ($\sigma = 10$ ms), representing cross-region or transcontinental tool hosting (100 ms nominal RTT).

The emulation runs directly in the bidirectional TLS transport relay using high-precision non-reordering FIFO delivery queues matching the behavior of Linux `tc netem`. Each real tool response across the WAN link gates the next action

TABLE S46. SERIAL SHADOW COVER FLUSH-PHASE DETECTION ACROSS NETWORK CONDITIONS (15,278 RECORDS FROM 276 TEST EPISODES). DIRECT TRANSFER APPLIES THE FROZEN LOOPBACK DETECTOR D_{LOOP}; ADAPTED FITS 5-FOLD CUSTOMER-GROUPED MODELS ON THE WAN WORKLOAD.

Network Condition	**Model**	**Flush Point BA**	**95% CI**	**AUROC**
Loopback (0 ms, 0 ms jitter)	Frozen D_{loop}	90.4%	[88.3, 92.5]	0.967
WAN Mild (20 ms, 5 ms jitter)	Transfer D_{loop}	89.4%	[87.2, 91.7]	0.962
	Adapted D_{WAN}	89.2%	[86.6, 91.5]	0.958
WAN Realistic (50 ms, 10 ms jitter)	Transfer D_{loop}	89.4%	[87.3, 91.7]	0.963
	Adapted D_{WAN}	89.0%	[86.5, 91.3]	0.958

release, faithfully preserving feedback synchronization against live database clones. We repeat measurements on the complete set of 276 customer-held-out test episodes across both prompt arms (standard baseline and system-prompt rules), yielding 1,104 new TLS sessions per WAN profile (2,208 additional measured sessions total) across raw transport, per-call constant framing, serial shadow cover, and 2 ms complete tunnels.

Resilience of flush detection. Table S46 assesses flush-phase detection across 15,278 ciphertext records from serial shadow cover sessions. We evaluate two attacker postures: (1) *Direct Transfer*, where the attacker applies the frozen loopback detector D_{loop} without network-specific recalibration, and (2) *Adapted 5-Fold CV*, where the attacker refits the 80-tree random forest on customer-grouped training folds under the observed WAN network condition.

Under both WAN conditions, flush detection exhibits remarkable resilience. With direct transfer of D_{loop}, the detector achieves 89.4% point BA (95% CI [87.2%, 91.7%], AUROC 0.962) under 20 ms/5 ms jitter and 89.4% point BA (95% CI [87.3%, 91.7%], AUROC 0.963) under 50 ms/10 ms jitter, falling by only 1.0 percentage point from the loopback baseline (90.4% BA, AUROC 0.967). Adapted retraining yields consistent performance (89.2% and 89.0% BA). This stability occurs because the flush phase transitions from feedback-gated tool intervals (punctuated by real database queries and network RTTs) to an un-gated burst of dummy requests. This macroscopic transition in inter-request density and duration creates a structural signature that easily cuts through 5–10 ms transit noise.

Intent classification under transit jitter. Table S47 compares full-suite intent Balanced Accuracy across loopback and WAN conditions. For raw transport and per-call constant framing, intent classification is dominated by packet size and sequence order distributions, which are network-delay invariant; hence cancellation BA remains 91.7% and 68.6%, respectively.

For serial shadow cover, however, the timing signal exploited on loopback is substantially attenuated. On loopback, residual execution time differences allowed the full attack union to elevate cancellation BA to 61.4% (11.4 points above size/order). Under WAN conditions, random transit jitter washes out microsecond-level tool service variations: cancellation BA falls to 50.0% (exact chance level) under 20 ms/5 ms jitter and 54.0% under 50 ms/10 ms jitter; return BA remains 50.7% and 54.6%. Thus, WAN network jitter acts as natural timing obfuscation against micro-timing intent inference, strengthening the intent-hiding property of serial cover while leaving macroscopic flush boundaries visible.

Operational limits: tunnel horizon exhaustion. The WAN extension reveals a critical operational vulnerability in fixed-horizon tunnel designs. The 2 ms full tunnel enforces a strict 133-slot horizon (266 ms) sized against local loopback execution. On loopback, all 276 sessions successfully completed all tool calls within this window.

When executed over WAN links, each sequential tool RPC incurs network RTT. Under 20 ms one-way delay (40 ms RTT per call), tasks requiring multiple tool round-trips exhaust the 133 slots before their responses arrive; 191 out of 276 sessions (69.2%) fail to complete due to horizon exhaustion (only 85/276 complete). Under 50 ms delay (100 ms RTT), horizon exhaustion surges to 98.2% (271 out of 276 sessions fail; only 5 single-call sessions complete). This highlights an intrinsic deployment hazard: fixed-rate constant-bandwidth tunnels require worst-case horizon provisioning over WAN links, which would drastically multiply byte overhead (e.g., thousands of slots), or else suffer catastrophic protocol failure whenever network latency fluctuates.

In contrast, serial shadow cover and per-call constant framing adapt naturally to network delays: all 276 sessions complete successfully with 100% response and database state equality across both WAN conditions. Their real-chain p95 latency expands predictably with network round-trips: from 26.2 ms on loopback to 855.4 ms under 20 ms WAN and 1999.5 ms under 50 ms WAN.

TABLE S47. TIMING DEFENSES UNDER LOOPBACK VERSUS WAN CONDITIONS (276 TEST EPISODES). FULL-SUITE INTENT BALANCED ACCURACY (BA) REPORTED FOR CANCELLATION / RETURN. INCOMPLETE/EXHAUSTED REFLECTS SESSIONS FAILING TO COMPLETE ALL REAL CALLS WITHIN THE FIXED HORIZON.

Condition	**Network Environment**	**Full-Suite BA**	**Real p95 ms**	**Dur p95 ms**	**Byte Mult.**	**Status / Exh.**
Raw transport	Loopback (0 ms, 0 ms jit.)	91.7% / 86.4%	15.4	15.5	1.0×	276/276
	WAN Mild (20 ms, 5 ms jit.)	91.7% / 86.4%	506.4	506.4	1.0×	276/276
	WAN Real. (50 ms, 10 ms jit.)	91.7% / 86.4%	1178.8	1178.8	1.0×	276/276
Per-call constant framing	Loopback (0 ms, 0 ms jit.)	68.6% / 62.7%	16.6	16.7	3.9×	276/276
	WAN Mild (20 ms, 5 ms jit.)	68.6% / 54.8%	504.8	504.9	3.9×	276/276
	WAN Real. (50 ms, 10 ms jit.)	68.6% / 54.8%	1172.6	1172.6	3.9×	276/276
Serial shadow cover	Loopback (0 ms, 0 ms jit.)	61.4% / 53.4%	26.2	46.6	15.1×	276/276
	WAN Mild (20 ms, 5 ms jit.)	50.0% / 50.7%	855.4	1314.9	15.3×	276/276
	WAN Real. (50 ms, 10 ms jit.)	54.0% / 54.6%	1999.5	3112.4	15.3×	276/276
Full tunnel (2 ms cell)	Loopback (0 ms, 0 ms jit.)	62.5% / 52.9%	56.4	270.4	125.7×	276/276
	WAN Mild (20 ms, 5 ms jit.)	54.4% / 56.8%	286.7	291.6	127.0×	85/276
	WAN Real. (50 ms, 10 ms jit.)	55.0% / 52.6%	269.5	325.9	127.0×	5/276

TABLE S48. E13 BASELINE PROMPT: INTENT BA [95% CUSTOMER-CLUSTER INTERVAL] AND AUROC. THE FULL SUITE UNIONS ALL AVAILABLE VIEWS.

Condition	View	Cancel BA [CI]	Return BA [CI]	Cancel AUC	Return AUC
Raw	Size/order	91.7 [84.7, 100.0]	86.4 [79.5, 97.2]	0.917	0.875
Raw	Timing/direction	53.8 [50.0, 72.5]	52.9 [50.0, 64.1]	0.531	0.570
Raw	Joint features	84.3 [72.5, 97.4]	80.5 [69.9, 93.6]	0.836	0.835
Raw	Full adaptive suite	91.7 [84.7, 100.0]	86.4 [79.5, 97.2]	0.917	0.875
Per-call padding	Size/order	68.6 [50.0, 92.8]	54.8 [50.0, 74.8]	0.678	0.514
Per-call padding	Timing/direction	54.6 [50.0, 63.0]	62.7 [53.8, 78.0]	0.500	0.657
Per-call padding	Joint features	51.0 [50.0, 61.4]	50.2 [50.0, 69.0]	0.569	0.545
Per-call padding	Full adaptive suite	68.6 [50.8, 92.8]	62.7 [54.2, 79.3]	0.678	0.657
Shadow, serial	Size/order	50.0 [50.0, 56.9]	50.7 [50.0, 62.5]	0.500	0.505
Shadow, serial	Timing/direction	61.4 [53.8, 67.7]	52.9 [50.0, 62.5]	0.625	0.500
Shadow, serial	Joint features	51.0 [50.0, 64.6]	50.0 [50.0, 57.7]	0.596	0.500
Shadow, serial	Estimated flush removed	57.8 [50.0, 70.7]	53.4 [50.0, 64.6]	0.583	0.530
Shadow, serial	Full adaptive suite	61.4 [54.6, 71.1]	53.4 [50.0, 65.8]	0.625	0.530
Shadow, two in flight	Size/order	51.7 [50.0, 68.3]	57.7 [51.5, 66.1]	0.509	0.569
Shadow, two in flight	Timing/direction	50.0 [50.0, 57.1]	50.0 [50.0, 59.9]	0.554	0.500
Shadow, two in flight	Joint features	50.0 [50.0, 57.1]	52.6 [50.0, 63.3]	0.531	0.522
Shadow, two in flight	Estimated flush removed	50.0 [50.0, 64.0]	50.0 [50.0, 56.7]	0.500	0.500
Shadow, two in flight	Full adaptive suite	51.7 [50.0, 69.1]	57.7 [53.4, 66.8]	0.554	0.569
Full tunnel, 2 ms	Size/order	62.5 [54.3, 70.1]	52.9 [50.0, 61.0]	0.594	0.521
Full tunnel, 2 ms	Timing/direction	53.1 [50.2, 59.3]	50.0 [50.0, 55.6]	0.641	0.500
Full tunnel, 2 ms	Joint features	56.2 [50.9, 63.9]	50.0 [50.0, 57.4]	0.656	0.569
Full tunnel, 2 ms	Full adaptive suite	62.5 [55.6, 70.2]	52.9 [50.1, 61.3]	0.656	0.569
Full tunnel, 5 ms	Size/order	51.4 [50.0, 59.0]	54.5 [50.0, 60.3]	0.500	0.513
Full tunnel, 5 ms	Timing/direction	53.1 [50.0, 71.9]	53.2 [50.0, 62.4]	0.575	0.548
Full tunnel, 5 ms	Joint features	52.4 [50.0, 61.8]	57.3 [53.1, 63.7]	0.534	0.555
Full tunnel, 5 ms	Full adaptive suite	53.1 [50.3, 71.9]	57.3 [54.2, 64.2]	0.575	0.555

TABLE S49. E13 RULES PROMPT: INTENT BA [95% CUSTOMER-CLUSTER INTERVAL] AND AUROC. THE FULL SUITE UNIONS ALL AVAILABLE VIEWS.

Condition	View	Cancel BA [CI]	Return BA [CI]	Cancel AUC	Return AUC
Raw	Size/order	94.4 [83.3, 100.0]	87.0 [78.0, 97.6]	0.944	0.905
Raw	Timing/direction	57.8 [50.0, 70.6]	56.1 [50.0, 63.9]	0.598	0.557
Raw	Joint features	84.7 [75.2, 98.2]	79.1 [67.8, 92.0]	0.851	0.822
Raw	Full adaptive suite	94.4 [83.3, 100.0]	87.0 [78.0, 97.6]	0.944	0.905
Per-call padding	Size/order	70.1 [50.0, 86.0]	54.8 [50.0, 74.7]	0.680	0.500
Per-call padding	Timing/direction	55.8 [50.0, 70.2]	50.6 [50.0, 62.4]	0.530	0.534
Per-call padding	Joint features	59.4 [50.0, 76.6]	53.6 [50.0, 71.0]	0.575	0.508
Per-call padding	Full adaptive suite	70.1 [50.0, 86.0]	54.8 [50.0, 74.8]	0.680	0.534
Shadow, serial	Size/order	58.2 [50.2, 71.2]	57.9 [50.7, 66.8]	0.581	0.597
Shadow, serial	Timing/direction	51.5 [50.0, 69.8]	57.7 [52.3, 68.8]	0.500	0.560
Shadow, serial	Joint features	51.1 [50.0, 64.4]	55.1 [52.2, 65.3]	0.519	0.555
Shadow, serial	Estimated flush removed	50.1 [50.0, 55.6]	60.7 [50.8, 70.9]	0.500	0.622
Shadow, serial	Full adaptive suite	58.2 [52.9, 72.2]	60.7 [56.3, 71.7]	0.581	0.622
Shadow, two in flight	Size/order	54.7 [50.0, 66.7]	60.2 [50.5, 73.8]	0.580	0.576
Shadow, two in flight	Timing/direction	50.0 [50.0, 57.8]	54.0 [50.0, 62.6]	0.556	0.585
Shadow, two in flight	Joint features	55.3 [50.0, 67.1]	52.3 [50.0, 61.0]	0.578	0.546
Shadow, two in flight	Estimated flush removed	50.8 [50.0, 61.7]	54.0 [50.0, 67.0]	0.541	0.528
Shadow, two in flight	Full adaptive suite	55.3 [50.8, 68.2]	60.2 [54.5, 73.9]	0.580	0.585
Full tunnel, 2 ms	Size/order	51.5 [50.0, 65.0]	53.7 [50.0, 59.7]	0.500	0.500
Full tunnel, 2 ms	Timing/direction	53.1 [50.0, 64.2]	53.1 [50.0, 60.9]	0.500	0.563
Full tunnel, 2 ms	Joint features	51.5 [50.0, 58.1]	59.1 [52.2, 66.1]	0.500	0.590
Full tunnel, 2 ms	Full adaptive suite	53.1 [50.0, 65.6]	59.1 [52.8, 66.1]	0.500	0.590
Full tunnel, 5 ms	Size/order	50.0 [50.0, 53.6]	55.4 [50.9, 62.7]	0.500	0.549
Full tunnel, 5 ms	Timing/direction	50.0 [50.0, 50.4]	56.3 [51.0, 61.6]	0.500	0.530
Full tunnel, 5 ms	Joint features	50.0 [50.0, 57.5]	52.3 [50.0, 59.5]	0.500	0.534
Full tunnel, 5 ms	Full adaptive suite	50.0 [50.0, 57.5]	56.3 [55.0, 63.3]	0.500	0.549

TABLE S50. E13 LIVE EXECUTION COSTS, BOTH PROMPT ARMS. MEDIANS AND 95TH PERCENTILES DESCRIBE OBSERVED LOCAL RPCS; THEY ARE NOT PRODUCTION-SERVICE LATENCY ESTIMATES.

Prompt	Condition	Bytes×	Calls×	First p50/p95	Last real p50/p95	Traffic p50/p95	RPC p50/p95
baseline	Raw	1.00	1.00	2.3/3.4	8.8/15.4	8.8/15.5	1.0/1.8
baseline	Per-call padding	3.92	1.00	2.3/3.7	8.5/16.6	8.5/16.7	1.0/1.8
baseline	Shadow, serial	15.11	3.83	3.0/5.0	12.5/26.2	27.8/46.6	0.9/1.7
baseline	Shadow, two in flight	15.11	3.83	2.7/4.5	11.3/20.8	23.6/37.4	1.2/2.4
baseline	Full tunnel, 2 ms	125.71	1.00	5.4/11.2	30.9/56.4	267.0/270.4	4.0/6.2
baseline	Full tunnel, 5 ms	125.71	1.00	11.3/12.4	71.4/112.3	666.1/666.6	10.0/10.9
rules	Raw	1.00	1.00	2.2/3.7	8.1/15.7	8.2/15.7	1.0/1.7
rules	Per-call padding	3.95	1.00	2.2/3.6	8.4/15.5	8.4/15.5	1.0/1.8
rules	Shadow, serial	15.47	3.88	3.0/5.2	12.7/23.9	28.9/41.8	1.0/1.7
rules	Shadow, two in flight	15.47	3.88	2.7/4.4	11.1/20.7	23.8/36.5	1.2/2.4
rules	Full tunnel, 2 ms	128.38	1.00	5.4/7.7	29.8/48.8	267.0/267.9	4.0/5.8
rules	Full tunnel, 5 ms	128.38	1.00	11.1/12.4	69.5/112.9	666.2/666.6	10.0/11.0

All time columns are milliseconds. The saved results additionally report queue/service times, a descriptive 100 ms per-call deadline count, cell lateness, failures, and paired mean last-response differences.

TABLE S51. E13 LEARNED TIMING/SHAPE DETECTORS ON HELD-OUT CUSTOMERS, POOLING BOTH PROMPT ARMS. FLUSH LABELS IDENTIFY RECORDS AFTER THE ONLINE FLUSH DECLARATION; COVER LABELS IDENTIFY ENCRYPTED REQUESTS. GROUND TRUTH IS USED ONLY FOR FITTING/SCORING, NEVER AS AN ATTACKER FEATURE.

Condition	Detector	BA [95% CI]	AUROC	Records/requests
Shadow, serial	flush	90.4 [88.3, 92.5]	0.967	15278
Shadow, serial	cover	82.5 [80.3, 84.7]	0.898	7639
Shadow, two in flight	flush	92.3 [90.6, 93.9]	0.979	15278
Shadow, two in flight	cover	96.1 [95.4, 96.7]	0.995	7639

S20. Stronger Attackers, Classical Traffic Controls and Power Planning (E14)

E14 is a post hoc sensitivity analysis of the existing cohort. It reuses E13's six measured conditions and adds seven offline transformations of its undefended captures, all on the same 414 training and 276 test trajectories. No new customer, model decision or packet measurement is produced by these transformations. The configuration and source hashes were frozen before fitting the new attacks; the source cohort and its earlier outcomes were already known. All configurations are reported.

S20.1. Defense-aware attacker expansion

The primary attacker is D_2: it knows the algorithm and parameter distribution and fits on training examples generated under the evaluated defense. Seven R7 models comprise a majority control, logistic regression, gradient boosting, random forest, extra trees, a k-FP-style forest-leaf nearest-neighbor attack [78], and a trainable 1D-CNN. This leaf adaptation uses 200 trees and Hamming-distance three-nearest-neighbor voting on training leaf fingerprints, not an exact reproduction of the original k-FP extractor. Tabular inputs contain full-trace counts, sizes, time summaries, directional bursts and sampled cumulative signed sizes. None includes tool names, labels, customer IDs, real/cover markers or test outcomes. The CNN uses direction, log length, log interarrival and a mask; its two convolutional layers have 16/32 channels, followed by mean/max pooling and a binary head. It is trained for 50 fixed epochs with Adam, learning rate .001 and class-weighted loss. The first/last halves of sequences longer than 512 records are retained; Table S52 reports truncation. Other learners preserve full-trace aggregate counts. Models and hashes are saved before test scoring. No test-guided hyperparameter selection is performed in this extension.

For measured conditions, the entire original E13 union remains included: size/order, timing, joint, training-derived profile decoding and learned flush stripping. The maximum therefore cannot decrease merely because a weaker new learner replaces an old one. Each of 2,000 common customer-bootstrap draws reselects the maximum. Draws missing a target class are excluded (counts are saved per cell). The intervals condition on fitted predictions and are not bounds against arbitrary future attackers. No expanded cell meets the original absolute criterion; the threshold is unchanged.

S20.2. Classical WF mechanisms adapted to RPC traces

Every offline adapter tracks abstract fragments derived from input record sizes and verifies no loss, reordering or scheduling before release. This checks metadata and byte accounting, not payload-content equality or actual delivery. Fixed cells use 512 plaintext bytes plus 24 TLS 1.2 AEAD bytes and the five-byte header. Fragment IDs are excluded from attack inputs. All controls use the same held-out trajectories and byte accounting; they are not Tor deployments and do not inherit a published implementation's accuracy or guarantees.

Tamaraw uses constant per-direction slots, 40/12 ms or 4/1.2 ms, and randomized geometric padding to multiples of 50 cells [26]. Input-order queueing propagates waits to subsequent real fragments. FRONT draws a uniform dummy budget and a Rayleigh schedule per direction and cancels outstanding dummies when the real trace ends [79]; the budget is the training median cell count and the window spans training-duration quartiles. The shifted calibration fits those values only on the shorter half of training customers; D2 attackers still use all defended training customers and every test customer remains included. This is an in-population calibration stress test, not evidence of cross-environment transfer.

The WTF-PAD mechanism adaptation retains send/receive triggers, burst/gap/wait states, token removal/refill and infinity outcomes [80]. It uses an empirical training interarrival histogram, burst/gap infinity probabilities .1/.25, real-event reset and a 64-dummy-per-gap cap, rather than reproducing the public simulator's fitted log-normal Tor configuration. Walkie-Talkie molds half-duplex burst lengths with a random training donor from another customer and adds donor-only tail bursts [81]. It has no sensitive/non-sensitive symmetric pairing oracle and therefore does not inherit that pairing's indistinguishability argument. These exact adaptations and their limits are documented with the executable source.

S20.3. Results and cost interpretation

The strongest learner depends on the operating point: the new CNN increases some tunnel BAs and supports inference on several classical controls; the forest-leaf attack improves the record-bucket return attack. The serial-shadow timing signal remains, so the earlier clock-boundary finding survives this expansion. The alternative WF points can reduce byte/tool-call cost substantially, precluding a claim that shadow cover is universally superior. The shorter-calibration FRONT result does not degrade monotonically: all points and both prompt arms are retained rather than selecting the direction expected by the hypothesis.

TABLE S52: E14: all prompt/defense cells, maximum over the expanded attack suite. Customer bootstrap 95% BA intervals; AUC is suite-maximum AUROC. All 52 cells fail the absolute criterion.

Prompt / condition	Target	BA [95% CI]	AUC	Attacks	CNN trunc.
Standard / Undefended	cancel	91.7 [84.7, 100.0]	0.917	19	0
Standard / Undefended	return	86.4 [79.5, 97.2]	0.875	19	0
Standard / Per-call constant	cancel	68.6 [51.4, 92.8]	0.678	19	0
Standard / Per-call constant	return	62.7 [54.2, 79.5]	0.657	19	0
Standard / Serial shadow	cancel	61.4 [54.6, 71.4]	0.625	22	0
Standard / Serial shadow	return	53.4 [50.3, 65.8]	0.530	22	0
Standard / Two-in-flight shadow	cancel	51.7 [50.0, 69.1]	0.554	22	0
Standard / Two-in-flight shadow	return	57.7 [56.6, 69.7]	0.596	22	0
Standard / Full tunnel, 2 ms	cancel	62.5 [57.6, 70.2]	0.708	17	0
Standard / Full tunnel, 2 ms	return	55.2 [51.9, 62.0]	0.569	17	0
Standard / Full tunnel, 5 ms	cancel	54.6 [51.4, 72.9]	0.618	17	0
Standard / Full tunnel, 5 ms	return	57.3 [54.2, 64.2]	0.555	17	0
Standard / Record buckets	cancel	53.1 [50.0, 78.5]	0.510	7	0
Standard / Record buckets	return	69.8 [50.0, 86.9]	0.689	7	0
Standard / Tamaraw 40/12 ms	cancel	55.3 [50.0, 68.7]	0.590	7	2
Standard / Tamaraw 40/12 ms	return	55.3 [51.1, 64.4]	0.594	7	2
Standard / Tamaraw 4/1.2 ms	cancel	57.8 [50.0, 71.8]	0.609	7	1
Standard / Tamaraw 4/1.2 ms	return	51.8 [50.0, 61.8]	0.525	7	1
Standard / FRONT	cancel	59.0 [50.0, 69.9]	0.623	7	0
Standard / FRONT	return	69.3 [58.3, 81.8]	0.713	7	0
Standard / WTF-PAD adaptation	cancel	51.9 [50.0, 64.1]	0.560	7	0
Standard / WTF-PAD adaptation	return	62.7 [50.0, 79.6]	0.635	7	0
Standard / WT random-donor	cancel	50.0 [50.0, 67.0]	0.526	7	0
Standard / WT random-donor	return	56.7 [51.7, 67.8]	0.607	7	0
Standard / FRONT, short calibration	cancel	52.4 [50.0, 61.3]	0.546	7	0
Standard / FRONT, short calibration	return	56.0 [50.0, 71.1]	0.618	7	0
Rules / Undefended	cancel	94.4 [83.3, 100.0]	0.944	19	0
Rules / Undefended	return	87.0 [78.4, 97.6]	0.905	19	0
Rules / Per-call constant	cancel	70.1 [50.0, 86.0]	0.680	19	0
Rules / Per-call constant	return	61.9 [50.8, 78.0]	0.563	19	0
Rules / Serial shadow	cancel	58.2 [53.8, 76.3]	0.589	22	0
Rules / Serial shadow	return	60.7 [57.1, 72.6]	0.626	22	0
Rules / Two-in-flight shadow	cancel	56.9 [51.9, 70.4]	0.580	22	0
Rules / Two-in-flight shadow	return	60.2 [55.4, 73.9]	0.639	22	0
Rules / Full tunnel, 2 ms	cancel	53.1 [51.7, 65.7]	0.504	17	0
Rules / Full tunnel, 2 ms	return	59.1 [55.3, 67.6]	0.590	17	0
Rules / Full tunnel, 5 ms	cancel	58.3 [50.0, 67.9]	0.539	17	0
Rules / Full tunnel, 5 ms	return	56.3 [55.7, 64.0]	0.595	17	0
Rules / Record buckets	cancel	56.7 [50.0, 81.2]	0.615	7	0
Rules / Record buckets	return	70.8 [51.2, 88.0]	0.703	7	0
Rules / Tamaraw 40/12 ms	cancel	58.6 [56.3, 70.8]	0.622	7	0
Rules / Tamaraw 40/12 ms	return	50.0 [50.0, 60.3]	0.500	7	0
Rules / Tamaraw 4/1.2 ms	cancel	52.5 [50.0, 61.6]	0.505	7	3
Rules / Tamaraw 4/1.2 ms	return	54.6 [51.4, 61.5]	0.534	7	3
Rules / FRONT	cancel	58.5 [50.0, 69.6]	0.544	7	0
Rules / FRONT	return	60.4 [50.0, 75.1]	0.635	7	0
Rules / WTF-PAD adaptation	cancel	57.2 [50.0, 74.2]	0.559	7	0
Rules / WTF-PAD adaptation	return	61.0 [50.0, 77.6]	0.625	7	0
Rules / WT random-donor	cancel	51.4 [50.0, 68.8]	0.606	7	0
Rules / WT random-donor	return	56.3 [50.0, 66.6]	0.558	7	0
Rules / FRONT, short calibration	cancel	51.2 [50.0, 70.1]	0.501	7	0
Rules / FRONT, short calibration	return	62.9 [50.0, 77.2]	0.655	7	0

S20.4. Matched-suite cost sensitivity

The strongest-available union contains 17–22 attacks for measured conditions and seven for offline controls. Maxima over these unequal unions cannot by themselves establish a matched-attacker dominance relation. We therefore recompute all cells using exactly the seven common R7 models, without new fitting (Table S53). Figure 2 uses this matched subset; its gray upper endpoints retain the larger available unions. For example, standard-prompt serial shadow changes from 61.4/53.4% under its full union to 50.0/52.9% under the common subset, while WT remains 50.0/56.7%. WT's lower byte cost then trades off against higher worst-target BA, rather than dominating serial shadow. The serial timing finding from the full suite remains unchanged. One subset cell (standard-prompt, two-in-flight shadow, cancellation) meets the gate, but its full-union cell does not; restricting the comparison suite does not certify the defense. This sensitivity separates matched cost comparisons from the strongest observed attack evidence.

TABLE S53: Matched-suite sensitivity: the same seven R7 models in every E14 condition. BA and AUROC maxima are independently reselected in every customer-bootstrap draw. This subset is used for the cost plot, not to replace the stronger available attack union or its absolute assessment.

Prompt / condition	Target	BA [95% CI]	AUROC	Valid draws	Subset gate
Standard / Undefended	cancel	84.3 [72.7, 97.4]	0.836	1988	Unmet
Standard / Undefended	return	76.9 [62.2, 90.4]	0.828	2000	Unmet
Standard / Per-call constant	cancel	53.2 [50.0, 69.5]	0.604	1988	Unmet
Standard / Per-call constant	return	59.1 [50.0, 73.8]	0.656	2000	Unmet
Standard / Serial shadow	cancel	50.0 [50.0, 65.0]	0.548	1988	Unmet
Standard / Serial shadow	return	52.9 [50.0, 60.3]	0.526	2000	Unmet
Standard / Two-in-flight shadow	cancel	50.0 [50.0, 59.7]	0.500	1988	Pass
Standard / Two-in-flight shadow	return	57.3 [55.0, 69.2]	0.596	2000	Unmet
Standard / Full tunnel, 2 ms	cancel	59.9 [54.6, 67.3]	0.708	1988	Unmet
Standard / Full tunnel, 2 ms	return	55.2 [50.1, 61.0]	0.567	2000	Unmet
Standard / Full tunnel, 5 ms	cancel	54.6 [50.0, 70.5]	0.618	1988	Unmet
Standard / Full tunnel, 5 ms	return	53.7 [51.3, 61.2]	0.551	2000	Unmet
Standard / Record buckets	cancel	53.1 [50.0, 78.5]	0.510	1988	Unmet
Standard / Record buckets	return	69.8 [50.0, 86.9]	0.689	2000	Unmet
Standard / Tamaraw 40/12 ms	cancel	55.3 [50.0, 68.7]	0.590	1988	Unmet
Standard / Tamaraw 40/12 ms	return	55.3 [51.1, 64.4]	0.594	2000	Unmet
Standard / Tamaraw 4/1.2 ms	cancel	57.8 [50.0, 71.8]	0.609	1988	Unmet
Standard / Tamaraw 4/1.2 ms	return	51.8 [50.0, 61.8]	0.525	2000	Unmet
Standard / FRONT	cancel	59.0 [50.0, 69.9]	0.623	1988	Unmet
Standard / FRONT	return	69.3 [58.3, 81.8]	0.713	2000	Unmet
Standard / WTF-PAD adaptation	cancel	51.9 [50.0, 64.1]	0.560	1988	Unmet
Standard / WTF-PAD adaptation	return	62.7 [50.0, 79.6]	0.635	2000	Unmet
Standard / WT random-donor	cancel	50.0 [50.0, 67.0]	0.526	1988	Unmet
Standard / WT random-donor	return	56.7 [51.7, 67.8]	0.607	2000	Unmet
Standard / FRONT, short calibration	cancel	52.4 [50.0, 61.3]	0.546	1988	Unmet
Standard / FRONT, short calibration	return	56.0 [50.0, 71.1]	0.618	2000	Unmet
Rules / Undefended	cancel	84.7 [75.2, 98.5]	0.851	1988	Unmet
Rules / Undefended	return	78.3 [63.3, 92.2]	0.818	2000	Unmet
Rules / Per-call constant	cancel	59.4 [50.0, 77.5]	0.594	1988	Unmet
Rules / Per-call constant	return	61.9 [50.0, 77.4]	0.563	2000	Unmet
Rules / Serial shadow	cancel	56.1 [50.0, 76.2]	0.589	1988	Unmet
Rules / Serial shadow	return	59.8 [53.8, 69.4]	0.626	2000	Unmet
Rules / Two-in-flight shadow	cancel	56.9 [50.0, 69.2]	0.566	1988	Unmet
Rules / Two-in-flight shadow	return	57.1 [53.8, 67.9]	0.639	2000	Unmet
Rules / Full tunnel, 2 ms	cancel	52.8 [51.2, 61.6]	0.504	1988	Unmet
Rules / Full tunnel, 2 ms	return	55.3 [51.7, 66.2]	0.577	2000	Unmet
Rules / Full tunnel, 5 ms	cancel	58.3 [50.0, 67.9]	0.539	1988	Unmet
Rules / Full tunnel, 5 ms	return	54.3 [51.6, 62.6]	0.595	2000	Unmet

Prompt / condition	Target	BA [95% CI]	AUROC	Valid draws	Subset gate
Rules / Record buckets	cancel	56.7 [50.0, 81.2]	0.615	1988	Unmet
Rules / Record buckets	return	70.8 [51.2, 88.0]	0.703	2000	Unmet
Rules / Tamaraw 40/12 ms	cancel	58.6 [56.3, 70.8]	0.622	1988	Unmet
Rules / Tamaraw 40/12 ms	return	50.0 [50.0, 60.3]	0.500	2000	Unmet
Rules / Tamaraw 4/1.2 ms	cancel	52.5 [50.0, 61.6]	0.505	1988	Unmet
Rules / Tamaraw 4/1.2 ms	return	54.6 [51.4, 61.5]	0.534	2000	Unmet
Rules / FRONT	cancel	58.5 [50.0, 69.6]	0.544	1988	Unmet
Rules / FRONT	return	60.4 [50.0, 75.1]	0.635	2000	Unmet
Rules / WTF-PAD adaptation	cancel	57.2 [50.0, 74.2]	0.559	1988	Unmet
Rules / WTF-PAD adaptation	return	61.0 [50.0, 77.6]	0.625	2000	Unmet
Rules / WT random-donor	cancel	51.4 [50.0, 68.8]	0.606	1988	Unmet
Rules / WT random-donor	return	56.3 [50.0, 66.6]	0.558	2000	Unmet
Rules / FRONT, short calibration	cancel	51.2 [50.0, 70.1]	0.501	1988	Unmet
Rules / FRONT, short calibration	return	62.9 [50.0, 77.2]	0.655	2000	Unmet

Measured delays start at the RPC measurement origin; simulated spans start at the first input record. Zero added delay in FRONT/WTF-PAD's trace transformation does not measure contention on a live link. Actual tool-call ratios remain one for cell-only controls; shadow rows include extra executions. Neither delay panel measures timing-dependent LLM replanning. Thus the byte/BA comparison is shared, while a complete same-scheduler online Pareto comparison remains unperformed.

S20.5. Prospective sample planning

The planning script varies positive/negative customer counts, assumed BA and attacker dependence using 20,000 binomial trials per setting. It assumes independent customer correctness with equal sensitivity and specificity and a normal 97.5% upper endpoint, omits the AUROC gate, and is not a power estimate for the observed heterogeneous bootstrap. One attacker also models perfect dependence; seven independent attackers supply a contrasting multiplicity scenario. Under chance-level population BA and three negatives per positive, the searched 80%-power points are 75 positive customers for one attacker and 150 for seven. Thirty positives is a feasibility floor, not a guaranteed pass. At BA .55, requirements are substantially larger. The planning grid extends the frozen description's 250-customer maximum to 400; this post hoc range extension is recorded in the artifact errata. New independent customer/template sampling must precede a population-level absolute-bound claim.

TABLE S54. PLANNING SENSITIVITY: SMALLEST SEARCHED POSITIVE-CLUSTER COUNT WITH AT LEAST 80% PROBABILITY OF AN UPPER BA BOUND BELOW .60. INDEPENDENT BERNOULLI CUSTOMER CORRECTNESS AND A NORMAL 97.5% UPPER ENDPOINT; THIS OMITS THE AUROC GATE AND IS NOT THE POWER OF THE OBSERVED HETEROGENEOUS BOOTSTRAP.

Negatives per positive	Population BA	Independent attacks	Positive clusters
1	0.50	1	100
1	0.50	7	200
1	0.55	1	400
1	0.55	7	>400 in grid
3	0.50	1	75
3	0.50	7	150
3	0.55	1	400
3	0.55	7	>400 in grid

The archived prospective protocol describes further environment/model versions, precollection customer splits, training-only calibration, the unchanged BA/AUROC gates and live planner utility measurements. Its proposed collections are separate from the measurements reported here; the subsequent completed extensions retain their own protocols and results.

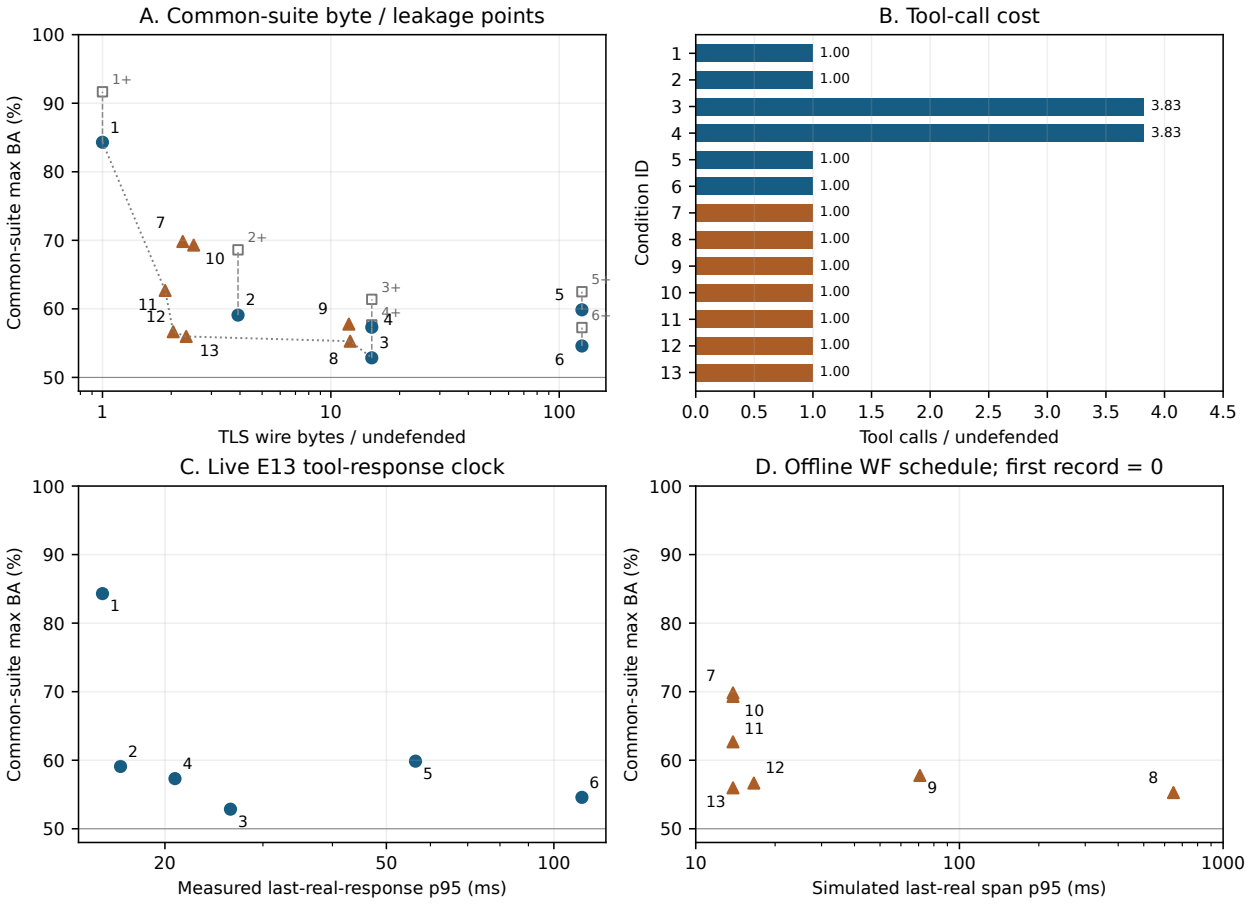


Figure 2. E14 standard-prompt operating points under the same seven attacks. Labels: 1 undefended, 2 per-call constant, 3 serial shadow, 4 two-in-flight shadow, 5/6 full 2/5 ms tunnels, 7 record buckets, 8/9 Tamaraw slow/fast, 10 FRONT, 11 WTF-PAD adaptation, 12 random-donor WT, 13 short-calibration FRONT. Gray squares with "+" labels in A show larger full-union maxima; dotted nondominance uses only matched colored points and is neither statistically certified nor globally optimal. Circles are measured E13 clocks; triangles are offline transformations. Panels C/D separate time semantics.

S21. Expanded Cross-Fitted Attacks and Oracle Localization

This extension reanalyzes the existing 690 Sol episodes with the same five customer folds as E10. It is post hoc: the cohort, defense design and prior results were already available. The new source/configuration freeze precedes the new fits, not the original test collection. All 53 customers are retained, with 14 cancellation-positive and 18 return-positive customers. Each prompt contributes 345 out-of-fold episodes. Fold sizes are 11/12/10/10/10 customers; a rare joint-label stratum has only three customers, so not every fold can contain that stratum. The saved fold assignment and customer/target counts make this restriction explicit.

S21.1. Complete attack union and fitting boundary

For each fold, joint count templates, cover donors, framing caps and size-to-tool decoders use only the other four folds. The extended union retains all nine prior network attacks: majority, undefended-data-trained gradient boosting, D2 logistic regression and gradient boosting, random forest, nearest neighbors, and three decoded-profile attacks. It adds the fixed trained 1D-CNN and k-FP-style leaf attack described in S20. CNN inputs are direction, log record length and a mask; the timing channel is identically zero because this analysis concerns length/order. No interarrival times are synthesized. The mapping from archived C/S direction strings to numeric signs is explicit and checked. An initial mapping error stopped before any prediction pack was produced; its failed protocol snapshot is retained separately.

All 60 fold/prompt/condition/target packs preserve their fitted learners, test scores and predictions. Out-of-fold rows are pooled only after every fold is evaluated. Each of 2,000 common customer-bootstrap draws retains a customer's tasks and repetitions together and reselects the full 11-attack maximum. Paired decreases use the same draw for undefended and defended conditions. These intervals condition on the fitted attackers and original cohort. The absolute gate remains an upper BA bound below .60 and suite-maximum AUROC below .65. Neither an attack's favorable point estimate nor an increase in the number of resampled trajectories replaces independent positive customers.

TABLE S55: Expanded E10: all 12 cells. Percent BA and paired undefended-minus-defended decrease, each with 95% customer-bootstrap intervals. AUROC is independently maximized over the same 11 attacks. Ratios are bytes/calls.

Prompt	Condition	Target	BA [95% CI]	AUROC	Decrease [95% CI]	B×/C×
Standard	Undefended	cancel	91.7 [82.1, 100.0]	0.979	–	1.00/1.00
Standard	Undefended	return	85.5 [76.7, 93.4]	0.897	–	1.00/1.00
Standard	Joint constant	cancel	60.5 [52.9, 72.4]	0.594	31.2 [15.3, 41.0]	19.76/5.04
Standard	Joint constant	return	55.5 [51.8, 64.2]	0.533	30.0 [14.8, 39.2]	19.76/5.04
Standard	Joint buckets	cancel	57.1 [50.0, 71.5]	0.579	34.6 [15.1, 46.4]	12.36/5.04
Standard	Joint buckets	return	52.7 [50.0, 61.7]	0.536	32.8 [20.8, 38.9]	12.36/5.04
Rules	Undefended	cancel	90.7 [81.5, 98.9]	0.978	–	1.00/1.00
Rules	Undefended	return	85.5 [76.6, 93.6]	0.880	–	1.00/1.00
Rules	Joint constant	cancel	58.8 [52.1, 71.6]	0.581	32.0 [13.1, 44.2]	19.88/5.04
Rules	Joint constant	return	59.4 [54.2, 68.8]	0.581	26.1 [11.1, 36.6]	19.88/5.04
Rules	Joint buckets	cancel	56.9 [50.2, 71.2]	0.573	33.8 [14.2, 45.8]	12.45/5.04
Rules	Joint buckets	return	52.7 [50.0, 65.0]	0.565	32.8 [19.0, 39.8]	12.45/5.04

S21.2. What changes under stronger attacks

In the standard-prompt arm, the CNN raises bucketed cancellation inference to 57.1%, and the leaf-fingerprint learner raises constant-framing return inference to 55.5%. The corresponding paired decreases from undefended transport remain positive: 34.6 points [15.1,46.4] and 30.0 points [14.8,39.2]. This exploratory comparison shows measured reduction with an unresolved absolute bound. It is not tied to the original 50% frozen-split realization. Both prompt arms and every evaluated operating point are retained, including rules-arm estimates that differ from the standard prompt. The main descriptive table reports the complete union; S15 retains the earlier suite and six-point selection analysis as historical context.

The extension saves 4,140 out-of-fold condition/target rows and checks all 45,540 attack scores against the fitted model archives. Input hashes preserve the original cohort and folds; a documented path map permits read-only verification after checkout relocation without changing frozen source paths or data hashes.

S21.3. Expanded oracle localization

A separate before-fit extension reuses the original E12 oracle filter and the same five folds. It retains or removes complete exchanges belonging to true state-changing calls, then exposes only network record directions and lengths to attackers. Tool identities choose the diagnostic subset but are not attack features. The full-view reference is the verified expanded E10 undefended output. All nine prior attacks plus the same fixed CNN and forest-leaf learners are fitted for each oracle view, yielding 40 additional fold packs and 2,760 out-of-fold rows. Exact model reloading and legacy refitting check 30,360 scores, customer separation, labels and the full-view reference. Empty filtered traces remain included.

TABLE S56: Expanded E12 oracle ablation, both prompts. Suite-maximum BA (percent) and AUROC, with customer-bootstrap 95% intervals. The full reference is the verified expanded E10 undefended prediction set.

Prompt	Target	Oracle view	BA [95% CI]	AUROC [95% CI]
Standard	cancel	Full trace	91.7 [82.1, 100.0]	0.979 [0.949, 1.000]
Standard	cancel	Writes removed	62.8 [54.6, 75.3]	0.611 [0.533, 0.768]
Standard	cancel	Writes only	91.7 [82.1, 100.0]	0.973 [0.936, 1.000]
Standard	return	Full trace	85.5 [76.7, 93.4]	0.897 [0.849, 0.965]
Standard	return	Writes removed	59.9 [52.2, 71.8]	0.657 [0.571, 0.789]
Standard	return	Writes only	84.9 [78.8, 94.0]	0.906 [0.864, 0.973]
Rules	cancel	Full trace	90.7 [81.5, 98.9]	0.978 [0.954, 0.999]
Rules	cancel	Writes removed	68.7 [59.4, 80.2]	0.690 [0.593, 0.822]
Rules	cancel	Writes only	90.7 [81.4, 98.9]	0.968 [0.930, 0.998]
Rules	return	Full trace	85.5 [76.6, 93.6]	0.880 [0.805, 0.955]
Rules	return	Writes removed	66.5 [56.5, 77.7]	0.664 [0.579, 0.807]
Rules	return	Writes only	85.1 [78.6, 93.7]	0.917 [0.862, 0.978]

For the standard prompt, write exchanges constitute 19.6% of real calls and 23.4% of TLS wire bytes, yet retain 91.7/84.9% BA versus 91.7/85.5% for full traces. Removing writes reduces the suite maximum by 28.9 points [10.7,41.9] for cancellation and 25.6 [11.0,35.4] for returns. The rules-arm decreases are 22.0 [6.0,35.1] and 19.0 [6.0,30.1] points. Read-only traces still carry measurable information; this is concentration of the main signal, not exclusive localization. The small full-minus-writes-only point difference does not constitute a formal equivalence test. Per-prompt read/write retained-call and byte fractions sum to one and are checked against the original trace projection.

S22. Additional Same-Provider Real-Agent Study (E15)

This supplementary study tests another model alias on actual tool decisions while keeping the original 690 Sol episodes unchanged. GPT-6 Luna with low reasoning effort serves as the planner and, in a separate context, the user simulator. The authenticated subscription interface returns structured actions; the harness executes them in the public retail sandbox and returns the actual observations. Requested and returned model identifiers, instructions, requests, raw events and decisions are archived. Ten pilot episodes are excluded from the 150 formal episodes.

S22.1. Sample, collection and analysis boundaries

The convenience sample contains 75 existing tasks: the first 29 inherited training tasks after excluding pilot IDs 0–4, plus all 46 inherited test tasks. Each has standard and rules prompts and one repetition. This gives 58 training and 92 test episodes from 10 and 24 nonoverlapping customers. The training customers include two cancellation-positive and seven return-positive customers; the test set includes five and seven. All customers and tasks were available during earlier manuscript development. The collection fingerprint fixes the model alias, effort, instructions, tool schema, code and customer split.

The appended dispatch/network algorithms and configuration were frozen after 44 formal episodes, before any formal test output. Defense parameters were fitted after the complete collection, using only the 58 training episodes; unlike E9, the fitted parameters were not frozen before test collection. Every completed formal episode remains included, irrespective of task reward, unknown tool attempts or protocol errors. Infrastructure failures are distinct from unsuccessful task outcomes. Table S57 reports reward, elapsed time and errors. These clocks include model interaction and local execution under the collection service, rather than estimating latency under cover.

TABLE S57. ADDITIONAL LUNA STUDY: ALL 75 RUNS PER PROMPT, INCLUDING TASK FAILURES. TOOL COUNTS ARE DISPATCHES TO REGISTERED TOOLS, INCLUDING FAILED CALLS; TOOL ERRORS ALSO INCLUDE UNKNOWN-NAME ATTEMPTS. WALL-CLOCK TIME COVERS MODEL INTERACTION AND LOCAL EXECUTION. THESE ARE UNDEFENDED PLANNER RUNS, NOT LIVE-COVER REPLANNING.

Prompt	Reward	Median / p95 (s)	Tool / model calls	Tool errors	Protocol errors	Infra / timeout / retry
Standard	32/75	92.5 / 159.0	498 / 1530	36	3	0 / 0 / 0
Rules	33/75	92.5 / 175.5	505 / 1542	37	4	0 / 0 / 0

The standard and rules arms obtain 32/75 and 33/75 native rewards. Median episode time is 92.5 s in both arms; the respective 95th percentiles are 159.0 and 175.5 s. There are three and four protocol-error episodes, retained in the analysis. The collection records zero infrastructure errors, timeouts or automatic retries; this does not imply all tasks succeeded. These are descriptive within-study outcomes, not a controlled performance ranking against the differently sampled Sol cohort.

S22.2. Dispatch diagnostic and local TLS replay

The dispatch diagnostic fits a majority control, logistic regression and gradient boosting to each prompt/view/target on its 29 training episodes. The all, writes-only and reads-only views expose tool names and response lengths. Reads-only retains names beginning with get/find/list; it excludes other non-write calls and is therefore different from the complete writes-removed complement in Sol E12. The archive records all retained/excluded names and counts. The inherited gradient-boosting minimum leaf size is 20: with 29 training rows it cannot split. This fixed model is retained and its constant predictions are disclosed; the suite maximum must be read in light of this small training set. Table S58 reports all 12 cells.

The network analysis pools only the training episodes to fit count templates, other-customer donor pools and framing caps. Each trajectory is replayed through undefended, per-call padding, shape cover and joint conditions, producing 600 actual TLS 1.2 loopback sessions. Unknown-tool error exchanges are retained; they are dispatch attempts rather than successful simulator-tool executions. The network projection contains directions, lengths and order, excluding provider traffic and arrival timing. The inherited R4 suite includes undefended-trained D_0, condition-refitted D_2 and training-derived decoder/count controls, evaluated per prompt. It is distinct from the expanded 11-attack Sol cross-fit suite. Table S59 retains all 16 network cells. Both analyses reselect the suite maximum in each of 2,000 customer-bootstrap draws and exclude draws missing a class. The intervals condition on the fitted predictions.

TABLE S58: Luna dispatch-view diagnostic: maximum over majority, D2 logistic regression and D2 gradient boosting. Each prompt uses 29 training and 46 test tasks. Percent BA and AUROC have customer-bootstrap 95% intervals. Read-only retains only get/find/list prefixes; it is not Sol E12's entire writes-removed complement.

Prompt	View	Target	BA [95% CI]	AUROC [95% CI]	Valid draws
Standard	all	cancel	78.3 [50.0, 96.2]	0.829 [0.619, 0.959]	1994/2000
Standard	all	return	63.2 [50.0, 78.4]	0.691 [0.502, 0.842]	2000/2000
Standard	writes-only	cancel	87.5 [81.1, 94.6]	0.925 [0.859, 1.000]	1994/2000
Standard	writes-only	return	66.6 [53.6, 80.5]	0.769 [0.582, 0.919]	2000/2000
Standard	reads-only	cancel	50.0 [50.0, 64.3]	0.500 [0.500, 0.722]	1994/2000
Standard	reads-only	return	57.0 [50.0, 74.2]	0.522 [0.500, 0.758]	2000/2000
Rules	all	cancel	73.8 [50.0, 98.0]	0.850 [0.513, 1.000]	1994/2000
Rules	all	return	67.5 [50.0, 82.2]	0.782 [0.575, 0.913]	2000/2000
Rules	writes-only	cancel	89.2 [67.1, 100.0]	0.917 [0.768, 1.000]	1994/2000
Rules	writes-only	return	77.8 [60.2, 92.6]	0.906 [0.769, 0.986]	2000/2000
Rules	reads-only	cancel	50.8 [50.0, 63.2]	0.500 [0.500, 0.774]	1994/2000
Rules	reads-only	return	57.3 [50.0, 75.2]	0.564 [0.500, 0.800]	2000/2000

TABLE S59: Luna local TLS replay: all 16 prompt/condition/target cells from 600 sessions. The inherited R4 suite includes undefended-trained D_0, condition-refitted D_2 and decoder/count controls; it differs from the expanded Sol union. Percent BA and AUROC have customer-bootstrap 95% intervals. B/C denotes wire-byte and dispatch-record ratios to the same prompt's undefended test replay. Records include unknown/error attempts and generated cover; C is not a successful-action ratio.

Prompt	Condition	Target	BA [95% CI]	AUROC [95% CI]	B×/C×	Gate
Standard	Undefended	cancel	64.2 [50.0, 100.0]	0.642 [0.500, 1.000]	1.00/1.00	Inconcl.
Standard	Padding	cancel	50.4 [50.0, 69.8]	0.567 [0.500, 0.781]	3.98/1.00	Inconcl.
Standard	Shape	cancel	58.3 [50.0, 70.0]	0.767 [0.500, 0.955]	4.25/3.91	Inconcl.
Standard	Joint	cancel	60.4 [50.0, 72.0]	0.623 [0.500, 0.724]	15.57/3.91	Inconcl.
Rules	Undefended	cancel	65.4 [58.3, 100.0]	0.667 [0.507, 1.000]	1.00/1.00	Inconcl.
Rules	Padding	cancel	57.5 [50.0, 81.1]	0.542 [0.500, 0.839]	3.95/1.00	Inconcl.
Rules	Shape	cancel	55.8 [50.0, 77.4]	0.579 [0.500, 0.878]	4.26/3.95	Inconcl.
Rules	Joint	cancel	59.2 [50.0, 72.5]	0.569 [0.500, 0.715]	15.58/3.95	Inconcl.
Standard	Undefended	return	84.9 [72.2, 100.0]	0.851 [0.728, 1.000]	1.00/1.00	Above
Standard	Padding	return	50.0 [50.0, 56.4]	0.531 [0.500, 0.702]	3.98/1.00	Pass
Standard	Shape	return	54.5 [50.0, 64.3]	0.500 [0.500, 0.643]	4.25/3.91	Inconcl.
Standard	Joint	return	57.0 [50.0, 71.4]	0.595 [0.500, 0.769]	15.57/3.91	Inconcl.
Rules	Undefended	return	90.9 [71.4, 100.0]	0.909 [0.714, 1.000]	1.00/1.00	Above
Rules	Padding	return	50.0 [50.0, 70.9]	0.500 [0.500, 0.669]	3.95/1.00	Inconcl.
Rules	Shape	return	59.1 [50.0, 75.0]	0.500 [0.500, 0.750]	4.26/3.95	Inconcl.
Rules	Joint	return	50.0 [50.0, 57.1]	0.500 [0.500, 0.617]	15.58/3.95	Pass

S22.3. Observed model sensitivity

Writes-only dispatch BA is 87.5/66.6% for standard and 89.2/77.8% for rules, compared with reads-only 50.0/57.0% and 50.8/57.3%. This reproduces the qualitative write-associated pattern in a smaller dispatch diagnostic. It does not estimate an equivalent Sol/Luna effect because the attack suite, read view, repetition count and cohort differ.

Undefended network BA is 64.2/84.9% for standard and 65.4/90.9% for rules. Joint shaping gives 60.4/57.0% and 59.2/50.0%, at 15.57/15.58× wire bytes and 3.91/3.95× dispatch records. Per-call padding alone gives 50.4/50.0% and 57.5/50.0%: joint shaping does not uniformly improve this smaller suite's point estimates. The original gate requires a BA upper 95% bound below .60 and maximum AUROC below .65. Two of 16 cells meet it (standard padding/return and rules joint/return); none of the conditions meets it for both targets in either prompt arm. The observed constant-score diagnostic covers every fitted gradient booster, including the inherited network and decoder variants, while logistic regression remains nonconstant. These findings motivate population- and attacker-specific calibration rather than a universal ranking of the program conditions.

S22.4. Validation and scope

The executable audit reconstructs native reward in fresh sandbox states and checks every model answer against the completed raw event, request-history prefix, subsequent assistant/tool record, arguments, response bytes and usage totals. A public raw archive redacts only the local home-directory prefix, retaining per-file original/public hashes. The same decision

alignment and reward reconstruction use this public archive and the pinned benchmark/tool sources in the package, without reading the original raw directories. Training-only recalibration, 600 TLS replays and 32 dispatch/network result cells reproduce the saved BA/AUROC estimates and intervals. A separate replay of the 12 dispatch-view cells exactly matches all 36 per-model prediction sets and suite statistics. A derived receipt separates dispatches to known tool names from all dispatch attempts; known-tool counts include failed calls. The original frozen verifier's receipt is retained with an explicit field-label correction. Network call ratios use all test dispatch records, including unknown/error attempts and, in shaped conditions, generated cover exchanges; they do not count successful actions.

This extension adds real model trajectories and a reusable evidence archive. Different cohort sampling, one repetition, the same provider/user simulator, few positive customers and weaker inherited attacks preclude a causal model ranking or an independent replication of Sol E12. Replayed program conditions preserve logical outcomes by construction. The study does not measure live-cover LLM replanning, a second environment, fixed open weights, another provider or new independent customers.

S23. R8 Construct validity and evidence calibration

This appendix applies the paper's evidence requirements to its own experiments. It adds deterministic sensitivity calculations on saved predictions, not new model runs, customers or defense deployments. The separate analysis plan and code hashes were frozen before these new calculations; the original test predictions were already available, so this remains post hoc.

S23.1. Customer composition and fixed-attacker sensitivity

The complete out-of-fold prediction collection contains 690 original model trajectories, re-used across conditions and targets. Each prompt-specific cell contains 345 trajectories (115 tasks, three repetitions), but only 53 customers. Cancellation has 54 positive trajectories from 18 tasks and 14 customers; return has 96 from 32 tasks and 18 customers. These full-cohort positive-customer counts differ from the original held-out subset's five/seven. Repetitions, prompts and transformed views do not create additional independent customers.

For each of the 11 saved attackers, episode-weight BA pools correctness within each true class. The sensitivity estimand first averages correctness within each customer/class stratum, then equally weights customers within each class and finally the two class rates. The four cancellation and eight return mixed-label customers contribute separately to both class rates; they are neither discarded nor forced to have one customer-level label. Suite maxima are independently reselected under both weightings. Thresholds and predictions remain fixed.

TABLE S60. FIXED-PREDICTION CLUSTER SENSITIVITY FOR THE COMPLETE OOF COHORT. BA AND LEAVE-ONE-POSITIVE-CUSTOMER-OUT RANGES ARE PERCENTAGES. EACH ROW HAS 345 TRAJECTORIES AND 53 CUSTOMERS. EVERY DELETION REMOVES THE CUSTOMER'S NEGATIVE TASKS AS WELL AS POSITIVE TASKS; ALL 14/18 DELETIONS REMAIN VALID FOR CANCEL/RETURN. RANGES ARE DETERMINISTIC INFLUENCE RANGES, NOT CONFIDENCE INTERVALS.

Prompt	Condition	Target	Episode BA	Customer-equal BA	LOO episode range	LOO customer-equal range
Standard	Undefended	cancel	91.67	94.05	90.62–96.67	93.59–96.15
Standard	Undefended	return	85.49	85.76	84.17–87.65	84.98–87.96
Standard	Joint bucket	cancel	57.07	54.80	54.30–61.59	53.70–57.55
Standard	Joint bucket	return	52.71	56.48	51.04–54.72	55.11–58.05
Standard	Joint constant	cancel	60.50	58.96	57.26–61.98	56.90–60.74
Standard	Joint constant	return	55.46	53.15	53.37–56.72	51.37–54.27
Rules	Undefended	cancel	90.74	92.86	89.58–95.56	92.31–94.87
Rules	Undefended	return	85.49	85.85	84.17–87.65	85.08–88.06
Rules	Joint bucket	cancel	56.90	54.71	54.12–61.42	53.61–57.45
Rules	Joint bucket	return	52.67	53.42	51.66–54.22	52.44–53.91
Rules	Joint constant	cancel	58.75	56.60	55.17–60.86	54.16–58.12
Rules	Joint constant	return	59.40	60.40	57.59–62.17	59.28–62.23

Weighting changes the estimand and moves different cells in different directions; it is not a universal correction in favor of either defense. Positive-customer deletion exposes concentration in particular customers. Neither calculation removes that customer's influence on models trained in other folds: no attacker is refitted. Thus these results condition on the saved fitted attacks and do not include retraining uncertainty, an independent population, or uncertainty induced by prior development access to the public tasks. We do not pool scores into one customer-level classifier because task labels can differ within a customer. The complete per-attack, per-customer/class and per-deletion values, including remaining denominators, are saved in `sp2027-r8-cluster-sensitivity/results.json`; an independent direct-weight verifier checks all 12 cells and 192 deletions.

S23.2. Applying R1–R9 to our own evidence

Table S61 distinguishes a bound measurement from an unverified extension. In particular, a structural planner-history invariant, measured fixed-action response/state equality, and live LLM task utility are different outcomes. Inherited native reward does not count as utility measured under the defended operating point.

E9/E10 program conditions inherit reward from reused trajectories. E13 additionally checks real responses and final live database states in all 4,140 local sessions, but the next action is specified by the saved trajectory rather than a new LLM decision. E14's offline WF adapters preserve abstract fragment identity/order and do not schedule fragments before input availability; they neither decrypt and compare payloads nor execute tools. Their copied native rewards are not independent utility measurements. E15's actual 150-task rewards and wall clocks describe the original collection service, not planner behavior under cover. Own live defended-LLM utility therefore remains unbound.

TABLE S61. SCOPED SELF-AUDIT OF E9/E10/E13/E14/E15 UNDER R1–R9. ENTRIES DESCRIBE EVIDENCE BOUNDARIES, NOT OVERALL SUCCESS GRADES. THE MISSING LIVE DEFENDED-LLM UTILITY BINDING IS RETAINED FOR OUR EXPERIMENTS UNDER THE SAME STANDARD APPLIED TO THE SOURCE CORPUS.

Item	Bound or partially bound observation	Unverified extension
R1: target/polarity	Privacy inference of cancel/return action-labelled task trajectories	Unobserved psychological intent or a general semantic secret
R2: receiver	E9/E10/E15 dispatch or TLS length/order; E13 local arrival timing	Production provider logs, routing leaks and unmeasured observers
R3: placement/execution	E13 executes tools on fixed saved actions; E14 separates measured and offline rows	Live LLM replanning under cover; executing WF transforms as production defenses
R4: defended fitting	Training-only calibration and condition-specific attack fitting; E15 fits only 58 training trajectories	Arbitrary attacker strength or prospective parameter freeze for E15
R5: matched adaptation	Defended-data attackers; common-seven condition comparison in E14	A fully matched clean-versus-adapted causal contrast from every D2/union difference
R6: workload shifts	Customer splits/cross-fitting, two prompts and same-provider Luna retail alias	New customers, new environment, independent vendor or blinded confirmation
R7: inference	Saved suite maxima, customer intervals, failures and negative outcomes retained	An absolute bound against unknown attackers or success inferred from a low point estimate
R8: utility	Original LLM rewards; E13 measured real-response/final-state equality on fixed actions	Live defended LLM task utility; timing-dependent replanning, quotas and failures
R9: cost	E13 local tool-chain cost; E14 matched byte/call accounting with separate simulated clocks	End-to-end deployed cost or a common live latency ranking across measured/simulated rows

S23.3. Matched-seven comparison and the limits of dominance

All rows in Table S62 use the same seven attack families: majority, logistic, gradient boosting, random forest, extra trees, leaf fingerprint and sequence CNN. BA and AUROC suite maxima are separately reselected in each customer-bootstrap draw. The measured conditions retain additional E13 attacks in the full-union assessment elsewhere; this common subset cannot replace that stronger assessment. The exact values, intervals, full-union values and cost denominators are retained in `provenance/r8-review/defense-self-audit.json`.

TABLE S62. BOTH PROMPTS UNDER THE MATCHED SEVEN-MODEL SUBSET. BA IS CANCEL/RETURN IN PERCENT. BYTES AND DISPATCH COUNTS USE THE CORRESPONDING PROMPT'S UNDEFENDED TOTALS; PER-PROMPT EVALUATION HAS 138 TRAJECTORIES AND 24 CUSTOMERS. WT IS THE RANDOM-TRAINING-DONOR RPC ADAPTATION, NOT THE ORIGINAL SYMMETRIC-PAIR GUARANTEE. NO ROW ESTABLISHES LIVE DEFENDED-LLM UTILITY.

Prompt	Condition	Cancel/return BA	Bytes×	Calls×	Execution
Standard	WT random donor	50.00/56.67	2.040	1.000	Offline
Standard	Serial shadow	50.00/52.86	15.106	3.827	Local tools
Standard	Two-in-flight shadow	50.00/57.32	15.106	3.827	Local tools
Standard	Full tunnel 2 ms	59.86/55.19	125.706	1.000	Local tools
Standard	Full tunnel 5 ms	54.58/53.68	125.706	1.000	Local tools
Rules	WT random donor	51.39/56.28	2.233	1.000	Offline
Rules	Serial shadow	56.11/59.78	15.468	3.882	Local tools
Rules	Two-in-flight shadow	56.94/57.06	15.468	3.882	Local tools
Rules	Full tunnel 2 ms	52.78/55.28	128.384	1.000	Local tools
Rules	Full tunnel 5 ms	58.33/54.33	128.384	1.000	Local tools

WT's lower byte and backend-call cost is a real advantage of the evaluated adaptation. For standard prompts it exchanges lower cost for higher return BA than serial shadow (56.67% versus 52.86%); neither point dominates the other. For rules, WT has lower point BA on both targets and lower byte/call cost than either shadow scheduler. It also weakly dominates the standard two-in-flight point in these four dimensions. These are descriptive point relations, not statistically certified or deployment dominance: they exclude AUROC and latency, intervals are wide, and the WT schedule is simulated while the shadow tool chain is measured. A claim including AUROC can change the ordering. Tunneled dummies execute no tools; the two fixed local tunnel operating points trade much larger byte cost for no extra backend calls, without establishing an optimal tunnel configuration. The evaluated shadow policy is therefore not presented as Pareto-superior to these controls.

What flush recognition establishes.. The E13 serial/parallel flush detectors achieve 90.45/92.31% BA on 15,278 record labels (8,902 after the declared flush start, 6,376 before), pooling 276 trajectories and 24 customers across prompts. This is schedule-phase recognition, not intent recovery. Separate real/cover request detection reaches 82.53/96.08% BA on 7,639 requests. These observations refute cover-indistinguishability for the measured scheduler/view. Already-saved attacks estimate a single flush boundary using a training-fitted detector, with customer-out-of-fold segmentation for intent training, then attack the remaining prefix. The highest such intent BA is 60.69% (rules serial return, 95% interval [50.83,70.95]); the

TABLE S63. R4, R5 AND R8 FAMILY DISPOSITIONS. REASONS ARE SCOPED TO THE ITEM, NOT JUDGMENTS OF OVERALL STUDY QUALITY.

Family	R4	R5	R8	Scope note
UI Traces	B	B	U	Offline attribution does not report objective task utility.
WRA	–	–	P	Proxy visibility and executable-decoy utility are not fully joined.
NetEcho	B	?	U	Topic prediction is not downstream task utility.
Early Bird	B	?	U	Semantic-cache timing is not end-to-end task utility.
Whisper Leak	P	U	U	Global fitting and defended results are described; condition-specific mapping is incomplete and matched adaptation unreported.
SpliceLeak	–	–	U	QCP component cost; no task-utility outcome.
Time Will Tell	P	U	U	Input-length threshold profiling is specified; instruction-specific reprofiling is unbound and matched adaptation unreported.
I Know What You Said	–	–	–	Zero-copy component control; family-level utility item inapplicable.
Speculative decoding	P	U	–	Profiling and mitigation outcomes are described; protected-profile mapping is incomplete and matched adaptation unreported.
Dangol workflow	B	B	U	Matched result is capture-scoped; objective workflow utility is unreported.
Cloud traffic benchmark	U	U	P	Fixed-configuration fingerprints under changed prompts/temperatures; protected refitting and matched adaptation unreported.
Remote timing	–	–	U	Fixed-rate analysis does not bind task utility.
TDXRay	–	–	B	Vocabulary coverage is component utility, not deployment-task utility.
SafeKV	–	–	U	Detection and serving performance use separate conditions.
AHB/Turing	P	U	B	Online task accuracy is bound; condition-specific code is partial run evidence; matched adaptation is unreported.
KV-Gov	–	–	U	Simulation, hardware gap and HMAC costs are distinct outcomes.
CachePrune	–	–	B	QA/summarization outcomes bind selective cache sharing.
OTRO	–	–	–	Mechanism microbenchmark has no scoped task-utility outcome.
SparLeak	P	U	B	§7.5/Table 4 pairs ASR/ROUGE; profiling-to-intensity mapping is incomplete and matched adaptation unreported.
Shape and Substance	U	U	U	Inspected scope reports analytic padding, not protected-fit/matched-test or task utility.

unstripped views remain in the complete suite. Recognizable flush does not imply 90% intent recovery, and the relatively low stripped-view points do not erase the stronger unstripped attacks or certify defense success.

S23.4. Scoped reuse audit: applicability and denominators

S23.5. What the nine items mean

R1–R9 are checks on whether an observation can be reused for a specified comparison. They are not a quality score for a paper, a safety grade, or a judgment that an original study ought to have performed every possible experiment. Bound means that at least one result record in a family supplies the requested binding; Partial retains both supported and unbound parts; NR-R means the inspected source scope does not report the requested item; NR-V means the source/run/version binding could not be verified; NA means the requested inference is outside the scoped claim. The 20-family matrix is a family-level summary, while result-level records remain distinct where mechanisms or test populations differ.

R4 concerns fitting/calibrating an attacker on protected data when adaptation is part of the reuse question. R5 concerns a matched clean/adapted contrast on the same defended test. Thus R5 does not demand a learned-attacker refit for fixed-rate analytical costs, deterministic namespace controls, or non-learning mechanism benchmarks. Each item has 11 applicable families. In the record-derived projection, R4 has 4 B, 5 P, 2 U and 9 inapplicable; R5 has 2 B, 2 ?, 7 U and 9 inapplicable. U denotes an unreported binding within the inspected scope, not a failed experiment.

B=Bound, P=Partial, U=NR-R, ?=NR-V, –=NA.

For R5, the applicable denominator is 11, with 2 B (2/11), 2 ? and 7 U. Under the narrower interpretation that only explicitly learned/calibrated adaptive-attack experiments call for this matched contrast, Shape and Substance may be recoded inapplicable: 2/10, with 2 ? and 6 U among the remaining entries. This is a denominator sensitivity, not a correction to the matrix. For R8, the broad item is task *or* component utility: 4 B, 2 P, 11 U, 3 inapplicable (4/17). If the estimand is restricted to deployment-task utility, TDXRay's vocabulary-coverage/lookup result is out of scope, yielding 3 B, 2 P, 11 U and 4 inapplicable (3/16). This alternative does not convert U into an experimental failure. CachePrune's joint R7–R9 binding remains a scoped positive example, not evidence of adaptive robustness.

Across the 20 audited families, the R1–R9 column distributions (B/P/U/?/inapplicable) are: R1 20/0/0/0/0; R2 17/3/0/0/0; R3 17/3/0/0/0; R4 4/5/2/0/9; R5 2/0/7/2/9; R6 17/3/0/0/0; R7 16/1/0/0/3; R8 4/2/11/0/3; R9 11/4/4/0/1. Each column sums to 20. S36 documents the record-derived projection and its 13 changes; the earlier source-scope reconciliation remains archived in S34.

S23.6. Same-operation semantic-attribute diagnostic (E16)

This exploratory analysis addresses a different target from cancel/return. We fix the observed operation to `return_delivered_order_items`, requesting exactly one item, and predict whether the selected initial-order product belongs to a frozen 15-name digital-device grouping. The benchmark supplies product names, not this broader taxonomy; the grouping is researcher-defined and is not validated as a real-world secret. It was fixed before fitting or inspecting traffic/target associations. The original 690-episode population was already available, so this is not a fresh confirmatory dataset.

Source, alignment and grouping. The 138 eligible calls occur in 114 episodes. Full-session reconstruction exactly matches the saved raw TLS socket records, including episode hashes, ordered calls and UTF-8 body lengths. We select exact per-call request/response record slices using known call boundaries. This is an oracle diagnostic; it does not demonstrate passive discovery of a call's location. Each feature vector contains record count, total, maximum and minimum length in each direction. No plaintext, identifiers or measured timing are used.

Repeated calls are averaged within prompt/customer/order/item. Each prompt then has 20 instances (9 positive, 11 negative), spanning 15 customers (8 with positives, 9 with negatives, 2 with both). The same 20 instances and aggregated features occur in both prompt arms; the two arms are not independent replications. We retain the original five customer folds. All training folds have both classes; one test fold lacks positives, so results pool OOF predictions rather than average fold BA.

Frozen fits and negative controls. The pre-fit protocol binds code, input and source hashes. Learners are LR ($C = 1$, balanced), RF (200 trees, depth 3, minimum leaf 2), and GB (50 stages, depth 1, learning rate 0.05), with no post-result tuning or orientation reversal. Operation-only and canonical constant-size inputs contain no varying information; a constant-class predictor gives exactly 50% BA. The order-item-count-only diagnostic retains a potential whole-order-size confounder. A fixed-seed label shuffle within the old folds is shared across repeats/prompts; it is one sanity check, not a permutation distribution or a significance test.

TABLE S64. SAME-OPERATION ATTRIBUTE DIAGNOSTIC. BOTH PROMPTS HAVE IDENTICAL FEATURES AND RESULTS. FIXED THREE-MODEL SUITE MAXIMUM BA AND 95% CUSTOMER-BOOTSTRAP INTERVAL; 20 INSTANCES, 15 CUSTOMERS PER PROMPT.

Observation / target	BA (%)	95% interval
Raw record summaries, original target	30.3	[16.7,52.7]
Constant-class reference	50.0	[50.0,50.0]
Whole-order item count only	51.5	[40.6,68.8]
Raw summaries, fixed shuffled target	65.2	[47.4,85.7]

The raw fixed-suite maximum is 30.3%; the original-direction AUROCs are 0.182/0.182/0.187 for LR/RF/GB. These below-chance scores do not imply no information: reverse associations may exist. We preserve the frozen orientation instead of selecting a more favorable attack after seeing the result. The high shuffled-label point further illustrates the instability of this tiny grouped cohort. These results neither establish a stable non-action-attribute attack nor prove absence of such leakage, and provide no evidence of defense effectiveness.

Residual confounding and reproducibility. Initial whole-order sizes 1/2/3/4/5 have 6/3/2/7/2 instances, with 3/0/1/5/0 positive labels. Fixing the requested count does not balance full-order size. No post-result matching is fitted. Intervals use 2,000 fixed-prediction customer bootstrap draws, omitting one-class draws and recording their count; they exclude refitting uncertainty and target-choice uncertainty. An independent verifier checks the source chain, grouping and repeat aggregation, reloads 150 saved models, reproduces 600 OOF scores and all BA intervals, and verifies positive-class score orientation. Full per-model BA/AUROC, every fold's scores, controls and support counts are retained in the attribute module. Protocol SHA-256: `c2c7c67363f80a89c5df947380e7f071619 6844ce4365a2b9770fd732d28ad59`.

S24. Additional Endpoint Model Replications

Two endpoint replications each execute 115 tasks once with the standard prompt, using the original 69/46 task and 29/24 customer split. All training trajectories precede defense calibration and test collection. Network interventions replay the same real model trajectories over local TLS; they do not measure defended online replanning. Model aliases are not immutable weight revisions. The deployment owner identifies MiniMax-M3 as a DeepSeek backend; this attribution is not independently verified.

MiniMax uses a 4,096-token output limit. GLM uses 16,384 after an interface preflight exposed truncation; all earlier-interface episodes, including successes, are archived and excluded. GLM requests disable parallel tool calls to match the single-action harness. Transport timeouts resume cached decisions and preserve failed attempts; model protocol failures remain terminal outcomes. The GLM transport waiting limit changed from 180 to 600 seconds without changing inference requests. This budget asymmetry permits within-model defense contrasts, not an equal-budget model ranking.

S24.1. MiniMax-M3

Native task success is 68/115, including 23 terminal protocol failures in the denominator. The archived verifier passes 5627 checks. These archived endpoint rows use the original six attacks; the subsequent transfer section adds total exchange count and supplies the expanded-suite figures retained in S32. Network BA intervals use 2,000 customer bootstrap draws, reselecting the suite maximum conditional on fitted attackers. The small positive customer counts remain five/seven for cancellation/return.

View / condition	Target	BA (%)	95% interval
raw	cancel	58.3	[50.0, 83.3]
shape_const	cancel	60.4	[50.0, 74.1]
raw	return	83.5	[68.2, 98.7]
shape_const	return	59.9	[50.0, 79.7]
all	cancel	58.3	[50.0, 83.3]
all	return	83.5	[68.2, 98.7]
writes-only	cancel	58.3	[50.0, 80.0]
writes-only	return	83.8	[72.5, 95.8]
writes-removed	cancel	52.9	[50.0, 70.5]
writes-removed	return	58.7	[50.0, 81.4]

S24.2. GLM-5.3-flash

Native task success is 83/115, including 16 terminal protocol failures in the denominator. The archived verifier passes 5023 checks. These archived endpoint rows use the original six attacks; the subsequent transfer section adds total exchange count and supplies the expanded-suite figures retained in S32. Network BA intervals use 2,000 customer bootstrap draws, reselecting the suite maximum conditional on fitted attackers. The small positive customer counts remain five/seven for cancellation/return.

View / condition	Target	BA (%)	95% interval
raw	cancel	100.0	[100.0, 100.0]
shape_const	cancel	50.0	[50.0, 79.1]
raw	return	86.6	[73.7, 97.4]
shape_const	return	67.3	[58.7, 89.5]
all	cancel	100.0	[100.0, 100.0]
all	return	86.6	[73.7, 97.4]
writes-only	cancel	100.0	[100.0, 100.0]
writes-only	return	78.1	[67.5, 92.4]
writes-removed	cancel	65.0	[50.0, 89.5]
writes-removed	return	63.0	[50.0, 84.6]

Interpretation. Both endpoints reproduce a return-operation signature and its reduction under joint shaping. MiniMax cancellation is weak before defense and does not improve; GLM return retains substantial residual inference. No endpoint establishes the joint absolute screening criterion for both targets. The result strengthens the evidence-binding contribution: protection claims require the model, target, receiver view and calibration population to be specified together. Write-selected TLS captures are oracle localization diagnostics; attackers receive only lengths, directions and order.

S25. Attacker and Calibration Transfer Boundaries

This exploratory extension retains the earlier small-suite results and adds the complete inherited endpoint suite plus an exchange-count rule: majority, clean-trained gradient boosting, defended-data logistic regression and gradient boosting, decoded-tool gradient boosting, decoded target-call count, and total exchange count. It is not the larger Sol timing/CNN suite. All thresholds and decoders fit training customers only. Each of 2,000 customer bootstrap draws reselects the maximum; single-class draws are excluded. Point metrics are independently recomputed from saved predictions. Model endpoints share the original task/customer partition; neither has a verified immutable weight revision.

S25.1. Attacker transfer

Rows train on the source model and test on held-out customers under the target model. Joint conditions retain target-local calibration, so this table isolates attacker transfer. Calibration transfer is a separate experiment below.

Source	Target	Condition	Target label	BA (%)	Bytes×	95% interval
MiniMax	MiniMax	Raw	cancel	58.3	–	[50.0, 83.3]
MiniMax	MiniMax	Raw	return	83.5	–	[68.4, 98.7]
MiniMax	MiniMax	Joint	cancel	60.4	–	[50.0, 74.1]
MiniMax	MiniMax	Joint	return	67.5	–	[57.7, 87.2]
MiniMax	GLM	Raw	cancel	75.0	–	[59.7, 100.0]
MiniMax	GLM	Raw	return	77.5	–	[66.0, 97.4]
MiniMax	GLM	Joint	cancel	50.0	–	[50.0, 50.0]
MiniMax	GLM	Joint	return	55.8	–	[50.0, 81.1]
GLM	MiniMax	Raw	cancel	65.4	–	[54.7, 85.2]
GLM	MiniMax	Raw	return	88.1	–	[76.5, 98.7]
GLM	MiniMax	Joint	cancel	67.1	–	[50.0, 87.0]
GLM	MiniMax	Joint	return	53.5	–	[50.0, 70.0]
GLM	GLM	Raw	cancel	100.0	–	[100.0, 100.0]
GLM	GLM	Raw	return	86.6	–	[73.7, 97.4]
GLM	GLM	Joint	cancel	50.0	–	[50.0, 80.8]
GLM	GLM	Joint	return	67.3	–	[58.7, 89.5]

S25.2. Calibration transfer and localized padding

Source training templates, donors and caps are applied to target trajectories, including target training trajectories used to refit attackers. All 690 sessions are actual local TLS captures. Write-only padding uses true tool identity at the trusted defense boundary; the adversary receives lengths and directions only. It introduces no cover calls. Byte ratios divide summed test wire bytes by target-model undefended test bytes. Test-call ratios (MiniMax/GLM) are 3.36/4.24× for local joint, 3.11/4.66× for imported joint and 1.00/1.00× for write padding. Raw totals are 510/332 calls; request-record counts agree with saved call arrays.

Source	Target	Condition	Target label	BA (%)	Bytes×	95% interval
MiniMax	MiniMax	Joint	cancel	60.4	20.06	[50.0, 74.1]
MiniMax	MiniMax	Joint	return	67.5	20.06	[57.7, 87.2]
MiniMax	MiniMax	Write padding	cancel	50.0	1.45	[50.0, 62.4]
MiniMax	MiniMax	Write padding	return	61.3	1.45	[52.6, 80.2]
MiniMax	GLM	Joint	cancel	70.8	18.65	[60.5, 90.2]
MiniMax	GLM	Joint	return	50.0	18.65	[50.0, 59.2]
GLM	MiniMax	Joint	cancel	50.8	17.48	[50.0, 69.2]
GLM	MiniMax	Joint	return	63.0	17.48	[50.0, 82.9]
GLM	GLM	Joint	cancel	50.0	15.95	[50.0, 80.8]
GLM	GLM	Joint	return	67.3	15.95	[58.7, 89.5]
GLM	GLM	Write padding	cancel	60.0	1.51	[50.0, 87.0]
GLM	GLM	Write padding	return	61.6	1.51	[53.4, 83.1]

Scope of the conclusion. Target-specific contrasts retain the original engineering screen. Expanded-suite MiniMax joint return BA is 67.5%, versus 59.9% previously: the added exchange-count attack changes the evaluation contract, not the trajectories. Write-padding comparisons likewise use the expanded-suite 61.3/61.6% return values.

Paired transfer uncertainty. Paired contrasts use the same customer draws and reselect each condition's suite maximum. Moving MiniMax calibration to GLM changes cancellation by $+20.8$ points (95% interval $[-12.6, 35.3]$) and return by -17.3 points ($[-38.2, -5.3]$). Moving GLM calibration to MiniMax changes cancellation by -9.6 ($[-21.1, 6.4]$) and return by -4.5 ($[-25.0, 12.3]$). These are historical nominal intervals, conditional on fitted predictions. They are superseded for directional inference by the eight-contrast permutation/Holm analysis in Section S30. The same section adds local recalibration and attacker-refit controls; no nominal interval alone establishes a model-source effect.

Request-field mechanism check. Exact serialization accounting identifies which request fields contribute variable length. Across 39 MiniMax and 40 GLM return calls, the item-list field contributes 28–70 bytes (means 40.2 and 37.8), whereas the order identifier contributes 25 bytes and payment-method field 39–44 bytes. This explains a source of length variability without identifying a semantic privacy attribute: the accounting omits fields only for diagnosis, not as a valid API intervention. Response size, call count and timing provide additional channels. The earlier same-operation product-category diagnostic remains inconclusive; operation localization cannot be promoted to attribute inference.

S26. Live Planner Utility with Prefix Shadow Execution

This exploratory paired study selects 50 existing task IDs by a frozen SHA-based ordering before new collection, with two fresh GLM planner episodes per task. Both agent and user contexts issue new model requests; cached decisions are used only to resume the same episode after transport failure. The 100 episodes use the served GLM alias, native single-tool actions, a 16,384-token budget and ten workers. They share known tasks with earlier studies and are not a new-customer confirmation. Arm order alternates by task parity.

Execution and deadline contract. The baseline executes real actions in the retail environment. The prefix-shadow arm additionally executes at most one training-template deficit call on isolated state after each real call; its response is hidden from the planner. Both arms delay each real response by 20 ms, with another 20 ms for cover. For task IDs divisible by five, the first tool response receives another 200 ms delay. The 120 ms deadline is evaluated against this configured delay, not OS scheduling noise. The real operation can already have executed when its response is replaced by a timeout warning; the next planner request sees that warning and can inspect or retry the state. This tests reaction to a controlled execution/observation mismatch. There is no terminal flush, constant TLS framing or actual WAN path in this utility diagnostic.

Outcome	Baseline	Prefix shadow
Episodes	50	50
Native successes	37	33
Terminal protocol failures	11	7
Real tool calls	360	390
Isolated cover calls	0	390
Injected response timeouts	12	12
Tool actions following timeout	96	93
Final-pass response-wait p95 (ms)	32.0	47.0

Interpretation and verification. The paired native-success difference is -8.0 percentage points, with a 2,000-draw customer-bootstrap interval [-23.9,6.5] over 37 customers. All protocol failures remain in the denominator. Subsequent actions demonstrate live reactions to timeout observations; their count alone does not establish correct recovery. The verifier rebuilds actual and reference tool states, checks output requirements and native rewards, matches model-request counts, and checks all live-metric rows. Recorded response waits are from the final execution pass: cache recovery deterministically replays completed action prefixes and their delays. The frozen Windows monotonic clock has 15.625 ms resolution, so sub-tick differences are not resolved. These waits exclude LLM generation and WAN transit, and are not original uninterrupted end-to-end latencies. Reward is measured on the actual model decisions, rather than inherited from an undefended trajectory.

Transport recovery provenance. An import-order bug selected the shared endpoint adapter, which honored a 180 s caller timeout despite the live protocol's intended 600 s transport wait. After 97 completed episodes, explicit adapter resolution selected the local 600 s implementation; request bodies, model settings, parsing and all cached decisions remained unchanged. The amendment preserves the old adapter, hashes, logs and no-progress stop, and permits bounded recovery after the identified code fix. This infrastructure wait is separate from the experimentally injected 120 ms tool-response deadline. Failed model outputs are terminal outcomes, not transport retries. The utility estimate applies to this prefix scheduler and fault model; it cannot supply a reward value for the full joint framing/flush replay rows in Exp 3.

S27. Complete Record-Mode Synthesis

Every record is classified once by its primary measurement mode. Executed collection/prototype includes separately collected conditions and does not imply paired full-workflow utility. A simulated or derived result remains in that category even when its parameters were measured. Two placement-unresolved records remain explicit. This exhaustive census supplies the denominator for the three explanatory lenses in the main paper; those lenses are retrospective illustrations, not a prevalence sample.

Record	Mode	Source locator
UI-delay-offline [57]	Trace transformation	6.1 Fig5;A.5
WRA-decoy-execution [40]	Executed collection or prototype	8.1-8.3 Tables8-11
NET-upstream-local [50]	Executed collection or prototype	IV-C-E TableIII-IV
NET-dummy-padding-simulation [50]	Trace transformation	VII TableIX
EARLY-K-calibration-oracle [47]	Simulation or derived estimate	V-A TableVI;IV-A
EARLY-semantic-deidentification [47]	Executed collection or prototype	V-B
WHISPER-randompadding-capture [58]	Executed collection or prototype	6.1 Tables3-4
WHISPER-batching-replay [58]	Trace transformation	6.2 Fig7
WHISPER-injection-replay [58]	Trace transformation	6.3 Table5
SPLICE-CTBF-signal [54]	Executed collection or prototype	VII-C TableII
SPLICE-QCP-cost [54]	Component or calibration measurement	VII-C TableII
TIME-length-instruction [56]	Executed collection or prototype	7.2
TIME-padding-estimate [56]	Simulation or derived estimate	7.3
SPEC-padding-REST [53]	Placement unresolved	6.1 Fig6-7 Table2
DANGOL-capture [23]	Trace transformation	9.8-9.9 Table6
DANGOL-livebinding [23]	Executed collection or prototype	9.7, OHTTP paragraph after Table 5
SIG-common-guardrail [62]	Executed collection or prototype	4.1-4.2;AppendixC
SAFE-post-trigger [52]	Executed collection or prototype	7.2.4 Fig12;7.3-7.4
AHB-online-executed [45]	Executed collection or prototype	5.1.2 Table1;E.1-E.3
AHB-offline [45]	Trace transformation	Table1 offline rows;E.1-E.3
KVGOV-partialflush-sim [49]	Simulation or derived estimate	5.1-5.3
KVGOV-hardware-gap [49]	Component or calibration measurement	5.7-5.8
KVGOV-HMAC-cost [49]	Component or calibration measurement	5.9
TRAFFIC-prompt-reformulation [35]	Executed collection or prototype	V-C1 Fig9a
TRAFFIC-temperature [35]	Executed collection or prototype	V-C1 Fig9a
CARLINI-fixed-rate [3]	Placement unresolved	VII Fig11-12
GAO-zero-copy-off [48]	Component or calibration measurement	7.1
TDXRAY-oblivious-map [55]	Component or calibration measurement	6.1 Fig8 Table3
AGENTTELL-fixed-instruction [59]	Executed collection or prototype	4.3.1;5 Finding1
FP-CLOUDFLARE [61]	Executed collection or prototype	6.4 Table5
BAR-co-running-noise [60]	Executed collection or prototype	7.2-7.3 Table8
RHYTHM-VPN [9]	Executed collection or prototype	VI-B;VII
DEB-filter-origin [27]	Executed collection or prototype	Threatmodel;4.2
CACHEPRUNE-selective [46]	Executed collection or prototype	3.1;5.1-5.6;Table1;AppendixA
CACHEPRUNE-context [46]	Executed collection or prototype	3.1;5.1-5.6;Table1;AppendixA
OTRO-unpadded [51]	Component or calibration measurement	III;V;VI;VII-C-E;TableIII
OTRO-epochpad [51]	Simulation or derived estimate	III;V;VI;VII-C-E;TableIII
SPARLEAK-replacement [37]	Executed collection or prototype	3.1;4-7;Table4
SHAPE-padding-estimate [36]	Simulation or derived estimate	IV;V;VII;VIII;IX

For OTRO-epochpad, the estimate category identifies the derived padding cost; it does not negate the reported conditional-leakage measurement. EARLY-K similarly combines measured calibration with a stochastic oracle result. These mixed records illustrate why the measured component must remain attached to the claim.

S28. Family-Level Holdout Evidence

This is a family-level discovery index, not a count of defense-transfer tests. The three model H entries concern attack/identification generalization; even an H in a mitigation family must be linked to its defended condition before reuse. The main RQ3 table separates those evidence objects, and S34 identifies every H entry. Tables S66–S68 cover all 82 indexed families. T: task/input/user; M: model; C: harness/topology; H: hardware; N: network/load; D: time. **H** records an explicitly reported held-out unit/domain, **V** a condition comparison or recalibration, **X** an explicit restriction or absent/unverified holdout in the inspected scope, and **?** an unresolved axis. A dash applies only to conceptual families. H does not certify all grouping details or successful transfer; V is not a holdout. Notes restrict each code. Core and adjacent families remain separate (C/A). The machine-readable matrix preserves the full source excerpts, locators and input hash; these tables synthesize that archived extraction rather than claim a new full-paper audit.

TABLE S66. HOLDOUT EVIDENCE, FAMILIES 1–28. CODES AND AXES ARE DEFINED IN APPENDIX S28.

Family	Scope	T	M	C	H	N	D	Source locator and qualification
F01 AgentPrint [31]	C	?	?	?	?	?	?	Random trace splits; prompt grouping unresolved. III; V; Appendix D.A–D.D; Fig. 11
F02 UI Traces [57]	C	V	V	V	?	?	?	Cross-task/site comparisons; exact-question overlap remains. 3–4; source index
F03 WRA [40]	C	H	?	?	?	?	?	Prompt tests and synthetic persona split; version reconciliation open. 3.1; 6.1; 8–9; Tables 1/8/10/11; Appendix C.1–C.3
F04 FP-Agent [61]	C	H	?	X	?	?	?	Separate task holdout; visitor/host separation unresolved. 4.1–4.7; Tables 2/4; 6.1–6.2; Appendix E/Table 8
F05 Prompt length [32]	C	H	?	V	?	V	V	OOD inputs; browser/API and day/night buffering controls. 5.1–5.3; Fig. 4; Tables 2–3
F06 PromptPeek [71]	C	V	?	?	?	V	?	Candidate-prior and simulated request-load sweeps. III.B; VI.A–VI.C; Table II
F07 NetEcho [50]	C	V	?	V	?	?	?	Deployment/frontend scenarios; independent split binding unresolved. III.A; Table III; VI.A–VI.B/Table VI
F08 Early Bird [47]	C	?	V	?	?	?	?	Provider observations and candidate calibration; grouped split unresolved. arXiv 2409.20002: II.C; IV.A; V.A; Tables IV–V
F09 InputSnatch [38]	C	H	?	?	?	?	?	Held-out and generated target prompts; candidate-prior dependence. arXiv 2411.18191: 3.2; 4.3–4.4; Tables 2/4
F10 Whisper Leak [58]	C	H	V	?	?	?	?	Unique target-prompt holdout; model/provider comparisons. 3.1; 3 classifier/split discussion; 4.1–4.2; Table 1; 6
F11 Spill the Beans [74]	C	V	?	?	?	?	?	Exposure/monitor sweeps; independent capture grouping unresolved. Threat model; 5/Table 1; 6.1
F12 SpliceLeak [54]	C	V	?	?	?	?	?	Bounded cloze tasks and chunk/candidate coverage. Threat model; VII.A–VII.C; Figs. 7–8
F13 Time Will Tell [56]	C	?	?	?	?	V	V	Concurrent network/load profiling; short-window recalibration. 3; 4.1–4.4; 6.2–6.5; Tables 3–4
F14 Prompt Echo [34]	C	H	V	?	?	?	?	Overlap-filtered prompts; models evaluated separately. 3.1; 5.1–5.3; Tables 2–3
F15 I Know What You Said [48]	C	H	V	V	V	?	?	Prompt overlap filtering; model/hardware/framework comparisons. Abstract; source index
F16 Speculation [53]	C	H	?	?	?	V	?	New traces and OOD paraphrases; remote/load conditions. v4: 4.3–4.8; Table 1; 6.1
F17 Behind Bars [60]	C	?	V	?	V	?	?	Model identification CV; additional hardware evaluation. Table1; 6.1–6.3; Table7 (printed5383/PDF12); 7.1/7.3; Open Science
F18 Prompt caching [33]	C	?	V	?	?	?	V	Provider survey over two months; historical exposure only. 3.1–3.2; 4.1–4.2; Tables 1–2
F19 Whose Agent [77]	C	?	?	?	?	?	?	Trial split; task/site/version independence unresolved. 4,6.1–6.3; Tables4–5; Fig7;7
F20 Workflow traffic [23]	C	?	?	H	?	?	?	Leave-composition-out; stored and live populations separated. 3,9.1,9.7–9.14;Table6
F21 GPU contention [2]	C	?	?	V	?	V	?	Serving-stack and interleaved-load controls. IV,IX,TableVII,X,XIV–XVII
F22 AgentTell [59]	C	V	?	?	?	V	?	Secret rotation; cold/loaded and equal-completability controls. 3–5;AppendixD.2–D.3,Table8;AppendixE
F23 Traffic benchmark [35]	C	?	?	H	?	?	?	Topology shift distinct from feature-view restriction. III,IV-A–C,V-A TablesI–III,V-B Fig8,VI
F24 Remote timing [3]	C	?	V	?	?	?	V	Month-long observations; method/model changes confounded. III,IV-B–C,V-D–E,VI-C,VII,Figs8,10,12
F25 Persona fingerprint [4]	C	?	?	X	?	?	?	Pooled sites; leave-site-out not verified. 2,3.1–3.4,4,5.1,5.5–5.8
F26 TDXRay [55]	C	?	?	?	?	?	?	Tokenizer-localized LLM case; AES result separate. 3;5.1–5.4;6.1–6.2
F27 Detokenization [5]	C	?	?	X	?	?	?	Explicit same-instance profiling restriction. 3.1 and 3.3: shared-code, SMT and same-instance profiling; source index
F28 Spillage [15]	A	V	?	?	?	?	?	Task-irrelevance and selected-site evaluation. 4.1–4.2 (evaluator inputs), 5.4, Table 4, 6.2

TABLE S67. HOLDOUT EVIDENCE, FAMILIES 29–56. CODES AND AXES ARE DEFINED IN APPENDIX S28.

Family	Scope	T	M	C	H	N	D	Source locator and qualification
F29 PlanTwin [82]	A	V	?	?	?	?	?	Template candidate pool; production transfer unresolved. 2.2,3.2,4.2–4.4,5.1–5.9,6.1
F30 SICC [83]	A	?	?	?	?	?	?	Authorized-view theorem; empirical shift binding unresolved. Theorem 1; Experiments Setup; Tables 3–4, 7–8
F31 TOPR [16]	A	V	?	?	?	?	?	Benchmark and alignment populations differ. 3.1–3.2,4,5.2,5.6,6,Limitations
F32 ClawGuard [12]	C	?	?	V	?	?	V	Carrier recollection and cycle drift; not zero-shot transfer. III-C,IV,VI,AppendixA
F33 SIGLEAK [62]	C	V	?	?	?	?	?	Paired prompts; not physical-skill removal. Definition3,3.2,4.1–4.2
F34 ProcGrep [8]	C	H	?	V	?	?	?	Task-grouped attribution under scaffold regimes. 4–5,Tables7–8
F35 SecEmb [17]	A	?	X	?	?	?	?	Single GPT-2 workload; DLRM result separate. III,V,VI-D;Figs14–15
F36 Mobile agents [18]	A	?	?	?	?	?	?	Direct leakage cases; holdout binding unresolved. IV-D3 log leakage; V-D2 deep-link leakage; V-D3 log leakage; Table II; VI-A
F37 Hidden risks [19]	A	?	?	?	?	?	?	Content exfiltration/access-control cases. 3.2 framework and risk scope; source index
F38 Browser privacy [20]	A	?	?	V	?	?	?	Browser/profile-state measurements; mixed leakage mechanisms. V-C1–3 browser/network/profile-state measurements; source index
F39 LLMs Have Rhythm [9]	C	?	V	?	?	H	V	New captures and cross-network tests; overlapping windows caveat. III-IV; V-C–G; VI-A/B; Table I; Figs10–16; VII
F40 MoEcho [6]	C	H	V	?	?	V	?	Cross-template/dataset tests; contention/context conditions. 3.1;4;6.2–6.6;Tables3,5–10;7.2;8.1–8.2
F41 LLM-Fin [21]	A	?	H	?	?	?	?	Unseen variants within known architecture families. III; IV-B/C; Tables II–III; IV-C/D; V-A; V-A versus Introduction/Fig.2; Table IV
F42 ML privacy channels [27]	C	?	V	?	?	?	?	Enumerative/filter probes across named models; no learned split. 4.1; 4.2.1; 4.2.2/Fig.7; 4.2.3; 4.2.3/Table2
F43 MoE stealing [7]	C	?	?	?	?	?	?	Constructive word extraction in controlled shared batches. 2.2; 4; 4; 4/Table2; 5/Fig.7
F44 Energon [10]	C	?	V	?	?	?	?	Known-candidate CV; exploratory hardware not transfer proof. III-A/B; IV-A/C; IV-B/Tables I,III; V; VII
F45 SafeKV [52]	C	?	V	?	?	?	?	Named serving models; calibration/test separation unreported. 3; 7.1; 7.2.1;7.2.3/Fig.12; 7.3/Table7; 7.4; 4.2.2 versus6/7.2.3
F46 InferNet [22]	A	?	V	?	V	?	?	Same-label mixed-model and platform comparisons. 5.2; source index
F47 ArchSnoop [11]	C	?	H	?	V	?	?	Random-weight training to official/fine-tuned models; power modes. 3.2;6.1; 6.1; Fig9, paperp58, visually checked; 6.2.3;6.3.1; 4.1;7.1-7.2
F48 Coding fingerprints [84]	A	X	?	X	?	?	X	PR CV; repository/time/human-mediated holdouts unreported. 2.1-2.4; Tables 1-2; 4; 6; replication reference
F49 AHB [45]	C	?	?	X	?	?	?	Offline/live humanization; website-transfer binding unresolved. 2.1-2.2; 3.1; 4.1-4.3; 5.1.2; Table 1 Trip Planning; E.1-E.3
F50 WildAuth [85]	A	H	?	X	?	?	?	User-disjoint cross-topic split; no verified cross-service test. 2.3; 4.1-4.3; 5.3-5.5; 6.1-6.2; Tables 1-3; Appendix D/Table4
F51 Cycles [13]	C	X	?	X	?	?	X	One application; no held-out split located. 1; 2; 2.1 CDCS/CDDAG versus CDSA/Hybrid; 3.1; Tables1-2; 3.2; 4
F52 LumiMAS [14]	C	?	V	V	?	?	?	Nonoverlap logs; backend/platform comparisons; task grouping open. 3;4.1-4.3;5.1.1-5.1.7;5.2;Table1;6.1-6.5;B.1/Table3;B.2/Table4;C.1.1
F53 SymDiag [86]	A	V	V	?	?	?	?	Domains/reasoners vary with inputs and feedback. 3.1-3.4;4.1.1-4.1.5;5.1;6.1-6.4;7
F54 LeakyLMs [66]	C	?	H	?	V	?	?	Held-out architecture configurations; known-GPU calibration. 3;4.4-4.5 Tables1-2;5.4-5.5 Tables3-6/Fig6;6;A.1;B.4-B.7
F55 KVGov [49]	C	?	?	?	V	?	?	Simulation, A100 timing and HMAC microcost separate. 2.2-2.3;4;5.1-5.3 Tables3-4;5.7-5.9 Tables8-9;7
F56 Data-centric survey [28]	A	-	-	-	-	-	-	Conceptual synthesis; empirical holdout not applicable. 1 Fig1;4-6;8-9 discussion/conclusions

TABLE S68. HOLDOUT EVIDENCE, FAMILIES 57–82. CODES AND AXES ARE DEFINED IN APPENDIX S28.

Family	Scope	T	M	C	H	N	D	Source locator and qualification
F57 Serving RCA [68]	C	?	?	X	?	?	?	Single serving deployment with injected faults. Sections3,4.1-4.3,5.1-5.2; Table1
F58 ZeroPrint [87]	A	?	V	?	?	?	?	Known-family provenance pairs; partition not re-extracted. Header;4.1-4.4;Figure1;5.1-5.3
F59 Endpoint fingerprints [88]	A	?	?	V	?	?	V	Controlled endpoint changes over matched time windows. Sections3,4.1-4.3,6.1-6.2/Table1,7-9
F60 StepFinder [89]	A	H	?	?	?	?	?	Task-disjoint subsets; denominator reconciliation unresolved. Header;Sections4,5.1-5.4,6;Tables4-5
F61 Web-agent detection [76]	C	?	V	?	?	?	?	Six agents with human/scraper controls. 3.1.4;3.3;6;7
F62 Cross-campaign [64]	C	V	?	V	?	?	?	Crossed-style and regenerated conditions; execution stress subset. AppendixB decomposition; source index
F63 AgentProv [63]	C	?	V	?	?	?	?	Substituted checkpoints; system-prompt perturbation. 3.1-3.4;4.1-4.4;A.5
F64 LiDAR [67]	C	?	V	V	?	?	?	Enrollment-only identity; two harnesses and perturbations. 2;5.1-5.3;6.1-6.5
F65 LLM-WFIN [69]	C	V	?	?	?	?	?	Site traces; loading versus query phases. 3.2-3.3;5.1-5.2;6.1-6.5;Tables2-5;publisher rendered primary article
F66 MIRAGE [1]	C	V	V	?	?	?	?	Encodings/models/scenarios; matched decoder-layer controls. 3.4;4.1-4.2;7.1-7.4;AppendixL.3-L.4
F67 Routing metadata [72]	C	V	?	V	?	?	?	Request domains and routers; exploratory sensitive-label gate. III;IV;V;VI;VII
F68 Shape and Substance [36]	C	V	V	V	V	?	?	Geometry/model/cache/hardware conditions; padding estimate. IV;V;VII;VIII;IX
F69 SplitDecoy [75]	C	?	?	?	?	?	?	Repeated seeds; forward/backward boundary, not a holdout. 2;5.1-5.3;Table7-8;A.2.2
F70 PAGE404 [70]	C	V	V	?	?	?	?	Architecture and prompt-length targets; transfer binding open. II-B;IV-D4-D6;V
F71 Edge EM [65]	C	H	?	?	?	?	?	Unseen-word test separate from profiled vocabulary recovery. IV;V;VI
F72 SparLeak [37]	C	V	V	V	?	?	?	Models/datasets/sparse mechanisms; per-phase outcomes. 3.1;4-7;Table4
F73 Sparseety [73]	C	V	V	V	?	?	?	Models/datasets/implementations; oracle-conditioned extraction. III;V;VI;VIII;IX
F74 OTRO [51]	C	?	?	?	?	?	?	Closed-corpus conditional leakage; epoch cost separately derived. III;V;VI;VII-C-E;TableIII
F75 CachePrune [46]	C	V	V	?	?	?	?	Models/workloads and detector-miss stress; separate fits. 3.1;5.1-5.6;Table1;AppendixA
F76 Privacy foundations [90]	A	-	-	-	-	-	-	Conceptual contract; cross-institution example illustrative. III;IV/TableIII;IV-A;VI
F77 DP-MemView [91]	A	?	?	?	?	?	?	Semantic response privacy; holdout binding not re-extracted. Problem Definition;DP-MemView;Experiments;AppendixB.5
F78 PriCon [92]	A	?	?	?	?	?	?	Semantic cloud consultation; holdout binding not re-extracted. PDF3;5.1-5.2;Tables2-4
F79 SCAAR [93]	A	?	?	?	?	?	?	LLM EM/length test; other classifier results separate. 7;LLM extension;TVLA experiment
F80 CacheSolidarity [94]	A	?	?	?	?	?	?	Candidate-prefix timing gap; no scored inference error. v1 title;5.2.3;Figure7
F81 Vision encoder [95]	A	?	?	?	?	?	?	Representation inversion; holdout binding not re-extracted. 2;3;5;7;8/Table5
F82 Split gradients [96]	A	?	?	?	?	?	?	Split-compute content inversion; holdout binding unresolved. 3;4.1-4.5;5;AppendixB

TABLE S69. WHY A BINDING IS NECESSARY FOR A SPECIFIED REUSE INFERENCE. THESE ARE ANALYST-DEFINED TRANSFER QUERIES, NOT REQUIREMENTS IMPOSED ON EVERY SOURCE PAPER.

Item	Query that activates the item	What the missing binding prevents	Scope that does not require the stronger claim
R1 Target	Reuse an inference outcome for a protected interest.	Accuracy for one property cannot establish protection of another, or the same authorization relation.	A diagnostic may retain its operator objective without being treated as a privacy defense.
R2 Receiver	Reuse at an observer with particular access.	A stronger/different view may expose a signal absent from the measured projection.	A result restricted to the measured view need not establish all observers' protection.
R3 Intervention	Reuse a transform at an execution boundary.	Stored-trace changes do not establish that live emission or state effects implement that transform.	An offline transformation can support a claim confined to stored traces.
R4 Protected fit	Claim resistance to fitting/calibration on protected observations.	A clean-trained attack alone leaves this attacker response unevaluated.	A fixed analytical rule or a claim restricted to a fixed clean-trained attacker need not include refitting.
R5 Matched adaptation	Attribute a change in attack success to adaptation.	Different test populations confound the clean/protected-fit contrast.	A defended-condition accuracy estimate alone need not estimate the effect of adaptation.
R6 Workload	Transfer to a changed customer, system or workload.	In-sample variation does not establish the specified held-out deployment axis.	A within-configuration result may remain valid without a generalization claim.
R7 Inference	Quantify protection against a scoped attack.	A mechanism description or resource measurement alone supplies no defended attack outcome.	A cost-only component record can answer a resource question without an inference estimate.
R8 Utility	Reuse protection with a claim of preserved functionality.	Quality or reward from a different state/operating point does not establish preservation under this intervention.	Component quality suffices for a component claim; a leakage-only result need not establish planner utility.
R9 Cost	Claim deployability at a resource budget or compare overhead.	Another primitive's cost or a different unit/baseline does not price this operating point.	A leakage result can be valid without an overhead claim; payload cost can remain a payload-only result.

Applicability is decided for the requested conclusion before evidence status. The joint table asks whether existing records support combined protection–utility–cost reuse; its eligible sources need not themselves have asserted that combination. Unreported, partial and unverified bindings remain distinct and do not establish defense failure. Source-reported claims and analyst-proposed extensions must remain separate when applying this table.

S29. Complete Record-Local Audit and Joint Bindings

The archived record cells incorporate the targeted source reconciliation in S34. Their measured-utility R8 convention predates the outcome-specific applicability rule in S44; current quality joins use that overlay. Supported B joins and record/pair membership are unchanged; S33 remains the labeled pre-reinspection sensitivity snapshot. This retrospective reanalysis applies R1–R9 to each of the 39 archived result records, yielding 351 cells. It retains the original source locators and hashes; source follow-up resolves the SparLeak utility correction and distinguishes CachePrune's shared default configuration and OTRO's padded/unpadded costs. The table below binds each judgment to a source locator and rationale.

Counting rule. For a requested join, a mitigation record is applicable if none of the required items is inapplicable. A join is bound only if every required item is B and the evidence describes the same intervention/workload configuration. U (unreported), ? (unverified) and P (partial) are not bound; they remain distinct in the cell table. The six contextual controls are excluded from all mitigation join denominators. Section S33 expands every query from the common 33-record starting set, including each excluded identity and triggering item. No family-level union of cells is used. Configuration-linked evidence need not use identical sampled requests and does not license a paired effect. In particular, the inspected CachePrune source has not established that its recovery and serving tables use the same requests or paired runs; the two target records reuse one source-described default setting, not two independent trials.

Positive joins and deduplication. Inference plus utility: AHB online execution, CachePrune direct recovery, CachePrune contextual recovery and SparLeak replacement (4/22 records, three families/configurations). Inference plus cost: speculative REST padding, Dangol capture, Dangol live binding, and the two CachePrune records (5/22, three families, four configurations). Matched protected fitting/adaptation plus inference: UI offline and Dangol capture (2/17). Inference–utility–cost: the two CachePrune records share one default configuration's quality/cost package (2/20, one family/configuration). All five items together: 0/14. These query-specific counts describe the inspected collection. The 2/20 row is supported by CachePrune alone; two records reuse one configuration package and provide no estimate of the prevalence of such evidence across the literature.

Cost-object discipline. The speculative REST record binds token-payload overhead, not verified live wire bytes; Dangol capture binds transformed capture cost, while its live record binds loopback wire/latency. CachePrune's secure default setting links recovery with generation quality and serving cost; its privacy-disabled cache-hit comparison is not borrowed. SparLeak's attack overhead is not a defense-cost observation. OTRO's unpadded TTFT does not price its separately derived epoch padding. These distinctions explain why cost cannot be copied by family name.

S29.1. Claim-conditioned necessity, not a universal checklist

The nine items operationalize our proposed reuse inferences. For each application, separate (i) the source's stated result, (ii) the analyst's intended extension, and (iii) the premise that extension requires. A missing premise blocks that extension; it need not contradict the original claim. Table S69 lists the consequence and an exemption for each item. The R19 codes and denominators are retained; this table explains the premises behind each requested inference.

TABLE S70: All 39 record-local judgments. B/P/U/?/– retain the main-paper meanings; six contextual controls (†) are not mitigation denominators.

Record	1	2	3	4	5	6	7	8	9	Source scope and join rationale
UI-delay-offline [57]	B	P	B	B	B	B	B	U	-	6.1 Fig5;A.5. Offline protected identity prediction; no live task outcome or executed delay cost.
WRA-decoy-execution [40]	B	P	B	-	-	P	B	P	U	8.1-8.3 Tables8-11. Decoy inference and separately judged usefulness are not a same-condition task-utility pair.
NET-upstream-local [50]	B	B	B	B	?	B	B	U	U	IV-C-E TableIII-IV. Scenario-fitted inference; matched clean/refitted protected-test rows not bound.
NET-dummy-padding-simulation [50]	B	B	B	?	?	B	B	U	P	VII TableIX. Replay inference; insertion budget is a partial cost object, not measured live transport.
EARLY-K-calibration-oracle [47]	B	B	B	B	?	B	B	U	P	V-A TableVI;IV-A. K-conditioned calibrated oracle recovery; query cost is not protected service cost.
EARLY-semantic-deidentification [47]	B	B	B	-	-	B	-	U	B	V-B. Semantic processor cost; no protected token-recovery outcome in this record.
WHISPER-randompadding-capture [58]	B	B	B	P	U	B	B	U	U	3.2;6.1 Tables3-4. New padded captures; global fitting is specified, its protected-condition mapping incomplete; matched adaptation unreported.
WHISPER-batching-replay [58]	B	B	B	P	U	B	B	U	U	6.2 Fig7. Grouped traces bind AUPRC, not executed delay or task utility.
WHISPER-injection-replay [58]	B	B	B	P	U	B	B	U	P	6.3 Table5. Synthetic volume estimate lacks a complete placement-specific resource binding.
SPLICE-CTBF-signal [54]	B	B	B	-	-	B	B	U	U	VII-C TableII. CTBF timing-gap outcome; QCP costs are a different component.
SPLICE-QCP-cost [54]	B	B	B	-	-	B	-	U	B	VII-C TableII. QCP token/TTFT cost; CTBF signal suppression is not attached to this record.
TIME-length-instruction [56]	B	B	B	P	U	B	B	U	U	V-B;VII-B. Input-conditioned threshold profiling is specified; the instruction-specific fit mapping is incomplete and matched adaptation unreported.
TIME-padding-estimate [56]	B	B	P	-	-	B	-	U	B	7.3. Analytical plaintext padding cost; no independently evaluated protected inference here.
SPEC-padding-REST [53]	B	B	P	P	U	B	B	-	B	6.1 Fig6-7 Table2. Accuracy and token-payload ratio share the REST padding condition; live/wire interpretation is unresolved.
DANGOL-capture [23]	B	B	B	B	B	B	B	U	B	9.8-9.9 Table6. Capture-scoped matched adaptation, accuracy and transformed byte cost.
DANGOL-livebinding [23]	B	B	B	?	?	B	B	U	B	9.7, OHTTP paragraph after Table 5. Live loopback recovery and wire/latency cost; capture refit cannot be inherited.
SIG-common-guardrail† [62]	B	B	B	-	-	B	B	P	U	4.1-4.2;AppendixC. Contextual common-instruction control; skill-use proxy is not a paired protected task outcome.
SAFE-post-trigger [52]	B	B	B	-	-	B	B	U	P	7.2.4 Fig12;7.3-7.4. Post-trigger inference and separately reported service costs have no matched operating-point join.
AHB-online-executed [45]	B	B	B	P	U	B	B	B	U	5.1.2 Table1;E.1-E.3. Online detectability and objective task accuracy share the executed wrapper; refit code alone is partial.
AHB-offline [45]	B	B	B	P	U	B	B	U	-	Table1 offline rows;E.1-E.3. Offline detectability; online task harm and execution cost do not carry to replay.
KVGOV-partialflush-sim [49]	B	B	B	-	-	B	B	U	U	5.1-5.3. Protected simulated inference; HMAC microcost is not its deployment cost.
KVGOV-hardware-gap [49]	B	B	B	-	-	B	-	U	-	5.7-5.8. Unprotected hardware timing calibration; no protected-inference or incremental defense cost claim.
KVGOV-HMAC-cost [49]	B	B	B	-	-	B	-	U	B	5.9. Key derivation microcost; neither protected recovery nor lost cache reuse is measured here.
TRAFFIC-prompt-reformulation [35]	B	P	B	U	U	P	B	P	U	V-C1 Fig9a. Prompt-intervention inference; semantic preservation asserted rather than task-scored.
TRAFFIC-temperature [35]	B	P	B	U	U	P	B	U	U	V-C1 Fig9a. Temperature changes test fixed-configuration fingerprints; protected fitting and matched adaptation are unreported in scope.
CARLINI-fixed-rate [3]	B	B	P	-	-	B	-	U	B	VII Fig11-12. Rate-curve bandwidth/per-token-delay cost, with live placement unresolved.
GAO-zero-copy-off [48]	B	B	B	-	-	P	-	-	B	7.1. Loading/memory cost; structural argument is not measured defended ASR.
TDXRAY-oblivious-map [55]	B	B	B	-	-	B	-	B	B	6.1 Fig8 Table3. Vocabulary coverage and lookup cost are component utility/cost; no measured defended recovery.
AGENTTELL-fixed-instruction† [59]	B	B	B	-	-	B	B	P	U	4.3.1;5 Finding1. Loaded/cold instruction context; task-completing controls are not a guardrail-on/off effect.
FP-CLOUDFLARE† [61]	B	B	B	-	-	B	B	B	U	6.4 Table5. Contextual access-control and task-completion result, not an audited privacy mitigation.
BAR-co-running-noise† [60]	B	B	B	-	-	B	B	U	U	7.2-7.3 Table8. Contextual channel-capacity result under interference; no task-cost frontier.

Record	1	2	3	4	5	6	7	8	9	Source scope and join rationale
RHYTHM-VPN† [9]	B	B	B	U	U	B	B	U	U	VI-B;VII. Contextual VPN test uses a baseline-trained attacker; route cost is unreported.
DEB-filter-origin† [27]	B	B	B	-	-	B	B	U	U	Threatmodel;4.2. Contextual filter-created membership view; no mitigation of that channel is evaluated.
CACHEPRUNE-selective [46]	B	B	B	-	-	B	B	B	B	3.1;5.1-5.6;Table1;AppendixA. Default selective-sharing configuration links direct recovery, local generation quality and serving cost.
CACHEPRUNE-context [46]	B	B	B	-	-	B	B	B	B	3.1;5.1-5.6;Table1;AppendixA. Contextual recovery uses the same selective-sharing defense; shared quality/cost evidence is counted once at configuration level.
OTRO-unpadded [51]	B	B	B	-	-	B	-	-	B	III;V;VI;VII-C-E;TableIII. Measured unpadded TTFT/memory; padded leakage is a different operating condition.
OTRO-epochpad [51]	B	B	B	-	-	B	B	-	P	III;V;VI;VII-C-E;TableIII. Conditional leakage with a derived per-epoch cost; full measured padded TTFT remains unbound.
SPARLEAK-replacement [37]	B	B	B	P	U	B	B	B	U	3.1;4-7;Table4. Table 4 pairs ASR and ROUGE at each model/intensity; attack overhead is not defense cost.
SHAPE-padding-estimate [36]	B	B	P	U	U	B	P	U	P	IV;V;VII;VIII;IX. Proposed constant-work defense with derived cost; protected-fit and matched adaptation unreported.

S30. Calibration Source and Training-Customer Sensitivity

Question and design. The opposite GLM point changes ($+20.8$ cancellation and -17.3 return BA points) describe a target-dependent operating point. They do not isolate a causal model-source effect: changing source calibration also changes a finite training sample's templates, caps and donors. We therefore add a local-recalibration baseline and an attacker-only baseline. This post hoc protocol fixed 100 replicate seeds and the complete seven-attack endpoint suite before the new results. The source inputs, plan and implementation are hashed. No model calls, new customers or new task outcomes are added.

Each replicate draws 29 training customers with replacement and retains all their tasks, using identical customer draws for the two source endpoints. All 24 original test customers remain disjoint. Duplicate customer selections weight episode-derived calibration lengths; the original donor deduplication and fixed cover random seeds are retained. Donor pool ordering follows the resampling order. The original calibration helper's allowed-task list remains distinct from the actual sampled tasks, which are archived explicitly. The resulting training sample can contain a different number of episode occurrences from the original 69 tasks.

Both calibrations are applied to all 115 saved trajectories from each endpoint, producing 460 fresh local TLS 1.2 captures per replicate and 46,000 captures overall. Attacks fit only target training trajectories. Each fit includes majority, clean-trained gradient boosting, defended-data logistic regression and gradient boosting, decoded-tool gradient boosting, decoded target-call count and total exchange count. The decoder and count thresholds are refitted. These are length/direction experiments, with no reconstructed timing or new live-planner measurements.

Three sensitivity regimes. *Calibration only* rebuilds the defense and refits attacks on the full original target training population. *Calibration plus attacker* also resamples training customers for attack fitting, independently of the calibration draw but paired across conditions. *Attacker only* refits attacks on the original frozen local-calibration captures. Single-class training fits retain the draw and use constant predictions. Fixed-test percentile ranges measure pipeline sensitivity; they are not population confidence intervals. Original results are preserved, and the new implementation reproduces both original calibrators and all 84 original seven-attacker prediction arrays exactly.

Conditional permutation tests and multiplicity. The original table has eight target-level contrasts: two endpoints, two interventions relative to local joint shaping, and two labels. For each contrast we perform 99,999 seeded customer-level swaps of the entire seven-attacker condition prediction vector; all tasks of a customer move together. We reselect each suite maximum and use the absolute difference as a two-sided statistic with the plus-one Monte Carlo correction. Holm adjustment covers all eight contrasts. The null is conditional exchangeability of prediction vectors within customers, not a randomized causal model-source intervention. A small clustered example is checked against complete swap enumeration.

TABLE S71. ORIGINAL CONTRASTS UNDER CUSTOMER-PAIRED PERMUTATION TESTS. Δ IS INTERVENTION MINUS LOCAL JOINT MAX-BA IN POINTS.

Target	Intervention	Label	Δ	Raw p	Holm p
MiniMax	Write padding	cancel	-10.4	0.20175	1.000
MiniMax	Imported joint	cancel	-9.6	0.12785	1.000
MiniMax	Write padding	return	-6.2	0.34301	1.000
MiniMax	Imported joint	return	-4.5	0.55133	1.000
GLM	Imported joint	cancel	+20.8	0.13382	1.000
GLM	Write padding	cancel	+10.0	0.48630	1.000
GLM	Imported joint	return	-17.3	0.17389	1.000
GLM	Write padding	return	-5.7	0.61884	1.000

The earlier return percentile interval excluded zero, whereas the paired permutation test gives $p = 0.174$ before adjustment. These are different inferential constructions for a nonsmooth maximum over fitted attacks in a small clustered sample. We retain the historical intervals for traceability and withdraw the earlier "resolved return reduction" interpretation. None of the eight contrasts is resolved by the corrected tests; this does not establish equivalence between the defenses.

TABLE S72. LOCAL-PIPELINE STABILITY ON THE FIXED 46-TASK TEST POPULATION. ENTRIES ARE MEDIAN BA [2.5TH, 97.5TH PERCENTILES] OVER 100 TRAINING-CUSTOMER RESAMPLES.

Endpoint	Label	Calibration only	Calibration + attacker	Attacker only
MiniMax	cancel	50.0 [50.0, 64.8]	52.1 [50.0, 65.4]	57.5 [50.0, 72.5]
MiniMax	return	67.5 [55.2, 73.4]	62.6 [50.0, 72.1]	63.9 [50.0, 67.5]
GLM	cancel	57.5 [50.0, 69.2]	55.8 [50.0, 68.1]	52.9 [50.0, 60.8]
GLM	return	52.5 [50.0, 67.9]	53.6 [50.0, 65.7]	66.4 [50.0, 67.3]

TABLE S73. IMPORTED-MINUS-LOCAL MAX-BA CONTRASTS AFTER PAIRED SOURCE-CALIBRATION RESAMPLING. FIXED-TEST CELLS ARE MEDIAN [95% SENSITIVITY SPREAD], IN POINTS. JOINT TRAIN/TEST CELLS ARE EXPLORATORY PERCENTILE INTERVALS WITH ATTACKER RETRAINING.

Endpoint	Label	Calibration only	Calibration + attacker	Joint train/test
MiniMax	cancel	+0.0 [-13.0, 11.1]	+0.0 [-14.6, 16.5]	[-26.1, 22.3]
MiniMax	return	-4.5 [-20.2, 6.1]	-1.2 [-11.7, 10.8]	[-18.7, 17.5]
GLM	cancel	+2.5 [-13.8, 17.1]	+2.9 [-14.4, 17.5]	[-21.7, 27.4]
GLM	return	+0.0 [-16.9, 18.4]	+0.3 [-11.2, 18.9]	[-19.9, 26.3]

For joint sampling intervals, each of the 100 training replicates supplies 20 common 24-customer test bootstrap draws. All tasks from a selected customer stay together and both suite maxima are reselected. Single-class test draws are counted and omitted without replacement. The 2,000 nested draws are not 2,000 independent training fits. These exploratory empirical-population intervals include calibration and attacker-refit variability but are neither multiplicity-adjusted confirmatory intervals nor evidence for unobserved customer populations. Invalid joint test draws (MiniMax cancel/return, GLM cancel/return): 8, 3, 8, 3.

TABLE S74. RESAMPLED-CALIBRATION COST: MEDIAN [95% SPREAD] OF SUMMED TEST COST DIVIDED BY RAW TARGET TOTALS.

Target	Calibration source	Bytes×	Calls×
MiniMax	MiniMax	17.35 [12.7, 20.0]	3.17 [2.8, 3.4]
GLM	MiniMax	16.06 [11.7, 18.6]	4.39 [3.9, 4.7]
MiniMax	GLM	15.58 [13.4, 17.5]	2.90 [2.6, 3.1]
GLM	GLM	13.97 [11.7, 16.0]	3.85 [3.4, 4.2]

Results and inferential scope. Local GLM return BA has median 52.5% and 95% calibration-sensitivity range 50.0–67.9%: a 17.9-point width comparable to the original 17.3-point source contrast. Under paired source resampling, the imported-minus-local return contrast has median 0.0 and range $[-16.9, 18.4]$ points; with attacker resampling its median is $+0.3$ and range $[-11.2, 18.9]$. The joint train/test interval is $[-19.9, 26.3]$. Hence the original directional attribution is not stable to local training variability. These ranges describe empirical sensitivity, not equivalence or a universal absence of source effects. These results quantify calibration and attacker-fitting sensitivity within the observed customer population. The 100 resamples add no independent customers. The corrected tests leave the source-calibration contrasts unresolved; their nonsignificance establishes neither equivalent defenses nor the validity of the audit framework. A local-recalibration comparator is justified by the design: a source swap changes both the generating endpoint and the finite sample used to fit templates, caps and donors. Comparing imported calibration with repeated local calibration under matched attacker fitting addresses that ambiguity. The literature audit supplies the SoK's comparison and joint-binding judgments; this case records the output of applying them. All original attack, reward and cost measurements retain their stated scopes.

S31. Detailed Operating Points for the Appendix Case Study

The appendix case study applies record-level comparison, configuration-level joint binding and deployment decisions to selected measurements. This section indexes seven detailed grids. The emitter grid already appears in Table S28; the other six follow with unchanged values, denominators and attack suites. The complete literature audit supplies the SoK findings; the case's tests assess its stated empirical contrasts.

Emitter and state boundaries (Exps 1–2). Table S28 prices buffered/streaming changes, defended refits and overflow on identical responses. Table S75 separates dispatch inference from live-state conformance.

TABLE S75. EXP 2 DISPATCH RESULTS: ALL 115 REFERENCE PROGRAMS, THREE SEEDS, KNOWN-POPULATION DEFENSE CALIBRATION; TASK-FOLDED ATTACKERS. MAXIMUM BA (CANCEL/RETURN); EXTRA CALLS PER REAL CALL; DATABASE EQUALITY WITH THE UNPROTECTED RUN.

Policy	BA (%)	Calls+	DB eq.
No cover	100.0/100.0	0.00	100%
Cancel, $p = .5$	86.1/100.0	.10	100%
Cancel, $p = 1$, shadow	100.0/100.0	.20	100%
Cancel, $p = 1$, live	100.0/100.0	.20	0%
Cancel count $\rightarrow 2$	76.9/100.0	.35	100%
One decoy program	91.9/85.9	1.01	100%
Two decoy programs	83.5/78.1	1.98	100%
Shape equalization	50.0/54.5	6.71	100%

Calibration and operation signatures (Exp 3). Table S76 retains both prompt arms and paired contrasts. Tables S77 and S78 retain exploratory cross-fitting and oracle localization under the 11-attack union. S30 separately reports the 100 calibration resamples and corrected contrasts on the existing customers.

TABLE S76. EXP 3, FROZEN CALIBRATION: OPERATION-CATEGORY INFERENCE ON REAL-AGENT TRAJECTORIES. TESTING COVERS 46 TASKS FROM 24 HELD-OUT CUSTOMERS, TWO PROMPTS AND THREE REPEATS (276 EPISODES). NETWORK BA IS THE MAXIMUM OVER THE FROZEN ATTACK SUITE, WITH CUSTOMER-CLUSTER 95% INTERVALS.

Condition	Cancel BA [95% CI]	Return BA [95% CI]	Original reward (%)	Bytes×	Calls×
Undefended	97.2 [90.0, 100.0]	94.0 [85.8, 100.0]	79.7	1.00	1.00
Rules prompt	97.2 [90.0, 100.0]	93.5 [85.0, 99.5]	76.1	1.00	1.00
Padding only	66.1 [50.0, 83.9]	55.5 [50.0, 73.0]	–	4.02	1.00
Shape only	60.8 [55.4, 71.3]	51.9 [50.5, 57.8]	–	4.16	3.83
Joint program	50.0 [50.0, 60.6]	50.0 [50.0, 65.5]	–	15.39	3.83
Rules + joint	50.0 [50.0, 65.1]	60.6 [50.0, 73.3]	–	15.72	3.88

Paired joint-minus-undefended BA differences (percentage points): standard-prompt cancellation -47.2 $[-50.0, -33.3]$, return -44.0 $[-48.9, -27.8]$. These post hoc contrasts use the same customer draws and reselect the attack-suite maximum in each condition. Joint = online shadow cover plus constant framing; padding/shape ablations use standard-prompt trajectories. Ratios use undefended transport within each prompt. Shadow isolation preserves the logical planner transcript; the four program conditions replay actual LLM trajectories over local TLS 1.2. Reward is reported only for original model executions; dashes denote no new live defended-planner reward measurement (§S31).

TABLE S77. EXP 3: EXPLORATORY CROSS-FIT COMPARISON UNDER THE 11-ATTACK UNION: ALL 690 EPISODES AND 53 CUSTOMERS. MAXIMUM NETWORK BA WITH CUSTOMER-BOOTSTRAP 95% INTERVALS. STANDARD PROMPT UNLESS MARKED RULES; BYTE/CALL RATIOS RELATIVE TO UNDEFENDED TRANSPORT.

Configuration	Cancel BA (%)	Return BA (%)	B×/C×
Undefended	91.7 [82.1, 100.0]	85.5 [76.7, 93.4]	1.0/1.0
Joint constant	60.5 [52.9, 72.4]	55.5 [51.8, 64.2]	19.8/5.0
Joint buckets	57.1 [50.0, 71.5]	52.7 [50.0, 61.7]	12.4/5.0
Rules + buckets	56.9 [50.2, 71.2]	52.7 [50.0, 65.0]	12.4/5.0

Joint constant is the frozen study's shadow-count plus constant-framing design, refit within each fold. Joint buckets changes framing to 1024-byte buckets. Every bootstrap draw reselects the complete maximum. Intervals condition on fitted predictions.

TABLE S78. EXP 3 ORACLE LOCALIZATION UNDER THE 11-ATTACK UNION. STANDARD PROMPT: 345 CROSS-FITTED EPISODES, 53 CUSTOMERS. MAXIMUM NETWORK BA (%) WITH CUSTOMER-BOOTSTRAP INTERVALS.

Target	Full trace	Writes removed	Writes only
Cancel	91.7 [82.1, 100.0]	62.8 [54.6, 75.3]	91.7 [82.1, 100.0]
Return	85.5 [76.7, 93.4]	59.9 [52.2, 71.8]	84.9 [78.8, 94.0]

Observation boundary (Exp 4). Table S79 composes learned suffix removal with operation inference; Table S80 retains all 13 measured/offline conditions, upper bounds and costs. S19–S20 supply implementation and attack details.

TABLE S79. EXP 4 INTENT BA AFTER LEARNED FLUSH REMOVAL (95% CUSTOMER-CLUSTER INTERVAL). EACH ROW: 138 TEST EPISODES, 24 CUSTOMERS. MAXIMUM OF THREE CLASSIFIERS PLUS MAJORITY; THE FULL ATTACK UNION ADDITIONALLY KEEPS UNSTRIPPED OBSERVATIONS.

Prompt	Schedule	Cancel	Return
Standard	Serial	57.8 [50.0, 70.7]	53.4 [50.0, 64.6]
Standard	Parallel	50.0 [50.0, 64.0]	50.0 [50.0, 56.7]
Rules	Serial	50.1 [50.0, 55.6]	60.7 [50.8, 70.9]
Rules	Parallel	50.8 [50.0, 61.7]	54.0 [50.0, 67.0]

TABLE S80. TIMING-AWARE EXP 4 COMPARISON: STANDARD PROMPT, 138 HELD-OUT EPISODES, 24 CUSTOMERS (5 CANCEL-POSITIVE, 7 RETURN-POSITIVE). EACH BA PAIR IS CANCEL/RETURN (%). THE COMMON SEVEN MODELS PERMIT A MATCHED-SUITE COMPARISON; THE FULL UNION ALSO RETAINS ALL EARLIER ATTACKS.

View	Condition	Common-seven BA	Full-union BA	Upper BA	Bytes×	Calls×	RPC p95 (ms)
M	Undefended	84.3/76.9	91.7/86.4	100.0/97.2	1.00	1.00	15.4
M	Per-call constant	53.2/59.1	68.6/62.7	92.8/79.5	3.92	1.00	16.6
M	Serial shadow	50.0/52.9	61.4/53.4	71.4/65.8	15.11	3.83	26.2
M	Two-in-flight shadow	50.0/57.3	51.7/57.7	69.1/69.7	15.11	3.83	20.8
M	Full tunnel, 2 ms	59.9/55.2	62.5/55.2	70.2/62.0	125.71	1.00	56.4
M	Full tunnel, 5 ms	54.6/53.7	54.6/57.3	72.9/64.2	125.71	1.00	112.3
S	Record buckets	53.1/69.8	53.1/69.8	78.5/86.9	2.25	1.00	–
S	Tamaraw 40/12 ms	55.3/55.3	55.3/55.3	68.7/64.4	12.15	1.00	–
S	Tamaraw 4/1.2 ms	57.8/51.8	57.8/51.8	71.8/61.8	11.99	1.00	–
S	FRONT	59.0/69.3	59.0/69.3	69.9/81.8	2.51	1.00	–
S	WTF-PAD adaptation	51.9/62.7	51.9/62.7	64.1/79.6	1.88	1.00	–
S	WT random-donor	50.0/56.7	50.0/56.7	67.0/67.8	2.04	1.00	–
S	FRONT, short calibration	52.4/56.0	52.4/56.0	61.3/71.1	2.33	1.00	–

M: measured local tool RPC on saved actions; S: offline traffic transformation. RPC p95 ends at the last real response, excluding LLM generation. S-row simulated spans are not RPC latency. Upper BA is the full-union 95% bound; every condition exceeds 60%. M unions contain 17–22 attacks; S unions contain the common seven. Ratios use prompt-specific undefended captures.

Scope of utility and intervals. Frozen-table rewards concern original model executions. Replay conformance preserves logical observations/state; S26 measures fresh prefix-scheduler utility. Original intervals condition on fitted predictions; S30 resamples calibration and attacker training. Each comparison retains its cohort, receiver, execution mode and cost denominator.

S32. Case-Study Cohort, Model and Calibration Register

These unchanged grids retain the populations, served-model comparisons and calibration-source/local-sensitivity results underlying the main appendix case. Section S30 retains all eight raw and adjusted tests plus the local, paired and joint train/test intervals. The tests concern those empirical contrasts; the SoK contributions rest on the literature audit. Historical Exp 3–4 labels are archive identifiers.

TABLE S81. COHORT, OBSERVATION AND COST DENOMINATORS FOR EXPS 3–4. STANDARD-PROMPT PAIRS: CANCEL/RETURN BA (%). BYTE RATIOS PRICE JOINT SHAPING AGAINST EACH STUDY'S OWN UNDEFENDED SERIALIZATION.

Study	Observation / attack suite	Undef. BA	Joint B×
Exp 3 frozen	Replay lengths/order; frozen suite	97.2/94.0	15.39
Exp 3 cross-fit	Analytic lengths/order; 11-attack union	91.7/85.5	19.76
Exp 4	New RPC envelope; lengths/order/time; full union	91.7/86.4	15.11

Frozen-split and Exp 4 rows: 276 test episodes, 24 customers, 5/7 positive customers. Exp 3 cross-fit: 690 episodes, 53 customers, 14/18 positive customers. Each prompt has half the episodes. Cross-fit fold calibration yields larger templates ($5.04\times$ calls); frozen-split and Exp 4 rows use $3.83\times$. Repeats measure execution variation within these customer groups.

TABLE S82. EXP 3 MODEL-SPECIFIC REPEATED EVALUATIONS: NETWORK SUITE-MAX BA (CANCEL/RETURN, %). EACH MODEL HAS 46 TEST TASKS AND 24 CUSTOMERS. ENDPOINT ROWS USE THE ENDPOINT SUITE; LUNA RETAINS THE FROZEN SUITE. LUNA USES 29 TRAINING TASKS; THE TWO ENDPOINT REPLICATIONS USE 69. CONTRASTS ARE WITHIN MODEL; ENDPOINT ALIASES AND INFERENCE BUDGETS DIFFER.

Model	Undefended	Joint
Luna	64.2 / 84.9	60.4 / 57.0
MiniMax-M3	58.3 / 83.5	60.4 / 67.5
GLM-5.3-flash	100.0 / 86.6	50.0 / 67.3

TABLE S83. CALIBRATION SOURCE, LOCALIZED PADDING AND TRAINING SENSITIVITY. TOP: ORIGINAL 46 TEST TASKS / 24 CUSTOMERS PER ENDPOINT; Δ IS THE SEVEN-ATTACK MAX-BA CONTRAST AGAINST TARGET-LOCAL JOINT SHAPING (POINTS). BOTTOM: 100 PAIRED TRAINING-CUSTOMER RESAMPLES, EVALUATED ON THE SAME FIXED TESTS.

Target	Calibration / intervention	Cancel BA	Return BA	Δ cancel	Δ return	Bytes×	Calls×
MiniMax	MiniMax joint	60.4	67.5	–	–	20.06	3.36
MiniMax	GLM joint	50.8	63.0	-9.6	-4.5	17.48	3.11
MiniMax	Write padding	50.0	61.3	-10.4	-6.2	1.45	1.00
GLM	GLM joint	50.0	67.3	–	–	15.95	4.24
GLM	MiniMax joint	70.8	50.0	$+20.8$	-17.3	18.65	4.66
GLM	Write padding	60.0	61.6	$+10.0$	-5.7	1.51	1.00

Target	Label	Local recalibration BA: 95% spread	Imported − local Δ: median [95% spread]	Same Δ with attacker resampling: median [95% spread]
MiniMax	cancel	[50.0, 64.8]	+0.0 [-13.0, 11.1]	+0.0 [-14.6, 16.5]
MiniMax	return	[55.2, 73.4]	-4.5 [-20.2, 6.1]	-1.2 [-11.7, 10.8]
GLM	cancel	[50.0, 69.2]	+2.5 [-13.8, 17.1]	+2.9 [-14.4, 17.5]
GLM	return	[50.0, 67.9]	+0.0 [-16.9, 18.4]	+0.3 [-11.2, 18.9]

Original eight customer-paired contrasts: 99,999 whole-suite swaps each, Holm-adjusted $p = 1.000$ throughout. Historical nominal intervals remain in the supplement. Resampling spreads measure training sensitivity, not confidence in a population effect. All seven attacks (including decoding and call counts) are refitted per condition. Costs are summed local TLS test bytes/calls relative to raw target totals; they exclude live planner utility and WAN delay.

S33. Audit Accounting and Judgment Sensitivity

This section preserves the pre-reinspection sensitivity snapshot; current source-reconciliation codes and queries are in S34. The deterministic reanalysis uses the archived 39-record/351-cell audit and all 741 pair judgments. It reuses the existing sources, model outputs and customers. The original labels, cohort definitions, source locators and rationales are unchanged. The released tables record both input hashes and every query membership. The goal is to expose which choices support a finite-corpus statement and which additional evidence would change it.

S33.1. Common starting set and query masks

There are 33 mitigation records from 20 families and six contextual controls. Every join starts from the same 33; a required inapplicable item excludes that record from that query only. Partial, unverified and unreported bindings stay eligible. All required items must be B to count as bound. Table S86 makes this rule visible record by record; the archived pre-reinspection cell file supplies its labels; Section S29 presents the current cells and S34 gives the exact changes. Inference–utility and inference–cost each admit 22 records but not the same set. Triple-join eligibility covers 20 records, 14 families and 19 archived configuration groups; its two positive records reduce to one CachePrune configuration. Configuration identifiers deduplicate the archived packages and do not imply common sampled requests.

S33.2. Status-rule stress checks

Table S84 varies accepted statuses while holding applicability, cohorts and records fixed. B is the supported result. The other columns deliberately weaken that standard: accepting P tests the effect of treating partial evidence as sufficient, and accepting ? treats an unverified binding as if satisfied. P may encode a known missing or mismatched operating-point link, so these stress counts are not additional supported joins or a recoding of sources. U remains unreported in these columns. If U is also mechanically accepted, every eligible row passes by construction; the released all-status column documents this vacuous extreme rather than presenting it as evidence.

TABLE S84. STATUS-RULE STRESS COUNTS, WITH APPLICABILITY HELD FIXED. ONLY THE B COLUMN DENOTES SUPPORTED JOINS.

Query	Eligible	B	B or P	B or ?	B or P or ?
Inference + utility	22	4	6	4	6
Inference + cost	22	5	11	5	11
Matched adaptation + inference	17	2	2	15	16
Inference + utility + cost	20	2	2	2	2
All five bindings	14	0	0	0	0

The triple join is unchanged under these relaxations, whereas matched adaptation–inference rises from 2 to 15 when ? is accepted, and to 16 when P is also accepted. The distinction matters operationally: the latter query needs resolution of fit/test bindings, while the triple join needs presently unreported outcome bindings. Stability to P/? does not authenticate the source interpretation or establish that additional unreported evidence does not exist.

S33.3. Pair scope and family dependence

The 133 related pairs belong to six mechanism-question cohorts (Table S85). The other 608 are outside those declared queries, not failed comparisons. Resolving all 11 uncertain related judgments as compatible gives 28 compatible pairs instead of 17; retaining them as unresolved leaves 17. The 105 known mismatches stay fixed. This is a conditional completion range within the stated cohorts, not robustness to every alternative cohort construction.

TABLE S85. EXHAUSTIVE RELATED-PAIR ACCOUNTING, STRATIFIED BY DECLARED QUESTION.

Cohort	Pairs	Scoped match	Different object	Unresolved
behavioral humanization	3	1	2	0
cache service	45	2	37	6
fixed instruction	1	0	0	1
generator fingerprint	3	3	0	0
local memory	15	0	14	1
traffic envelope	66	11	52	3

The 15 same-family pairs arise from 11 families, and their eight mismatches occur in six families. Pairs share records; families with more records contribute more pairs. Table S87 removes every family and all its incident pairs, one family at a time. Related-pair compatibility spans 7.8–15.2%; the same-family mismatch fraction spans 5/12–8/12. Deleting KVGov removes three mismatches, while deleting Whisper Leak removes nine compatible related pairs (17 to 8), including three within-family pairs. These percentages describe record weighting under the declared split, without a high/low reference interpretation. The coarsening analysis in S37 also changes the comparison units. Neither operation adds independent observations. We report descriptive counts and dependency checks rather than binomial confidence intervals over dependent pairs.

For the triple join, removing CachePrune leaves zero bound records among 18 eligible records. Removing any other mitigation family retains the two records and their single configuration. The archived `joint-leave-family-out.csv` gives all 100 query-by-family deletions, and `query-membership.csv` gives all 165 query-by-record decisions. These diagnostics show which queries depend on a particular family and how changing the acceptance rule changes the reported joins.

TABLE S86. ALL 33 MITIGATION RECORDS AND THEIR FIVE QUERY MEMBERSHIPS. B: ELIGIBLE AND BOUND; E: ELIGIBLE BUT NOT BOUND; X WITH ITEM NUMBERS: EXCLUDED BECAUSE THOSE REQUIRED ITEMS ARE INAPPLICABLE. IU=R7/R8; IC=R7/R9; AI=R4/R5/R7; IUC=R7/R8/R9; ALL=R4/R5/R7/R8/R9. SOURCE RATIONALES ARE IN SECTION S29.

Record	IU	IC	AI	IUC	All
UI-delay-offline	E	x9	B	x9	x9
WRA-decoy-execution	E	E	x4,5	E	x4,5
NET-upstream-local	E	E	E	E	E
NET-dummy-padding-simulation	E	E	E	E	E
EARLY-K-calibration-oracle	E	E	E	E	E
EARLY-semantic-deidentification	x7	x7	x4,5,7	x7	x4,5,7
WHISPER-randompadding-capture	E	E	E	E	E
WHISPER-batching-replay	E	E	E	E	E
WHISPER-injection-replay	E	E	E	E	E
SPLICE-CTBF-signal	E	E	x4,5	E	x4,5
SPLICE-QCP-cost	x7	x7	x4,5,7	x7	x4,5,7
TIME-length-instruction	E	E	E	E	E
TIME-padding-estimate	x7	x7	x4,5,7	x7	x4,5,7
SPEC-padding-REST	x8	B	E	x8	x8
DANGOL-capture	E	B	B	E	E
DANGOL-livebinding	E	B	E	E	E
SAFE-post-trigger	E	E	x4,5	E	x4,5
AHB-online-executed	B	E	E	E	E
AHB-offline	E	x9	E	x9	x9
KVGOV-partialflush-sim	E	E	x4,5	E	x4,5
KVGOV-hardware-gap	x7	x7,9	x4,5,7	x7,9	x4,5,7,9
KVGOV-HMAC-cost	x7	x7	x4,5,7	x7	x4,5,7
TRAFFIC-prompt-reformulation	E	E	E	E	E
TRAFFIC-temperature	E	E	E	E	E
CARLINI-fixed-rate	x7	x7	x4,5,7	x7	x4,5,7
GAO-zero-copy-off	x7,8	x7	x4,5,7	x7,8	x4,5,7,8
TDXRAY-oblivious-map	x7	x7	x4,5,7	x7	x4,5,7
CACHEPRUNE-selective	B	B	x4,5	B	x4,5
CACHEPRUNE-context	B	B	x4,5	B	x4,5
OTRO-unpadded	x7,8	x7	x4,5,7	x7,8	x4,5,7,8
OTRO-epochpad	x8	E	x4,5	x8	x4,5,8
SPARLEAK-replacement	B	E	E	E	E
SHAPE-padding-estimate	E	E	E	E	E

TABLE S87. ALL 26 LEAVE-ONE-FAMILY-OUT PAIR INVENTORIES. M: SCOPED-MATCH RELATED PAIRS; D: SAME-FAMILY PAIRS WITH A DIFFERENT-OBJECT WITNESS. COUNTS REUSE REMAINING RECORDS AND ARE NOT SAMPLING REPLICATES.

Removed family	Related pairs	M	Same-family pairs	D
KVGov [49]	109	16	12	5
LLMs Have Rhythm [9]	131	15	15	8
Remote timing [3]	122	17	15	8
SparLeak [37]	128	17	15	8
SafeKV [52]	124	16	15	8
Dangol workflow [23]	112	16	14	8
ML privacy channels [27]	133	17	15	8
Early Bird [47]	116	16	14	7
I Know What You Said [48]	128	17	15	8
SIGLEAK [62]	132	17	15	8
Behind Bars [60]	133	17	15	8
Shape and Substance [36]	128	17	15	8
TDXRay [55]	128	17	15	8
WRA [40]	122	17	15	8
OTRO [51]	124	17	14	7
NetEcho [50]	112	10	14	8
Cloud traffic benchmark [35]	130	14	14	8
AgentTell [59]	132	17	15	8
SpliceLeak [54]	116	17	14	7
Time Will Tell [56]	112	17	14	7
UI Traces [57]	131	17	15	8
FP-Agent [61]	133	17	15	8
Speculative decoding [53]	122	17	15	8
Whisper Leak [58]	103	8	12	8
CachePrune [46]	116	16	14	7
AHB [45]	130	16	14	8

S34. Source Reconciliation, Record Boundaries and Transfer Objects

This targeted reporting audit revisits all 11 formerly unverified family R4/R5 cells and their affected record cells. It retrieves the official versioned HTML of six papers and records content hashes, section/table locators, old/new status and a rationale. Three further versioned papers support the positive reuse and holdout examples. It adds no model calls, customers or behavioral experiments. The remaining questions concern what the published evaluations bind; a new Sol run cannot establish the training provenance of somebody else's reported result.

S34.1. A bounded reading rule

B requires the requested connection to be explicit. P applies when relevant training/profiling and defended results are described but their condition-specific connection is incomplete. U means the requested connection is not reported in the inspected methods/result passages; it does not mean the authors never performed it. A remains inaccessible source, and ? remains a binding not adjudicated within the reading scope. For R5, an undefended-versus-defended score or two detector architectures do not establish clean-trained versus protected-trained attacks on identical defended tests. We do not infer B from an implementation alone without its result binding.

TABLE S88: All 11 family-cell rereads. Each old code is ?. Source versions are pinned in the receipt file.

Family	New R4/R5	Inspected locator	Evidence and remaining link
Whisper Leak [58]	P/U	3.2; 6.1-6.3; Tables 3-5; Figure 7	Training and prompt splits are specified globally; padding uses new captures and batching/injection transform traces. These passages do not map protected training partitions to each mitigation result, or give clean/protected fits on identical defended tests.
Time Will Tell [56]	P/U	V-B; VII-B	The attack fits an input-length-dependent token-count threshold. The fixed-word instruction is evaluated, but its threshold reprofiling and a paired clean/protected-fit contrast are not specified in these passages.
Speculative decoding [53]	P/U	4.3-4.4; 6.1; Appendix D	Fingerprint profiling and padded/aggregated attack outcomes are described. The mitigation result does not identify whether reference profiles are rebuilt on protected traces, nor compare both fits on a shared defended test population.

Family	New R4/R5	Inspected locator	Evidence and remaining link
Cloud traffic benchmark [35]	U/U	III-C; IV; V-C1; Figure 9(a)	The mitigation discussion evaluates fingerprints learned at a fixed configuration under rewritten prompts and changed temperatures. Protected-condition refitting and a matched clean/protected-fit contrast are not reported in this scope.
SparLeak [37]	P/U	6; 7.1; 7.5; Table 4	Profiling-based attacks and replacement-intensity ASR/ROUGE outcomes are specified. The mitigation table does not bind the attack-training distribution to each intensity or report matched clean/protected-fit tests.
AHB [45]	P (retained)/U	4.2.2; 5.1.2; Table 1; E.1-E.3	Different detectors on a humanization row are a detector comparison, not a clean/protected-fit contrast. The inspected paper scope and archived static call-chain review do not report that matched contrast; condition-specific code fitting remains partial run evidence.

In the archived source-scope reconciliation, four family cells became P and seven U; no additional B was inferred. S36 documents the subsequent record-derived family projection, which preserves the six unresolved record bindings rather than inheriting these broader judgments. Nineteen record cells are updated consistently, including AHB's offline fitting as partial under the same code-versus-run distinction. Six other record-level ? cells remain in NetEcho, Early Bird and Dangol scopes, distinct from the 11 family cells reviewed here. Current record cells appear in Section S29; the artifact records the 19 exact changes. These targeted decisions resolve the reporting category, not the source's unpublished execution history.

S34.2. Effect on the quantitative synthesis

All supported joins remain unchanged: inference–utility 4/22, inference–cost 5/22, matched adaptation–inference 2/17, triple join 2/20, and all five bindings 0/14. Query masks still start from the same 33 mitigation records. The triple package remains CachePrune's two outcomes at one shared configuration. The pair census remains 741 pairs, 133 related, with 17 descriptive matches, 105 mismatches and 11 unresolved pair contracts. Family-cell reporting and pair-contract status are separate objects.

TABLE S89. CURRENT QUERY STRESS COUNTS AFTER SOURCE REREADING. ONLY B IS SUPPORTED; WEAKER COLUMNS REMAIN COUNTERFACTUAL. S33 PRESERVES THE EARLIER SNAPSHOT.

Query	Eligible	B	B/P	B/?	B/P/?
Inference + utility	22	4	6	4	6
Inference + cost	22	5	11	5	11
Matched adaptation + inference	17	2	2	6	6
Inference + utility + cost	20	2	2	2	2
All five bindings	14	0	0	0	0

S34.3. Recordization and extraction coverage

We preserve a hierarchy: family/version → comparison record → model/workload/parameter coordinates → measured outcomes. A comparison record splits when the primitive, execution mode, receiver/target contract or measured cost object changes. Multiple numerical cells within a common record are not additional independent observations. A join must match its configuration coordinates; two endpoints can share a quality/cost package without becoming a paired effect. The extraction frame is the 20 mitigation families' scoped evaluations and linked methods/outcomes plus six contextual controls. It is not an enumeration of all numeric cells or all attack experiments in the 82-family map.

- Early Bird: split the prefix-recovery oracle from semantic-cache processor cost (V-A versus V-B).
- Dangol: split transformed captures from a live OHTTP loopback (9.8–9.9 versus 9.7); a common workflow target permits description, not a common execution denominator.
- SparLeak: retain model/intensity coordinates within the random-replacement record; Table 4's ASR and ROUGE stay paired by row.
- CachePrune: split direct and contextual recovery targets, but deduplicate the shared default quality/cost package.
- OTRO: keep measured unpadded execution separate from a padded operating point and its derived cost.

These rules articulate the existing ledger retrospectively; they are not a claim of preregistered extraction. The released record-scope file gives every one of the 39 units, its source locator, retained parameter coordinates, configuration package and question cohort.

TABLE S90: Declared question groups. Full record identities are in the released scope file.

Question group	Records	Pairs	Membership rule
traffic envelope	12	66	Network/API destination, size, timing, count or delivery shaping; includes WRA destination cover, excludes cache probes and content-only identity.
cache service	10	45	Cache hit/reuse probes, cache policy and its component costs; excludes local memory observation without a service-cache probe.
local memory	6	15	Memory access/loading/sparsity/obliviousness and associated component costs; excludes network metadata and service-cache timing probes.

Question group	Records	Pairs	Membership rule
behavioral humanization	3	3	Site/mobile behavioral-trajectory edits or executed GUI humanization; excludes network-destination cover such as WRA.
generator fingerprint	3	3	Prompt/decoding or route changes with model-identity outcomes; excludes topic/content recovery.
fixed instruction	2	1	Fixed instruction/guardrail contrasts; source target and observer still require alignment.
access control	1	0	FP-Agent contextual gate; singleton, no related pair.
contention noise	1	0	Behind Bars contextual channel; singleton, no related pair.
filter created view	1	0	ML privacy channels contextual filter; singleton, no related pair.

Question assignment uses the primary extracted intervention/receiver surface, then the intended comparison. Network destination cover belongs to traffic-envelope even when it is implemented with a decoy agent; site/mobile behavioral edits belong to humanization. Fixed instructions and generator-identity countermeasures retain their declared narrower questions. Cost-only records stay with the mechanism they would otherwise be used to price. A record that could answer another question needs an explicit alternate-cohort audit; outside-cohort status is not evidence of incompatibility. The six nonsingleton cohorts generate all 133 related pairs; three singleton contextual groups generate none. The full cross-product remains available for alternative queries.

S34.4. What the holdout index evaluates

The 19 H entries across six axes locate reported separations in the archived extraction. A family-level H does not itself bind a defense to that split. Table S91 lists the entries and whether the family also has a mitigation record. The three model entries concern attack/identification generalization outside the 20 mitigation families. The two topology entries likewise concern attacker generalization: Dangol's new-composition recognition and the traffic benchmark's graph/star classifier transfer. Neither is a demonstrated source-calibration transfer of the defense. The other 14 entries retain their source-qualified split descriptions; classifying their defended-condition linkage requires the relevant record, rather than a blanket attack-only or defense-transfer label.

Defense-conditioned variation is still useful. Time Will Tell's shared fixed-length instruction has model-dependent outcomes; Whisper Leak's mitigation results remain provider/condition-specific. In these cases retain the tested rule and residual attack outcome, rather than relabel multiple conditions as a held-out defense-calibration experiment. For actual calibration transfer, specify source training, frozen parameters, target testing, local recalibration and attacker fitting separately. The artifact's H-entry ledger carries each original locator and qualification; the 82-family discovery matrix remains unchanged.

TABLE S91: Source-qualified H entries. The three model and two topology splits evaluate attacker generalization; other entries retain the indexed split description. The H flag alone does not establish a defended holdout or calibration transfer. M indicates mitigation-audit membership.

Family	Axis	M	Archived qualification
WRA [40]	Task/input/user	yes	Prompt tests and synthetic persona split; version reconciliation open
FP-Agent [61]	Task/input/user	no	Separate task holdout; visitor/host separation unresolved
Prompt length [32]	Task/input/user	no	OOD inputs; browser/API and day/night buffering controls
InputSnatch [38]	Task/input/user	no	Held-out and generated target prompts; candidate-prior dependence
Whisper Leak [58]	Task/input/user	yes	Unique target-prompt holdout; model/provider comparisons
Prompt Echo [34]	Task/input/user	no	Overlap-filtered prompts; models evaluated separately
I Know What You Said [48]	Task/input/user	yes	Prompt overlap filtering; model/hardware/framework comparisons
Speculation [53]	Task/input/user	yes	New traces and OOD paraphrases; remote/load conditions
Workflow traffic [23]	Topology	yes	Leave-composition-out; stored and live populations separated
Traffic benchmark [35]	Topology	yes	Topology shift distinct from feature-view restriction
ProcGrep [8]	Task/input/user	no	Task-grouped attribution under scaffold regimes
LLMs Have Rhythm [9]	Network/load	no	New captures and cross-network tests; overlapping windows caveat
MoEcho [6]	Task/input/user	no	Cross-template/dataset tests; contention/context conditions
LLM-Fin [21]	Model	no	Unseen variants within known architecture families
ArchSnoop [11]	Model	no	Random-weight training to official/fine-tuned models; power modes
WildAuth [85]	Task/input/user	no	User-disjoint cross-topic split; no verified cross-service test
LeakyLMs [66]	Model	no	Held-out architecture configurations; known-GPU calibration
StepFinder [89]	Task/input/user	no	Task-disjoint subsets; denominator reconciliation unresolved
Edge EM [65]	Task/input/user	no	Unseen-word test separate from profiled vocabulary recovery

S34.5. Positive reuse and changed decisions

Early Bird's prefix record and CachePrune's direct-recovery record are an existing descriptive match. Both can inform a cache-reuse threat shortlist; they do not support a numerical ranking across oracle and serving measurements. CachePrune supplies recovery, quality and cost evidence linked to its source-described default setting for scoped feasibility analysis; the inspected tables do not establish identical sampled requests or paired runs. The semantic processor benchmark in Early Bird instead leaves the prefix candidate's serving cost open. This produces a positive retention decision and a specific next measurement. The main decision table also distinguishes read-cover state effects, final-emitter captures and workflow utility, extending the established WF evaluation discipline to LLM/agent execution objects.

S35. External Retrospective and Time-Ordered Internal Checks

S35.1. B: retrospective design and complete frame

The external frame includes both defenses in Deep Fingerprinting, all three implementations in WFDefProxy, and all nine defenses in Mathews et al.'s Table II (Standard mode). DynaFlow's live prototype is separate because its execution, workload and cost objects differ: 15 records, 14 defense–study pairs, three follow-up studies. The 82-family/39-record LLM/agent audit is unchanged.

The protocol preceded ledger assembly after abstracts and search previews had exposed outcomes. Extraction is retrospective and outcome-informed. We report neither prediction accuracy nor a random-item significance test. Source URLs, version/hash receipts, locators, inclusion rules and comparator outputs are archived. Papers are not redistributed.

Mechanism-name carryover omits measurements at changed objects; an always-failure rule wrongly discards the favorable B02/B03/B04/B11 controls. The 11 protected-fit/view requirements are B01–B02 and B06–B14; the four execution/cost requirements are B03–B05 and B15. The four favorable controls B02, B03, B04 and B11 overlap those categories. This routing matches established WF practice. These are worked examples of evidence routing. Different accuracy/TPR populations and cost denominators remain separate; Random-WT is not symmetric Walkie-Talkie. Additional security modes and layered configurations are not counted as independent events.

S35.2. C: ordering and corrected outcomes

The seven case-study judgments and their reported conditions are retained below; the artifact preserves the ordering receipt.

R3 verification recomputes 184 saved prediction sets, 16 dispatch and 24 network cells with 2,000 task-bootstrap draws. P1–P4 support the specified counterexamples; P5's count/decoy source lacked protection and is excluded from favorable evidence; P6 is partial; P7 fails the prewritten joint screen. The current tally is therefore four supported, one source-premise failure, one partial and one not established. Original labels, full costs and intervals remain in S13 and Appendix E. S30 retains the later Holm-adjusted calibration tests; their nonsignificance is not used to validate the framework. No new model calls or customers are introduced.

TABLE S92: Complete external retrospective frame. DF: Sirinam et al.; WFDP: WFDefProxy; M: Mathews et al. Locators refer to the follow-up publication. R items identify requested bindings, not recoded scores for the original defense paper.

ID	Defense / source locator	Published outcome	Binding and measurement decision
B01	WTF-PAD; DF [97], 5.4 / Table 3	DF accuracy 90.7%; strongest prior comparator 69.0% in the same table.	R4,R7: Fit the stronger attacker on the protected distribution; retain each attack-specific result.
B02	Walkie-Talkie; DF [97], 5.4 / Table 3; 5.8	DF accuracy 49.7% under symmetric collisions; asymmetric collisions give 87.2%.	R2,R4,R7: Keep the symmetric-pairing condition; a new attacker alone does not imply failure.
B03	FRONT; WFDP [98], 6.2 / Table 10	Same-parameter data overhead 52% to 75%; DF TPR 72.06% to 42.79% (FPR 2.16% to 2.49%).	R3,R9: Measure restart behavior and live cost; preserve the improved live security outcome.
B04	Tamaraw; WFDP [98], 6.1 / Table 9	Optimistic/pessimistic/live data overhead 207/224/179%; time 19/26/21%; DF TPR 2.59/2.74/6.43%.	R3,R9: Bind the soft-stop implementation and costs; retain the low-TPR live result.
B05	Random-WT; WFDP [98], 6.3 / Table 11; 4.4	DF TPR 87.47% to 83.41%; data overhead 96% to 88%; time overhead 44% to 23%.	R3,R9: Price the live burst schedule; Random-WT is not symmetric Walkie-Talkie.
B06	DynaFlow; M [99], IV-F; Table II	Closed-world attack accuracy 38.3%.	R2,R4,R6: Test IAT-switch features; keep the workload and subpage composition.
B07	BiMorphing; M [99], IV-A; Table II	Closed-world attack accuracy 81.0%.	R4,R6: Refit a deep attacker; do not pool different datasets.
B08	DFD; M [99], IV-D; Table II	Closed-world attack accuracy 94.2%.	R2,R4,R5: Match perturbation variation in protected training/testing and include timing.
B09	FRONT; M [99], IV-G; Table II	Closed-world attack accuracy 81.8%.	R4,R6: Fit with defended variants and the staged training schedule.
B10	HyWF; M [99], IV-B; Table II	Closed-world attack accuracy 56.8%.	R2,R4,R6: Bind one-split visibility and augmented protected fitting.
B11	TrafficSliver; M [99], IV-B; Table II	Closed-world attack accuracy 5.4%.	R2,R3,R4,R6: Retain BE-TS one-split results; do not inherit them for an all-split observer.
B12	Spring; M [99], IV-C; Table II	Closed-world attack accuracy 80.0%.	R2,R4: Include rounded timing and multiple protected training variants.
B13	Interspace; M [99], IV-C; Table II	Closed-world attack accuracy 76.1%.	R2,R4: Include rounded timing and multiple protected training variants.
B14	BANP; M [99], IV-E; Table II	Closed-world attack accuracy 89.6%.	R4,R5: Refit after generating perturbations with the accessible defense generator.
B15	DynaFlow; M [99], Appendix B	Live accuracy 63.6%; mean per-site bandwidth/time overhead 123%/379%; different URLs and crawler.	R3,R6,R9: Keep a separate live record; simulation and live numbers are not a matched causal contrast.

S36. Record-Derived Family Summaries and Publication-Stratum Sensitivity

S36.1. One record ledger, one family projection

The 39-record ledger and its 351 item judgments are the source of truth for the current family display. For each family/item, the projection returns B if any scoped record is B, otherwise P if any is P, then ?, U, A, and finally inapplicable. Thus B has an explicit record witness; P preserves known partial support; ? preserves a still-unverified record binding when no stronger support is available. The ordering summarizes available evidence, not overall study quality. Every record state remains available even when another record supplies a family's B. In particular, Dangol's stored-capture R5 witness does not resolve the live record's R5.

The projection changes 13 displayed family cells, without changing any record judgment or joined outcome (Table S93). Earlier family displays mixed broader source descriptions with record-specific scopes. NetEcho and Early Bird therefore retain ? in R5 under the new projection, although the earlier source-scope reread classified their broader descriptions as partial. The six record-specific ? cells remain six; the current family summary has two. A bound cost in an unpadded or component record can appear in a family summary while remaining unavailable to a protected-inference join. The witness file lists all 180 family/item projections, their contributing record IDs, states and original source locators.

R6 now has 17 bound families among 20 applicable; R7 has 16 among 17. R7 is inapplicable for the three families represented here only by a structural/cost or component mitigation record: I Know What You Said, Remote Timing and TDXRay. The unchanged record-local joint results are 4/22 for inference with utility, 5/22 with cost, 2/17 for matched adaptation with inference, 2/20 for the triple join and 0/14 for all five bindings. The two triple-join records share CachePrune's one configuration package.

TABLE S93. FAMILY-SUMMARY RECONCILIATION. RECORD-LOCAL JUDGMENTS AND ALL JOINT QUERIES ARE UNCHANGED. THE OLD COLUMN IS THE PRE-PROJECTION DISPLAY.

Family	Item	Old	Current	Record-local evidence retained
NetEcho [50]	R5	P	?	`NET-upstream-local`: ?; `NET-dummy-padding-simulation`: ?
NetEcho [50]	R9	U	P	`NET-upstream-local`: U; `NET-dummy-padding-simulation`: P
Early Bird [47]	R5	P	?	`EARLY-K-calibration-oracle`: ?; `EARLY-semantic-deidentification`: –
Whisper Leak [58]	R3	P	B	`WHISPER-randompadding-capture`: B; `WHISPER-batching-replay`: B; `WHISPER-injection-replay`: B
Whisper Leak [58]	R9	B	P	`WHISPER-randompadding-capture`: U; `WHISPER-batching-replay`: U; `WHISPER-injection-replay`: P
I Know What You Said [48]	R6	B	P	`GAO-zero-copy-off`: P
I Know What You Said [48]	R7	U	–	`GAO-zero-copy-off`: –
Remote timing [3]	R3	B	P	`CARLINI-fixed-rate`: P
Remote timing [3]	R7	B	–	`CARLINI-fixed-rate`: –
Remote timing [3]	R9	P	B	`CARLINI-fixed-rate`: B
TDXRay [55]	R7	U	–	`TDXRAY-oblivious-map`: –
SafeKV [52]	R9	B	P	`SAFE-post-trigger`: P
OTRO [51]	R9	P	B	`OTRO-unpadded`: B; `OTRO-epochpad`: P

S36.2. Bibliography-stratified sensitivity

The strata follow the frozen cited bibliography, retaining the consulted source version. Four mitigation families have a linked journal/conference publication: Early Bird (TIFS), I Know What You Said (USENIX Security 2025), TDXRay (S&P 2026) and WRA (USENIX Security 2026) [47], [48], [55], [40]. Early Bird's inspected source is its author-posted v5, which records acceptance; venue linkage does not certify that every consulted PDF is the final typeset version. The remaining 16 families are cited through preprint/web versions. This partition does not assert that they have never appeared at a peer-reviewed venue.

Table S94 applies the same five queries and applicability rules to both strata. Venue-linked families contribute five mitigation records, but only one or two records are eligible for each query. Their zero bound joins therefore cannot estimate a field-wide or venue-quality difference. The useful sensitivity result is more specific: the sole observed triple package depends on a cited preprint configuration, while the full ledger still identifies reusable component outcomes in both strata. Publication status is a provenance attribute, not a substitute for checking the configuration, receiver and cost object.

TABLE S94. BIBLIOGRAPHY-STRATIFIED JOINT QUERIES: BOUND/ELIGIBLE RECORDS, USING IDENTICAL APPLICABILITY RULES IN BOTH STRATA. VENUE-LINKED: FOUR FAMILIES/FIVE RECORDS; OTHER CITED VERSIONS: 16 FAMILIES/28 RECORDS.

Query	All records	Venue-linked	Other cited versions
Inference + utility	4/22	0/2	4/20
Inference + cost	5/22	0/2	5/20
Matched adaptation + inference	2/17	0/1	2/16
Inference + utility + cost	2/20	0/2	2/18
All five bindings	0/14	0/1	0/13

S36.3. Executable consistency contract

Before compilation, the released generator rebuilds both summaries from the fixed ledger. Its checker verifies identities, counts, all 180 witnesses, five joint queries and 15 stratified cells; missing query rows fail. The witness file binds the record-table hash. Original source locators remain the basis of each judgment.

Complete indexed-family citation inventory. The following citation groups enumerate all 82 indexed families. Categories are core eligibility and origin, not mutually exclusive threat objectives. The machine-readable index binds each key to its source version and locators. Core (58 families): [31], [57], [40], [61], [32], [71], [50], [47], [38], [58], [74], [54], [56], [34], [48], [53], [60], [33], [77], [23], [2], [59], [35], [3], [4], [55], [5], [12], [62], [8], [9], [6], [27], [7], [10], [52], [11], [45], [13], [14], [66], [49], [68], [76], [64], [63], [67], [69], [1], [72], [36], [75], [70], [65], [37], [73], [51], [46].

Adjacent (24 families): [15], [82], [83], [16], [17], [18], [19], [20], [21], [22], [84], [85], [86], [28], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96].

TABLE S95. SENSITIVITY TO THE COMPARISON UNIT. M/D/U: UNANIMOUS SCOPED MATCH/DIFFERENT OBJECT/UNRESOLVED; MIX: CONSTITUENT JUDGMENTS DIFFER. HIDDEN: ORIGINAL WITHIN-QUESTION PAIRS ABSORBED INSIDE A UNIT. NO NEW PAIR IS ADJUDICATED.

Unit	Units	M	D	U	Mix	Hidden
Scoped result records	39	17	105	11	0	0
Shared configuration package	38	16	98	9	1	1
Family and measurement mode	35	10	83	9	1	4
Family only	26	2	31	3	8	15

TABLE S96. FORMAL GUARANTEES AND THE RUNTIME EVIDENCE THEY REQUIRE WHEN INSTANTIATED IN LLM SYSTEMS. THE GUARANTEES HAVE DIFFERENT QUANTIFIERS AND ARE NOT RANKED BY ONE ATTACK SCORE.

Foundation	Protected object / guarantee	LLM-system correspondence	What must remain in scope
Differential privacy [100]	Neighboring inputs/databases and a released mechanism; likelihood-ratio bound.	Training privacy or a randomized output/metadata release, depending on the stated adjacency.	State adjacency, budget, composition and the complete observer transcript. A guarantee for model weights alone does not cover an unmodeled serving trace.
Quantitative information flow [101]	A secret, prior, observation channel and loss; Shannon information or guessing vulnerability.	Quantify prompt/model information in timing, length or access traces.	Specify the prior and leakage measure. A finite trained attack supplies an achieved score, not a universal leakage upper bound.
Constant-time verification [102]	Trace noninterference relative to permitted public outputs and a leakage model.	Fixed-work kernels or oblivious lookup components, e.g. TDXRay.	Preserve compiler/hardware assumptions and the released outputs; variable token counts and scheduling remain separate channels.
Oblivious memory [103]	Indistinguishability of physical access patterns for equal-length logical access sequences.	Protect cache/memory lookup addresses, as in the OTRO design.	Address obliviousness does not itself hide the number/timing of logical accesses. Price padded and unpadded implementations separately.

S37. Granularity, Evidence Status and Worked Use

The synthesis preserves all 39 source records, 351 judgments and historical pair decisions. Comparability counts are conditional inventory results, not a field-level defect rate. The family aggregation corrections in S36 were deterministic projection corrections, not an estimate of extraction error.

S37.1. Alternative units

Projection combines records at a shared configuration, a family/measurement mode, or a whole family. An edge is unanimous only if all original judgments agree; mixed edges retain all reasons. Original within-unit edges are hidden rather than resolved. This sensitivity projects the existing judgments without rereading or changing them. A random regrouping cannot calibrate correctness because the original 608 cross-question pairs were not adjudicated. Leave-family-out percentages in S33 quantify record weighting only.

S37.2. Reconstruction and evidence status

The released data distinguish bibliographic publication status from the exact inspected source version, execution/measurement mode, author-assigned binding and computational consistency. A published record can still lack a requested deployment binding; a preprint can report a precisely located measurement whose scientific replication remains unestablished. Missing reports remain unavailable evidence for that request, not inferred omissions of experiments. The atlas does not assign a scalar paper-quality score.

The supplementary tables and released ledger provide the positive joins, query memberships, and family inference/cost crosswalk. The cell rationales and pair CSV allow deeper inspection. The reporting-card template and filled Early Bird, TDXRay, and CachePrune examples appear in the supplement.

S37.3. Scope of the applications

The external 15-record, three-study WF frame is a retrospective illustration of established evaluation practice. The internal judgments are author-designed examples; P5's source-premise failure gives no favorable evidence for the framework. The implementation case study is compressed in Appendix E, with its full statistics retained in S11–S32. Neither a nonsignificant contrast nor a time-ordered protocol establishes procedure accuracy.

S37.4. Source-version ledger

The frozen bibliography lists venue publications for 21 families and preprints for 61. This metadata describes the cited bibliography, not certification that the exact inspected version was peer reviewed or that its findings were replicated. The ledger separately identifies source URLs, measurement modes, available bindings and unresolved fields for all 39 records. InputSnatch has a publisher citation year of 2027 and a first-online date in 2026; both are retained rather than treating the year difference as a typo.

TABLE S97: All record-level publication and evidence-scope entries. Venue means a venue-linked family in the cited bibliography; the inspected version may remain an author preprint. Modes are defined in the record index. I/U/C are inference/utility/cost bindings, not quality grades.

Record	Cited publication	Mode; I/U/C	Inspected version / source locator
UI-delay-offline [57]	Preprint	Tr; B/U/–	2605.14786v1; 6.1 Fig5;A.5
WRA-decoy-execution [40]	Venue-linked	Ex; B/P/U	Source URL in ledger; 8.1-8.3 Tables8-11
NET-upstream-local [50]	Preprint	Ex; B/U/U	2510.25472v1; IV-C-E TableIII-IV
NET-dummy-padding-simulation [50]	Preprint	Tr; B/U/P	2510.25472v1; VII TableIX
EARLY-K-calibration-oracle [47]	Venue-linked	Si; B/U/P	2409.20002v5; V-A TableVI;IV-A
EARLY-semantic-deidentification [47]	Venue-linked	Ex; –/U/B	2409.20002v5; V-B
WHISPER-randompadding-capture [58]	Preprint	Ex; B/U/U	2511.03675v1; 6.1 Tables3-4
WHISPER-batching-replay [58]	Preprint	Tr; B/U/U	2511.03675v1; 6.2 Fig7
WHISPER-injection-replay [58]	Preprint	Tr; B/U/P	2511.03675v1; 6.3 Table5
SPLICE-CTBF-signal [54]	Preprint	Ex; B/U/U	2606.21842v1; VII-C TableII
SPLICE-QCP-cost [54]	Preprint	Co; –/U/B	2606.21842v1; VII-C TableII
TIME-length-instruction [56]	Preprint	Ex; B/U/U	2412.15431v1; 7.2
TIME-padding-estimate [56]	Preprint	Si; –/U/B	2412.15431v1; 7.3
SPEC-padding-REST [53]	Preprint	Un; B/–/B	2411.01076v4; 6.1 Fig6-7 Table2
DANGOL-capture [23]	Preprint	Tr; B/U/B	2606.07150v3; 9.8-9.9 Table6
DANGOL-livebinding [23]	Preprint	Ex; B/U/B	2606.07150v3; 9.7, OHTTP paragraph after Table 5
SIG-common-guardrail [62]	Preprint	Ex; B/P/U	2607.25560v1; 4.1-4.2;AppendixC
SAFE-post-trigger [52]	Preprint	Ex; B/U/P	2508.08438v2; 7.2.4 Fig12;7.3-7.4
AHB-online-executed [45]	Preprint	Ex; B/B/U	2604.09574v1; 5.1.2 Table1;E.1-E.3
AHB-offline [45]	Preprint	Tr; B/U/–	2604.09574v1; Table1 offline rows;E.1-E.3
KVGOV-partialflush-sim [49]	Preprint	Si; B/U/U	2608.09225v1; 5.1-5.3
KVGOV-hardware-gap [49]	Preprint	Co; –/U/–	2608.09225v1; 5.7-5.8
KVGOV-HMAC-cost [49]	Preprint	Co; –/U/B	2608.09225v1; 5.9
TRAFFIC-prompt-reformulation [35]	Preprint	Ex; B/P/U	2609.31877v1; V-C1 Fig9a
TRAFFIC-temperature [35]	Preprint	Ex; B/U/U	2609.31877v1; V-C1 Fig9a
CARLINI-fixed-rate [3]	Preprint	Un; –/U/B	2410.17175v1; VII Fig11-12
GAO-zero-copy-off [48]	Venue-linked	Co; –/–/B	Source URL in ledger; 7.1
TDXRAY-oblivious-map [55]	Venue-linked	Co; –/B/B	Source URL in ledger; 6.1 Fig8 Table3
AGENTTELL-fixed-instruction [59]	Preprint	Ex; B/P/U	2609.32915v1; 4.3.1;5 Finding1
FP-CLOUDFLARE [61]	Preprint	Ex; B/B/U	2605.01247v1; 6.4 Table5
BAR-co-running-noise [60]	Venue-linked	Ex; B/U/U	Source URL in ledger; 7.2-7.3 Table8
RHYTHM-VPN [9]	Venue-linked	Ex; B/U/U	2502.20589v1; VI-B;VII
DEB-filter-origin [27]	Venue-linked	Ex; B/U/U	2309.05610v2; Threatmodel;4.2
CACHEPRUNE-selective [46]	Preprint	Ex; B/B/B	2605.23640v1; 3.1;5.1-5.6;Table1;AppendixA
CACHEPRUNE-context [46]	Preprint	Ex; B/B/B	2605.23640v1; 3.1;5.1-5.6;Table1;AppendixA
OTRO-unpadded [51]	Preprint	Co; –/–/B	2606.17358v1; III;V;VI;VII-C-E;TableIII
OTRO-epochpad [51]	Preprint	Si; B/–/P	2606.17358v1; III;V;VI;VII-C-E;TableIII
SPARLEAK-replacement [37]	Preprint	Ex; B/B/U	2609.38830v1; 3.1;4-7;Table4
SHAPE-padding-estimate [36]	Preprint	Si; P/U/P	2603.25403v1; IV;V;VII;VIII;IX

TABLE S98. REPORTING ASYMMETRY ACROSS THE 20 MITIGATION FAMILIES. EACH DENOMINATOR CONTAINS FAMILIES TO WHICH THAT ITEM APPLIES; B IS EXPLICIT, P PARTIAL. THE COMMON-POPULATION ROW REMOVES DENOMINATOR DIFFERENCES.

Reported binding	B / applicable	(B+P) / applicable
Inference outcome (R7)	16/17	17/17
Cost (R9)	11/19	15/19
Utility (R8)	4/17	6/17
Protected fitting (R4)	4/11	9/11
Matched adaptation (R5)	2/11	2/11
Same 14 families, inference / cost / utility:		
Explicit B	13/14 6/14 3/14	
B or P	14/14 10/14 5/14	

Expanding R4/R5 applicability to all 20 gives 4/20 and 2/20 (B), or 9/20 and 2/20 (B+P). Source judgments stay fixed. Removal checks and member identities accompany the census (main Appendix F; S38).

TABLE S99. SENSITIVITY OF REPORTING PATTERNS. FRACTIONS GIVE EXPLICIT / APPLICABLE FAMILY COUNTS; BRACKETED NUMERATORS ALSO ACCEPT P. REMOVING FINAL-SEARCH ADDITIONS EXCLUDES SIX RECORDS FROM FOUR FAMILIES; CONTEXTUAL CONTROLS ARE EXCLUDED IN EVERY ROW. IC SEPARATE MEANS NO SINGLE RECORD CARRIES BOTH EXPLICIT INFERENCE AND COST.

Inventory	R7	R9	R8	R4	R5	Agent R8	IC separate
All mitigation	16/17 [17]	11/19 [15]	4/17 [6]	4/11 [9]	2/11 [2]	0/4 [2]	5/8
Without CachePrune	15/16 [16]	10/18 [14]	3/16 [5]	4/11 [9]	2/11 [2]	0/4 [2]	5/7
Without final-search records	13/13 [13]	9/15 [12]	2/14 [4]	4/9 [8]	2/9 [2]	0/4 [2]	4/6

S38. Reporting Patterns and Sensitivity Census

This historical census recomputes six synthesis findings from the frozen record judgments and map, using the extended 20-family inventory that includes AHB's Type III evasion records. The current privacy/attribution cohort and its 19-family projections are in S41. No source cell, pair judgment or experimental observation is changed. The executable census `tools/reporting_patterns.py` saves every numerator/denominator member and input hash in `data/reporting-patterns.json`. Family availability uses the same B, P, ?, U, A, inapplicable precedence as the main matrix. Fractions describe reports in the inspected source frame. The sensitivity operations change conventions or inventory membership, not source facts or coding reliability.

S38.1. Inference, utility, fitting and matched adaptation

The common population contains 14 families: KVGov, SparLeak, SafeKV, Dangol, Early Bird, Shape and Substance, WRA, NetEcho, the cloud benchmark, SpliceLeak, Time Will Tell, Whisper Leak, CachePrune and AHB. On these same families, explicit inference/cost/utility counts are 13/6/3, increasing to 14/10/5 when P is accepted. This matched denominator supports the descriptive cost–utility asymmetry without dividing percentages from differently eligible populations.

Protected fitting has five partial bindings, so its count changes from 4/11 to 9/11 under P acceptance. Matched adaptation has two B, two ?, seven U, and no P. Calling fitting itself generally absent would erase relevant partial evidence. Even accepting the two unresolved family R5 bindings optimistically yields only 4/11; this is a bound on the current uncertainty, not an assignment of B. The broad applicability check temporarily treats all nine inapplicable R4/R5 family entries as U: counts become 4/20 and 2/20, or 9/20 and 2/20 with P. This deliberately stricter denominator is not a revised requirement on non-learning attacks or formal guarantees.

S38.2. Agent utility and polarity

TABLE S100. AGENT PRIVACY/ATTRIBUTION SUBSET: P5 GENERATION AND TYPE I OR II. EVERY APPLICABLE FAMILY IS LISTED.

Family	R8	Objective	Located utility evidence
UI Traces [57]	U	Attribution	Offline delay; task outcome unbound.
WRA [40]	P	Privacy	Separately judged usefulness; no same-condition task pair.
Dangol [23]	U	Privacy	Capture/loopback outcome; workflow utility unbound.
Cloud benchmark [35]	P	Privacy/attribution	Semantic preservation asserted; task score unbound.

The subset rule is P5 generation and Type I or II. Its zero explicit task-utility result becomes two of four under P acceptance, so the zero is specific to explicit binding. WRA's separately rated usefulness and the cloud benchmark's semantic-preservation statement remain partial evidence. AHB's task accuracy is explicitly bound but belongs to Type III detector evasion. TDXRay, CachePrune and SparLeak supply component utility in different settings. Removing CachePrune, the six final-search records, or both leaves the four-member agent subset unchanged. Contextual records never enter this denominator.

TABLE S101. CORE-MAP COVERAGE BY SOURCE-LAYER BLOCK. NUMERATORS ARE FAMILIES WITH AN EVALUATED MITIGATION RECORD; DENOMINATORS ARE THE EMPIRICAL CORE MAP, NOT ALL DEFENSE RESEARCH.

Generating-layer block	Mitigation	Core
P1 only	1	8
P1/P2	4	8
P2 only	10	22
Orchestration/workflow	4	15
Other cross-layer	1	5

Across overlapping purpose labels, Type II has 2/13 mitigation families (UI Traces and the cloud benchmark). C5 and the scoped evaluation rule define this coverage frame.

S38.3. Coverage of the empirical map

The five blocks partition all 58 core families. The numerator requires a scoped evaluated mitigation record, and the denominator requires C1–C5. Thus hardware coverage of 1/8 does not mean only one hardware defense exists; adjacent proof-only and defense-only studies are outside this empirical frame. Holding the 58-family map fixed but excluding final-search mitigation records changes the five block numerators from 1/4/10/4/1 to 1/1/9/4/1; the service-only block remains the largest and the Type II numerator remains two. This removal checks dependence on added mitigation evidence, not completeness of the map. The Type II count is an overlapping purpose-label incidence, not a sixth block. The machine-readable census gives every family identity and its generating/observing layer set.

S38.4. Inference/cost location and measurement mode

TABLE S102. SENSITIVITY TO ACCEPTING PARTIAL INFERENCE AND COST EVIDENCE. A FAMILY IS COUNTED ONLY IF IT HAS BOTH REQUESTED ITEMS UNDER THE SPECIFIED RULE.

Inventory	Both B	Separate	Both B/P	Separate
All mitigation	8	5	12	3
Without CachePrune	7	5	11	3
Without final-search records	6	4	9	3

Under strict B, the five separated families are Early Bird, SpliceLeak, Time Will Tell, KVGov and OTRO. The result is five of seven after removing CachePrune and four of six after removing the final-search additions. If P also counts, only SpliceLeak, Time Will Tell and KVGov remain separated among 12 eligible families (three of 11 without CachePrune, three of nine without the final-search records). This is why the source-scoped reconstruction names explicit bindings and does not claim separation for most partially specified evidence. Full record identities are retained in the census and the supplementary crosswalk (Table S126).

The 39-record mode distribution is Ex20/Tr6/Si5/Co6/Un2. All six contextual records are Ex, making the mitigation-only distribution Ex14/Tr6/Si5/Co6/Un2 (33 total). Ex denotes executed collection/prototype, not necessarily end-to-end production execution. The network receiver subset, defined by P3 in the receiving-layer set, has 12 records: WRA; two NetEcho; three Whisper Leak; SPEC; two Dangol; two cloud benchmark; and remote timing. Its six Ex, four Tr and two Un records do not support a claim that network evidence is mainly replay. The final-search exclusion leaves Ex11/Tr6/Si3/Co5/Un2 (27 mitigation records). Six Co records include five cost-bearing components and KVGov's unprotected hardware calibration; none automatically prices an entire protected deployment.

S38.5. Holdout evidence objects

The archived matrix has 13 task/input, three model, two topology and one network/load H entries; hardware and time have no H entry in that extraction. Four task H families belong to the mitigation audit. The three model H entries (LLM-Fin, ArchSnoop, LeakyLMs) evaluate attack/identification generalization outside the mitigation families. The two topology H entries (Dangol and the cloud benchmark) likewise evaluate workflow recognition. The source reconciliation in S34 gives their exact locators and objects. No frozen source-to-target defense calibration is established by those five entries. This scoped conclusion does not recode the other 14 H entries as uniformly attack-only or treat unresolved hardware/time cells as evidence that no experiment exists. Accepting variation V as coverage still requires identifying what was held fixed or locally refitted. Time Will Tell and Whisper Leak remain useful examples of defense-condition variation.

S39. Forward-Citation Context Pilot

We inspected whether citing sources transfer Early Bird's approximately 6 ms/4% semantic-cache de-identification cost to a different prefix defense, or KVGov's approximately $1.3\,\mu$s HMAC microcost to a complete defended deployment. A positive finding requires a source attribution and numerical transfer to a different configuration or cost object. Citations that do not use the number cannot establish this kind of misuse.

On 27 October 2026 a convenience search located five citing works and six checked citation edges: KVGov, Time Will Tell, PrefixWall and an autonomous-driving roadmap cite Early Bird; an LLM contention study cites both Early Bird and KVGov [49], [56]. Primary contexts and locators are listed in the released pilot ledger, including https://arxiv.org/html/2609.06853v1, https://arxiv.org/abs/2603.10726 and https://doi.org/10.1145/3807455. No target cost transfer was found in the inspected passages. KVGov's Early Bird citation has a version/metric ambiguity that is not sufficient to establish an error. One roadmap passage was obtained from an indexed excerpt rather than a complete PDF; its full context is unresolved.

An OpenAlex metadata query resolved identifiers for all 20 families of the extended inventory and returned eight indexed citing works: seven for the merged Early Bird preprint/journal record and one for a remote-timing paper. The index returned none for KVGov, while a separately located paper cites it. Seven of the eight indexed contexts remain unretrieved; they are unknown, not negative cases. After merging with targeted search, 12 candidate or verified citing works are retained; five have inspected passages. The index is incomplete, the pilot is not a random sample, and no misuse rate is estimated. A proposed deployment comparison remains an example of a measurement decision, rather than an observed attribution error.

S40. Three Object Joins: Source and Measurement Trace

The following examples instantiate established adaptive-evaluation and deployment principles in particular serving and agent objects. They are applications of those principles, not claims that the principles originated here. Juarez et al. examined how shifts in browsing habits, location and browser version affect website-fingerprinting attacks; Mathews et al. evaluated nine defenses with stronger attacks and an on-network DynaFlow implementation. Tramèr et al. and Carlini et al. give broader adaptive and robustness-evaluation guidance [104], [99], [105], [106].

Prefix isolation versus semantic-cache processing. Early Bird studies both a prefix-recovery oracle and a semantic-cache attack. Its average 6 ms processing delay and approximately 4% overhead describe de-identification on a GPTCache semantic-cache hit, whose unprotected mean is approximately 0.14 s [47, V-A–B]. They do not price its K-prefix isolation rule. For contrast, CachePrune reports a default serving configuration with QASPER generation quality 0.28 versus 0.30 and time-to-first-token 273 versus 797 ms for no sharing [46, Table 2, 5.2–5.6]. Its Table 1 recovery and Table 2 serving results are linked through the source-described default configuration and workload, but the inspected source does not establish identical requests or paired runs across those tables. The two papers' numbers do not form a controlled cross-system comparison. Early Bird's prefix results use a simulated oracle with measured hit/miss classifiers; the source does not thereby establish a deployed K-prefix serving cost.

Discarded reads versus executed writes. In a replay of reference programs for 97 eligible retail tasks, each repeated across three cover selections, the read-only and isolated-shadow conditions preserve reference database state in all 291 units at each insertion position. In contrast, a same-item self-write inserted before the reference program changes state in 173/291 units: the tool can change order status and append a refund entry, even when the item identifier and price do not change (Table S21, S9). These are simulator outcomes for the specified APIs and states. They show why discarding a response does not by itself establish that an executed write has no effect; they are not an estimate of mutation prevalence across real services.

Component quality versus planner outcome. CachePrune's QASPER score above measures generation quality in its serving evaluation, not an acting agent's task completion. A separate GLM experiment obtains new model decisions for 50 previously known retail tasks in each arm. It records 37/50 native successes without cover and 33/50 with a prefix-shadow scheduler, a paired difference of -8.0 percentage points with customer-bootstrap interval $[-23.9, 6.5]$ over 37 customers (S26). This interval does not resolve a success-rate difference. The measured scheduler has injected response delays and deadline warnings, and omits the full joint framing and terminal flush used in the trace replay. Its result therefore binds utility to this particular live diagnostic; it cannot be assigned to that other defense configuration or interpreted as a WAN latency result.

TABLE S103. PRIVACY/ATTRIBUTION COHORT AND SENSITIVITY TO RECORD REMOVAL. FRACTIONS ARE EXPLICIT B / APPLICABLE FAMILIES; BRACKETED NUMERATORS ALSO ACCEPT P. AHB AND CONTEXTUAL RECORDS ARE EXCLUDED THROUGHOUT.

Inventory	Inference R7	Cost R9	Utility R8	Protected fit R4	Matched adaptation R5
All 19 families	15/16 [16]	11/18 [15]	3/16 [5]	4/10 [8]	2/10 [2]
Without CachePrune	14/15 [15]	10/17 [14]	2/15 [4]	4/10 [8]	2/10 [2]
Without final-search records	12/12 [12]	9/14 [12]	1/13 [3]	4/8 [7]	2/8 [2]

TABLE S104. SOURCE-SCOPED EXAMPLES OF WHAT AN INDEXED INTERVENTION ESTABLISHES AND WHAT REMAINS OBSERVABLE. ALL ROWS IDENTIFY AN EMPIRICAL OR MODELLED RESULT IN THE INSPECTED RECORD; WHETHER THE SOURCE ALSO ESTABLISHES A FORMAL GUARANTEE IS NOT ADJUDICATED BY THIS INVENTORY.

Family / record	Intervention and evidence	Residual result or surviving observation	Source scope and deployment implication
WRA [40]	Concurrent decoy research queries; interleaved-domain evaluation.	Reported one-decoy E_{func} remains 0.610, down from 0.770.	Sections 8.1–8.3, Tables 8–11; usefulness is separately judged, not a paired planner task score.
OTRO [51]	Oblivious access plus epoch-padding estimate; conditional-information calculation.	The reported access-count information falls from 3.94 to 0.04 bits in the stated model.	Sections III, V–VI, VII-C–E and Table III; unpadded timing and padded access-count estimates are distinct records.
Early Bird [47]	K-specific timing calibration and stochastic oracle simulation; separate semantic-cache prototype.	The simulation concerns its chosen oracle; the semantic prototype's cost does not measure the simulated prefix policy.	Sections IV-A and V-A/B; retain distinct policies, receivers and measured cost objects.
NetEcho [50]	Protected upstream traces and local serving, plus synthetic padding/insertion analysis.	Collected padded traces retain an inference signal; downstream re-emission exposes a separate receiving view.	Sections IV-C–E, Tables III–IV and VII/Table IX; synthetic packet transformation and collected execution are separate.

S41. Archived Mitigation Reporting in Attack-Evaluated Studies

This section retains the pre-R41 measured-utility accounting; S44 supplies the superseding R8 applicability and denominators. In the C1–C5 attack-evaluated source frame, the Type I/II union contains 52 core families, including 19 with a qualifying mitigation record and 33 without one linked in this extraction. The archive preserves 39 records: 31 Type I/II mitigation records, two Type III evasion records and six contextual records. The R34 tables retain the earlier, explicitly broader 20-family/33-record inventory; this section projects the unchanged source judgments onto the narrower cohort.

For the 13 families eligible for inference, utility and cost, explicit bindings are 12, two and six, respectively; accepting P gives 13, four and ten. Expanding R4/R5 applicability to all 19 families gives 4/19 and 2/19 explicit, or 8/19 and 2/19 with P. The removal checks in Table S103 expose dependence on CachePrune and the final-search records. Every included and excluded record and family is listed in `data/r35-privacy-census.json`; the generator records hashes of the frozen inputs.

The five record-level queries give inference/utility 3/20, inference/cost 5/21, matched adaptation/inference 2/15, the three-way query 2/19 and all five items 0/13. Accepting P yields five, 11, two, two and zero bound records, respectively. The two three-way records are CachePrune outcomes for one configuration. Removing that family leaves 0/17; removing final-search records leaves 0/15. These numbers describe bindings at the inspected source scope.

The 31 records comprise 13 executed, five transformed, five simulated/estimated, six component/calibration and two unresolved records. The broader 33-record R34 inventory additionally contains AHB's executed and transformed evasion records. Table S104 presents representative interventions, their source-scoped measured outcomes and residual observations. A measurement result is not automatically a formal guarantee; formal proof status for the full matrix remains unaudited.

TABLE S105. TWO FIT-APPLICABILITY AND COST-PLACEMENT INTERPRETATIONS OVER THE SAME 31 PRIVACY/ATTRIBUTION RECORDS. ENTRIES ARE BOUND/ELIGIBLE RECORDS. BROADENING MAPS FIT-INAPPLICABLE RECORDS TO UNKNOWN, NOT TO A VERIFIED ABSENCE OF ADAPTATION.

Query	Fit scope	Broader unknown
Matched adaptation + inference	2/15	2/22
All five bindings	0/13	0/19
Inference + cost	5/21	4/21[†]

[†] Requires explicit R3 placement as well as R7/R9, or treats SPEC's payload-only cost as P for a wire-level query. The same four records remain, with distinct cost objects. Source cells and original query labels remain unchanged.

S42. Applicability and Source-Bound Sensitivity

S42.1. Two adaptation questions and heterogeneous costs

R4/R5 refer to protected-data fitting and a matched training-fit contrast. For a non-learning attack the historical "–" describes that specific contrast only; an adaptive strategy that adjusts probes, auxiliary information or operating conditions can still be relevant. The broadened sensitivity projection converts the historical "–" cells to unknown (?), not to U or a verified missing experiment. The historical record labels remain unchanged, and a separate source review would be needed to recode broad adaptive evaluation. Table S105 retains the historical broad R8 utility applicability, under which the all-five query changes from 0/13 to 0/19. Table S130 in the current reconstruction uses the later outcome-specific R8 applicability, giving 0/5 and 0/8 for that query. The different denominators reflect different quality-applicability rules, not changed source observations. The full item memberships and hashes are in `data/r36-sensitivity.json`.

SPEC's REST padding record reports an inference result and a payload-size ratio at its evaluated configuration (inspected v4, Section 6.1 and Appendix D Table 2; https://arxiv.org/html/2411.01076v4). It receives R9 B for that stated payload object, while R3 is P because placement/execution binding is incomplete. It does not measure wire bytes or end-to-end cost. Requiring R3 B, or treating payload-only cost as partial for a wire-cost question, removes SPEC from the inference-plus-cost positives; the four remaining records still refer to distinct cost objects. The five original positives are not a common cost ranking.

A leave-one-family-out computation reruns each query over the 19 Type I/II mitigation families. For the triple query the two CachePrune record positives both disappear when CachePrune is omitted, and remain for each of the other 18 omissions. They represent two target outcomes in one configuration. Removing all final-search additions gives 0/15 eligible triple records; all queries retain their own declared eligibility denominators. The projection preserves the frozen source cells and does not establish that unreviewed evidence is absent.

S42.2. Search frame and boundary cases

The 615 occurrences and 565 deduplicated records in Section S10 are a completed, finite retrieval update, not a PRISMA census of all venues. Its disposition table and query strings give the reproducible route; 392 records were routed to the two stated exclusions and the remaining 173 to the other reported categories. The initial seeds and venue screening overlap that update and must not be added as independent PRISMA stages. Hardware and architecture venues including ISCA, MICRO, HPCA, ASPLOS and CHES/TCHES were not comprehensively screened. The October 21 closure is a snapshot, not a continuously updated claim.

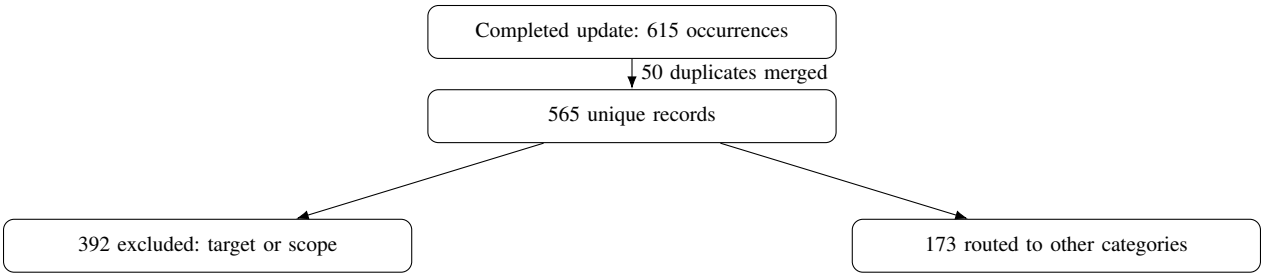


Figure 3. Finite retrieval update only. Other routes include already indexed studies, new and adjacent records, background and unresolved availability; full counts appear in Table S19. Overlapping seed and venue routes are excluded.

Three adjacent decisions illustrate the operational boundary. CacheSolidarity v1 (https://arxiv.org/html/2603.10726v1) has an auxiliary timing/cache target and a candidate-prefix experiment in Section 5.2.3, Figure 9. The archived family index incorrectly points to Figure 7, which is a benign-workload figure; Figure 9 is the corrected locator. The inspected v1 presents attack traces and qualitative interpretation; whether that meets C5 when a scored inference result is required is a boundary decision, not proof of no attack. Inclusion under a broader C5 requires a separate re-audit and count update. TOPR v4 (https://arxiv.org/html/2512.16310v4) studies information exposed through response/reasoning content; under the current semantic-only exclusion its inspected analysis does not isolate the auxiliary-only C3 condition. SecEmb (https://bearhw.ece.vt.edu/content/dam/bearhw_ece_vt_edu/publications/2025_HPCA_SecEmb.pdf) pairs its attack demonstration with DLRM embedding operations; its GPT-2 measurements concern protection cost/quality rather than a paired GPT-2 attacker recovery result. These are scoped selection explanations, not claims that the studies lack security value. Other adjacent defense decisions remain open. SICC (https://arxiv.org/html/2609.26076v1, Sections 3 and 5) studies private-state-dependent message form in addition to content; DP-MemView (https://arxiv.org/html/2608.03130v1, Appendix B.5) includes an internal-score side-channel diagnostic. Whether their evaluated receiver and isolated signal satisfy C3–C5 requires a record-level adjudication, not a blanket semantic-only exclusion. PlanTwin's re-identification experiment (https://arxiv.org/html/2603.18377v2, Section 5.6) uses disclosed schema fields; its relation to the auxiliary-only C3 boundary also needs explicit adjudication. PriCon's accessible HTML (https://arxiv.org/html/2609.32642v1) supplies only an abstract; its detailed eligibility remains unresolved. No complete adjacent-defense exclusion ledger is claimed.

S43. Attacker Capability as an Evidence-Reuse Boundary

Matched evidence from the controlled transport examples. The question is whether a protection estimate transfers from the evaluated attacker to one fitted on defended data. The clean-trained attacker (D_0) and defended-trained attacker (D_2) receive identical defended test outputs. Table S106 extracts existing rows from S11/S12; no new captures or model requests were made for this revision. Accuracy is brief-versus-expanded classification, rather than the retail study's balanced accuracy. There are 24 held-out outputs from 12 topics; topic repetition and padding draws do not add independent topics.

TABLE S106. EXISTING PAIRED ATTACKER COMPARISONS ON A FIXED DEFENDED VIEW. PERCENT ACCURACY.

Defended configuration	D_0	D_2	Location
Buffered, 1024-byte buckets	50.0	70.8	S11, matched fitting
Streaming, constant eight-token batches	50.0	100.0	S12, E5
Streaming, constant events and count cap	50.0	50.0	S12, E5

The buffered adapted-accuracy interval is 58.3–83.3%; the streaming table reports point estimates. These examples support a specific reuse decision: retain the tested attacker and configuration with a protection estimate. Training on defended data recovers signal for the bucket and batch schemes; the count cap retains the measured reduction in this projection. The comparison illustrates the established adaptive-evaluation principle in emitter-specific settings. It establishes neither universal recovery nor a model-independent privacy guarantee.

WRA-motivated GLM pilot: residual identification with incomplete outputs. WRA evaluates decoy research activity against prompt and trait inference [40]. An exploratory adaptation used 20 previously inspected public topics, five GLM-generated decoys per topic and 120 saved search-result responses. A trace here is a list of result-link hostnames: the real query's saved response is reused across the no-decoy and five-decoy conditions, and the latter appends the five decoy result lists before seeded shuffling. These hostnames describe returned search results, not a live research agent's visited destinations or timed packets.

The attacker uses the served `glm-5.3-flash` alias and sees all 20 candidate topics, including the true one without its answer label. It is told that some hosts may be unrelated decoys. No attacker training or matched weaker-attacker condition was collected. Accordingly, this pilot estimates residual identification under this candidate-list attack, not an adaptation gain. Generation, search and identification records remain separate from the author's archived WRA experiments and from the retail cohort.

TABLE S107. TWO SEPARATELY RECORDED GLM ATTACK ROUNDS. FRACTIONS COUNT CORRECT GUESSES OVER VALID OUTPUTS; MISSING PREDICTIONS REMAIN IN THE PLANNED 20 PER CONDITION.

Output budget	Clean	Five decoys	Missing	Missing IDs
16,384	19/20	17/18	2	1, 10; defended
32,768, post hoc	18/20	15/18	2	10, 18; defended

Both rounds are incomplete under their complete-pair analysis rules. Missing outputs exhaust the output budget without a valid tool call. The higher-budget sensitivity reruns every condition and preserves the first round. Assigning all missing guesses as incorrect or correct gives full-cohort defended-accuracy bounds of 85–95% and 75–85%, respectively; these are missing-value bounds, not confidence intervals. The two rounds do not provide an attack-adaptation comparison. They do show residual topic recoverability among valid responses in this restricted view, motivating the requirement to bind a protection estimate to its observation and attacker.

Each topic issued one real query and five decoy queries. This accounting does not measure bytes, agent delay, task completion or answer quality. `data/r40-wra-glm-summary.json` publishes derived outcomes, failure types and source hashes without redistributing third-party prompts or response content. The paper's attacker-recovery example uses the complete matched transport comparisons above; it does not infer recovery or online WRA effectiveness from this incomplete pilot.

S44. Source-Frame and Functional-Outcome Applicability

Selection consequence. All reporting statistics concern mitigation results in studies admitted through C1–C5, including an empirical inference evaluation. They are not a census of all defense research. SafeKV, CachePrune and OTRO also demonstrate why this is not a claim that every included study is attack-first. Adjacent defense-oriented work remains outside the denominator when qualification is unresolved or a required criterion is not established.

Revised R8 rule. A quality measurement is applicable when the indexed intervention can change prompts, model content, sampling, approximate computation or component coverage. A lossless wrapper, fixed-trace transformation or exact logical computation can instead support a source-scoped preservation claim. Such a record is exempt from the content/function-quality denominator; its timing, resource use and end-to-end task outcomes still require their own evidence. Padding an image's patch grid and inserting null prompt tokens are not automatically content-preserving: they change model processing, unlike wire padding.

The 31 records comprise 11 potentially quality-changing interventions, 18 scoped preservation witnesses and two standalone cost/calibration units without an output-quality comparison. Projection gives eight quality-applicable families: three B, one P and four U; eleven families are exempt from this narrow query. Thus the old 3/16 and 5/16 counts are superseded by 3/8 and 4/8 for functional quality. These describe the selected source frame and the named outcome. The old source cells remain archived; `data/r41-utility-scopes.json` provides every new applicability judgment, original status, locator and reason.

Agent subset. The map-selected four-family inventory contains six records. UI timestamp transformations and Dangol capture/live transport preserve saved actions/messages for the narrow functional object. WRA returns only the real output despite parallel decoy execution, so it also supports a narrow real-answer-content preservation claim. The quality-applicable agent-associated subset is therefore the cloud benchmark alone (0/1 B, 1/1 B+P family; 0/2 B, 1/2 B+P records). Its prompt/temperature conditions measure fingerprinting, not MARBLE task completion. Added execution is a separate outcome: WRA supplies no matched end-to-end task score for the decoy condition (0/1), and no explicit same-condition task score is linked for any of the four mapped families (0/4). The latter is a broad inventory, not four execution-changing cover implementations.

Positive functional routes. The measured-quality joins remain CachePrune's two target outcomes and SparLeak's intensity-specific quality result. Separately, SPEC's payload framing and Dangol's capture/live records combine scoped preservation with bound inference/cost objects. These records support their stated payload, trace or message claims, rather than an unmeasured task/deadline guarantee. Timing and placement limitations of their costs remain unchanged.

TABLE S108. FUNCTIONAL-QUALITY DENOMINATOR AFTER APPLICABILITY AND SOURCE-REMOVAL CHECKS. PRESERVATION AND COMPONENT-ONLY UNITS ARE EXEMPT.

Source inventory	Families	Quality B/applicable	Quality B+P/applicable
All source families	19	3/8	4/8
Without CachePrune	18	2/7	3/7
Without final-search records	15	1/5	2/5

TABLE S109: Complete R8 applicability overlay for all 31 Type I/II records. Relevant = functional quality applies; preserved = scoped content/action/function witness; component = no output-quality unit. Archived source statuses remain available.

Record	Scope	Old/new R8	Source locator	Outcome-specific reason
CACHEPRUNE-context [46]	Relevant	B/B	2.2.2; 4.2; 5.3; 5.6	Position-independent KV reuse approximates attention dependencies; generation quality depends on recomputation.
CACHEPRUNE-selective [46]	Relevant	B/B	2.2.2; 4.2; 5.3; 5.6	Same approximate segment-reuse configuration as contextual recovery; linked F1/ROUGE results apply.
CARLINI-fixed-rate [3]	Preserved	U/–	VII; Figures11-12	Fixed-rate delivery queues existing tokens and emits empty cover messages; narrow token-content preservation excludes delay and live-placement claims.
DANGOL-capture [23]	Preserved	U/–	9.8-9.9; Table6	Shaping transforms a saved capture with fixed application messages; no new output-quality experiment is needed for those fixed messages.
DANGOL-livebinding [23]	Preserved	U/–	9.7; OHTTP paragraph after Table5	Encapsulation and traffic cadence preserve application message bodies; task deadlines and end-to-end workflow completion remain separate outcomes.
EARLY-K-calibration-oracle [47]	Preserved	U/–	V-A; TableVI	Prefix-sharing granularity and oracle calibration do not edit model input or the logical prefix KV computation; this is narrower than an online function measurement.
EARLY-semantic-deidentification [47]	Relevant	U/U	V-B; Figure12	PII is replaced before semantic-cache processing and restored afterwards; content-sensitive processing needs functional evidence.
GAO-zero-copy-off [48]	Preserved	–/–	7.1	Disabling zero-copy changes loading and memory sharing while retaining the logical model; loading/memory costs remain applicable.

Record	Scope	Old/new R8	Source locator	Outcome-specific reason
KVGOV-HMAC-cost [49]	Component	U/–	5.9	Key-derivation microcost has no model-output/function comparison unit; its resource cost is retained.
KVGOV-hardware-gap [49]	Component	U/–	5.7-5.8	Unprotected TTFT calibration is not a modified model-output condition; its timing result is not a quality measurement.
KVGOV-partialflush-sim [49]	Preserved	U/–	4.1-4.2; 5.1-5.3	Namespace salting and cache reuse/eviction policy change cache visibility and recomputation, not prompt content or the logical prefix computation; scope is the simulated policy.
NET-dummy-padding-simulation [50]	Preserved	U/–	VII; TableIX	Synthetic packet insertion and padding operate on saved traces, leaving their stored application content fixed.
NET-upstream-local [50]	Preserved	U/–	IV-B; IV-C-E; TablesIII-IV; Figure5	Padding and batching alter framing/grouping of the token stream. The exemption concerns that content projection, not every frontend or online deployment behavior.
OTRO-epochpad [51]	Preserved	–/–	V-C-D; VI; VII-C-E	Dummy memory accesses and internal scheduling preserve the logical token sequence; padded execution cost is still distinct from unpadded cost.
OTRO-unpadded [51]	Preserved	–/–	V-C; VII-C-D	Internal oblivious-tokenization scheduling preserves token IDs relative to the non-ORAM baseline; only its own measured costs may be reused.
SAFE-post-trigger [52]	Preserved	U/–	4.2.3; 4.3.1; 7.2-7.4	Visibility bits and exact-prefix cache lookup govern reuse without rewriting prompts or approximate KV substitution; this exempts narrow generation quality, not service performance.
SHAPE-padding-estimate [36]	Relevant	U/U	IX-B1	Forces a worst-case image grid/patch configuration; unlike wire padding this changes model preprocessing, so quality preservation is not automatic.
SPARLEAK-replacement [37]	Relevant	B/B	7.5; Table4	Random token replacement changes generated content; source reports intensity-matched ROUGE.
SPEC-padding-REST [53]	Preserved	–/–	6.1; Figures6-7; Table2	Padding/aggregation wraps already generated token payloads; live/wire placement and interactive delay remain separately scoped.
SPLICE-CTBF-signal [54]	Relevant	U/U	VI-B; VII-C; TableII	Changes selective boundary recomputation in non-prefix approximate fusion; unchanged generation quality is not established by a fixed FLOP budget.
SPLICE-QCP-cost [54]	Relevant	U/U	VI-A; VII-C; TableII	Appends null tokens to a private prefix; no output-invariance or attention-mask witness is bound in the inspected scope.
TDXRAY-oblivious-map [55]	Relevant	B/B	6.1; Figure8; Table3	Truncated token-frequency/vocabulary mapping has a component coverage outcome; it remains a functional-quality query.
TIME-length-instruction [56]	Relevant	U/U	VII-B	A fixed-length explanation instruction changes the requested answer, unlike transport padding.
TIME-padding-estimate [56]	Preserved	U/–	VII-C	System-level padding delays and frames existing output tokens; it is separate from the answer-length instruction.
TRAFFIC-prompt-reformulation [35]	Relevant	P/P	V-C1; Figure9a	Prompt wrapping changes model input. Semantic preservation is asserted rather than matched task-scored.
TRAFFIC-temperature [35]	Relevant	U/U	V-C1; Figure9a	Temperature changes decoding and can change generated content.
UI-delay-offline [57]	Preserved	U/–	6.1; Figure5; AppendixA.5	Offline timestamp edits retain saved action types and structure; the exemption does not establish live task or timeout preservation.
WHISPER-batching-replay [58]	Preserved	U/–	6.2; Figure7	Grouping stored packet features leaves saved model text fixed; streaming response time remains an unmeasured deployment outcome.
WHISPER-injection-replay [58]	Preserved	U/–	6.3; Table5	Synthetic packet injection is a trace transformation rather than model-content editing.
WHISPER-randompadding-capture [58]	Preserved	U/–	5; 6.1; Tables3-4	Additional random fields change serialized event length, not decoded token content.
WRA-decoy-execution [40]	Preserved	P/–	8.1-8.3; Tables8-11	Real and decoy requests execute concurrently; only the real output is returned. This preserves the real-answer content projection, not task completion under added execution.

S45. Defense-First Extension: Sources, Outcomes and Boundaries

This purposive challenge set inspects four named seeds—CacheSolidarity/PrefixWall, PlanTwin, SICC and DP-MemView—plus two selected additions, OCELOT and AirGapAgent. Its purpose is to test scope interpretations against specific defense-oriented sources. The six family cards remain separate from the original 19-family/31-record C1–C5 cohort, which contains both attack-oriented and defense-oriented studies selected through empirical attack evaluation. Neither source frame estimates reporting prevalence across defense research.

Search history and incomplete coverage. The retained materials record prior reconnaissance, nine logged queries and a cap of four seeds plus at most two additional relevant primary sources. Queries include exact identities, mitigation/isolation, padding/cache partitioning, agent privacy and a USENIX-focused check. The protocol is dated October 27, but has no independently verifiable preregistration timestamp. We report the cap as documentation of this bounded exercise, without claiming a prospectively registered stopping rule or inferring when it was chosen.

The search returns were not fully screened, and citation neighborhoods were not exhausted. The admission log retains unassessed candidates including AgentPrivArena and *An AI Agent Execution Environment to Safeguard User Data* (GAAP); PriCon retains its prior abstract-only uncertainty. ISCA, MICRO, HPCA, ASPLOS and CHES/TCHES were not systematically covered by either search. SecEmb illustrates a consequence of the original eligibility rule: its GPT-2 quality/performance evidence remained adjacent because matched GPT-2 attack recovery was not established in the inspected scope [17]. This is a selection-boundary example, not a completed hardware-defense search. The six cards therefore challenge particular interpretations without repairing omissions or selection bias.

A further hardware boundary check. Two selected source checks illustrate evidence that the original empirical-attack selection need not admit. SecEmb argues protection for its specified embedding-access threat and reports GPT-2 quality and overhead, while its inspected attack demonstration uses a separate DLRM setting [17, III–VI]. Those results do not establish a matched GPT-2 attack-recovery evaluation. Palermo (https://arxiv.org/html/2411.05400v1) analyzes a GPT-2/OpenORCA workload in a modeled memory-system setting against an off-chip observer. Its binary response-time statistic concerns the specified stash/tree distinction; it is neither prompt recovery nor a general bound on all leakage. Its source describes simulated performance and controller area/power, while output quality and identical runs across these measurements were not established in the inspected scope. The old index includes SecEmb as adjacent; an exact-name and identifier search did not locate Palermo, which does not establish why it was absent or exclude an alias. These selected cases do not alter any archived cells or denominators and do not complete the extended search.

Source and outcome dictionary. Each card records the exact inspected arXiv version, receiving permissions, intervention, attack/test condition, functional object, cost and transfer scope. R4 asks about protected-data attacker fitting; R5 requires clean-data and protected-data fits tested on the same defended population. A transcript-conditioned adaptive adversary is a separate capability. Functional quality applies to changed content or behavior; exact logical preservation supports only its stated object, leaving latency and task deadlines separate.

What the selected cards support. All five quality-applicable cards locate a measured functional outcome; CacheSolidarity instead has narrow logical-preservation evidence. These positive examples do not form a homogeneous or randomly selected comparison with the original R8 count of 3/8 (4/8 accepting P). No population difference in reporting rates can be inferred from these counts. Four cards contain quantified empirical protection for their named receivers and two retain partial protection bindings. A functional score alone does not establish a same-configuration protection/function/cost package; the cards retain each binding separately. Semantic-disclosure, transcript-inference and auxiliary-channel outcomes also remain distinct. No complete R5 fit contrast was located (four inapplicable, two unreported), while SICC and OCELOT explicitly evaluate broader adaptive attacks. This source-scoped R5 status must not be described as absence of adaptive evaluation.

Receiver-specific admission. SICC isolates form-only length and structure inference, which warrants a distinct auxiliary-channel record candidate [83, Table 3]. DP-MemView's main endpoint is a semantic response transcript; its separate exposed internal-score diagnostic reports AUC 0.976 while the compliant response paths are fixed [91, RQ2, after Table 4]. That diagnostic must not be substituted for an attack against the compliant interface. PlanTwin's unrestricted schema re-identification is separate from budgeted quality; OCELOT's semantic trajectory guarantee is calibration-relative; AirGapAgent evaluates contextual disclosure rather than passive metadata [82], [107], [108].

Versions and runnable artifacts. CacheSolidarity v1 is renamed PrefixWall in arXiv v2. Both inspected preprints retain a timing demonstration rather than a scored prompt-recovery estimate. The author-linked CacheSolidarity repository describes a richer Middleware artifact and a SafeKV reimplementation; its README expected results were not executed or substituted for the preprint [94]. PlanTwin's default heuristic description and SLM-backed timing table leave a configuration join to resolve [82, 5.1,5.4]. The source cards retain these uncertainties.

TABLE S110: Complete source-scoped R1–R9 coding for the six defense-first cards (54 cells) and the separately scoped DP-MemView auxiliary-score diagnostic (nine cells). B = explicit binding, P = partial, U = unreported in inspected scope, ? = unverified, – = inapplicable to the named outcome.

Card / source	Item	Code	Source locator / required scope
DF01 CacheSolidarity / PrefixWall [94]	R1	B	2.3;4
	R2	B	2.2-2.3
	R3	B	3.2-3.3;5.1
	R4	–	5.2.3:non-learned demonstration
	R5	–	5.2.3:no fit contrast
	R6	B	5.1/Table2
	R7	P	5.2.3/Fig9:timing trace, not scored recovery
	R8	–	3.2:narrow logical preservation
	R9	B	5.2.4/Table3:component cost; retain object
DF02 PlanTwin [82]	R1	B	3.2
	R2	B	3.1-3.2
	R3	B	4.1-4.4
	R4	–	5.6:field-overlap matcher
	R5	–	5.6:no fitted-attack contrast
	R6	B	5.1
	R7	P	5.2:exposure metric;5.6:unbudgeted inference
	R8	B	5.2/Table5:planning quality only
	R9	P	5.1 default heuristic versus5.4 SLM timing
DF03 SICC [83]	R1	B	Task and Threat Model
	R2	B	Task and Threat Model
	R3	B	Operational Compiler
	R4	P	Setup:train/test stated; defended-fit allocation unclear
	R5	U	Setup/Adaptive Attacks:no matched clean/protected fit contrast
	R6	B	Setup;Tables7-9
	R7	B	Tables3,5,6,8
	R8	B	Tables7-9:condition-matched completion
	R9	P	Operational cost:operation description only
DF04 DP-MemView [91]	R1	B	Problem Definition
	R2	B	Problem Definition/interface contract
	R3	B	DP-MemView;Table1
	R4	P	C.5:fit/calibration split; per-interface fitting unclear
	R5	U	Experiments/C.5:no clean/protected fit contrast
	R6	B	C.1-C.3/Table5
	R7	B	Table2;C.5:empirical transcript AUC
	R8	B	Table2;C.5:matched model-judged quality
	R9	U	Experiments/C.5:no operational cost measurement
DF05 OCELOT [107]	R1	B	III-C-D
	R2	B	III-B-C
	R3	B	IV;Algorithm1
	R4	–	V-A:fixed LLM proxy pools;defense calibration separate
	R5	–	V-A,V-E:no clean/protected fitted-attacker comparison
	R6	B	V-A;AppendixD
	R7	B	V-B/TableII;V-E/TableV
	R8	B	V-B/TableII;V-F/TableVI
	R9	B	V-F/TableVII
DF06 AirGapAgent [108]	R1	B	2.2-2.3
	R2	B	2.3
	R3	B	4.1-4.2
	R4	–	6.1:prompted models,not learned attacker fitting
	R5	–	5.3;6:no clean/protected fitted-attacker comparison
	R6	B	6.1
	R7	B	5.3;Tables3-6:explicit-disclosure protection
	R8	B	Tables3-5:matched answer utility
	R9	U	6:cost not located
DF04-score [91]	R1	B	RQ2 Interface Safeguards, after Table4;B.5
	R2	B	RQ2 Interface Safeguards, after Table4;B.5
	R3	B	RQ2 Interface Safeguards, after Table4;B.5
	R4	–	RQ2 Interface Safeguards, after Table4;B.5
	R5	–	RQ2 Interface Safeguards, after Table4;B.5
	R6	P	RQ2 Interface Safeguards, after Table4;B.5; exact diagnostic denominator is partial
	R7	B	RQ2 Interface Safeguards, after Table4;B.5; exposed top-score AUC 0.976
	R8	–	RQ2 Interface Safeguards, after Table4;B.5; compliant response paths fixed; narrow preservation only
	R9	U	RQ2 Interface Safeguards, after Table4;B.5

The artifact directory `data/r42-defense-first/` retains the protocol, search returns, admission decisions, source-read scopes, cards and deterministic projection. This extension records source-reported results; new model and defense executions are documented separately.

S46. R8 Sensitivity and Supplemental Atlas

The table preserves the archived broad R8 interpretation alongside the outcome-specific functional-quality interpretation, source removal and partial evidence. Its members and source hashes are computed by `tools/r42_sensitivity.py`. These projections show how coding choices and source membership change this inventory. They do not test representativeness or add independent literature observations. The retained family-level Wilson calculations are binomial reference arithmetic, not sampling-based uncertainty estimates for defense literature; they cannot account for purposive selection, omitted venues or coding dependence. The separate original task-score inventory of zero in four families has a Wilson reference upper limit of 48.99% under that reference model. Neither the count nor that calculation establishes a field-wide absence. No main-text conclusion relies on these limits.

S46.1. Current family projection and original map

The current family projection uses outcome-specific R8. The landscape and mechanism maps retain the original C1–C5 frame, rather than pooling semantic/privacy extension cards. SICC's auxiliary form view and DP-MemView's separate internal-score diagnostic are reassessed in the defense-first extension; their old adjacent placement must not be read as a current blanket semantic-only judgment.

S46.2. External worked applications: retain, separate, remeasure

We apply the procedure to a fixed external frame: both defenses in Deep Fingerprinting, all three in WFDefProxy, and all nine in Mathews et al.'s reevaluation, with DynaFlow's live implementation kept as a separate record [97], [98], [99]. The resulting 15 measurement records represent 14 defense–study pairs in three studies. The ledger retains favorable outcomes alongside weaknesses, fixes the Standard setting for the nine-defense table, and records a locator and a concrete measurement decision for every row (S35).

The worked output separates decisions that a mechanism name conflates (Table S117). Stronger protected-data fitting requires revisiting WTF-PAD's attack result, while Walkie-Talkie's symmetric-pairing result remains a valid scoped outcome. FRONT's live protection improves even as its bandwidth cost changes. A record-level audit therefore preserves the security observation and replaces the cost estimate separately. DynaFlow's live result needs a separate execution/workload record, preventing a simulated cost from being attached to a live attack result.

The 11 protected-fit/view requirements cover B01–B02 and B06–B14; the four execution/cost requirements cover B03–B05 and B15. Favorable controls overlap both groups: symmetric Walkie-Talkie B02 and one-split TrafficSliver B11, and live FRONT B03 and Tamaraw B04. Each row produces a retain, separate or remeasure decision. These retrospective examples encode established WF practice; their visible outcomes make them worked applications rather than a prediction-accuracy evaluation.

S46.3. Internal case-study illustrations

Seven pre-specified judgments illustrate how changed receivers, attacks and source premises affect a comparison. Four have scoped counterexamples, one is partial, one has an unmet source premise and one remains unestablished. The main PDF's case-study appendix and S35 retain their conditions and outcomes. These examples illustrate the decision procedure; they do not estimate its accuracy.

The artifact separates *analysis reconstruction* from *new model collection*. Archived requests, responses, traces, calibration hashes and predictions permit the reported calculations to be reconstructed. Fresh calls to mutable Sol/Luna/MiniMax/GLM aliases, with documented unequal budgets, cannot reproduce an identical generator. Those archived alias-based examples make no fixed-weight replication claim; the separate pinned-artifact E4 study is described in Section S47. Detailed protocols and model identities for the earlier examples remain explicit in S14–S32.

TABLE S111. R8 INTERPRETATION AND SOURCE-REMOVAL SENSITIVITY IN THE SAME ATTACK-EVALUATED SOURCE FRAME. EACH PANEL HAS THE SAME SIX INVENTORY VIEWS. A USES ARCHIVED BROAD UTILITY; B USES OUTCOME-SPECIFIC FUNCTIONAL QUALITY AND RETAINS WRA'S REAL-ANSWER PRESERVATION EXEMPTION.

Panel A: R8 family aggregation. Entries: bound/applicable [Wilson reference limits in percent].

Scope	Inventory	Families/records	B	B+P
A	All source families	19/31	3/16 [6.6, 43.0]	5/16 [14.2, 55.6]
A	Without CachePrune	18/29	2/15 [3.7, 37.9]	4/15 [10.9, 52.0]
A	Without final-search records	15/25	1/13 [1.4, 33.3]	3/13 [8.2, 50.3]
B	All source families	19/31	3/8 [13.7, 69.4]	4/8 [21.5, 78.5]
B	Without CachePrune	18/29	2/7 [8.2, 64.1]	3/7 [15.8, 75.0]
B	Without final-search records	15/25	1/5 [3.6, 62.4]	2/5 [11.8, 76.9]

Panel B: record-level joint queries. Entries: B/applicable [numerator accepting P]; no record-level binomial interval.

Scope	Inventory	Inference+ quality	Inference+ cost	Adaptation+ inference	Inference+ quality+cost	All five
A	All source families	3/20 [5]	5/21 [11]	2/15 [2]	2/19 [2]	0/13 [0]
A	Without CachePrune	1/18 [3]	3/19 [9]	2/15 [2]	0/17 [0]	0/13 [0]
A	Without final-search records	0/16 [2]	3/16 [7]	2/13 [2]	0/15 [0]	0/11 [0]
B	All source families	3/8 [4]	5/21 [11]	2/15 [2]	2/8 [2]	0/5 [0]
B	Without CachePrune	1/6 [2]	3/19 [9]	2/15 [2]	0/6 [0]	0/5 [0]
B	Without final-search records	0/4 [1]	3/16 [7]	2/13 [2]	0/4 [0]	0/3 [0]

Wilson limits are descriptive binomial references after aggregation by family. This purposive finite inventory is not a probability sample; the limits do not estimate population prevalence or coding reliability. Joint-query records can share a family and configuration, so they receive no binomial intervals. In Panel B, quality means the broader archived R8 outcome in A and the narrower functional-quality outcome in B; adaptation requires R4/R5/R7, and all five requires R4/R5/R7/R8/R9. Source-removal occurs before aggregation. Member IDs and source hashes are released with the table.

TABLE S112. MITIGATION-SOURCE BINDINGS IN THE ATTACK-EVALUATION FRAME. R8 EXEMPTS SCOPED PRESERVATION; COLUMNS CAN REFER TO DIFFERENT RECORDS.

Family	R1	R2	R3	R4	R5	R6	R7	R8	R9
UI Traces [57]	**B**	**P**	**B**	**B**	**B**	**B**	**B**	–	–
WRA [40]	**B**	**P**	**B**	–	–	**P**	**B**	–	U
NetEcho [50]	**B**	**B**	**B**	**B**	?	**B**	**B**	–	**P**
Early Bird [47]	**B**	**B**	**B**	**B**	?	**B**	**B**	U	**B**
Whisper Leak [58]	**B**	**B**	**B**	**P**	U	**B**	**B**	–	**P**
SpliceLeak [54]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
Time Will Tell [56]	**B**	**B**	**B**	**P**	U	**B**	**B**	U	**B**
I Know What You Said [48]	**B**	**B**	**B**	–	–	**P**	–	–	**B**
Speculative decoding [53]	**B**	**B**	**P**	**P**	U	**B**	**B**	–	**B**
Dangol workflow [23]	**B**	**B**	**B**	**B**	**B**	**B**	**B**	–	**B**
Cloud traffic benchmark [35]	**B**	**P**	**B**	U	U	**P**	**B**	**P**	U
Remote timing [3]	**B**	**B**	**P**	–	–	**B**	–	–	**B**
TDXRay [55]	**B**	**B**	**B**	–	–	**B**	–	**B**	**B**
SafeKV [52]	**B**	**B**	**B**	–	–	**B**	**B**	–	**P**
KVGov [49]	**B**	**B**	**B**	–	–	**B**	**B**	–	**B**
CachePrune [46]	**B**	**B**	**B**	–	–	**B**	**B**	**B**	**B**
OTRO [51]	**B**	**B**	**B**	–	–	**B**	**B**	–	**B**
SparLeak [37]	**B**	**B**	**B**	**P**	U	**B**	**B**	**B**	U
Shape and Substance [36]	**B**	**B**	**P**	U	U	**B**	**P**	U	**P**
Bound	19	16	16	4	2	16	15	3	11
Applicable	19	19	19	10	10	19	16	8	18

B: bound; P: partial; U: unreported; ?: unverified; –: inapplicable to the named outcome. For R8, – supports only the stated functional object, not timeliness or task completion. Current record codes: the main PDF's reconstruction appendix; source rationales: S44.

TABLE S113. COMPLETE MAP OF THE 58 CORE STUDY FAMILIES. EACH FAMILY APPEARS ONCE; LAYER AND TYPE LABELS MAY BE MULTIPLE AND DO NOT IMPLY ONE-TO-ONE CAUSAL PATHS. THE BLOCKS GROUP SOURCE LAYERS, NOT EVIDENTIAL QUALITY.

Core family	$G \to O$	Type
hardware generation, P1 (8)		
Edge EM leakage [65]	$1 \to 1$	I
Energon [10]	$1 \to 1$	II
MoEcho [6]	$1 \to 1$	I
I Know What You Said [48]	$1 \to 1$	I
Behind Bars [60]	$1 \to 1$	I
PAGE404 [70]	$1 \to 1$	I
Spill the Beans [74]	$1 \to 1$	I
ArchSnoop [11]	$1 \to 1$	II
hardware and service, P1/P2 (8)		
SparLeak [37]	$1,2 \to 1$	I
Shape and Substance [36]	$1,2 \to 1$	I
TDXRay [55]	$1,2 \to 1$	I
SparSEEty [73]	$1,2 \to 1$	I
OTRO [51]	$1,2 \to 1$	I
Detokenization Leaks [5]	$1,2 \to 1$	I
LLM-WFIN [69]	$1,2 \to 1,3$	I
LLM-serving RCA [68]	$1,2 \to 2$	IV
service generation, P2 (first 13 of 22)		
SplitDecoy [75]	$2 \to 2$	I
KVGov [49]	$2 \to 2,4$	I
LLMs Have Rhythm [9]	$2 \to 3$	II
Remote timing [3]	$2 \to 3$	I
NetEcho [50]	$2 \to 3$	I
Prompt Echo [34]	$2 \to 3$	I
Speculative decoding [53]	$2 \to 3$	I
Remote keylogging [32]	$2 \to 3$	I
Whisper Leak [58]	$2 \to 3$	I
Cache contention [2]	$2 \to 4$	I
Routing metadata [72]	$2 \to 4$	I
SafeKV [52]	$2 \to 4$	I
ML privacy channels [27]	$2 \to 4$	I

Core family	$G \to O$	Type
service generation, P2 (remaining 9 of 22)		
Early Bird [47]	$2 \to 4$	I
Prompt-cache audit [33]	$2 \to 4$	I
InputSnatch [38]	$2 \to 4$	I
Leaky Language Models [66]	$2 \to 4$	I
PromptPeek [71]	$2 \to 4$	I
SpliceLeak [54]	$2 \to 4$	I
Time Will Tell [56]	$2 \to 4$	I
CachePrune [46]	$2 \to 4$	I
MoE prompt stealing [7]	$2 \to 4$	I
cross-layer sources (5)		
ClawGuard [12]	$1,5 \to 1$	IV
MIRAGE [1]	$2,5 \to 2$	IV
Cloud traffic benchmark [35]	$2,5,8 \to 3$	I, II
Web-agent unmasking [76]	$3,7 \to 3,7$	II, III
LumiMAS [14]	$4,5 \to 4,6$	IV
orchestration/workflow sources (15)		
AgentPrint [31]	$5 \to 3$	I, II
Cycle detection [13]	$5 \to 4,6$	IV
SIGLEAK [62]	$5,6 \to 4,6$	I
ProcGrep [8]	$5,6 \to 4,6$	II
AgentProv [63]	$5,6 \to 6$	II
Lidar [67]	$5,6 \to 6$	II
Dangol workflow [23]	$5,6,8 \to 3,6,8$	I
Campaign attribution [64]	$5,6,8 \to 6,8$	II
WRA [40]	$5,7 \to 3$	I
PersonaFingerprint [4]	$5,7 \to 3$	I
Whose Agent? [77]	$5,7 \to 7$	II
AgentTell [59]	$5,7 \to 7$	I
UI Traces [57]	$5,7 \to 7$	II
FP-Agent [61]	$5,7 \to 7$	II
AHB [45]	$5,7 \to 7$	III

G denotes signal-generation layers and O the receiving observation layers. Layer indices: 1 hardware; 2 serving/runtime; 3 transport; 4 application/API; 5 orchestration; 6 tools; 7 browser/environment; 8 multi-agent communication. Types: I confidentiality/privacy; II attribution; III human–agent distinction/evasion; IV authorized diagnostics. These are objectives and observer roles, not mutually exclusive security guarantees. Privileged operator receivers and physical receivers are marked by the layer they inspect. Source versions, locators and normalization of prose-only layer descriptions are in the artifact.

TABLE S114. PRIVACY AND ATTRIBUTION INTERVENTIONS INDEXED BY MECHANISM AND SIGNAL-GENERATION LAYER. A CELL NAMES FAMILIES WITH A QUALIFYING MITIGATION RECORD IN THE C1–C5 ATTACK-EVALUATED SOURCE FRAME; MULTIPLE MECHANISMS MAY OCCUR IN ONE FAMILY. LETTERS DENOTE THE EVIDENCE PRESENT IN THE INDEXED RECORD, NOT A UNIVERSAL SECURITY GUARANTEE: E = EMPIRICAL PROTECTED-VIEW OR SIMULATED ATTACK OUTCOME; C = COMPONENT COST OR PROPOSAL/ESTIMATE WITHOUT A BOUND ATTACK OUTCOME IN THAT INDEXED RECORD. THE TABLE DOES NOT ADJUDICATE WHETHER EACH SOURCE ALSO PROVES A FORMAL GUARANTEE.

Mechanism	P1 only	P1/P2	P2 only	Workflow	Other cross-layer
Traffic framing / rate	–	–	NetEcho, Whisper, SPEC, Remote, Time [50], [58], [53], [3], [56] (E/C)	Dangol [23] (E)	–
Cache isolation, policy or timing	–	–	Early Bird, SpliceLeak, SafeKV, KV-Gov, CachePrune [47], [54], [52], [49], [46] (E/C)	–	–
Memory access / isolation	Gao [48] (C)	TDXRay, OTRO [55], [51] (C/E)	–	–	–
Model computation or output shaping	–	SparLeak, Shape [37], [36] (E/C)	–	–	–
Decoys, timing edits, or input changes	–	–	–	WRA, UI Traces [40], [57] (E)	Cloud benchmark [35] (E)

Cells concern the mapped generating layer, not necessarily the intervention layer. Residual observations are receiver-specific: grouped or re-emitted traffic can remain after framing; timing can remain after a cache-policy change; access counts can remain after address hiding; and action types/order can remain after timestamp edits. A dash means no matching indexed record in this collection, not that the mechanism cannot apply or that no external defense exists. Type III evasion and Type IV authorized diagnostics are boundary cases, not privacy-protection results.

TABLE S115. SEVEN PRIVACY AND ATTRIBUTION INTERVENTION CATEGORIES ORGANIZED BY RECEIVING VIEW AND SUPPORTED DEPLOYMENT JUDGMENT. TYPE I PROTECTS CONFIDENTIALITY; TYPE II ADDRESSES PROVENANCE OR IDENTITY ATTRIBUTION.

Intervention & polarity	Evaluated view, target and source	Reported evidence and applicable boundary
Padding / batching (Type I: confidentiality)	Network packet lengths and groups; topic inference. NetEcho and Whisper Leak [50], [58].	Collected padding traces retain inference signal; packet-group batching is evaluated by replay. A protected upstream length view does not cover downstream re-emission or residual grouping. Live incremental cost remains U.
Constant-rate / transport cover (Type I: confidentiality)	Network workflow fingerprint; Dangol's OHTTP testbed [23].	The co-located loopback setting yields 0.30 recovery at about $13\times$ bytes and $11\times$ latency. Transformed-capture results are separate; neither result supplies objective agent-task utility.
Concurrent query decoys (Type I: confidentiality)	Destination/domain observer; private prompt and trait inference. WRA [40].	Real/decoy research prompts execute in parallel as read-only queries whose decoy answers are discarded locally; interleaved-domain evaluation reduces one-decoy E_{func} from 0.770 to 0.610. Real output is retained; task completion under decoy load and execution cost remain unbound.
Cache isolation / keying (Type I: confidentiality)	Co-tenant prefix timing; SafeKV, KVGov and CachePrune [52], [49], [46].	SafeKV reports post-trigger AUC near 0.5 and separate TTFT/throughput comparisons. KVGov's HMAC median $1.3\,\mu$s prices key computation, excluding lost reuse. CachePrune associates marked-token protection and serving outcomes with a described default configuration; sample-level pairing is not established, and contextual reconstruction remains.
Cache timing equalization (Type I: confidentiality)	Co-tenant cache-fusion timing; SpliceDefense [54].	CTBF reports nonzero signed residuals in selected local tests; indistinguishability and adaptive recovery remain unverified. The 15-task/75-step scope needs reconciliation. QCP's < 5ms/$< 1\%$ prefill report is a separate cost object, not combined deployment cost.
Timestamp perturbation (Type II: attribution)	Recorded website action metadata; model/agent attribution. UI Traces [57].	Fitting on delayed traces (D_2) recovers substantial attribution performance under offline timestamp edits that retain action types/order. Live execution delay and task outcomes are not measured by the trace transformation.
Oblivious / unshared memory (Type I: confidentiality)	Host or local resource observations; TDXRay, Gao et al. and OTRO [55], [48], [51].	TDXRay measures oblivious-map lookup cost; Gao measures loading/memory cost after disabling zero-copy. OTRO v1 states an equal-access-count DRAM-trace guarantee. Closed-world leakage, unpadded TTFT and epoch-padding estimates are separate objects; textual cost discrepancies and padding-inclusive deployment remain unresolved.

Authorized operator diagnostics (Type IV: ClawGuard, cycle detection, LumiMAS, LLM-serving RCA and MIRAGE) are baseline telemetry rather than privacy attacks or evasion defenses, and are cataloged separately.

TABLE S116. FILLED REPORTING CARDS FROM SOURCE RESULTS. EARLY BIRD IS VENUE-LINKED (INSPECTED AUTHOR VERSION); TDXRAY IS AN S&P PAPER; CACHEPRUNE IS A CITED PREPRINT AND SERVES AS THE RUNNING CONFIGURATION EXAMPLE. NO NEW BENCHMARK IS IMPLIED.

Field	Early Bird [47, V-A–B]	TDXRay [55, 6.1, Fig. 8, Table 3]	CachePrune [46, 5.1–5.6]
Receiver/target	Timing oracle; sensitive prefix recovery.	Memory-access view; oblivious lookup defense component.	Cache-reuse probes; direct and contextual token recovery.
Intervention	Prefix-granularity oracle; semantic-cache prototype separately.	Oblivious mapping microbenchmark.	Selective cache sharing at default configuration.
Attacker/test	Granularity-specific calibration; matched adaptation unresolved.	Component cost, no bound protected end-to-end inference row.	Recovery evaluation; default quality/cost package; matched training-fit contrast inapplicable; broader adaptive attacks unassessed.
Function	Exact-prefix content preserved; serving/task outcomes unmeasured.	Vocabulary coverage measured; full-inference task outcome unbound.	QA/summarization quality at stated configuration.
Cost	Semantic-cache overhead belongs to a different primitive.	Lookup microcost; not complete serving cost.	Serving throughput/cost; preserve default package.
Extension/action	Retain prefix candidate; measure its own serving cost.	Retain component cost; measure protected service and its function.	Retain scoped package; test changed workload/hardware.

TABLE S117. EXTERNAL APPLICATIONS: RETAIN A SCOPED RESULT, SEPARATE AN OUTCOME PACKAGE, OR REQUEST A SPECIFIC MEASUREMENT. FULL 15-RECORD LEDGER: S35. ACCURACY AND TPR RETAIN THEIR ORIGINAL POPULATIONS.

Published contrast	Evidence-reuse decision
B01 WTF-PAD [97]	Refit: DF accuracy 90.7%; prior attackers in the same table reach at most 69.0%.
B02 Walkie-Talkie [97]	Retain symmetric-pairing scope: DF accuracy 49.7%; do not inherit it for asymmetric pairing.
B03 FRONT [98]	Separate outcomes: live DF TPR improves, 72.06% to 42.79%, while data overhead rises, 52% to 75%.
B04 Tamaraw [98]	Retain low live TPR, 6.43%; use live bandwidth overhead 179%, rather than simulated 207/224%.
B15 DynaFlow [99]	Separate live record: 379% mean per-site time overhead; changed URLs/crawler prevent a matched simulation–live effect.

S47. Fixed-Weight Replay and Online State-Boundary Results

This section gives the measurements behind the paper's receiver, state and attacker/workload boundaries. The fixed-weight replay and the online planner stress test are separate studies. The former measures a passive receiver on saved model output sent through a timed TLS loopback; the latter invokes a served planner and simulated user in a native retail sandbox. The sandbox study does not measure WRA's privacy protection.

S47.1. Local model generation and timed TLS replay

Three pinned local weight artifacts generated five templates for each of 100 synthetic topics (1,500 outputs), with a 128-token output budget. The primary binary target is medical versus nonmedical category, with 50 topics per class. Each of five test folds contains ten topics of each class and all their templates; chance BA is 50%. The secondary task predicts 100 known topic labels with one template held out per fold; chance BA is 1%. This second task is not new-topic generalization. Saved native chunks are replayed through timed TLS loopback. Attackers observe response application-record lengths and timing; content, requests and handshakes are excluded. No cloud-service, WAN, real-user-topic or online defended-generation outcome is measured.

For each defended test view, D0 learns on clean training data and D2 on defended training data. The matched-three comparison separately takes the highest OOF BA among logistic regression, random forest and gradient boosting on the same five folds used for reporting. This nonnested winner selection can be optimistic. Full12 (primary) and Full9 (secondary) add the declared attacks; their suite maxima are exploratory, not unbiased performance estimates. Bootstrap intervals resample the finite topic inventory and recompute maxima over saved predictions, without refitting attacks or recalibrating the defense. This does not remove winner-selection bias.

The Qwen 8B base-origin raw matched-three reference is 79.0% BA. At local batch point 1 (four native chunks per batch), matched D0 and D2 are 50.0% and 73.2%. These three values use the same prespecified learner classes with separate winner selection in each condition, not a paired effect estimate or interval for one fixed classifier. Full12 is 73.4% and must not be substituted for the matched D2 value. D0 logistic regression has AUROC 0.678272 despite BA 50.0%, so the thresholded chance result does not show zero information. On the same 500 replays over 100 topics, response application TLS bytes are 470,239 versus 1,234,823 paired raw bytes ($0.380815\times$). Paired session completion differences have median $+0.1524$ ms and 95th percentile $+2.3814$ ms; original content-chunk endpoint delays have median $+27.5909$ ms and 95th percentile $+65.7301$ ms. Aggregation reduces framing/record overhead; the ratio is not total request traffic. The full costs for both 81-row grids are printed in §S51 with their own raw origins.

At the default source comparisons, all 45 source–target–fold combinations share the same 32-byte bucket quantum, four-chunk batching parameter, or 32-byte cell/50-ms tick parameters. Batching is source-independent by design; source-training quantiles for buckets and cells round to identical values. Their zero imported-minus-local differences are identical-parameter controls and provide no evidence of calibration transfer. Some primary hard-cap source/fold comparisons differ, but Qwen 8B and Qwen 0.6B both use 87 events in fold IDs 1 and 3 and differ only in the other three folds; Llama 1B uses 128 in every fold. The six primary contrasts are listed in Table S120. Intervals condition on fitted attackers and omit recalibration/refitting uncertainty. Secondary fixed-observed-winner sign-flip tests assume sign symmetry or exchangeability and do not correct winner selection. Holm adjustment covers two imported-source comparisons per target only; it is not global across targets, controls and points. The reported analysis lacks a local fitting-variability control to distinguish a source effect from finite-training-sample calibration variation.

The original 35,000-session queue stopped with 34,440 completed sessions and one started session without a saved outcome. That outcome remains unknown. A separate continuation collected the 559 original jobs that had not started and a complete 500-session repetition of the affected Qwen 8B constant-rate arm. The repeat was selected after the interruption. It uses separate session identities and a new local TLS runtime/batch; the collector still calls the original local TLS implementation. Four overlapping continuation jobs are retained separately. Of 600 planned fold evaluations, ten affected folds use the whole repeated arm; the others use original completed captures or independently collected never-started jobs. Reconstruction of 35,000 routed captures and 600 evaluations does not make the original run complete or retrain the classifiers.

For the 495 old/new matched sessions, response bytes and application-record counts agree in every pair. The median across topic means of receiver completion differences is -0.461 ms; their empirical 5th–95th percentile range is $[-1.398, 0.185]$ ms. This describes observed timing differences, not a confidence interval, exchangeability test or proof that batch effects vanished. The proxy-to-direct path and concurrency changes in the following online study belong to R44, not E4. All original outcomes, repeats and continuation identities remain preserved.

S47.2. Online planner: nonrandom replacement cohort

The original served GLM run stopped after 347 of 900 planned episodes because of transport failures. At interruption, 113 task–repeat blocks had all three arms; the other 187 blocks were selected using terminal-file existence, not measured utility. The complete blocks and all eight additional GLM terminal episodes remain in the artifact. For each of the 187 incomplete blocks, a new `qwen3.8-flash-next` planner and simulated user were run in all three arms on independently cloned states: no cover, one extra read, and one same-ID item write. This gives 561 fresh episodes from 98 tasks and 46 customers. The served model name is an alias; fixed model weights are not claimed. The interrupted selection is nonrandom, and comparisons are within this model and cohort; no arm is paired across model services.

The read or write is triggered at most once immediately after the first planner-originated order-details lookup, and its reply is not delivered to the planner. A valid candidate is chosen from the initial fixture without reading the task's reference actions. Of 187 episodes per cover arm, the extra tool executed in 97 and was not exposed in 90. The write arm recorded 97 write attempts and 97 state effects, each consuming the native pending-item modification eligibility. All unexposed episodes remain in the denominator. The write is a controlled state/utility stress condition, not a claim of a deployed privacy defense or discovered authorization bypass.

The generated table reports raw success counts out of 187 separately from task-weighted means: 29 tasks contribute one selected repetition, 49 contribute two and 20 contribute three. For each contrast, selected repetitions are first averaged within task, and the 98 task means are then equally weighted. The predeclared task-cluster bootstrap and sign-flip use 20,000 draws; Holm covers the two displayed paired tests. A customer-cluster sensitivity retains tasks within each of 46 customers. Its 95% intervals for read minus no cover and write minus no cover are $[-3.65, 4.74]$ and $[-61.39, -32.04]$ percentage points, respectively; the corresponding Holm-adjusted sign-flip values are 0.8211 and 9.9995×10^{-5}. The intervals describe this nonrandom cohort; the sign-flip tests additionally require sign symmetry or exchangeability. They do not establish equivalence of read and no cover.

The verifier reconstructed all 561 saved conversations and 4,176 tool events; 9,801 model decisions were checked against and decoded from their saved requests and provider receipts. No completed response was intentionally resampled. Across no-cover, read-cover and write-cover arms respectively, real planner tool calls were 1,319, 1,323 and 1,340, with 61, 52 and 90 recorded real-tool failures. Cover-tool failures were zero. Model request attempts were 3,359, 3,353 and 3,206, including 35, 34 and 38 recorded infrastructure failures. The 10 interrupted requests without a locally saved response or error are preserved separately: their server-side completion and waiting times are unknown and are not included in recorded transport-time totals. Summed per-call waiting times overlap under concurrency, and cached reconstruction can affect final-visit wall time; neither is a complete end-to-end deployment cost.

The frozen inference request used a 16,384-token output budget, 600-second timeout and default certificate verification. During collection, the target domain was moved from the system-proxy path to a direct connection, and the effective scheduling limit changed from ten to two concurrent workers. The request body and source/configuration fingerprints stayed

TABLE S118. POST-INTERRUPTION RECOVERY. FOUR CONTROLS, HELD-OUT-TOPIC BINARY. CELLS: FULL12 BA AND FROZEN PAIRED IMPORTED-MINUS-LOCAL PERCENTAGE-POINT DIFFERENCE. SOURCE ROWS/TARGET COLUMNS; LOCAL IDENTITY DIAGONAL. BUCKET, BATCH AND FIXED-CELL DEFAULTS HAVE IDENTICAL PARAMETERS ACROSS SOURCES; THEIR ZERO DIFFERENCES DO NOT TEST TRANSFER. COMPLETE ABSOLUTE/PAIRED INTERVALS AND ALL CONTRASTS RETAINED IN ACCOMPANYING TABLES/CSV/JSON. QWEN 8B FIXED-CELL POINT 2 USES A SEPARATELY REPEATED ARM; THE ORIGINAL SESSION IS MISSING.

Control	Source calibration	qwen3:8b	qwen3:0.6b	llama3.2:1b
Buckets	qwen3:8b	58.6 (local)	58.4 (+0.0)	60.4 (+0.0)
Buckets	qwen3:0.6b	58.6 (+0.0)	58.4 (local)	60.4 (+0.0)
Buckets	llama3.2:1b	58.6 (+0.0)	58.4 (+0.0)	60.4 (local)
Chunk batches	qwen3:8b	73.4 (local)	73.2 (+0.0)	71.8 (+0.0)
Chunk batches	qwen3:0.6b	73.4 (+0.0)	73.2 (local)	71.8 (+0.0)
Chunk batches	llama3.2:1b	73.4 (+0.0)	73.2 (+0.0)	71.8 (local)
Fixed cells*	qwen3:8b	57.2 (local)	58.4 (+0.0)	60.8 (+0.0)
Fixed cells*	qwen3:0.6b	57.2 (+0.0)	58.4 (local)	60.8 (+0.0)
Fixed cells*	llama3.2:1b	57.2 (+0.0)	58.4 (+0.0)	60.8 (local)
True hard cap	qwen3:8b	76.0 (local)	72.6 (+0.6)	69.4 (+0.2)
True hard cap	qwen3:0.6b	74.8 (-1.2)	72.0 (local)	70.2 (+1.0)
True hard cap	llama3.2:1b	79.4 (+3.4)	75.8 (+3.8)	69.2 (local)

TABLE S119. POST-INTERRUPTION RECOVERY. FOUR CONTROLS, KNOWN-TOPIC, HELD-OUT-TEMPLATE100WAY. CELLS: FULL9 BA AND FROZEN PAIRED IMPORTED-MINUS-LOCAL PERCENTAGE-POINT DIFFERENCE. SOURCE ROWS/TARGET COLUMNS; LOCAL IDENTITY DIAGONAL. BUCKET, BATCH AND FIXED-CELL DEFAULTS HAVE IDENTICAL PARAMETERS ACROSS SOURCES; THEIR ZERO DIFFERENCES DO NOT TEST TRANSFER. COMPLETE ABSOLUTE/PAIRED INTERVALS AND ALL CONTRASTS RETAINED IN ACCOMPANYING TABLES/CSV/JSON. QWEN 8B FIXED-CELL POINT 2 USES A SEPARATELY REPEATED ARM; THE ORIGINAL SESSION IS MISSING.

Control	Source calibration	qwen3:8b	qwen3:0.6b	llama3.2:1b
Buckets	qwen3:8b	2.4 (local)	1.8 (+0.0)	1.6 (+0.0)
Buckets	qwen3:0.6b	2.4 (+0.0)	1.8 (local)	1.6 (+0.0)
Buckets	llama3.2:1b	2.4 (+0.0)	1.8 (+0.0)	1.6 (local)
Chunk batches	qwen3:8b	14.0 (local)	9.4 (+0.0)	7.8 (+0.0)
Chunk batches	qwen3:0.6b	14.0 (+0.0)	9.4 (local)	7.8 (+0.0)
Chunk batches	llama3.2:1b	14.0 (+0.0)	9.4 (+0.0)	7.8 (local)
Fixed cells*	qwen3:8b	1.6 (local)	2.0 (+0.0)	1.8 (+0.0)
Fixed cells*	qwen3:0.6b	1.6 (+0.0)	2.0 (local)	1.8 (+0.0)
Fixed cells*	llama3.2:1b	1.6 (+0.0)	2.0 (+0.0)	1.8 (local)
True hard cap	qwen3:8b	68.4 (local)	50.4 (+1.0)	35.4 (+1.6)
True hard cap	qwen3:0.6b	66.8 (-1.6)	49.4 (local)	34.0 (+0.2)
True hard cap	llama3.2:1b	67.8 (-0.6)	46.8 (-2.6)	33.8 (local)

fixed. Both paths passed task-free interface probes, but this does not locate the cause of the earlier connection failures. These operational changes and the unknown waits limit cross-run latency and cost comparisons.

The 113 completed GLM blocks are retained as a separate, post-interruption exploratory description over 80 tasks. Their success counts were 83, 87 and 28 out of 113 for no cover, read cover and write cover. Equal-task-weighted paired differences were $+3.1$ percentage points for read and -45.6 for write, with task-cluster 95% intervals $[-6.5, 12.7]$ and $[-57.9, -32.7]$, respectively. These completed-before-interruption blocks are not a random sample, are not paired with Qwen arms, and are not an independent test of privacy protection. The eight additional terminal GLM episodes are preserved but do not form complete three-arm blocks.

The artifact retains the respective protocols, saved decisions, interruption and transport amendments, selection manifests, verification receipts, prediction outputs and result projections under `experiments/sp2027-r42-open-weights/additions/interrupted-recovery/`, `experiments/sp2027-r43-agent-utility/` and `experiments/sp2027-r44-qwen-agent-utility/`. The Qwen statistics and counts above are reconstructed from the verified 561-episode projection; the GLM description is reconstructed from the verified 347-episode partial receipt and 113-block projection. The archived records support separate reconstruction rather than substituting one model's observations for the other's.

TABLE S120. ALL SIX DEFAULT HARD-CAP IMPORTED-MINUS-LOCAL PRIMARY CONTRASTS. DIFFERENCES AND 95% TOPIC-BOOTSTRAP INTERVALS ARE IN BA PERCENTAGE POINTS, CONDITIONAL ON FITTED ATTACKERS. THE SECONDARY FIXED-WINNER SIGN-FLIP p_{H} ADJUSTS ONLY THE TWO IMPORTS PER TARGET; IT OMITS REFITTING AND WINNER-SELECTION UNCERTAINTY. NO LOCAL FITTING-VARIABILITY CONTROL IS REPORTED.

Imported source	Target	Difference	Conditional 95% interval	p_{H}
Qwen 0.6B	Qwen 8B	-1.2	$[-2.6, 1.2]$	0.2378
Llama 1B	Qwen 8B	$+3.4$	$[0.8, 5.4]$	0.0112
Qwen 8B	Qwen 0.6B	$+0.6$	$[-2.4, 1.8]$	0.8122
Llama 1B	Qwen 0.6B	$+3.8$	$[0.0, 6.0]$	0.1224
Qwen 8B	Llama 1B	$+0.2$	$[-3.0, 4.4]$	1.0000
Qwen 0.6B	Llama 1B	$+1.0$	$[-2.2, 4.8]$	1.0000

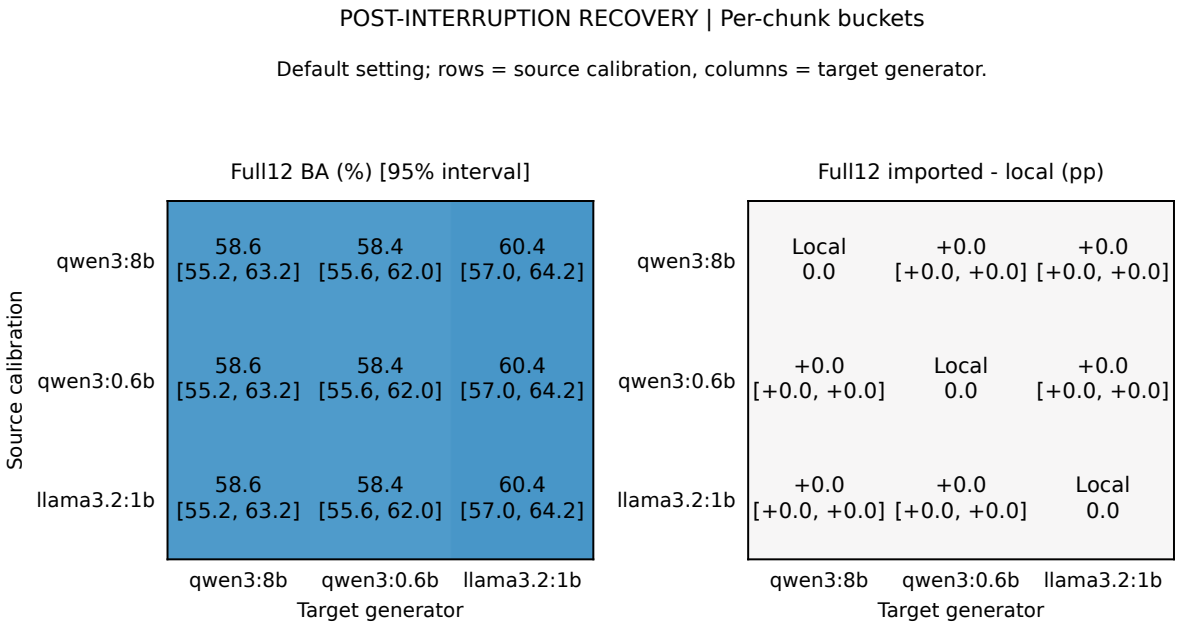


Figure 4. Archived default bucket matrix from the post-interruption replay projection. All source folds use the same 32-byte quantum: the zero differences compare identical parameters and do not measure transfer. The figure's historical calibration-transfer label denotes the comparison layout, not evidence of portability. Conditional intervals and repeated-arm provenance retain their stated limitations.

TABLE S121. R44 VERIFIED THREE-ARM UTILITY. EACH ARM HAS 187 EPISODES FROM 98 TASKS; THE SELECTED BLOCKS ARE NONRANDOM. TASK-WEIGHTED SUCCESS, PAIRED DIFFERENCE, AND INTERVALS WEIGHT EACH TASK EQUALLY AFTER AVERAGING ITS AVAILABLE REPEATS. THE PAIRED TESTS ASSUME TASK-LEVEL SIGN SYMMETRY; HOLM COVERS THE TWO SHOWN CONTRASTS. THIS IS A SANDBOX UTILITY STRESS TEST.

Condition	Successes /187	Task mean (%) [95%]	Difference (pp) [95%]	Holm p
No cover	154/187	82.7 [75.5, 89.3]	–	–
Read cover	156/187	83.3 [76.5, 89.6]	+0.7 [-4.9, +6.1]	0.859
Write cover	72/187	35.7 [26.7, 45.1]	-46.9 [-56.8, -36.9]	< 0.001

S48. Audit Trail and Current Record Reconstruction

The artifact preserves the 82-family map, 39-record ledger, 180 family cells and 351 record cells with source versions and locators. The archived query-membership and sensitivity checks are retained in S33. Case-study protocols, observations and uncertainty calculations retain their original cohorts and provenance.

S48.1. Result-record index

Table S122 indexes all 39 records by measurement mode and source locator. The numbered identities connect to the cell and pair index below. Marks distinguish six contextual controls (‡) and six final-search additions (†); all enter the pair inventory. The ledger retains full locators, versions and hashes.

S48.2. Current-code reconstruction

Table S123 defines the claim-conditioned items. Table S125 supplies the current record-level R1–R9 codes and question assignments. Its R8 column incorporates the preservation exemptions; the other cells retain their source coding. To reproduce a joint query, retain the 31 Type I/II mitigation records (exclude both AHB records and six contextual controls), exclude records with an inapplicable required item, and count records with B for every requested item. The required items are R7/R8 for inference/utility, R7/R9 for inference/cost, R4/R5/R7 for matched fitting/inference, R7/R8/R9 for the triple, and R4/R5/R7/R8/R9 for all five. Table S126 identifies each family inference/cost witness. S44 gives the source-linked rationale for the R8 exemptions; task scores for added execution remain separate.

Current selected-source counts and positive joins. The 31 mitigation records comprise 11 potentially quality-changing interventions, 18 scoped preservation witnesses and two standalone cost/calibration units. Eight families have a quality-applicable intervention: three B, one P and four U; eleven families are exempt. Thus explicit quality evidence is 3/8, or 4/8 accepting P. Table S128 retains the distinct applicability populations for inference, cost and fitting. The R5 family count of 2/10 concerns only clean-versus-protected fitting on the same defended test population: the other eight are six U and two ?. Neither state establishes that the source performed no adaptive evaluation.

Eight families have explicit inference and explicit cost somewhere in their evaluations. Under the declared record boundaries, five have no individual record carrying both: Early Bird (prefix policy versus semantic processing), SpliceLeak (background-load mitigation versus QCP), Time Will Tell (length instruction versus padding estimate), KVGov (partial flushing versus HMAC), and OTRO (padded versus unpadded ORAM). The separation is five of seven after removing CachePrune and four of six after removing final-search additions; accepting P gives three of 12. These are projections of the coding and split rules, not independent observations of deficient papers. The source objects, rather than the separation rate, determine whether a proposed join is justified.

Table S129 retains every positive joint-query record. SPEC, Dangol and CachePrune supply the five inference/cost records; CachePrune's two targets share one source-described default quality/cost configuration, without evidence in the inspected source that the protection and serving tables use identical requests or paired runs; SparLeak binds inference/quality by replacement intensity. Table S130 reports the current-R8 fit-applicability sensitivity: mapping fit-inapplicable cells to unknown changes the matched-fitting/inference query from 2/15 to 2/22, and the all-five query from 0/5 to 0/8. The historical broad-R8 table in Section S42 (Table S105) instead gives 0/13 and 0/19 for the all-five query; its quality-applicable population differs. Requiring explicit placement or a wire-level cost excludes SPEC's payload-cost record and changes inference/cost from 5/21 to 4/21. Removing CachePrune removes both measured-quality triple records; preservation-backed packages remain a separate functional route. These source-specific queries do not rank defenses or estimate a population rate.

Pair inventory and granularity. Table S127 covers all 133 within-question pair decisions. The 105 different-object pairs and eight of 15 within-family pairs reflect the declared split and grouping rules; a normal difference in research scope is not a source-paper defect. These pair counts are an index of the audit's decisions, not an independent validation of the distinctions used to create the records. Table S95 projects the unchanged judgments onto coarser units. Mixed edges retain all constituent statuses, and within-unit edges disappear from the index without being resolved. Family-only indexing yields 44 edges, eight mixed, and hides all 15 within-family pairs. Table S125 reconstructs the 133 labels; the pair CSV supplies the per-pair reasons and locators. Table S99 gives the archived measured-utility reporting-pattern sensitivity to removing CachePrune and final-search additions. The latter removes six records from four families; six contextual controls are excluded throughout. Removing both CachePrune and all final-search additions equals the latter row.

Case-study artifact index. The artifact index maps the case to experiment keys; S11–S26 provides the associated methods and results. These keys preserve original task/customer memberships; S30 retains the corrected permutation/Holm tests and calibration resampling.

S49. Mapped Privacy Families without a Linked Mitigation Record

Within the frozen corpus, the union of Type I and II contains 52 families; 19 have a scoped mitigation record. The remaining 33 families are enumerated below so that this coverage statement can be audited against the core map. This is a statement about the record-selection rules and inspected sources: several have contextual guardrails, noise, filters, decoy boundaries, or unbound claims. It does not assert that none of their papers or subsequent work evaluates any defense.

The Type I families without a linked qualifying mitigation record are Edge EM leakage, MoEcho, Behind Bars, PAGE404, Spill the Beans, SparSEEty, Detokenization Leaks, LLM-WFIN, SplitDecoy, Prompt Echo, remote keylogging, cache contention, routing metadata, ML privacy channels, prompt-cache audit, InputSnatch, Leaky Language Models, PromptPeek, MoE prompt stealing, AgentPrint, SIGLEAK, PersonaFingerprint, and AgentTell [65], [6], [60], [70], [74], [73], [5], [69], [75], [34], [32], [2], [72], [27], [33], [38], [66], [71], [7], [31], [62], [4], [59].

The Type II families without a linked qualifying mitigation record are Energon, ArchSnoop, LLMs Have Rhythm, Web-agent unmasking, AgentPrint, ProcGrep, AgentProv, Lidar, Campaign attribution, Whose Agent?, and FP-Agent [10], [11], [9], [76], [31], [8], [63], [67], [64], [77], [61]. AgentPrint belongs to both lists and is counted once in the union.

S50. Illustrative Case Study and Reconstruction Scope

The archived implementation examples instantiate distinctions used in the literature atlas: final receiver versus upstream trace, executed state versus discarded responses, and online planner utility versus reused trajectories. They are illustrative applications, not validation data for the comparison rules. The complete numerical grids, negative outcomes, intervals and corrected tests remain in S11–S32.

The earlier transport example has 48 saved responses on 24 topics (12 train/12 test). The public *retail* setting has 115 tasks from 53 customers; its frozen split contains 46 test tasks from 24 customers, including 5 cancellation-positive and 7 return-positive customers. The separate live planner example has 100 episodes: 37/50 baseline and 33/50 defended successes, a difference of -8.0 points with customer interval $[-23.9, 6.5]$. All eight calibration contrasts have Holm-adjusted $p = 1.000$; the original absolute protection screen remains unmet. Neither these unresolved contrasts nor resampling the same customers validates the framework.

Attacker-dependent recovery. On the same defended test outputs, buffered 1024-byte buckets give $D_0 = 50.0\%$ and $D_2 = 70.8\%$ accuracy (adapted accuracy interval 58.3–83.3%; S11). Constant-sized eight-token batches give 50.0% versus 100.0%, while a constant-event/count cap gives 50.0% for both (S12). The comparison fixes the outputs and defended view and changes fitting; it demonstrates configuration-specific recovery, without implying that adaptation defeats every defense. The held-out sample contains 24 outputs from 12 topics.

TABLE S122. COMPLETE INDEX OF THE 39 RESULT RECORDS (26 FAMILIES). BOLD NUMBERS 01–39 BESIDE EACH BOUNDARY LINK TO THE CELL AND PAIR INDEX IN SECTION S48. MODE IS THE PRIMARY MEASUREMENT MODE: EX EXECUTED COLLECTION/PROTOTYPE (20), TR TRACE TRANSFORMATION (6), SI SIMULATION/DERIVED ESTIMATE (5), CO COMPONENT/CALIBRATION MEASUREMENT (6), UN PLACEMENT UNRESOLVED (2). ‡: CONTEXTUAL RECORD (6); †: ADDED BY THE FINAL SEARCH (6). T=TABLE, F=FIGURE, APP.=APPENDIX; LOCATORS REFER TO THE CITED SOURCE.

Record ID / family	Mode	Boundary; source locator
UI-delay-offline [57]	Tr	**01**. Offline delay; no live task outcome. 6.1 F5; A.5
WRA-decoy-execution [40]	Ex	**02**. Executed decoys; usefulness proxy only. 8.1–8.3 T8–11
NET-upstream-local [50]	Ex	**03**. Protected deployment/local serving; topology scoped. IV-C–E TIII–IV
NET-dummy-padding-simulation [50]	Tr	**04**. Synthetic insert/pad; no live execution. VII TIX
EARLY-K-calibration-oracle [47]	Si	**05**. K-specific fit then oracle; not live KV trial. V-A TVI; IV-A
EARLY-semantic-deidentification [47]	Ex	**06**. GPTCache prototype cost; not KV-granularity cost. V-B
WHISPER-randompadding-capture [58]	Ex	**07**. New padded captures; rounded medians. 6.1 T3–4
WHISPER-batching-replay [58]	Tr	**08**. Offline batching; model-dependent result. 6.2 F7
WHISPER-injection-replay [58]	Tr	**09**. Synthetic packets; no live zero-delay claim. 6.3 T5
SPLICE-CTBF-signal [54]	Ex	**10**. Prototype background load; tested signal only. VII-C TII
SPLICE-QCP-cost [54]	Co	**11**. QCP component cost; excludes combined CTBF. VII-C TII
TIME-length-instruction [56]	Ex	**12**. Length instruction; unequal outputs remain. 7.2
TIME-padding-estimate [56]	Si	**13**. Analytic plaintext cost; not network bytes. 7.3
SPEC-padding-REST [53]	Un	**14**. Padding/aggregation; payload cost, placement unverified. 6.1 F6–7 T2
DANGOL-capture [23]	Tr	**15**. Capture transform; distinct from live population. 9.8–9.9 T6
DANGOL-livebinding [23]	Ex	**16**. Live loopback binding; do not join capture accuracy. 9.7, OHTTP paragraph after T5
SIG-common-guardrail‡ [62]	Ex	**17**. Common instruction; not a causal guardrail effect. 4.1–4.2; App. C
SAFE-post-trigger [52]	Ex	**18**. Post-trigger probes; detector cost not service TTFT. 7.2.4 F12; 7.3–7.4
AHB-online-executed [45]	Ex	**19**. Executed UI edits; task/state changes possible. 5.1.2 T1; E.1–E.3
AHB-offline [45]	Tr	**20**. Offline touch edits; no online task damage claim. T1 offline rows; E.1–E.3
KVGOV-partialflush-sim [49]	Si	**21**. Deterministic simulation; not live protected recovery. 5.1–5.3
KVGOV-hardware-gap [49]	Co	**22**. Hardware gap calibration; not protected ASR. 5.7–5.8
KVGOV-HMAC-cost [49]	Co	**23**. HMAC microcost; not cache reuse/task loss. 5.9
TRAFFIC-prompt-reformulation [35]	Ex	**24**. Collected prompt rewrites; task utility unmeasured. V-C1 F9a
TRAFFIC-temperature [35]	Ex	**25**. Collected decoding changes; no monotone benefit. V-C1 F9a
CARLINI-fixed-rate [3]	Un	**26**. Rate/replayed cost; per-token, not request delay. VII F11–12
GAO-zero-copy-off [48]	Co	**27**. Loader component costs; not end-to-end TTFT. 7.1
TDXRAY-oblivious-map [55]	Co	**28**. Lookup microbenchmark; not full inference cost. 6.1 F8 T3
AGENTTELL-fixed-instruction‡ [59]	Ex	**29**. Live local browser tasks; loaded/cold context contrast. 4.3.1; 5 Finding 1
FP-CLOUDFLARE‡ [61]	Ex	**30**. Live access control; not user confidentiality. 6.4 T5
BAR-co-running-noise‡ [60]	Ex	**31**. MIG noise test; not an LLM defense trial. 7.2–7.3 T8
RHYTHM-VPN‡ [9]	Ex	**32**. Live VPN shift; baseline-trained attacker only. VI-B; VII
DEB-filter-origin‡ [27]	Ex	**33**. Filter creates attack view; not mitigation of it. Threat model; 4.2
CACHEPRUNE-selective† [46]	Ex	**34**. Selective cache sharing; direct token recovery/quality. 3.1; 5.1–5.6; T1; App. A
CACHEPRUNE-context† [46]	Ex	**35**. Contextual guessing; distinct from direct recovery. 3.1; 5.1–5.6; T1; App. A
OTRO-unpadded† [51]	Co	**36**. ORAM TTFT/memory; access-count padding excluded. VII-C–E; TIII
OTRO-epochpad† [51]	Si	**37**. Epoch-padding estimate; not measured padded TTFT. VII-C–E; TIII
SPARLEAK-replacement† [37]	Ex	**38**. Sparse replacement/quality; attack cost is separate. 3.1; 4–7; T4
SHAPE-padding-estimate† [36]	Si	**39**. Constant-work proposal/estimate; no executed pair. IX-B1

TABLE S123. CLAIM-CONDITIONED REUSE RULES. EACH ROW NAMES THE REQUESTED INFERENCE AND THE EVIDENCE NEEDED TO SUPPORT IT.

Item	Reuse query → required binding (**B**)
R1 Target	Same protected interest → explicit hidden property, inference goal and authorization relation.
R2 Receiver	Same observer protection → permissions, released view and auxiliary inputs.
R3 Intervention	Same executed mechanism → placement/mode: stored data, wire emission or actions.
R4 Protected fit	Resistance to protected-data fitting → identified protected training/calibration and testing.
R5 Matched adaptation	Effect of adaptation → clean/protected fits tested on the same defended population.
R6 Workload	Changed workload/system → tasks, configuration and the relevant holdout axis.
R7 Inference	Quantified protection → metric, denominator and scoped defended evaluation condition.
R8 Quality	Changed generation/computation → matched quality; a preservation witness exempts only that scoped quality query.
R9 Cost	Resource-budget feasibility → units, baseline and cost object at that operating point.

B: bound; **P**: partial binding; U: unreported in inspected scope; A: source unavailable; ?: binding unverified; –: inapplicable. For R4/R5, – means that a training-fit contrast is inapplicable; it does not establish that non-learning adaptive attacks are irrelevant or absent. For task timeliness or completion, content preservation alone is insufficient. First determine applicability, then assign evidence status. Family **B** means at least one scoped record satisfies the item, not that every result does.

TABLE S124. MITIGATION-SOURCE BINDINGS IN THE ATTACK-EVALUATION FRAME. R8 EXEMPTS SCOPED PRESERVATION; COLUMNS CAN REFER TO DIFFERENT RECORDS.

Family	R1	R2	R3	R4	R5	R6	R7	R8	R9
UI Traces [57]	**B**	**P**	**B**	**B**	**B**	**B**	**B**	–	–
WRA [40]	**B**	**P**	**B**	–	–	**P**	**B**	–	U
NetEcho [50]	**B**	**B**	**B**	**B**	?	**B**	**B**	–	**P**
Early Bird [47]	**B**	**B**	**B**	**B**	?	**B**	**B**	U	**B**
Whisper Leak [58]	**B**	**B**	**B**	**P**	U	**B**	**B**	–	**P**
SpliceLeak [54]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
Time Will Tell [56]	**B**	**B**	**B**	**P**	U	**B**	**B**	U	**B**
I Know What You Said [48]	**B**	**B**	**B**	–	–	**P**	–	–	**B**
Speculative decoding [53]	**B**	**B**	**P**	**P**	U	**B**	**B**	–	**B**
Dangol workflow [23]	**B**	**B**	**B**	**B**	**B**	**B**	**B**	–	**B**
Cloud traffic benchmark [35]	**B**	**P**	**B**	U	U	**P**	**B**	**P**	U
Remote timing [3]	**B**	**B**	**P**	–	–	**B**	–	–	**B**
TDXRay [55]	**B**	**B**	**B**	–	–	**B**	–	**B**	**B**
SafeKV [52]	**B**	**B**	**B**	–	–	**B**	**B**	–	**P**
KVGov [49]	**B**	**B**	**B**	–	–	**B**	**B**	–	**B**
CachePrune [46]	**B**	**B**	**B**	–	–	**B**	**B**	**B**	**B**
OTRO [51]	**B**	**B**	**B**	–	–	**B**	**B**	–	**B**
SparLeak [37]	**B**	**B**	**B**	**P**	U	**B**	**B**	**B**	U
Shape and Substance [36]	**B**	**B**	**P**	U	U	**B**	**P**	U	**P**
Bound	19	16	16	4	2	16	15	3	11
Applicable	19	19	19	10	10	19	16	8	18

B: bound; P: partial; U: unreported; ?: unverified; –: inapplicable to the named outcome. For R8, – supports only the stated functional object, not timeliness or task completion. Current record codes: Section S48; source rationales: S44.

TABLE S125. COMPLETE CURRENT RECORD-CELL AND PAIR INDEX. R8 USES THE OUTCOME-SPECIFIC APPLICABILITY RULE; THE ORIGINAL 351 CELLS REMAIN ARCHIVED. NUMBERS FOLLOW SECTION S48.1, READING THE LEFT BLOCK (01–20) THEN THE RIGHT BLOCK (21–39). EACH NINE-CHARACTER STRING GIVES R1–R9 IN ORDER; B/P/U/A/?/– USE TABLE S123. THE SINGLE HYPHEN IN A STRING MEANS INAPPLICABLE. COHORTS T TRAFFIC, C CACHE, L LOCAL MEMORY, B BEHAVIORAL, G GENERATOR, F FIXED INSTRUCTION, A ACCESS GATE, N CONTENTION, V FILTER-CREATED VIEW.

No.	Group	R1–R9	No.	Group	R1–R9	No.	Group	R1–R9	No.	Group	R1–R9
01	B	BPBBBBB-	11	C	BBB-B-UB	21	C	BBB-BB-U	31	N	BBB-BBUU
02	T	BPB-PB-U	12	T	BBBPUBBUU	22	C	BBB-B--	32	G	BBBUUBBUU
03	T	BBBB?BB-U	13	T	BBP-B-B	23	C	BBB-B-B	33	V	BBB-BBUU
04	T	BBB??BB-P	14	T	BBPPUBB-B	24	G	BPBUUPBPU	34	C	BBB-BBBB
05	C	BBBB?BB-P	15	T	BBBBBBB-B	25	G	BPBUUPBUU	35	C	BBB-BBBB
06	C	BBB-B-UB	16	T	BBB??BB-B	26	T	BBP-B-B	36	L	BBB-B-B
07	T	BBBPUBB-U	17	F	BBB-BBPU	27	L	BBB-P-B	37	L	BBB-BB-P
08	T	BBBPUBB-U	18	C	BBB-BB-P	28	L	BBB-B-BB	38	L	BBBPUBBBU
09	T	BBBPUBB-P	19	B	BBBPUBBBU	29	F	BBB-BBPU	39	L	BBPUUBPUP
10	C	BBB-BBUU	20	B	BBBPUBBU-	30	A	BBB-BBBU			

R1 target, R2 receiver, R3 intervention, R4 protected fitting, R5 matched fitting, R6 workload, R7 inference, R8 utility, R9 cost; B bound, P partial, U unreported, A unavailable, ? unverified, - inapplicable.

Within each nonsingleton group, every unordered pair is included. The 17 scoped-match pairs are: 03/04, 03/07, 03/08, 03/09, 04/07, 04/08, 04/09, 05/34, 07/08, 07/09, 08/09, 15/16, 18/21, 19/20, 24/25, 24/32, 25/32. The 11 unresolved pairs are: 02/12, 02/14, 05/18, 05/21, 12/14, 17/29, 18/34, 18/35, 21/34, 21/35, 37/38. The remaining 105 within-group pairs have a different-object witness; the three singleton groups yield no pairs. Match refers to the specified object, not universal equality of costs, populations or contracts.

The eight different-object pairs within a family are: 05/06, 10/11, 12/13, 21/22, 21/23, 22/23, 34/35, 36/37. All 351 cells and 133 within-group labels are recoverable here; individual source rationales remain in Supplement S29 and S34.

TABLE S126. EVERY FAMILY WITH B IN BOTH INFERENCE AND COST COLUMNS OF TABLE S124. SEPARATE COLUMN WITNESSES DO NOT AUTOMATICALLY FORM ONE ROW-LEVEL RESULT. IDS EXPAND IN SECTION S48.1.

Family	Inference record(s)	Cost record(s)	Same-record inference + cost
[47]	EARLY-K-calibration-oracle	EARLY-semantic-deidentification	–
[54]	SPLICE-CTBF-signal	SPLICE-QCP-cost	–
[56]	TIME-length-instruction	TIME-padding-estimate	–
[53]	SPEC-padding-REST	SPEC-padding-REST	SPEC-padding-REST
[23]	DANGOL-capture; DANGOL-livebinding	DANGOL-capture; DANGOL-livebinding	DANGOL-capture; DANGOL-livebinding
[49]	KVGOV-partialflush-sim	KVGOV-HMAC-cost	–
[46]	CACHEPRUNE-selective; CACHEPRUNE-context	CACHEPRUNE-selective; CACHEPRUNE-context	CACHEPRUNE-selective; CACHEPRUNE-context
[51]	OTRO-epochpad	OTRO-unpadded	–

TABLE S127. THE 133 WITHIN-QUESTION INDEX LABELS; 608 OTHER PAIRS HAVE NO DECLARED COMPARISON. A DIFFERENT-OBJECT WITNESS LIMITS A SPECIFIED COMPARISON, NOT ALL COMPARISONS BETWEEN THE STUDIES.

Question	Scoped match	Different object	Unresolved
behavioral humanization	1	2	0
cache service	2	37	6
fixed instruction	0	0	1
generator fingerprint	3	0	0
local memory	0	14	1
traffic envelope	11	52	3
Total	17	105	11

TABLE S128. REPORTING IN THE C1–C5 ATTACK-EVALUATED SOURCE FRAME. R8 COUNTS ONLY FAMILIES WITH A POTENTIALLY CHANGED GENERATION, COMPUTATION OR COMPONENT OUTCOME; ELEVEN FAMILIES ARE EXEMPT FROM THAT NARROW QUALITY QUERY.

Requested evidence	B / applicable	B+P / applicable
Inference outcome (R7)	15/16	16/16
Cost (R9)	11/18	15/18
Functional quality (R8)	3/8	4/8
Protected fitting (R4)	4/10	8/10
Matched adaptation (R5)	2/10	2/10

TABLE S129. JOINT RESULTS IN THE SAME 31-RECORD SOURCE FRAME AFTER R8 APPLICABILITY IS RESOLVED. QUALITY QUERIES INCLUDE ONLY POTENTIALLY QUALITY-CHANGING RECORDS; A SCOPED PRESERVATION WITNESS SUPPLIES A SEPARATE FUNCTIONAL ROUTE.

Requested evidence	Bound records	Interpretation
Inference + measured quality (3/8)	CACHEPRUNE-context; CACHEPRUNE-selective; SPARLEAK-replacement	CachePrune shares one default package; SparLeak binds quality by intensity.
Inference + cost (5/21)	CACHEPRUNE-context; CACHEPRUNE-selective; DANGOL-capture; DANGOL-livebinding; SPEC-padding-REST	Payload, capture, loopback and serving costs remain distinct.
Adaptation + inference (2/15)	DANGOL-capture; UI-delay-offline	Matched clean/protected fitting on defended tests.
Inference + measured quality + cost (2/8)	CACHEPRUNE-context; CACHEPRUNE-selective	Two CachePrune outcomes, one configuration.
All five applicable checks (0/5)	None	No bound record for this scoped query.
Narrow preservation + inference/cost	DANGOL-capture; DANGOL-livebinding; SPEC-padding-REST	Fixed message/token content; no implied deadline or end-to-end task guarantee.

SPEC binds a payload-size ratio, not live wire cost; requiring explicit placement still leaves four of the 21 inference/cost-eligible records. A structural function witness and an empirical quality score are separate evidence types.

TABLE S130. FIT-SCOPE SENSITIVITY WITH THE CURRENT R8 QUALITY APPLICABILITY. BROADENING FIT-INAPPLICABLE CELLS GIVES UNKNOWN EVIDENCE.

Query	Fit scope	Broader unknown
Matched adaptation + inference	2/15	2/22
All five bindings	0/5	0/8
Inference + cost	5/21	4/21[†]

[†] Requires explicit R3 placement, or a wire-level cost rather than SPEC's payload ratio. R8 exemptions do not change this inference/cost query.

S51. Complete E4 Operating-Point Cost Accounting

Tables S131–S134 print all 162 rows of the saved operating-point projection: 81 for the topic-held-out binary study and 81 for the known-topic, template-held-out study. Each study has six raw-reference rows and 15 rows for each of five policies (three local points per target and two imported defaults per target). The two splits can fit different calibrations, so their cost grids are kept separate. Each row uses 500 native-output replay units from 100 topics, with five templates per topic; these are not 500 independent topics. These tables reformat saved results without new captures, attacker fits or statistical tests. Full-precision values, per-attacker scores and pairwise contrasts remain in the recovery-display CSV/JSON outputs.

Keys and provenance. Q8, Q0.6 and L1 denote qwen3:8b, qwen3:0.6b and llama3.2:1b. Each source/target pair names defense calibration and output generator. Origin B uses the original/continuation recovery mapping and its base raw reference; H uses separately collected hard-cap captures and that study's independent raw reference. A star marks the Q8 fixed-cell point-2 row routed to the separate post hoc 500-session arm. The original run remains incomplete. Row IDs P01–P81 and S01–S81 join the two cost panels within each study.

The zero-based point IDs preserve the machine output: BU uses bucket multipliers 0.5, 1 and 2 of the source-training frame-size $p95$, rounded upward to a 16-byte quantum; BA groups 2, 4 or 8 native chunks; FR emits cells on 20, 50 or 100 ms ticks with source-training $p95$ cell size rounded upward to 16 bytes. FR retains a variable stopping time. MP is minimum event-count padding at source-training quantiles 0.90, 0.95 and 0.99, retaining overflow; HC is the separate hard application-event cap, which is not a cap on TLS record fragments. Imports use point 1. Raw is the corresponding unshaped reference. Identical default BU/BA/FR source parameters yield no calibration-transfer test.

Measured quantities and units. Panel A gives summed response application TLS bytes B, the paired raw sum B_0, their ratio, receiver completion percentiles C_{50}, C_{95}, and percentiles $\Delta C_{50}, \Delta C_{95}$ of per-session completion differences against paired raw. Byte counts include TLS record headers and ciphertext overhead; request and handshake traffic is excluded. Completion is elapsed from the replay epoch immediately after the request carrier is received, with the saved schedule aligned to the first native chunk. It is not online model time-to-first-token. A difference of separately computed completion percentiles is not the paired-difference statistic. Negative differences can reflect timing noise across separate captures.

Panel B gives the receiver first-to-last-record span S_{95}, original content-chunk endpoint count N_e, endpoint-delay percentiles $\Delta R_{50}, \Delta R_{95}$ against the same byte endpoints in paired raw captures, planned queue-tail percentiles T_{50}, T_{95}, and summed sender blocking W. All times are milliseconds except W, which is seconds. Endpoint percentiles pool chunk endpoints, whereas completion percentiles use sessions. Planned tails are schedule quantities, not measured latency. Summed blocking is not end-to-end runtime. Decimal values are rounded for display; the machine CSV retains precision. Local replay timing does not establish online defended-generation, cloud-service or WAN cost, and unchanged decoded content does not establish online task utility.

TABLE S131: Complete primary E4 cost grid, panel A: response byte totals and receiver completion (ms). Keys, units and recovery limits are defined above; all rows retain their own raw reference.

ID	Origin	Source/target	Policy/point	B	B_0	B/B_0	C_{50}	C_{95}	ΔC_{50}	ΔC_{95}
P01	B	Q8/Q8	Raw/0	1234823	1234823	1.0000	1295.423	1829.320	0.000	0.000
P02	B	Q0.6/Q0.6	Raw/0	1137455	1137455	1.0000	320.535	615.328	0.000	0.000
P03	B	L1/L1	Raw/0	1817075	1817075	1.0000	692.232	908.777	0.000	0.000
P04	B	Q8/Q8	BU/0	1450536	1234823	1.1747	1293.777	1830.308	0.008	1.893
P05	B	Q8/Q8	BU/1	1804952	1234823	1.4617	1295.587	1829.407	0.023	2.269
P06	B	Q8/Q8	BU/2	2335160	1234823	1.8911	1295.075	1829.235	0.014	2.327
P07	B	Q8/Q8	BA/0	725059	1234823	0.5872	1294.610	1829.882	0.148	2.194
P08	B	Q8/Q8	BA/1	470239	1234823	0.3808	1294.175	1830.504	0.152	2.381
P09	B	Q8/Q8	BA/2	342922	1234823	0.2777	1295.306	1829.428	0.256	2.397
P10	B	Q8/Q8	FR/0	1942056	1234823	1.5727	1363.322	1903.405	87.626	211.617
P11	B	Q8/Q8	FR/1	1823040	1234823	1.4764	3250.907	4351.173	1957.286	2594.619
P12*	B	Q8/Q8	FR/2	1817694	1234823	1.4720	6500.899	8700.875	5181.888	6896.897
P13	B	Q8/Q8	MP/0	1476840	1234823	1.1960	1580.039	1926.689	337.938	578.010
P14	B	Q8/Q8	MP/1	1560571	1234823	1.2638	1693.060	2037.512	443.552	683.964
P15	B	Q8/Q8	MP/2	1693282	1234823	1.3713	1860.572	2199.910	620.893	860.122
P16	B	Q0.6/Q8	BU/1	1804952	1234823	1.4617	1295.587	1829.407	0.023	2.269
P17	B	Q0.6/Q8	BA/1	470239	1234823	0.3808	1294.175	1830.504	0.152	2.381
P18	B	Q0.6/Q8	FR/1	1823040	1234823	1.4764	3250.907	4351.173	1957.286	2594.619
P19	B	Q0.6/Q8	MP/1	1572072	1234823	1.2731	1722.010	2061.848	460.678	701.282
P20	B	L1/Q8	BU/1	1804952	1234823	1.4617	1295.587	1829.407	0.023	2.269
P21	B	L1/Q8	BA/1	470239	1234823	0.3808	1294.175	1830.504	0.152	2.381
P22	B	L1/Q8	FR/1	1823040	1234823	1.4764	3250.907	4351.173	1957.286	2594.619
P23	B	L1/Q8	MP/1	2191545	1234823	1.7748	2501.756	2839.471	1261.563	1501.588
P24	B	Q8/Q0.6	BU/1	1667306	1137455	1.4658	320.270	618.676	0.027	2.147
P25	B	Q8/Q0.6	BA/1	432236	1137455	0.3800	321.274	615.270	0.243	2.217
P26	B	Q8/Q0.6	FR/1	1679400	1137455	1.4765	2950.815	4351.674	2598.551	3916.426
P27	B	Q8/Q0.6	MP/1	1546562	1137455	1.3597	892.849	1218.958	578.112	943.546
P28	B	Q0.6/Q0.6	BU/0	1338538	1137455	1.1768	321.288	615.274	0.016	1.997
P29	B	Q0.6/Q0.6	BU/1	1667306	1137455	1.4658	320.270	618.676	0.027	2.147
P30	B	Q0.6/Q0.6	BU/2	2156730	1137455	1.8961	321.299	615.208	-0.013	2.206
P31	B	Q0.6/Q0.6	BA/0	667309	1137455	0.5867	320.641	615.441	0.086	2.179
P32	B	Q0.6/Q0.6	BA/1	432236	1137455	0.3800	321.274	615.270	0.243	2.217
P33	B	Q0.6/Q0.6	BA/2	314560	1137455	0.2765	321.129	616.094	0.357	2.991
P34	B	Q0.6/Q0.6	FR/0	1687770	1137455	1.4838	1180.883	1742.329	855.608	1321.031
P35	B	Q0.6/Q0.6	FR/1	1679400	1137455	1.4765	2950.815	4351.674	2598.551	3916.426
P36	B	Q0.6/Q0.6	FR/2	1678860	1137455	1.4760	5900.661	8702.107	5493.071	8328.051

Continued on next page

TABLE S131 continued

ID	Origin	Source/target	Policy/point	B	B_0	B/B_0	C_{50}	C_{95}	ΔC_{50}	ΔC_{95}
P37	B	Q0.6/Q0.6	MP/0	1483818	1137455	1.3045	806.204	1137.179	480.457	859.217
P38	B	Q0.6/Q0.6	MP/1	1558838	1137455	1.3705	908.853	1247.212	585.866	960.802
P39	B	Q0.6/Q0.6	MP/2	1803273	1137455	1.5854	1225.973	1565.247	919.783	1280.435
P40	B	L1/Q0.6	BU/1	1667306	1137455	1.4658	320.270	618.676	0.027	2.147
P41	B	L1/Q0.6	BA/1	432236	1137455	0.3800	321.274	615.270	0.243	2.217
P42	B	L1/Q0.6	FR/1	1679400	1137455	1.4765	2950.815	4351.674	2598.551	3916.426
P43	B	L1/Q0.6	MP/1	2173196	1137455	1.9106	1712.972	2043.229	1382.089	1760.152
P44	B	Q8/L1	BU/1	2675540	1817075	1.4724	692.325	908.943	-0.036	1.954
P45	B	Q8/L1	BA/1	677019	1817075	0.3726	693.477	909.000	0.269	2.474
P46	B	Q8/L1	FR/1	2687094	1817075	1.4788	4803.182	6401.016	4133.387	5495.639
P47	B	Q8/L1	MP/1	1854244	1817075	1.0205	742.145	911.919	0.327	280.523
P48	B	Q0.6/L1	BU/1	2675540	1817075	1.4724	692.325	908.943	-0.036	1.954
P49	B	Q0.6/L1	BA/1	677019	1817075	0.3726	693.477	909.000	0.269	2.474
P50	B	Q0.6/L1	FR/1	2687094	1817075	1.4788	4803.182	6401.016	4133.387	5495.639
P51	B	Q0.6/L1	MP/1	1857871	1817075	1.0225	748.050	924.939	0.519	300.033
P52	B	L1/L1	BU/0	2132516	1817075	1.1736	693.267	908.796	0.048	2.280
P53	B	L1/L1	BU/1	2675540	1817075	1.4724	692.325	908.943	-0.036	1.954
P54	B	L1/L1	BU/2	3463700	1817075	1.9062	693.183	908.590	-0.032	2.536
P55	B	L1/L1	BA/0	1056986	1817075	0.5817	693.753	908.383	0.135	2.534
P56	B	L1/L1	BA/1	677019	1817075	0.3726	693.477	909.000	0.269	2.474
P57	B	L1/L1	BA/2	487299	1817075	0.2682	693.224	909.089	0.332	2.642
P58	B	L1/L1	FR/0	2688768	1817075	1.4797	1922.339	2560.925	1244.967	1655.937
P59	B	L1/L1	FR/1	2687094	1817075	1.4788	4803.182	6401.016	4133.387	5495.639
P60	B	L1/L1	FR/2	2687040	1817075	1.4788	9603.091	12801.128	8982.591	11895.140
P61	B	L1/L1	MP/0	2248750	1817075	1.2376	1284.654	1601.880	599.147	1059.158
P62	B	L1/L1	MP/1	2274015	1817075	1.2515	1318.023	1637.724	639.696	1096.149
P63	B	L1/L1	MP/2	2274015	1817075	1.2515	1318.023	1637.724	639.696	1096.149
P64	H	Q8/Q8	Raw/0	1234823	1234823	1.0000	1294.373	1831.319	0.000	0.000
P65	H	Q0.6/Q0.6	Raw/0	1137455	1137455	1.0000	320.551	615.135	0.000	0.000
P66	H	L1/L1	Raw/0	1817075	1817075	1.0000	693.457	908.990	0.000	0.000
P67	H	Q8/Q8	HC/0	1466145	1234823	1.1873	1580.311	1928.678	336.041	579.044
P68	H	Q8/Q8	HC/1	1556045	1234823	1.2601	1692.538	2037.463	442.281	683.558
P69	H	Q8/Q8	HC/2	1692445	1234823	1.3706	1860.567	2201.509	620.735	860.399
P70	H	Q0.6/Q8	HC/1	1568445	1234823	1.2702	1721.794	2066.016	460.672	701.724
P71	H	L1/Q8	HC/1	2191545	1234823	1.7748	2502.074	2838.978	1261.416	1503.480
P72	H	Q8/Q0.6	HC/1	1537696	1137455	1.3519	894.208	1218.811	577.889	943.325
P73	H	Q0.6/Q0.6	HC/0	1469496	1137455	1.2919	806.650	1136.270	480.121	859.207
P74	H	Q0.6/Q0.6	HC/1	1550096	1137455	1.3628	906.981	1246.250	581.941	960.336
P75	H	Q0.6/Q0.6	HC/2	1801196	1137455	1.5835	1225.473	1567.093	919.592	1281.846
P76	H	L1/Q0.6	HC/1	2173196	1137455	1.9106	1712.330	2044.914	1383.144	1759.913
P77	H	Q8/L1	HC/1	1638515	1817075	0.9017	740.999	911.919	0.937	280.521
P78	H	Q0.6/L1	HC/1	1650915	1817075	0.9086	748.201	925.883	1.068	295.629
P79	H	L1/L1	HC/0	2246115	1817075	1.2361	1284.388	1601.052	598.696	1060.263
P80	H	L1/L1	HC/1	2274015	1817075	1.2515	1318.254	1638.009	638.519	1099.758
P81	H	L1/L1	HC/2	2274015	1817075	1.2515	1318.254	1638.009	638.519	1099.758

TABLE S132: Complete primary E4 cost grid, panel B: receiver span, chunk delivery, planned tail (ms), and sender blocking (s). Keys, units and recovery limits are defined above; all rows retain their own raw reference.

ID	S_{95}	N_e	ΔR_{50}	ΔR_{95}	T_{50}	T_{95}	W (s)
P01	1829.117	33138	0.000	0.000	0.000	0.000	9.115
P02	615.067	30589	0.000	0.000	0.000	0.000	7.739
P03	908.490	49260	0.000	0.000	0.000	0.000	12.500
P04	1829.981	33138	-0.011	2.118	0.000	0.000	8.879
P05	1828.941	33138	-0.008	2.166	0.000	0.000	8.888
P06	1829.118	33138	-0.018	2.039	0.000	0.000	8.983
P07	1786.747	33138	3.354	23.748	0.000	0.000	4.494
P08	1733.796	33138	27.591	65.730	0.000	0.000	2.453
P09	1651.834	33138	66.356	141.359	0.000	0.000	1.273
P10	1902.783	33138	38.348	126.055	87.589	211.466	9.771
P11	4350.777	33138	971.764	2052.567	1957.568	2594.453	9.122
P12*	8700.601	33138	2599.329	5430.918	5183.798	6898.355	10.646
P13	1926.345	33138	-0.008	2.121	340.000	580.000	11.093
P14	2037.094	33138	-0.017	2.209	440.000	681.000	11.788

Continued on next page

TABLE S132 continued

ID	S_{95}	N_e	ΔR_{50}	ΔR_{95}	T_{50}	T_{95}	W (s)
P15	2199.573	33138	-0.004	2.152	620.000	860.000	12.918
P16	1828.941	33138	-0.008	2.166	0.000	0.000	8.888
P17	1733.796	33138	27.591	65.730	0.000	0.000	2.453
P18	4350.777	33138	971.764	2052.567	1957.568	2594.453	9.122
P19	2061.589	33138	-0.014	2.282	460.000	700.000	12.083
P20	1828.941	33138	-0.008	2.166	0.000	0.000	8.888
P21	1733.796	33138	27.591	65.730	0.000	0.000	2.453
P22	4350.777	33138	971.764	2052.567	1957.568	2594.453	9.122
P23	2839.104	33138	-0.044	2.026	1260.000	1500.000	17.591
P24	618.232	30589	0.014	1.964	0.000	0.000	7.235
P25	576.648	30589	7.508	22.289	0.000	0.000	2.276
P26	4351.429	30589	1328.791	3097.641	2597.828	3915.933	8.327
P27	1218.567	30589	-0.002	1.963	580.000	941.000	10.710
P28	615.023	30589	0.014	1.956	0.000	0.000	7.631
P29	618.232	30589	0.014	1.964	0.000	0.000	7.235
P30	615.011	30589	-0.003	1.954	0.000	0.000	7.397
P31	589.996	30589	2.256	8.854	0.000	0.000	4.139
P32	576.648	30589	7.508	22.289	0.000	0.000	2.276
P33	553.723	30589	17.497	47.719	0.000	0.000	1.155
P34	1742.112	30589	425.312	1017.965	855.923	1321.829	8.411
P35	4351.429	30589	1328.791	3097.641	2597.828	3915.933	8.327
P36	8701.612	30589	2832.336	6569.634	5493.302	8327.522	8.415
P37	1136.720	30589	0.016	2.181	480.000	860.000	10.988
P38	1246.932	30589	0.020	2.047	580.000	960.000	10.924
P39	1564.854	30589	0.004	1.974	920.000	1280.000	13.085
P40	618.232	30589	0.014	1.964	0.000	0.000	7.235
P41	576.648	30589	7.508	22.289	0.000	0.000	2.276
P42	4351.429	30589	1328.791	3097.641	2597.828	3915.933	8.327
P43	2043.095	30589	0.006	1.925	1380.000	1760.000	16.387
P44	908.448	49260	0.006	2.092	0.000	0.000	12.582
P45	881.325	49260	10.607	23.258	0.000	0.000	3.409
P46	6400.732	49260	2097.878	4430.952	4130.402	5495.072	13.908
P47	911.535	49260	-0.015	2.012	0.000	280.000	13.905
P48	908.448	49260	0.006	2.092	0.000	0.000	12.582
P49	881.325	49260	10.607	23.258	0.000	0.000	3.409
P50	6400.732	49260	2097.878	4430.952	4130.402	5495.072	13.908
P51	924.638	49260	0.016	2.090	0.000	300.000	13.634
P52	908.417	49260	0.001	2.219	0.000	0.000	12.892
P53	908.448	49260	0.006	2.092	0.000	0.000	12.582
P54	908.183	49260	0.013	2.064	0.000	0.000	12.502
P55	896.956	49260	2.989	9.118	0.000	0.000	6.977
P56	881.325	49260	10.607	23.258	0.000	0.000	3.409
P57	854.283	49260	24.017	50.806	0.000	0.000	1.775
P58	2560.610	49260	626.775	1336.190	1244.206	1657.096	13.470
P59	6400.732	49260	2097.878	4430.952	4130.402	5495.072	13.908
P60	12800.843	49260	4550.378	9601.888	8982.094	11894.334	13.146
P61	1601.507	49260	0.013	2.009	600.000	1060.000	16.316
P62	1637.454	49260	-0.019	1.842	640.000	1100.000	16.607
P63	1637.454	49260	-0.019	1.842	640.000	1100.000	16.607
P64	1831.069	33138	0.000	0.000	0.000	0.000	8.452
P65	615.033	30589	0.000	0.000	0.000	0.000	6.988
P66	908.584	49260	0.000	0.000	0.000	0.000	11.744
P67	1928.371	33138	0.039	2.560	340.000	580.000	10.273
P68	2037.127	33138	-0.002	2.549	440.000	681.000	10.667
P69	2201.181	33138	-0.017	2.381	620.000	860.000	11.695
P70	2065.790	33138	0.015	2.543	460.000	700.000	10.927
P71	2838.726	33138	0.043	2.489	1260.000	1500.000	16.327
P72	1218.624	30589	-0.014	2.173	580.000	941.000	9.901
P73	1136.157	30589	0.004	2.552	480.000	860.000	9.423
P74	1246.067	30589	-0.004	2.252	580.000	960.000	10.028
P75	1566.726	30589	-0.018	2.088	920.000	1280.000	12.147
P76	2044.521	30589	-0.013	2.116	1380.000	1760.000	14.882
P77	911.612	49260	0.115	131.225	0.000	280.000	10.036
P78	925.665	49260	0.114	126.153	0.000	300.000	10.262
P79	1600.957	49260	-0.016	2.340	600.000	1060.000	14.890
P80	1637.710	49260	-0.016	2.267	640.000	1100.000	15.273
P81	1637.710	49260	-0.016	2.267	640.000	1100.000	15.273

TABLE S133: Complete secondary E4 cost grid, panel A: response byte totals and receiver completion (ms). Keys, units and recovery limits are defined above; all rows retain their own raw reference.

ID	Origin	Source/target	Policy/point	B	B_0	B/B_0	C_{50}	C_{95}	ΔC_{50}	ΔC_{95}
S01	B	Q8/Q8	Raw/0	1234823	1234823	1.0000	1295.423	1829.320	0.000	0.000
S02	B	Q0.6/Q0.6	Raw/0	1137455	1137455	1.0000	320.535	615.328	0.000	0.000
S03	B	L1/L1	Raw/0	1817075	1817075	1.0000	692.232	908.777	0.000	0.000
S04	B	Q8/Q8	BU/0	1450536	1234823	1.1747	1293.777	1830.308	0.008	1.893
S05	B	Q8/Q8	BU/1	1804952	1234823	1.4617	1295.587	1829.407	0.023	2.269
S06	B	Q8/Q8	BU/2	2335160	1234823	1.8911	1295.075	1829.235	0.014	2.327
S07	B	Q8/Q8	BA/0	725059	1234823	0.5872	1294.610	1829.882	0.148	2.194
S08	B	Q8/Q8	BA/1	470239	1234823	0.3808	1294.175	1830.504	0.152	2.381
S09	B	Q8/Q8	BA/2	342922	1234823	0.2777	1295.306	1829.428	0.256	2.397
S10	B	Q8/Q8	FR/0	1942056	1234823	1.5727	1363.322	1903.405	87.626	211.617
S11	B	Q8/Q8	FR/1	1823040	1234823	1.4764	3250.907	4351.173	1957.286	2594.619
S12*	B	Q8/Q8	FR/2	1817694	1234823	1.4720	6500.899	8700.875	5181.888	6896.897
S13	B	Q8/Q8	MP/0	1478359	1234823	1.1972	1599.728	1944.713	339.744	602.360
S14	B	Q8/Q8	MP/1	1555611	1234823	1.2598	1698.198	2048.799	440.384	701.457
S15	B	Q8/Q8	MP/2	1679022	1234823	1.3597	1857.490	2193.598	601.644	862.762
S16	B	Q0.6/Q8	BU/1	1804952	1234823	1.4617	1295.587	1829.407	0.023	2.269
S17	B	Q0.6/Q8	BA/1	470239	1234823	0.3808	1294.175	1830.504	0.152	2.381
S18	B	Q0.6/Q8	FR/1	1823040	1234823	1.4764	3250.907	4351.173	1957.286	2594.619
S19	B	Q0.6/Q8	MP/1	1566709	1234823	1.2688	1739.204	2066.977	457.443	719.974
S20	B	L1/Q8	BU/1	1804952	1234823	1.4617	1295.587	1829.407	0.023	2.269
S21	B	L1/Q8	BA/1	470239	1234823	0.3808	1294.175	1830.504	0.152	2.381
S22	B	L1/Q8	FR/1	1823040	1234823	1.4764	3250.907	4351.173	1957.286	2594.619
S23	B	L1/Q8	MP/1	2191545	1234823	1.7748	2501.756	2839.471	1261.563	1501.588
S24	B	Q8/Q0.6	BU/1	1667306	1137455	1.4658	320.270	618.676	0.027	2.147
S25	B	Q8/Q0.6	BA/1	432236	1137455	0.3800	321.274	615.270	0.243	2.217
S26	B	Q8/Q0.6	FR/1	1679400	1137455	1.4765	2950.815	4351.674	2598.551	3916.426
S27	B	Q8/Q0.6	MP/1	1538719	1137455	1.3528	880.137	1234.519	559.818	960.371
S28	B	Q0.6/Q0.6	BU/0	1338538	1137455	1.1768	321.288	615.274	0.016	1.997
S29	B	Q0.6/Q0.6	BU/1	1667306	1137455	1.4658	320.270	618.676	0.027	2.147
S30	B	Q0.6/Q0.6	BU/2	2156730	1137455	1.8961	321.299	615.208	-0.013	2.206
S31	B	Q0.6/Q0.6	BA/0	667309	1137455	0.5867	320.641	615.441	0.086	2.179
S32	B	Q0.6/Q0.6	BA/1	432236	1137455	0.3800	321.274	615.270	0.243	2.217
S33	B	Q0.6/Q0.6	BA/2	314560	1137455	0.2765	321.129	616.094	0.357	2.991
S34	B	Q0.6/Q0.6	FR/0	1687770	1137455	1.4838	1180.883	1742.329	855.608	1321.031
S35	B	Q0.6/Q0.6	FR/1	1679400	1137455	1.4765	2950.815	4351.674	2598.551	3916.426
S36	B	Q0.6/Q0.6	FR/2	1678860	1137455	1.4760	5900.661	8702.107	5493.071	8328.051
S37	B	Q0.6/Q0.6	MP/0	1485151	1137455	1.3057	822.327	1174.713	500.032	900.432
S38	B	Q0.6/Q0.6	MP/1	1556420	1137455	1.3683	909.761	1285.201	599.782	1004.551
S39	B	Q0.6/Q0.6	MP/2	1788579	1137455	1.5724	1226.888	1583.471	918.192	1300.856
S40	B	L1/Q0.6	BU/1	1667306	1137455	1.4658	320.270	618.676	0.027	2.147
S41	B	L1/Q0.6	BA/1	432236	1137455	0.3800	321.274	615.270	0.243	2.217
S42	B	L1/Q0.6	FR/1	1679400	1137455	1.4765	2950.815	4351.674	2598.551	3916.426
S43	B	L1/Q0.6	MP/1	2173196	1137455	1.9106	1712.972	2043.229	1382.089	1760.152
S44	B	Q8/L1	BU/1	2675540	1817075	1.4724	692.325	908.943	-0.036	1.954
S45	B	Q8/L1	BA/1	677019	1817075	0.3726	693.477	909.000	0.269	2.474
S46	B	Q8/L1	FR/1	2687094	1817075	1.4788	4803.182	6401.016	4133.387	5495.639
S47	B	Q8/L1	MP/1	1856321	1817075	1.0216	745.348	924.261	0.396	280.610
S48	B	Q0.6/L1	BU/1	2675540	1817075	1.4724	692.325	908.943	-0.036	1.954
S49	B	Q0.6/L1	BA/1	677019	1817075	0.3726	693.477	909.000	0.269	2.474
S50	B	Q0.6/L1	FR/1	2687094	1817075	1.4788	4803.182	6401.016	4133.387	5495.639
S51	B	Q0.6/L1	MP/1	1860537	1817075	1.0239	751.845	929.916	0.432	320.119
S52	B	L1/L1	BU/0	2132516	1817075	1.1736	693.267	908.796	0.048	2.280
S53	B	L1/L1	BU/1	2675540	1817075	1.4724	692.325	908.943	-0.036	1.954
S54	B	L1/L1	BU/2	3463700	1817075	1.9062	693.183	908.590	-0.032	2.536
S55	B	L1/L1	BA/0	1056986	1817075	0.5817	693.753	908.383	0.135	2.534
S56	B	L1/L1	BA/1	677019	1817075	0.3726	693.477	909.000	0.269	2.474
S57	B	L1/L1	BA/2	487299	1817075	0.2682	693.224	909.089	0.332	2.642
S58	B	L1/L1	FR/0	2688768	1817075	1.4797	1922.339	2560.925	1244.967	1655.937
S59	B	L1/L1	FR/1	2687094	1817075	1.4788	4803.182	6401.016	4133.387	5495.639
S60	B	L1/L1	FR/2	2687040	1817075	1.4788	9603.091	12801.128	8982.591	11895.140
S61	B	L1/L1	MP/0	2245681	1817075	1.2359	1276.205	1630.154	580.461	1079.246
S62	B	L1/L1	MP/1	2274015	1817075	1.2515	1318.023	1637.724	639.696	1096.149
S63	B	L1/L1	MP/2	2274015	1817075	1.2515	1318.023	1637.724	639.696	1096.149
S64	H	Q8/Q8	Raw/0	1234823	1234823	1.0000	1294.373	1831.319	0.000	0.000
S65	H	Q0.6/Q0.6	Raw/0	1137455	1137455	1.0000	320.551	615.135	0.000	0.000

Continued on next page

TABLE S133 continued

ID	Origin	Source/target	Policy/point	B	B_0	B/B_0	C_{50}	C_{95}	ΔC_{50}	ΔC_{95}
S66	H	L1/L1	Raw/0	1817075	1817075	1.0000	693.457	908.990	0.000	0.000
S67	H	Q8/Q8	HC/0	1459945	1234823	1.1823	1599.442	1944.733	339.127	600.627
S68	H	Q8/Q8	HC/1	1546745	1234823	1.2526	1699.962	2049.898	439.988	701.765
S69	H	Q8/Q8	HC/2	1676945	1234823	1.3580	1857.273	2193.011	602.001	861.035
S70	H	Q0.6/Q8	HC/1	1562245	1234823	1.2652	1739.884	2066.989	458.103	719.233
S71	H	L1/Q8	HC/1	2191545	1234823	1.7748	2502.074	2838.978	1261.416	1503.480
S72	H	Q8/Q0.6	HC/1	1528396	1137455	1.3437	879.492	1235.151	560.224	960.887
S73	H	Q0.6/Q0.6	HC/0	1457096	1137455	1.2810	823.027	1174.287	499.948	899.929
S74	H	Q0.6/Q0.6	HC/1	1543896	1137455	1.3573	908.562	1287.710	599.640	1004.156
S75	H	Q0.6/Q0.6	HC/2	1785696	1137455	1.5699	1225.799	1583.646	918.593	1300.705
S76	H	L1/Q0.6	HC/1	2173196	1137455	1.9106	1712.330	2044.914	1383.144	1759.913
S77	H	Q8/L1	HC/1	1629215	1817075	0.8966	746.786	926.098	0.942	282.112
S78	H	Q0.6/L1	HC/1	1644715	1817075	0.9051	751.150	929.332	1.049	320.401
S79	H	L1/L1	HC/0	2239915	1817075	1.2327	1277.044	1630.239	579.650	1078.577
S80	H	L1/L1	HC/1	2274015	1817075	1.2515	1318.254	1638.009	638.519	1099.758
S81	H	L1/L1	HC/2	2274015	1817075	1.2515	1318.254	1638.009	638.519	1099.758

TABLE S134: Complete secondary E4 cost grid, panel B: receiver span, chunk delivery, planned tail (ms), and sender blocking (s). Keys, units and recovery limits are defined above; all rows retain their own raw reference.

ID	S_{95}	N_e	ΔR_{50}	ΔR_{95}	T_{50}	T_{95}	W (s)
S01	1829.117	33138	0.000	0.000	0.000	0.000	9.115
S02	615.067	30589	0.000	0.000	0.000	0.000	7.739
S03	908.490	49260	0.000	0.000	0.000	0.000	12.500
S04	1829.981	33138	-0.011	2.118	0.000	0.000	8.879
S05	1828.941	33138	-0.008	2.166	0.000	0.000	8.888
S06	1829.118	33138	-0.018	2.039	0.000	0.000	8.983
S07	1786.747	33138	3.354	23.748	0.000	0.000	4.494
S08	1733.796	33138	27.591	65.730	0.000	0.000	2.453
S09	1651.834	33138	66.356	141.359	0.000	0.000	1.273
S10	1902.783	33138	38.348	126.055	87.589	211.466	9.771
S11	4350.777	33138	971.764	2052.567	1957.568	2594.453	9.122
S12*	8700.601	33138	2599.329	5430.918	5183.798	6898.355	10.646
S13	1944.352	33138	0.000	2.220	340.000	600.000	10.976
S14	2048.434	33138	-0.036	1.966	440.000	700.000	11.494
S15	2193.155	33138	-0.028	2.129	600.000	860.000	12.630
S16	1828.941	33138	-0.008	2.166	0.000	0.000	8.888
S17	1733.796	33138	27.591	65.730	0.000	0.000	2.453
S18	4350.777	33138	971.764	2052.567	1957.568	2594.453	9.122
S19	2066.699	33138	-0.015	2.100	460.000	720.000	11.887
S20	1828.941	33138	-0.008	2.166	0.000	0.000	8.888
S21	1733.796	33138	27.591	65.730	0.000	0.000	2.453
S22	4350.777	33138	971.764	2052.567	1957.568	2594.453	9.122
S23	2839.104	33138	-0.044	2.026	1260.000	1500.000	17.591
S24	618.232	30589	0.014	1.964	0.000	0.000	7.235
S25	576.648	30589	7.508	22.289	0.000	0.000	2.276
S26	4351.429	30589	1328.791	3097.641	2597.828	3915.933	8.327
S27	1234.409	30589	-0.001	1.844	560.000	960.000	10.756
S28	615.023	30589	0.014	1.956	0.000	0.000	7.631
S29	618.232	30589	0.014	1.964	0.000	0.000	7.235
S30	615.011	30589	-0.003	1.954	0.000	0.000	7.397
S31	589.996	30589	2.256	8.854	0.000	0.000	4.139
S32	576.648	30589	7.508	22.289	0.000	0.000	2.276
S33	553.723	30589	17.497	47.719	0.000	0.000	1.155
S34	1742.112	30589	425.312	1017.965	855.923	1321.829	8.411
S35	4351.429	30589	1328.791	3097.641	2597.828	3915.933	8.327
S36	8701.612	30589	2832.336	6569.634	5493.302	8327.522	8.415
S37	1174.381	30589	0.002	2.056	500.000	900.000	10.286
S38	1284.728	30589	0.016	2.036	600.000	1001.000	11.197
S39	1583.193	30589	0.013	2.107	920.000	1300.000	13.116
S40	618.232	30589	0.014	1.964	0.000	0.000	7.235
S41	576.648	30589	7.508	22.289	0.000	0.000	2.276
S42	4351.429	30589	1328.791	3097.641	2597.828	3915.933	8.327
S43	2043.095	30589	0.006	1.925	1380.000	1760.000	16.387

Continued on next page

TABLE S134 continued

ID	S_{95}	N_e	ΔR_{50}	ΔR_{95}	T_{50}	T_{95}	W (s)
S44	908.448	49260	0.006	2.092	0.000	0.000	12.582
S45	881.325	49260	10.607	23.258	0.000	0.000	3.409
S46	6400.732	49260	2097.878	4430.952	4130.402	5495.072	13.908
S47	924.035	49260	0.012	2.162	0.000	280.000	13.273
S48	908.448	49260	0.006	2.092	0.000	0.000	12.582
S49	881.325	49260	10.607	23.258	0.000	0.000	3.409
S50	6400.732	49260	2097.878	4430.952	4130.402	5495.072	13.908
S51	929.602	49260	0.004	2.016	0.000	320.000	13.486
S52	908.417	49260	0.001	2.219	0.000	0.000	12.892
S53	908.448	49260	0.006	2.092	0.000	0.000	12.582
S54	908.183	49260	0.013	2.064	0.000	0.000	12.502
S55	896.956	49260	2.989	9.118	0.000	0.000	6.977
S56	881.325	49260	10.607	23.258	0.000	0.000	3.409
S57	854.283	49260	24.017	50.806	0.000	0.000	1.775
S58	2560.610	49260	626.775	1336.190	1244.206	1657.096	13.470
S59	6400.732	49260	2097.878	4430.952	4130.402	5495.072	13.908
S60	12800.843	49260	4550.378	9601.888	8982.094	11894.334	13.146
S61	1629.767	49260	0.006	1.992	580.000	1080.000	16.475
S62	1637.454	49260	-0.019	1.842	640.000	1100.000	16.607
S63	1637.454	49260	-0.019	1.842	640.000	1100.000	16.607
S64	1831.069	33138	0.000	0.000	0.000	0.000	8.452
S65	615.033	30589	0.000	0.000	0.000	0.000	6.988
S66	908.584	49260	0.000	0.000	0.000	0.000	11.744
S67	1944.339	33138	0.015	2.843	340.000	600.000	9.855
S68	2049.640	33138	0.024	2.675	440.000	700.000	10.676
S69	2192.849	33138	-0.030	2.458	600.000	860.000	11.574
S70	2066.633	33138	-0.011	2.537	460.000	720.000	10.594
S71	2838.726	33138	0.043	2.489	1260.000	1500.000	16.327
S72	1234.771	30589	-0.020	2.300	560.000	960.000	9.773
S73	1174.121	30589	0.004	2.890	500.000	900.000	9.333
S74	1287.379	30589	-0.016	2.273	600.000	1001.000	9.619
S75	1583.518	30589	0.004	2.187	920.000	1300.000	12.141
S76	2044.521	30589	-0.013	2.116	1380.000	1760.000	14.882
S77	926.026	49260	0.119	142.312	0.000	280.000	10.007
S78	928.968	49260	0.126	135.138	0.000	320.000	10.382
S79	1629.658	49260	-0.003	2.340	580.000	1080.000	15.273
S80	1637.710	49260	-0.016	2.267	640.000	1100.000	15.273
S81	1637.710	49260	-0.016	2.267	640.000	1100.000	15.273

S52. Frozen R47 Source Map and Remaining Limits

Acquisition and screening are frozen for this revision; the freeze does not retroactively complete the protocol. The discovery interval remains January 1, 2018 through October 30, 2026, 13:15:46 UTC, with earlier foundations as background. Notes recorded later do not themselves establish availability at that cutoff. The extension is retrospective, nonblind and model-assisted, with prior results and reviews visible; no preregistration or independent human coding is claimed.

Separate eligibility from evidence type and goal. Direct defenses (D) explicitly address auxiliary-observation confidentiality or unlinkability in LLM inference, serving or agent execution. Designs, formal analyses and implementations can qualify without an empirical attack-success score. Relevant generic hardware, ORAM, constant-time and traffic mechanisms remain M unless the source explicitly analyzes that setting. Semantic-only privacy, general model safety and secondary sources without the relevant auxiliary-observation claim remain A; unresolved cases stay pending. User/task privacy, model/service secrets, attribution and audit/detection integrity are separate protected interests. A defense of audit randomness therefore does not supply a user-privacy success or failure sample.

The original C5 empirical inference-evaluation condition, 82-family map, 19-family privacy/attribution sample and historical record codes remain intact. The six-source defense-first challenge also retains its prior purposive scope. R47 does not redefine either set or supply a finalized D-family denominator. Version-traceable papers, study families, retrieval routes and configurations are distinct units; repeated routes or paper versions do not become independent studies. Formal statements, proposed mechanisms and measurements retain their respective assumptions and limitations.

Selected direct-source examples beyond a required attack score. Opal v1 has a recorded source-version D judgment for personal-AI memory access outside trusted enclaves. Its stated privacy/integrity theorem is conditional on a public leakage function, enclave/cryptographic assumptions, a durable client counter and no ORAM stash overflow. Operation type and public trace shape remain visible; enclave-internal side channels are excluded. Ten-run latency and annual infrastructure-cost extrapolation are distinct observations, not paired residual-attack, task-quality and cost measurements. Proof correctness, publication identity and family relationships remain pending.

The eFPGA LLM design v1 also has a recorded source-version D judgment. It proposes equalization, masking, perturbation, execution shuffling and constant-shape traffic for separately described observers and assets. Qualitative comparison marks are neither measured protection nor an end-to-end theorem. Other integrity and supply-chain goals remain separate. Final family inclusion and independent semantic review remain pending. These deliberately selected sources illustrate the broader rule; they do not form a complete or deduplicated defense sample.

CachePrune: a configuration association is not a paired trial. The retained v1 review links the source to existing family F75. Marked-token protection uses an abstract binary reuse oracle and depends on detection; contextual reconstruction remains possible. Protection and serving evaluations are separate studies, without established identical requests or sample-level pairing. The Section 5.7 fixed-chunk comparison disables privacy filtering. Historical triple joins retain their configuration-grouping interpretation and numerical cells; no stricter pairing rule has been silently substituted or used to recompute a rate.

SpliceDefense: signed residuals and component separation. The retained v1 readings distinguish CTBF timing cases from QCP costs. Defended median differences, including a bulk -9 ms, establish neither distributional indistinguishability nor successful adaptive exploitation. The longer extraction examples deliberately include the true token among 21 candidates. The relationship between 15 tasks/75 steps and separate ten-token targets is unresolved. Neither QCP's component cost nor the selected timing observations establish a combined protection/task-quality/cost trial.

OTRO: conditional security and unreconciled costs. The retained v1 notes preserve the equal-access-count DRAM-trace theorem conditions, public length/count observations and excluded implementation channels. The uniform 20,000-prompt closed-world exercise is not a universal leakage bound. Plotted TTFT excludes access-count padding. The abstract's at-most-4.5% and body average of 4.59% remain unreconciled. The stated 2.8 ms padding estimate divided by the named 116 ms rebuild latency is about 2.4%, whereas the text says less than 0.4%; this arithmetic check is not a corrected measured result. Pool/memory text conflicts, figures, proof/implementation correspondence and padding-inclusive multi-client costs remain pending. The separately recorded v3 observations are not substituted for v1's algorithms, leakage or measurements.

Time Will Tell: ideal observation and selected tasks. The retained v1 follow-up limits the 60-word prompt experiment to three selected high-bias tasks, unchanged three-shot examples and perfect output-token-count observation. Reported mean ASR changes from 88.2% to 61.9% for GPT-4o and from 79.3% to 80.3% for Gemma2-9b-IT; the former difference is 26.3 percentage points. Residual inference remains. The remote attack setup and separate model/language padding estimates do not establish a paired adapted-defense, quality and cost evaluation for the prompt intervention.

Frozen limitations. Venue/index/citation coverage, uncollected tails, inaccessible sources, final eligibility, cutoff availability and unresolved publication/version/family identities remain limitations. A pass over saved program positions is not complete proceedings coverage, unique-paper accounting or full-text review. Title-only entries, secondary abstracts, project descriptions and separate theses retain their recorded scope. Source bytes and figures were not always saved or independently checked. Local hash/pointer validation checks bookkeeping, not source semantics or proof correctness. No missing observation is recast as defense failure, and no source-version D judgment implies completed independent review.

This revision adds no experiments: the installed R46 replay, calibration, operating-cost and retail-sandbox results remain unchanged. Historical statistics are preserved, while the revised prose makes their configuration-level and selection-conditioned interpretation explicit.

References

[1] P. Revankar, K. Chauhan, J. Kim, S. N. Nur, V. Siu, and C.-G. Wang, "Mirage: A polarity-flipping encoding subspace in LLM agents," *arXiv preprint arXiv:2606.10304v1*, 2026, consulted 2606.10304v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2606.10304v1

[2] R. A. Bakar, "Characterizing contention-induced reliability collapse in KV-cache timing side channels for multi-tenant LLM serving," *arXiv preprint arXiv:2609.06853v1*, 2026. [Online]. Available: https://arxiv.org/abs/2609.06853v1

[3] N. Carlini and M. Nasr, "Remote timing attacks on efficient language model inference," *arXiv preprint arXiv:2410.17175v1*, 2024. [Online]. Available: https://arxiv.org/abs/2410.17175v1

[4] C. Song, H. Wang, and R. Martin, "Personafingerprint: Measuring persona inference on modern websites with LLM-driven browsing," *arXiv preprint arXiv:2605.15962*, 2026. [Online]. Available: https://arxiv.org/abs/2605.15962

[5] R. Weiss, B. Konstantinov, E. Sheetrit, T. Simon, and Y. Mirsky, "Detokenization leaks: Reconstructing local LLM outputs from cache traces," *arXiv preprint arXiv:2609.06674v1*, 2026. [Online]. Available: https://arxiv.org/abs/2609.06674v1

[6] R. Ding, T. Xu, X. Shen, A. A. Ding, and Y. Fei, "MoEcho: Exploiting side-channel attacks to compromise user privacy in mixture-of-experts LLMs," in *ACM SIGSAC Conference on Computer and Communications Security*, 2025, pp. 2159–2173. [Online]. Available: https://arxiv.org/abs/2508.15036v1

[7] I. Yona, I. Shumailov, J. Hayes, and N. Carlini, "Stealing user prompts from mixture of experts," *arXiv preprint arXiv:2410.22884v1*, 2024. [Online]. Available: https://arxiv.org/abs/2410.22884v1

[8] H. Oderinwale, "Agent trajectories as programs: fingerprinting and programming coding-agent behavior," *arXiv preprint arXiv:2606.16988v1*, 2026. [Online]. Available: https://arxiv.org/abs/2606.16988v1

[9] S. Alhazbi, A. Hussain, G. Oligeri, and P. Papadimitratos, "LLMs have rhythm: Fingerprinting large language models using inter-token times and network traffic analysis," *IEEE Open Journal of the Communications Society*, vol. 6, pp. 5050–5071, 2025. [Online]. Available: https://arxiv.org/abs/2502.20589v1

[10] A. Chaudhuri, S. Shukla, S. Bhattacharya, and D. Mukhopadhyay, "Energon: Unveiling transformers from GPU power and thermal side-channels," in *IEEE/ACM International Conference on Computer-Aided Design*, 2025. [Online]. Available: https://arxiv.org/abs/2508.01768v1

[11] H. Weng, R. Zhou, Y. Qu, X. Ji, and W. Xu, "ArchSnoop: LLM architecture snooping via electromagnetic side-channel on edge devices," in *Proceedings of the 19th ACM Conference on Security and Privacy in Wireless and Mobile Networks*, 2026, pp. 51–62. [Online]. Available: https://dl.acm.org/doi/10.1145/3765613.3811673

[12] L. L. Gan, J. Wu, L. Ge, L. Yang, Y. Song, J. Zhang, H. Jin, W. Wang, and G. Xue, "ClawGuard: Out-of-band detection of LLM agent workflow hijacking via EM side channel," *arXiv preprint arXiv:2605.06205v1*, 2026. [Online]. Available: https://arxiv.org/abs/2605.06205v1

[13] F. George, H. Kumar, D. Pathak, K. Ray, M. Verma, and P. Moogi, "Unsupervised cycle detection in agentic applications," *arXiv preprint arXiv:2511.10650v1*, 2025. [Online]. Available: https://arxiv.org/abs/2511.10650v1

[14] R. Solomon, Y. Y. Levi, L. Vaknin, E. Aizikovich *et al.*, "LumiMAS: A comprehensive framework for real-time monitoring and enhanced observability in multi-agent systems," *arXiv preprint arXiv:2508.12412v2*, 2026. [Online]. Available: https://arxiv.org/abs/2508.12412v2

[15] J. Roh, E. Bagdasarian, H. Haddadi, and A. S. Shamsabadi, "SPILLage: Agentic oversharing on the web," *arXiv preprint arXiv:2602.13516v1*, 2026. [Online]. Available: https://arxiv.org/abs/2602.13516v1

[16] Y. Qiao, D. Liu, H. Yang, W. Zhou, and S. Hu, "Agent tools orchestration leaks more: Dataset, benchmark, and mitigation," *arXiv preprint arXiv:2512.16310v4*, 2026, version 4. [Online]. Available: https://arxiv.org/abs/2512.16310v4

[17] M. Umar, A. P. Marathe, M. D. Gupta, S. J. Ghosh, G. E. Suh, and W. Xiong, "Efficient memory side-channel protection for embedding generation in machine learning," in *IEEE International Symposium on High-Performance Computer Architecture*, 2025. [Online]. Available: https://bearhw.ece.vt.edu/content/dam/bearhw_ece_vt_edu/publications/2025_HPCA_SecEmb.pdf

[18] L. Wu, C. Wang, T. Liu, Y. Zhao, and H. Wang, "From assistants to adversaries: Exploring the security risks of mobile LLM agents," *arXiv preprint arXiv:2505.12981v1*, 2025, version 1. [Online]. Available: https://arxiv.org/abs/2505.12981

[19] M. Mudryi, M. Chaklosh, and G. M. Wójcik, "The hidden dangers of browsing AI agents," *arXiv preprint arXiv:2505.13076v1*, 2025, version 1. [Online]. Available: https://arxiv.org/abs/2505.13076

[20] A. Ukani, H. Haddadi, A. S. Shamsabadi, and P. Snyder, "Privacy practices of browser agents," *arXiv preprint arXiv:2512.07725*, 2025. [Online]. Available: https://arxiv.org/abs/2512.07725

[21] N. Nazari, F. Xiang, C. Fang, H. M. Makrani, A. Puri, K. Patwari, H. Sayadi, S. Rafatirad, C.-N. Chuah, and H. Homayoun, "LLM-FIN: Large language models fingerprinting attack on edge devices," in *International Symposium on Quality Electronic Design*, 2024. [Online]. Available: https://web.cs.ucdavis.edu/~srafatir/pages/documents/llm_fingerprint.pdf

[22] R. H. Anwar, J. O. Weiss, T. Alves, S. Kundu, and M. T. Raza, "InferNet: Exploiting aggregate GPU profiles as side-channel for DNN architecture inference," *ACM Transactions on AI Security and Privacy*, 2026. [Online]. Available: https://rhasnainanwar.me/assets/pdf/2304.03388v2.pdf

[23] B. Dangol, "From privacy to workflow integrity: Communication-graph metadata in autonomous agent interoperability," *arXiv preprint arXiv:2606.07150v3*, 2026, consulted arXiv:2606.07150v3; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2606.07150v3

[24] S. Banerjee, P. Sahu, M. Luo, A. Vahldiek-Oberwagner, N. J. Yadwadkar, and M. Tiwari, "SoK: A systems perspective on compound AI threats and countermeasures," *arXiv preprint arXiv:2411.13459v1*, 2024, consulted arXiv:2411.13459v1; inspected 20 October 2026. [Online]. Available: https://arxiv.org/abs/2411.13459v1

[25] J. Forough, M. Kogias, and H. Haddadi, "When agents handle secrets: A survey of confidential computing for agentic AI," *arXiv preprint arXiv:2605.03213v2*, 2026, consulted arXiv:2605.03213v2; inspected 20 October 2026. [Online]. Available: https://arxiv.org/abs/2605.03213v2

[26] X. Cai, R. Nithyanand, T. Wang, R. Johnson, and I. Goldberg, "A systematic approach to developing and evaluating website fingerprinting defenses," in *ACM CCS*, 2014, pp. 227–238. [Online]. Available: https://doi.org/10.1145/2660267.2660362

[27] E. Debenedetti, G. Severi, N. Carlini, C. A. Choquette-Choo, M. Jagielski, M. Nasr, E. Wallace, and F. Tramèr, "Privacy side channels in machine learning systems," in *33rd USENIX Security Symposium*. USENIX Association, 2024. [Online]. Available: https://www.usenix.org/conference/usenixsecurity24/presentation/debenedetti

[28] N. Lahjouji and A. G. Colaco, "Agents that know too much: A data-centric survey of privacy in LLM agents," *arXiv preprint arXiv:2606.26627*, 2026, consulted arXiv:2606.26627v1; archived access 19 October 2026. [Online]. Available: https://arxiv.org/abs/2606.26627v1

[29] J. Kim, W. Guo, and D. Song, "SoK: Attack and defense landscape of agentic AI systems," in *35th USENIX Security Symposium*, 2026. [Online]. Available: https://www.usenix.org/conference/usenixsecurity26/presentation/kim-juhee-agentic

[30] P. Horváth, D. Lauret, Z. Liu, and L. Batina, "SoK: Neural network extraction through physical side channels," in *33rd USENIX Security Symposium*, 2024. [Online]. Available: https://www.usenix.org/conference/usenixsecurity24/presentation/horvath

[31] Y. Zhang, X. Deng, Z. Gu, Y. Chen, K. Xu, Q. Li, and J. Wu, "Exposing LLM user privacy via traffic fingerprint analysis: A study of privacy risks in LLM agent interactions," *arXiv preprint arXiv:2510.07176*, 2025. [Online]. Available: https://arxiv.org/abs/2510.07176

[32] R. Weiss, D. Ayzenshteyn, G. Amit, and Y. Mirsky, "What was your prompt? a remote keylogging attack on AI assistants," in *33rd USENIX Security Symposium*. USENIX Association, 2024, pp. 3367–3384. [Online]. Available: https://www.usenix.org/conference/usenixsecurity24/presentation/weiss

[33] C. Gu, X. L. Li, R. Kuditipudi, P. Liang, and T. Hashimoto, "Auditing prompt caching in language model apis," in *International Conference on Machine Learning*, vol. 267, 2025, pp. 20 477–20 496. [Online]. Available: https://proceedings.mlr.press/v267/gu25b.html

[34] S. Li, T. Cui, M. Chen, X. Lin, Z. Gu, X. Deng, K. Xu, and Q. Li, "From length to content: Token-length side-channel attacks on LLM api merged outputs," in *35th USENIX Security Symposium*, 2026, pp. 2187–2206. [Online]. Available: https://www.usenix.org/conference/usenixsecurity26/presentation/li-sijia

[35] S. Pouryousef, J. Lopez, S. R. Nowmi, M. M. Kamol, M. Hossain, M. Tran, and M. S. Rahman, "A large-scale benchmark and risk assessment of traffic analysis attacks on cloud LLM services," *arXiv preprint arXiv:2609.31877v1*, 2026, consulted arXiv:2609.31877v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2609.31877v1

[36] E. Hadad and M. Guri, "Shape and substance: Dual-layer side-channel attacks on local vision-language models," *arXiv preprint arXiv:2603.25403v1*, 2026, consulted 2603.25403v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2603.25403v1

[37] F. Chen, L. Du, J. Zhou, P. Li, and Z. Su, "Sparleak: Privacy leakage from sparse attention in LLM inference on shared gpus," *arXiv preprint arXiv:2609.38830v1*, 2026, consulted 2609.38830v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2609.38830v1

[38] X. Zheng, H. Han, S. Shi, Q. Fang, Z. Du, X. Hu, and Q. Guo, "Inputsnatch: Stealing input in LLM services via cache-sharing timing side-channel attacks," in *Advanced Parallel Processing Technologies (APPT 2026)*, 2027, pp. 483–497, first online August 24, 2026; publisher citation year 2027. [Online]. Available: https://link.springer.com/chapter/10.1007/978-981-92-4805-6_32

[39] C. Hopps, "Aggregation and Fragmentation Mode for Encapsulating Security Payload (ESP) and Its Use for IP Traffic Flow Security (IP-TFS)," RFC Editor, Tech. Rep. RFC 9347, Jan. 2023. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9347

[40] H. Jeong, M. Teymoorianfard, A. Kumar, A. Houmansadr, and E. Bagdasarian, "Network-level prompt and trait leakage in local research agents," in *35th USENIX Security Symposium*. USENIX Association, 2026, pp. 6485–6504. [Online]. Available: https://www.usenix.org/conference/usenixsecurity26/presentation/jeong

[41] Kubernetes Authors, "Kubernetes API Concepts: Dry-run," https://kubernetes.io/docs/reference/using-api/api-concepts/#dry-run, accessed October 24, 2026.

[42] Stripe, "Sandboxes," https://docs.stripe.com/sandboxes, accessed October 24, 2026.

[43] PostgreSQL Global Development Group, "PostgreSQL 18 Documentation: ROLLBACK," https://www.postgresql.org/docs/18/sql-rollback.html, accessed October 24, 2026.

[44] ——, "PostgreSQL 18 Documentation: Sequence Manipulation Functions," https://www.postgresql.org/docs/18/functions-sequence.html, accessed October 24, 2026.

[45] J. Zhu, L. Yang, R. Shan, C. Zheng, Z. Zheng, W. Liu, Y. Yu, W. Zhang, and J. Lin, "Turing test on screen: A benchmark for mobile GUI agent humanization," *arXiv preprint arXiv:2604.09574v1*, 2026, consulted arXiv:2604.09574v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/html/2604.09574v1

[46] G. Wu, Z. Li, Y. Zhang, Z. Zhang, J. Niu, Y. Wu, and Y. Zhang, "Cacheprune: Privacy-aware and fine-grained KV cache sharing for efficient LLM inference," *arXiv preprint arXiv:2605.23640v1*, 2026, consulted 2605.23640v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2605.23640v1

[47] L. Song, Z. Pang, W. Wang, Z. Wang, X. Wang, H. Chen, W. Song, Y. Jin, D. Meng, and R. Hou, "The early bird catches the leak: Unveiling timing side channels in LLM serving systems," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 11 431–11 446, 2025, consulted arXiv:2409.20002v5; archived access 18 October 2026. [Online]. Available: https://ieeexplore.ieee.org/document/11206380/

[48] Z. Gao, J. Hu, F. Guo, Y. Zhang, Y. Han, S. Liu, H. Li, and Z. Lv, "I know what you said: Unveiling hardware cache side-channels in local large language model inference," in *34th USENIX Security Symposium*, 2025, pp. 1649–1668. [Online]. Available: https://www.usenix.org/conference/usenixsecurity25/presentation/gao-zibo

[49] T. C. Addagada, "Governing the KV cache: Preventing timing side-channel leakage in multi-tenant LLM inference," *arXiv preprint arXiv:2608.09225*, 2026, consulted arXiv:2608.09225v1; archived access 19 October 2026. [Online]. Available: https://arxiv.org/abs/2608.09225v1

[50] Z. Zhang, G. Wu, S. Deng, S. Wang, and Y. Zhang, "Netecho: From real-world streaming side-channels to full LLM conversation recovery," *arXiv preprint arXiv:2510.25472*, 2025, consulted arXiv:2510.25472v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2510.25472v1

[51] J. Lee, Y. Wang, R. Rajat, D. Wong, M. Li, and M. Annavaram, "Otro: Oblivious tokenization path with square-root oram," *arXiv preprint arXiv:2606.17358v1*, 2026, consulted 2606.17358v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2606.17358v1

[52] K. Chu, Z. Lin, D. Xiang, Z. Shen, J. Su, C. Chu, Y. Yang, W. Zhang, W. Wu, and W. Zhang, "Selective KV-cache sharing to mitigate timing side-channels in LLM inference," *arXiv preprint arXiv:2508.08438v2*, 2026, consulted arXiv:2508.08438v2; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2508.08438v2

[53] J. Wei, A. Abdulrazzag, T. Zhang, A. Muursepp, and G. Saileshwar, "When speculation spills secrets: Side channels via speculative decoding in llms," *arXiv preprint arXiv:2411.01076*, 2024, consulted arXiv:2411.01076v4; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2411.01076v4

[54] H. Sun, S. Liu, S. Ma, J. Li, M. Xiao, and W. Jiang, "Agent-assisted side-channel attacks on non-prefix KV cache in rag," *arXiv preprint arXiv:2606.21842*, 2026, consulted arXiv:2606.21842v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2606.21842v1

[55] T. Hornetz, H. Yavarzadeh, A. Cheu, A. Gascon, L. Gerlach, D. Moghimi, P. Schoppmann, M. Schwarz, and R. Zhang, "Tdxray: Microarchitectural side-channel analysis of Intel TDX for real-world workloads," in *IEEE Symposium on Security and Privacy*, 2026. [Online]. Available: https://tdxray.cpusec.org/assets/tdxray_sp26.pdf

[56] T. Zhang, G. Saileshwar, and D. Lie, "Time will tell: Timing side channels via output token count in large language models," *arXiv preprint arXiv:2412.15431*, 2024, consulted arXiv:2412.15431v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2412.15431v1

[57] W. Lugoloobi, S. Marro, J. Magomere, J. Wright, and C. Russell, "Known by their actions: Fingerprinting LLM browser agents via UI traces," *arXiv preprint arXiv:2605.14786*, 2026, consulted arXiv:2605.14786v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2605.14786v1

[58] G. McDonald and J. B. Or, "Whisper leak: a side-channel attack on large language models," *arXiv preprint arXiv:2511.03675*, 2025, consulted arXiv:2511.03675v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2511.03675v1

[59] A. Shahriar, M. N. Rahman, S. Ahmed, F. Sadeque, and M. R. Parvez, "AgentTell: Behavioural side-channel leakage in browser-use agents," *arXiv preprint arXiv:2609.32915v1*, 2026, consulted arXiv:2609.32915v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2609.32915v1

[60] C. Gu, R. Levine, Z. Zhang, T. Sorensen, and Y. Guo, "Behind bars: A side-channel attack on NVIDIA MIG cache partitioning using memory barriers," in *35th USENIX Security Symposium*. USENIX Association, 2026, pp. 5373–5392. [Online]. Available: https://www.usenix.org/conference/usenixsecurity26/presentation/gu-cheng

[61] E. Wang, Z. Shafiq, and Y. Vekaria, "Fp-agent: Fingerprinting AI browsing agents," *arXiv preprint arXiv:2605.01247*, 2026, consulted arXiv:2605.01247v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2605.01247v1

[62] J. Geng, R. He, Z. Fei, B. Yi, R. Wang, Z. Liu, X. Hu, X. Wu, and Q. Zeng, "Agent skills matter: Inferring proprietary skills from execution trajectories," *arXiv preprint arXiv:2607.25560v1*, 2026, consulted arXiv:2607.25560v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2607.25560v1

[63] X. Wang, B. Zhao, M. Backes, F. Boenisch, and A. Dziedzic, "Agentprov: Auditing agentic LLM api providers via tool-use policy probes," *arXiv preprint arXiv:2609.00052v1*, 2026, consulted 2609.00052v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2609.00052v1

[64] S.-J. Park, M. Choi, J. Kim, and M.-S. Kang, "Cross-agent campaign attribution: Linking asynchronous attacks across LLM agents," *arXiv preprint arXiv:2607.18826v1*, 2026, consulted 2607.18826v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2607.18826v1

[65] N. Alhalawani, G. Acar, Z. Liu, and L. Batina, "Physical side-channel attacks on edge LLMs: Extracting output tokens and user prompts," *arXiv preprint arXiv:2610.14215v1*, 2026, consulted 2610.14215v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2610.14215v1

[66] S. Majidi, N. Mireshghallah, and K. Taram, "Leaky language models: Stealing architecture and inference optimizations via per-token timing," *arXiv preprint arXiv:2607.20723*, 2026, consulted arXiv:2607.20723v1; archived access 19 October 2026. [Online]. Available: https://arxiv.org/abs/2607.20723v1

[67] C.-Y. Wang, X.-H. Xie, T.-Z. Wang, F.-C. Luo, and Y. Cui, "Who is behind the harness? fingerprinting LLMs through agentic behavior," *arXiv preprint arXiv:2609.28559v1*, 2026, consulted 2609.28559v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2609.28559v1

[68] D. Scheinert, A. Acker, T. Wittkopp, S. Becker, H. Yous, K. Reddy, I. Farhat, H. Hacid, and O. Kao, "Beyond microservices: Testing web-scale RCA methods on GPU-driven LLM workloads," 2026, version 1, 2 March 2026. [Online]. Available: https://arxiv.org/html/2603.02057v1

[69] J.-J. Jiao, H. Yang, and R. Wen, "Llm-wfin: A fine-grained large language model (LLM)-oriented website fingerprinting attack via fusing interrupt trace and network traffic," *Electronics*, vol. 14, no. 7, p. 1263, 2025, consulted publisher version 2025-03-23; source-scoped access 21 October 2026. [Online]. Available: https://doi.org/10.3390/electronics14071263

[70] R. Nazaraliyev, Y. Zhang, and N. Abu-Ghazaleh, "Page404: Breaking GPU isolation through hardware-assisted page migration," *arXiv preprint arXiv:2610.20962v1*, 2026, consulted 2610.20962v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2610.20962v1

[71] G. Wu, Z. Zhang, Y. Zhang, W. Wang, J. Niu, Y. Wu, and Y. Zhang, "I know what you asked: Prompt leakage via KV-cache sharing in multi-tenant LLM serving," in *Network and Distributed System Security Symposium (NDSS)*, 2025. [Online]. Available: https://www.ndss-symposium.org/ndss-paper/i-know-what-you-asked-prompt-leakage-via-kv-cache-sharing-in-multi-tenant-llm-serving/

[72] T.-R. Chen, "Sensitive-topic leakage through LLM routing metadata: Measurement and mitigation," *arXiv preprint arXiv:2610.09981v1*, 2026, consulted 2610.09981v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2610.09981v1

[73] Y. Jo, J. Park, E. Lee, and D. Song, "Sparseety: Extracting tokens from sparsity-exploiting LLM serving systems via deterministic side channels," *arXiv preprint arXiv:2608.02995v1*, 2026, consulted 2608.02995v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2608.02995v1

[74] A. Adiletta and B. Sunar, "Spill the beans: Exploiting CPU cache side-channels to leak tokens from large language models," *arXiv preprint arXiv:2505.00817*, 2025. [Online]. Available: https://arxiv.org/abs/2505.00817

[75] G. Politis and E. Pappas, "Privacy failure in split-LLM training, the returned gradient nullifies the decoys," *arXiv preprint arXiv:2609.04382v1*, 2026, consulted 2609.04382v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2609.04382v1

[76] I. Fayolle, S. Bouhenniche, S. Pélissier, P. Laperdrix, C. Maurice, and W. Rudametkin, "On the internet, nobody knows you're an LLM bot: Unmasking web agents with multi-layer fingerprinting," *arXiv preprint arXiv:2606.30119v1*, 2026, consulted 2606.30119v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2606.30119v1

[77] D. Kang, H. Jeong, J. Sheffey, P. Datta, and A. Houmansadr, "Whose agent are you? multi-layer fingerprinting and attribution of autonomous web agents," *arXiv preprint arXiv:2606.20910v1*, 2026. [Online]. Available: https://arxiv.org/abs/2606.20910v1

[78] J. Hayes and G. Danezis, "k-fingerprinting: A robust scalable website fingerprinting technique," in *25th USENIX Security Symposium*, 2016, pp. 1187–1203. [Online]. Available: https://www.usenix.org/conference/usenixsecurity16/technical-sessions/presentation/hayes

[79] J. Gong and T. Wang, "Zero-delay lightweight defenses against website fingerprinting," in *29th USENIX Security Symposium*. USENIX Association, 2020, pp. 717–734. [Online]. Available: https://www.usenix.org/conference/usenixsecurity20/presentation/gong

[80] M. Juarez, M. Imani, M. Perry, C. Diaz, and M. Wright, "Toward an efficient website fingerprinting defense," in *ESORICS*, 2016, pp. 27–46. [Online]. Available: https://arxiv.org/abs/1512.00524

[81] T. Wang and I. Goldberg, "Walkie-talkie: An efficient defense against passive website fingerprinting attacks," in *26th USENIX Security Symposium*, 2017. [Online]. Available: https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/wang-tao

[82] G. Yu, Q. Wang, R. Lang, S. Su, and X. Wang, "PlanTwin: Privacy-preserving planning abstractions for cloud-assisted LLM agents," *arXiv preprint arXiv:2603.18377v2*, 2026. [Online]. Available: https://arxiv.org/abs/2603.18377v2

[83] J. Xu, L. Fan, Z. Wang, X. Li, and H. Liu, "Selection-invariant communication compilers for privacy-aware multi-agent LLM workflows," *arXiv preprint arXiv:2609.26076v1*, 2026. [Online]. Available: https://arxiv.org/abs/2609.26076v1

[84] T. A. Ghaleb, "Fingerprinting AI coding agents on GitHub," *arXiv preprint arXiv:2601.17406v1*, 2026, consulted arXiv:2601.17406v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2601.17406v1

[85] W. Wang, Z. Liu, H. Kou, X. Liu, and J. Shen, "The chatbot knows it's you: Dialogue attribution in unauthenticated human–LLM sessions," in *Proceedings of the ACM Web Conference 2026*, 2026, pp. 8797–8807. [Online]. Available: https://dl.acm.org/doi/10.1145/3774904.3793048

[86] W. Cui, H. Zhang, Y. Huang, Q. Li *et al.*, "SymDiag: Explainable diagnosis for LLM reasoning via neuro-symbolic verification," in *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*, 2026, consulted arXiv:2608.08786v1; archived access 18 October 2026. [Online]. Available: https://arxiv.org/abs/2608.08786v1

[87] S. Shao, Y. Li, H. Yao, Y. Chen, Y. Yang, and Z. Qin, "Reading between the lines: Towards reliable black-box LLM fingerprinting via zeroth-order gradient estimation," 2026, version 2, 21 January 2026. [Online]. Available: https://arxiv.org/html/2510.06605v2

[88] J. Leshin, M. Shah, I. Timmis, and D. Kang, "Behavioral fingerprints for LLM endpoint stability and identity," 2026, version 1, 19 March 2026. [Online]. Available: https://arxiv.org/html/2603.19022v1

[89] T. Zhu, Y. Wu, W. Jin, Y. Li, and G. Huang, "StepFinder: A temporal semantic framework for failure attribution in multi-agent systems," 2026, version 1, 2 June 2026. [Online]. Available: https://arxiv.org/html/2606.03467v1

[90] O. Kotevska, S. Jha, A. Bellet, R. Hu, N. D. Bastian, R. F. da Silva, R. Madduri, and K. Kim, "Privacy foundations for multi-institutional scientific artificial intelligence," *arXiv preprint arXiv:2609.39787v1*, 2026, consulted 2609.39787v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2609.39787v1

[91] J. W. Kim, B. Min, K. Edemacu, Y. Choi, S.-H. Cho, and B. Jang, "DP-MemView: A memory interface for attribute-level transcript privacy in long-term LLM agents," *arXiv preprint arXiv:2608.03130v1*, 2026, consulted 2608.03130v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2608.03130v1

[92] Y.-M. Wang, Y.-X. Li, S.-L. Fan, Y.-H. Zheng, and T.-H. Chang, "Ask without telling: Local slms consult cloud LLMs without revealing task intent," *arXiv preprint arXiv:2609.32642v1*, 2026, consulted 2609.32642v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2609.32642v1

[93] Z. Liu, S. van Hoek, P. Horváth, D. Lauret, X. Xu, and L. Batina, "Real-world edge neural network implementations leak private interactions through physical side channel," *arXiv preprint arXiv:2501.14512v1*, 2025, consulted 2501.14512v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2501.14512v1

[94] P. G. Pennas, K. Papaioannou, M. Guarnieri, and T. D. Doudali, "CacheSolidarity: Preventing prefix caching side channels in multi-tenant LLM serving systems," *arXiv preprint arXiv:2603.10726v1*, 2026, original map inspected v1 on 21 October 2026; defense-first extension also inspected v2, retitled PrefixWall, on 27 October 2026. [Online]. Available: https://arxiv.org/html/2603.10726v1

[95] C.-Y. Zhou, Q.-L. Jiang, S.-N. Wu, and X.-H. Zhou, "The vision encoder as a privacy boundary: Visual-token side channels in encoder-free vision-language models," *arXiv preprint arXiv:2606.14783v1*, 2026, consulted 2606.14783v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2606.14783v1

[96] G. Politis and E. Pappas, "What gradients add to text leakage in split language models, counted per token and per document," *arXiv preprint arXiv:2610.04128v1*, 2026, consulted 2610.04128v1; source-scoped access 21 October 2026. [Online]. Available: https://arxiv.org/html/2610.04128v1

[97] P. Sirinam, M. Imani, M. Juarez, and M. Wright, "Deep fingerprinting: Undermining website fingerprinting defenses with deep learning," in *ACM CCS*, 2018, pp. 1928–1943.

[98] J. Gong, W. Zhang, C. Zhang, and T. Wang, "WFDefProxy: Modularly implementing and empirically evaluating website fingerprinting defenses," arXiv:2111.12629v1, 2021. [Online]. Available: https://arxiv.org/abs/2111.12629v1

[99] N. Mathews, J. K. Holland, S. E. Oh, M. S. Rahman, N. Hopper, and M. Wright, "SoK: A critical evaluation of efficient website fingerprinting defenses," in *IEEE Symposium on Security and Privacy*, 2023.

[100] C. Dwork and A. Roth, "The algorithmic foundations of differential privacy," *Foundations and Trends in Theoretical Computer Science*, vol. 9, no. 3–4, pp. 211–407, 2014. [Online]. Available: https://www.cis.upenn.edu/~aaroth/Papers/privacybook.pdf

[101] G. Smith, "On the foundations of quantitative information flow," in *Foundations of Software Science and Computational Structures*, ser. LNCS, vol. 5504, 2009, pp. 288–302. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-642-00596-1_21

[102] J. B. Almeida, M. Barbosa, G. Barthe, F. Dupressoir, and M. Emmi, "Verifying Constant-Time implementations," in *25th USENIX Security Symposium*, 2016, pp. 53–70. [Online]. Available: https://www.usenix.org/conference/usenixsecurity16/technical-sessions/presentation/almeida

[103] E. Stefanov, M. van Dijk, E. Shi, C. Fletcher, L. Ren, X. Yu, and S. Devadas, "Path ORAM: An extremely simple oblivious RAM protocol," in *ACM SIGSAC Conference on Computer and Communications Security*, 2013, pp. 299–310. [Online]. Available: https://doi.org/10.1145/2508859.2516660

[104] M. Juarez, S. Afroz, G. Acar, C. Diaz, and R. Greenstadt, "A critical evaluation of website fingerprinting attacks," in *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, 2014, pp. 263–274.

[105] F. Tramèr, N. Carlini, W. Brendel, and A. Madry, "On adaptive attacks to adversarial example defenses," in *Advances in Neural Information Processing Systems*, vol. 33, 2020. [Online]. Available: https://arxiv.org/abs/2002.08347v2

[106] N. Carlini, A. Athalye, N. Papernot, W. Brendel, J. Rauber, D. Tsipras, I. Goodfellow, A. Madry, and A. Kurakin, "On evaluating adversarial robustness," *arXiv preprint arXiv:1902.06705*, 2019. [Online]. Available: https://arxiv.org/abs/1902.06705v2

[107] J. Xie and S. Li, "OCELOT: Inference-leakage budgets for privacy-preserving LLM agents," *arXiv preprint arXiv:2606.12341v1*, 2026, consulted primary v1 on 27 October 2026. [Online]. Available: https://arxiv.org/abs/2606.12341v1

[108] E. Bagdasarian, R. Yi, S. Ghalebikesabi, P. Kairouz, M. Gruteser, S. Oh, B. Balle, and D. Ramage, "AirGapAgent: Protecting privacy-conscious conversational agents," *arXiv preprint arXiv:2405.05175v2*, 2024, consulted primary v2; arXiv metadata identifies CCS 2024; exact publisher version not compared. [Online]. Available: https://arxiv.org/abs/2405.05175v2